Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-11 13:17:45
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0005
min_learning_rate...................... 1e-09
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 10
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_64x128/best_model.pt
input_data_path........................ data/processed_64x128/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_64x128/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_64x128/std_params.nc
  trend_params_path..... data/processed_64x128/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

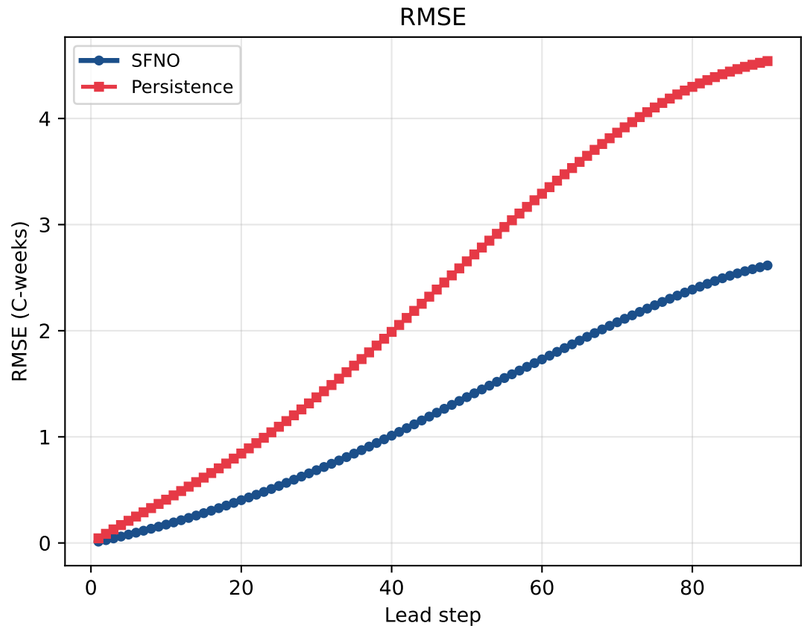
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

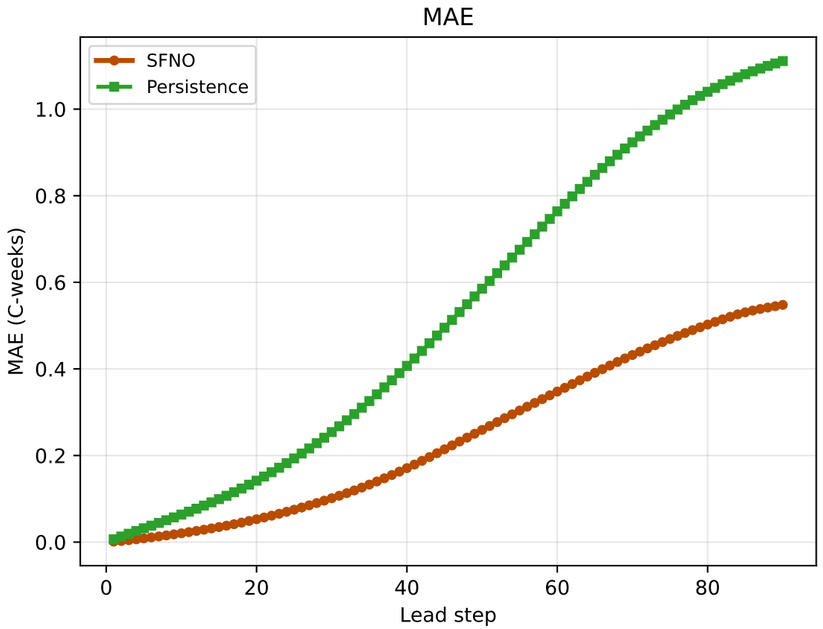
DATASETS
  Test          19 chains x 90 leads x 64 lat x 128 lon
  Val           16 chains x 90 leads x 64 lat x 128 lon
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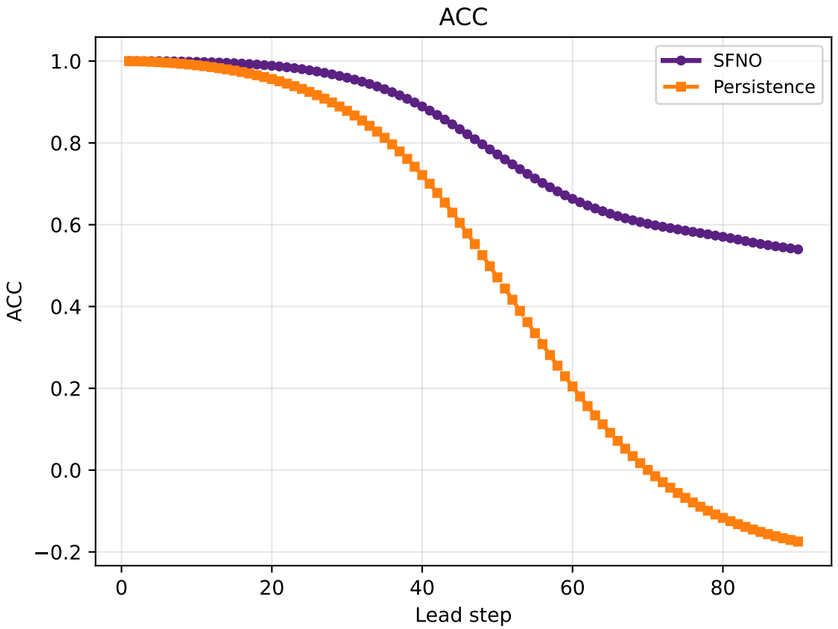
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

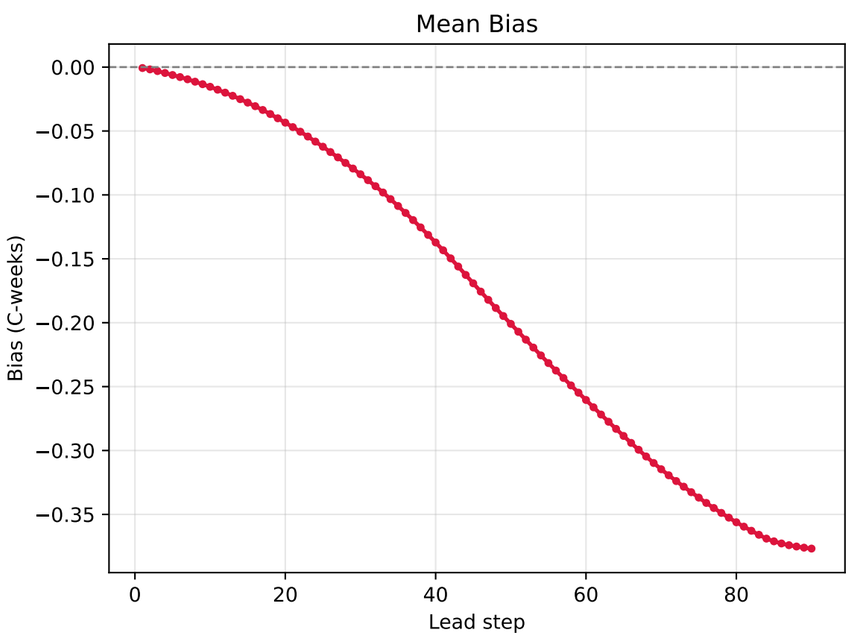
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0127	0.0437	70.9%	0.0011	0.0065	-0.0007	1.0000	0.9999	1.0000
2	0.0268	0.0862	69.0%	0.0027	0.0130	-0.0018	1.0000	0.9996	0.9999
3	0.0431	0.1278	66.2%	0.0045	0.0193	-0.0031	0.9999	0.9990	0.9998
4	0.0607	0.1687	64.0%	0.0065	0.0257	-0.0046	0.9998	0.9983	0.9996
5	0.0789	0.2091	62.3%	0.0087	0.0319	-0.0062	0.9996	0.9973	0.9994
6	0.0971	0.2491	61.0%	0.0109	0.0381	-0.0078	0.9994	0.9962	0.9990
7	0.1154	0.2888	60.0%	0.0131	0.0443	-0.0095	0.9992	0.9949	0.9986
8	0.1341	0.3285	59.2%	0.0155	0.0507	-0.0114	0.9989	0.9934	0.9981
9	0.1534	0.3680	58.3%	0.0179	0.0571	-0.0134	0.9986	0.9917	0.9976
10	0.1731	0.4079	57.6%	0.0204	0.0635	-0.0154	0.9981	0.9898	0.9969
11	0.1936	0.4484	56.8%	0.0231	0.0702	-0.0177	0.9977	0.9877	0.9961
12	0.2144	0.4893	56.2%	0.0258	0.0772	-0.0200	0.9971	0.9853	0.9951
13	0.2360	0.5310	55.6%	0.0286	0.0844	-0.0225	0.9965	0.9827	0.9940
14	0.2584	0.5733	54.9%	0.0316	0.0918	-0.0251	0.9957	0.9798	0.9928
15	0.2810	0.6158	54.4%	0.0348	0.0993	-0.0278	0.9949	0.9767	0.9914
16	0.3043	0.6590	53.8%	0.0380	0.1071	-0.0306	0.9939	0.9733	0.9898
17	0.3281	0.7031	53.3%	0.0414	0.1152	-0.0335	0.9928	0.9695	0.9880
18	0.3526	0.7484	52.9%	0.0450	0.1237	-0.0367	0.9915	0.9654	0.9859
19	0.3778	0.7951	52.5%	0.0489	0.1326	-0.0401	0.9901	0.9609	0.9836
20	0.4034	0.8429	52.1%	0.0528	0.1419	-0.0435	0.9885	0.9560	0.9811
21	0.4288	0.8913	51.9%	0.0569	0.1515	-0.0470	0.9868	0.9507	0.9783
22	0.4546	0.9406	51.7%	0.0611	0.1614	-0.0505	0.9849	0.9450	0.9753
23	0.4812	0.9912	51.5%	0.0655	0.1717	-0.0543	0.9827	0.9387	0.9719
24	0.5088	1.0427	51.2%	0.0701	0.1824	-0.0583	0.9803	0.9319	0.9681
25	0.5372	1.0951	50.9%	0.0749	0.1933	-0.0623	0.9776	0.9246	0.9639
26	0.5666	1.1486	50.7%	0.0799	0.2046	-0.0665	0.9746	0.9167	0.9592
27	0.5962	1.2029	50.4%	0.0850	0.2163	-0.0706	0.9713	0.9081	0.9541
28	0.6257	1.2585	50.3%	0.0903	0.2285	-0.0750	0.9676	0.8989	0.9485
29	0.6558	1.3152	50.1%	0.0957	0.2411	-0.0794	0.9637	0.8889	0.9425
30	0.6858	1.3721	50.0%	0.1013	0.2541	-0.0838	0.9594	0.8783	0.9361
31	0.7159	1.4298	49.9%	0.1071	0.2675	-0.0885	0.9547	0.8669	0.9292
32	0.7461	1.4883	49.9%	0.1131	0.2813	-0.0932	0.9497	0.8547	0.9218
33	0.7771	1.5482	49.8%	0.1193	0.2956	-0.0981	0.9441	0.8416	0.9137
34	0.8092	1.6090	49.7%	0.1260	0.3104	-0.1033	0.9380	0.8274	0.9050
35	0.8420	1.6707	49.6%	0.1329	0.3257	-0.1087	0.9313	0.8123	0.8955
36	0.8749	1.7328	49.5%	0.1400	0.3414	-0.1141	0.9240	0.7963	0.8854
37	0.9083	1.7960	49.4%	0.1474	0.3573	-0.1197	0.9162	0.7792	0.8746
38	0.9421	1.8599	49.3%	0.1550	0.3737	-0.1254	0.9077	0.7609	0.8632
39	0.9766	1.9244	49.3%	0.1628	0.3902	-0.1313	0.8987	0.7417	0.8510
40	1.0112	1.9890	49.2%	0.1708	0.4070	-0.1372	0.8891	0.7214	0.8383
41	1.0462	2.0544	49.1%	0.1791	0.4241	-0.1433	0.8789	0.7001	0.8251
42	1.0816	2.1205	49.0%	0.1877	0.4416	-0.1497	0.8682	0.6776	0.8114
43	1.1177	2.1870	48.9%	0.1964	0.4592	-0.1560	0.8570	0.6541	0.7971
44	1.1544	2.2540	48.8%	0.2053	0.4771	-0.1626	0.8453	0.6297	0.7826
45	1.1910	2.3209	48.7%	0.2143	0.4951	-0.1692	0.8334	0.6045	0.7679
46	1.2276	2.3878	48.6%	0.2233	0.5132	-0.1757	0.8212	0.5787	0.7532
47	1.2644	2.4546	48.5%	0.2323	0.5313	-0.1821	0.8089	0.5524	0.7384
48	1.3012	2.5213	48.4%	0.2413	0.5494	-0.1885	0.7965	0.5255	0.7236
49	1.3378	2.5876	48.3%	0.2503	0.5673	-0.1948	0.7841	0.4985	0.7090
50	1.3742	2.6535	48.2%	0.2592	0.5852	-0.2009	0.7718	0.4712	0.6946
51	1.4102	2.7188	48.1%	0.2681	0.6031	-0.2071	0.7596	0.4439	0.6806
52	1.4466	2.7840	48.0%	0.2772	0.6211	-0.2133	0.7476	0.4166	0.6667
53	1.4827	2.8487	48.0%	0.2862	0.6391	-0.2195	0.7357	0.3892	0.6531
54	1.5185	2.9131	47.9%	0.2951	0.6572	-0.2256	0.7241	0.3620	0.6399
55	1.5543	2.9772	47.8%	0.3040	0.6752	-0.2316	0.7128	0.3348	0.6271
56	1.5895	3.0409	47.7%	0.3127	0.6931	-0.2374	0.7020	0.3079	0.6148
57	1.6246	3.1042	47.7%	0.3214	0.7110	-0.2432	0.6915	0.2813	0.6030
58	1.6598	3.1668	47.6%	0.3302	0.7287	-0.2490	0.6815	0.2552	0.5915
59	1.6949	3.2292	47.5%	0.3389	0.7463	-0.2548	0.6720	0.2296	0.5805
60	1.7299	3.2912	47.4%	0.3476	0.7638	-0.2605	0.6631	0.2045	0.5701
61	1.7652	3.3528	47.4%	0.3564	0.7813	-0.2662	0.6548	0.1801	0.5603
62	1.8007	3.4139	47.3%	0.3651	0.7985	-0.2718	0.6470	0.1565	0.5509
63	1.8363	3.4741	47.1%	0.3737	0.8155	-0.2775	0.6397	0.1338	0.5420
64	1.8716	3.5329	47.0%	0.3823	0.8320	-0.2831	0.6330	0.1121	0.5336
65	1.9069	3.5910	46.9%	0.3909	0.8483	-0.2886	0.6268	0.0913	0.5257
66	1.9423	3.6481	46.8%	0.3994	0.8641	-0.2940	0.6211	0.0715	0.5183
67	1.9774	3.7045	46.6%	0.4078	0.8795	-0.2994	0.6157	0.0524	0.5112
68	2.0122	3.7596	46.5%	0.4160	0.8945	-0.3047	0.6109	0.0343	0.5046
69	2.0462	3.8136	46.3%	0.4242	0.9091	-0.3098	0.6064	0.0169	0.4984
70	2.0797	3.8659	46.2%	0.4321	0.9233	-0.3147	0.6023	0.0005	0.4925
71	2.1127	3.9166	46.1%	0.4398	0.9370	-0.3194	0.5985	-0.0149	0.4870
72	2.1451	3.9656	45.9%	0.4474	0.9503	-0.3239	0.5949	-0.0294	0.4818
73	2.1771	4.0132	45.8%	0.4547	0.9631	-0.3283	0.5917	-0.0431	0.4768
74	2.2085	4.0591	45.6%	0.4619	0.9755	-0.3326	0.5885	-0.0558	0.4721
75	2.2400	4.1036	45.4%	0.4691	0.9875	-0.3368	0.5855	-0.0678	0.4674
76	2.2711	4.1466	45.2%	0.4761	0.9991	-0.3410	0.5825	-0.0790	0.4627
77	2.3014	4.1876	45.0%	0.4829	1.0101	-0.3450	0.5796	-0.0895	0.4582
78	2.3308	4.2264	44.9%	0.4896	1.0206	-0.3488	0.5765	-0.0993	0.4536
79	2.3596	4.2630	44.7%	0.4961	1.0305	-0.3525	0.5735	-0.1084	0.4489
80	2.3876	4.2978	44.4%	0.5025	1.0400	-0.3561	0.5704	-0.1168	0.4443
81	2.4149	4.3305	44.2%	0.5086	1.0491	-0.3596	0.5671	-0.1248	0.4395
82	2.4420	4.3612	44.0%	0.5145	1.0576	-0.3628	0.5637	-0.1321	0.4344
83	2.4686	4.3898	43.8%	0.5204	1.0657	-0.3659	0.5600	-0.1388	0.4291
84	2.4944	4.4167	43.5%	0.5261	1.0733	-0.3689	0.5562	-0.1452	0.4238
85	2.5181	4.4418	43.3%	0.5307	1.0805	-0.3711	0.5530	-0.1511	0.4191
86	2.5401	4.4649	43.1%	0.5347	1.0873	-0.3727	0.5501	-0.1565	0.4147
87	2.5608	4.4865	42.9%	0.5384	1.0937	-0.3741	0.5473	-0.1616	0.4105
88	2.5799	4.5062	42.7%	0.5417	1.0998	-0.3751	0.5447	-0.1664	0.4066
89	2.5979	4.5244	42.6%	0.5448	1.1055	-0.3760	0.5422	-0.1708	0.4028
90	2.6151	4.5408	42.4%	0.5479	1.1109	-0.3767	0.5396	-0.1748	0.3990

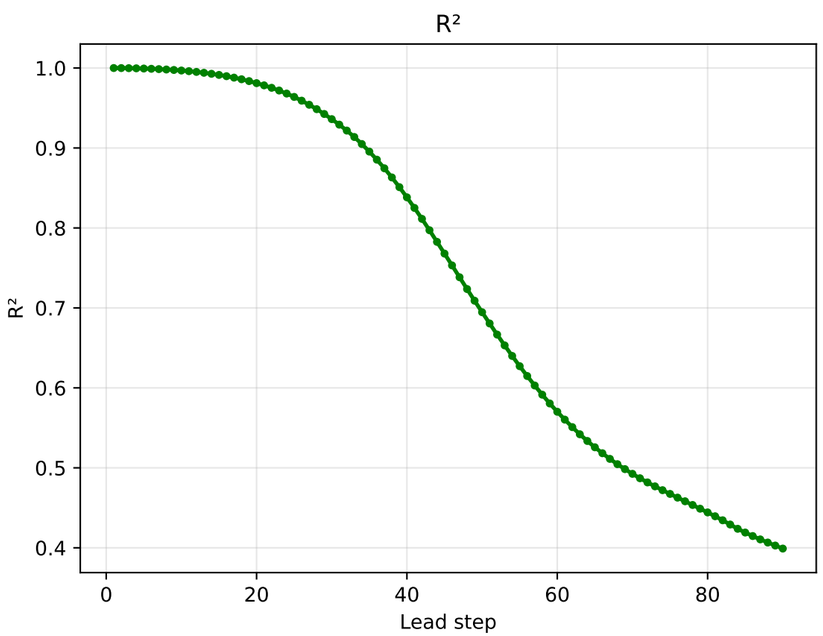
DHW Forecast Skill vs Lead Step - Test

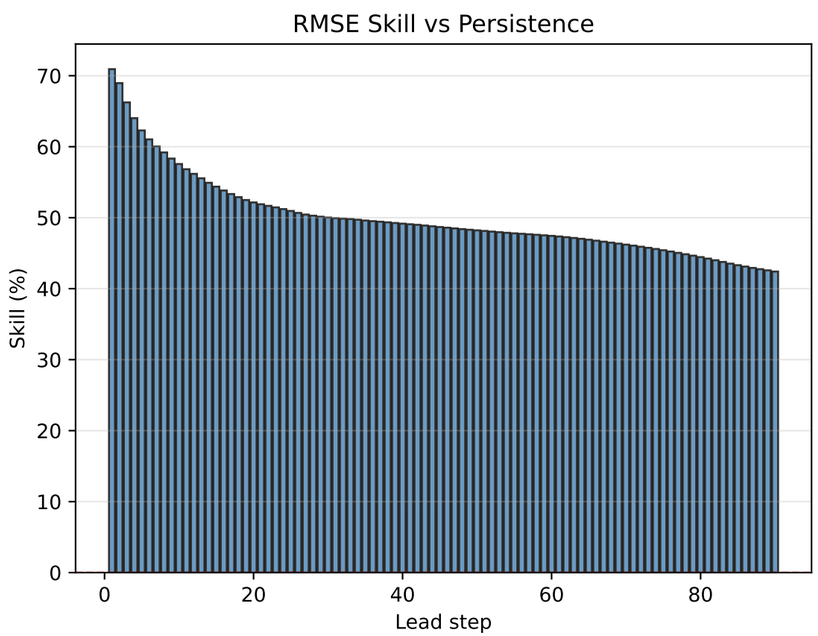


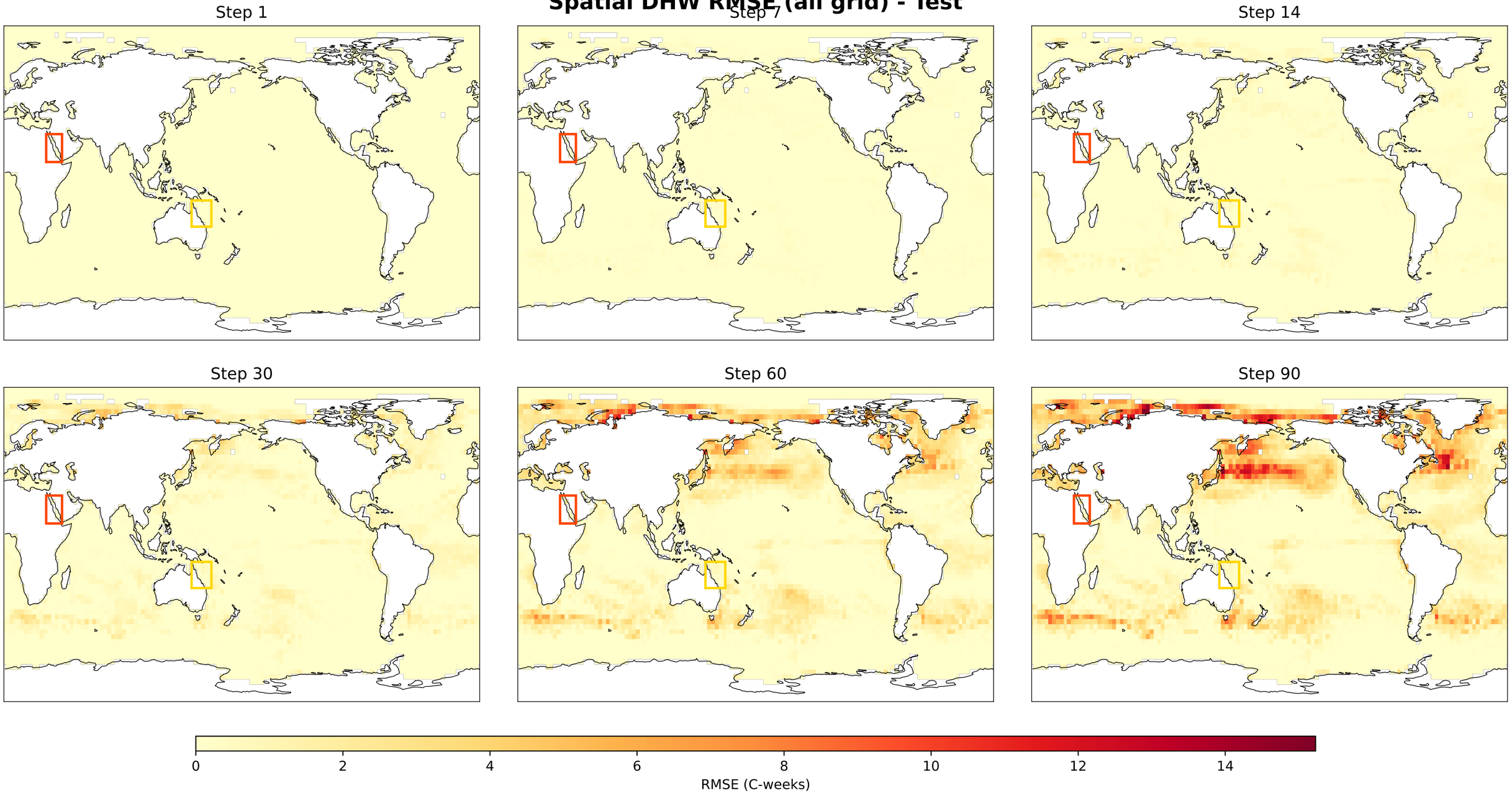


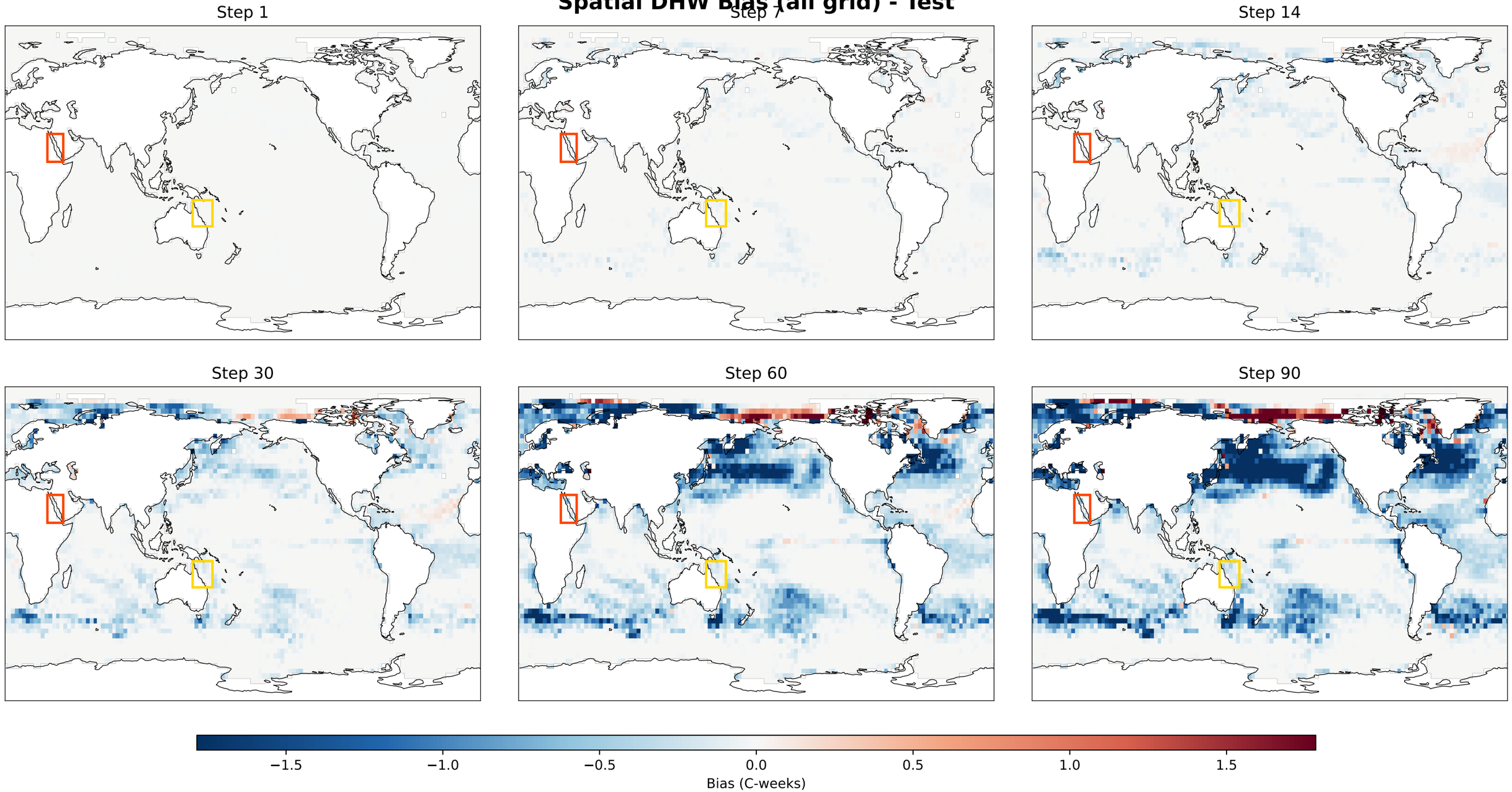


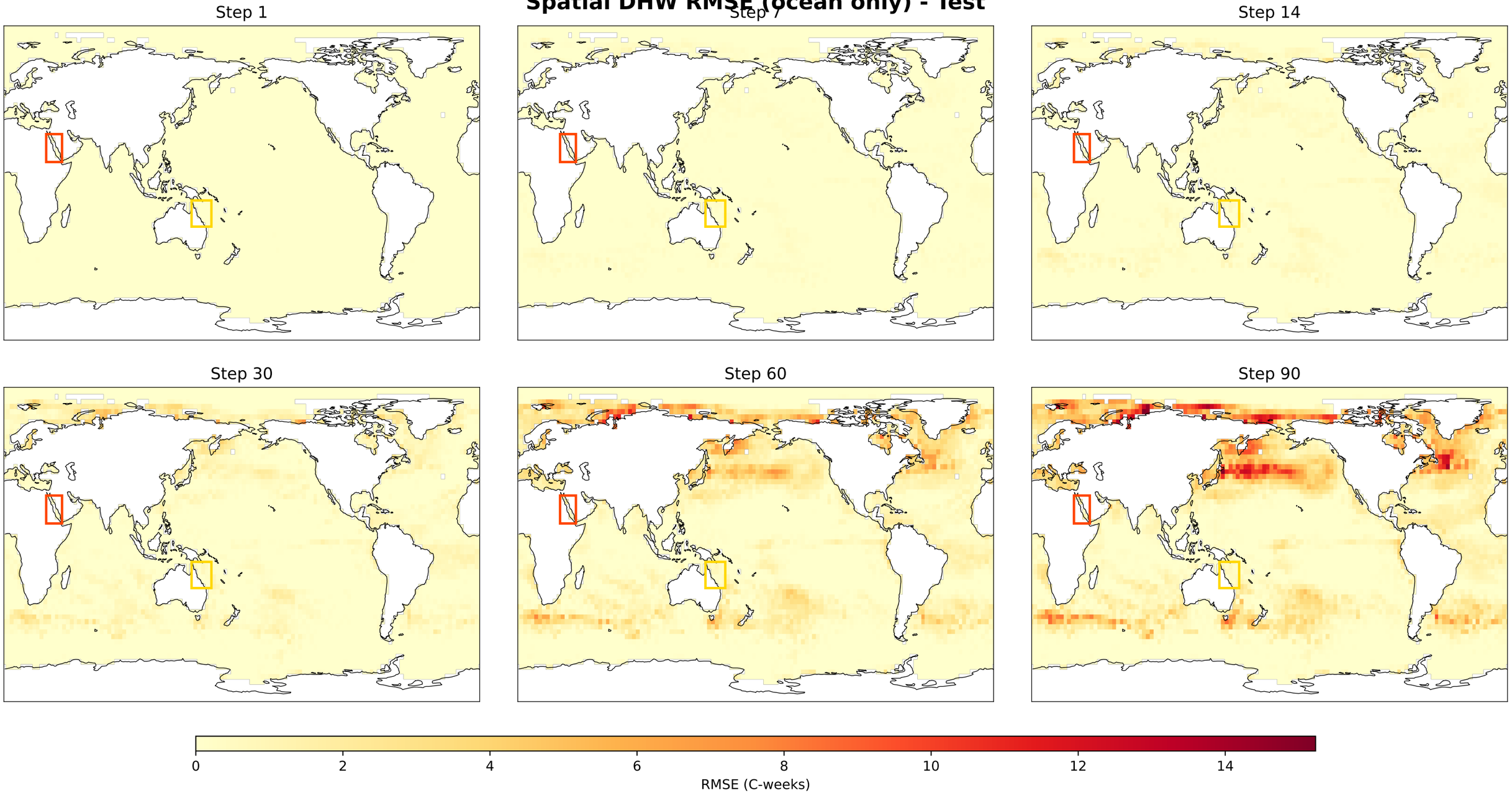


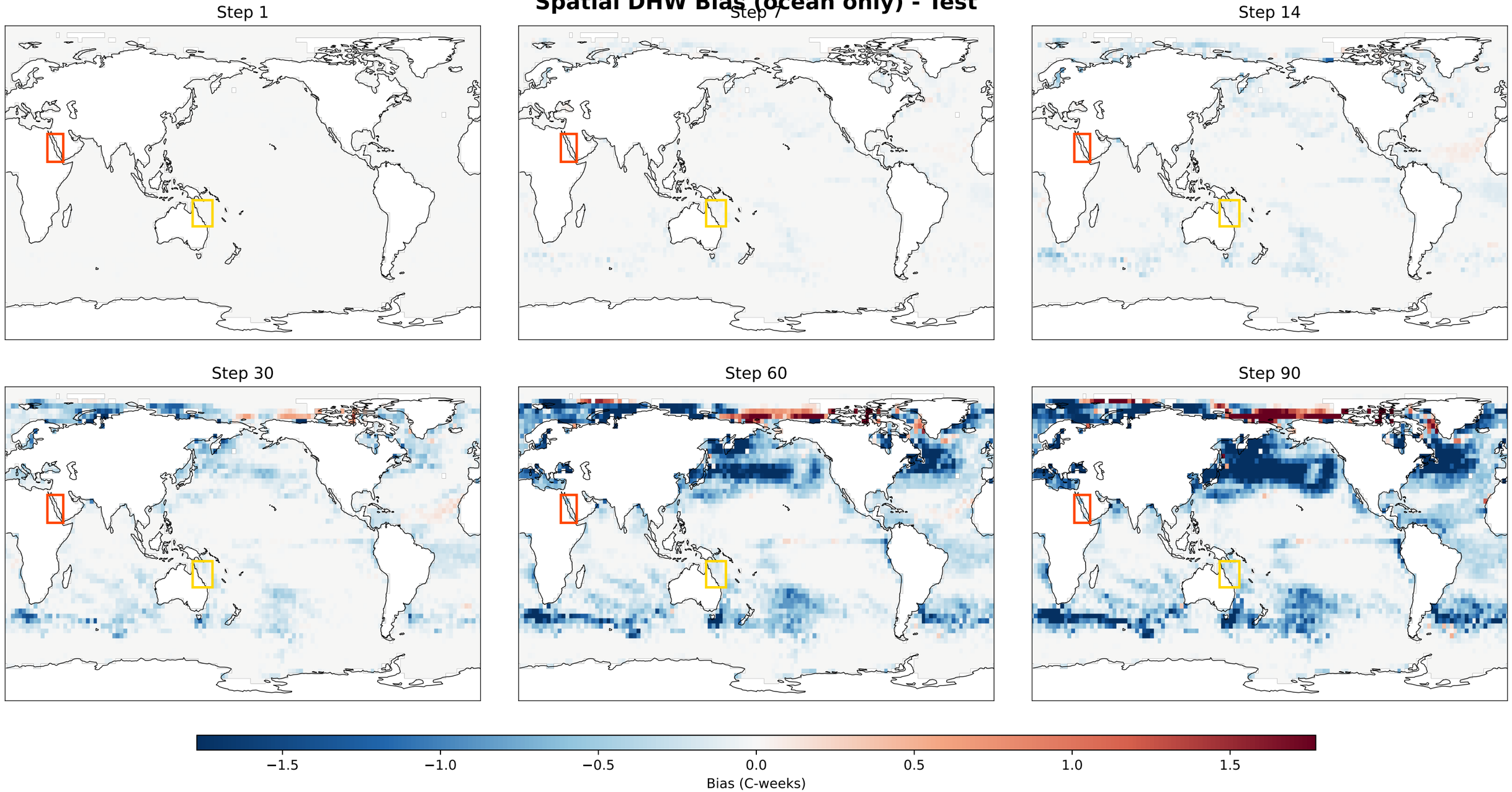





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
RMSE (C-weeks)

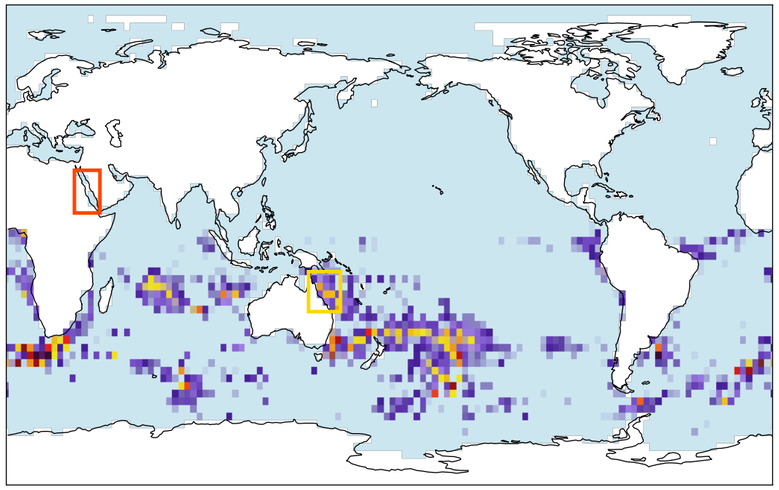

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
RMSE (C-weeks)

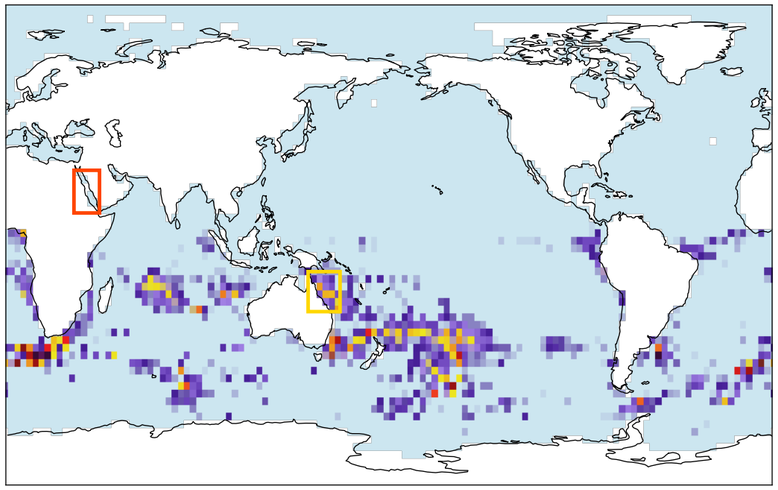

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

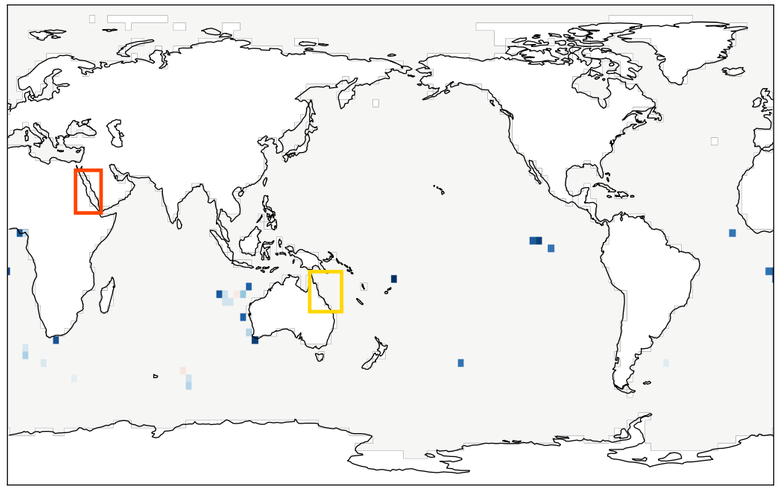
Actual DHW - Step 1



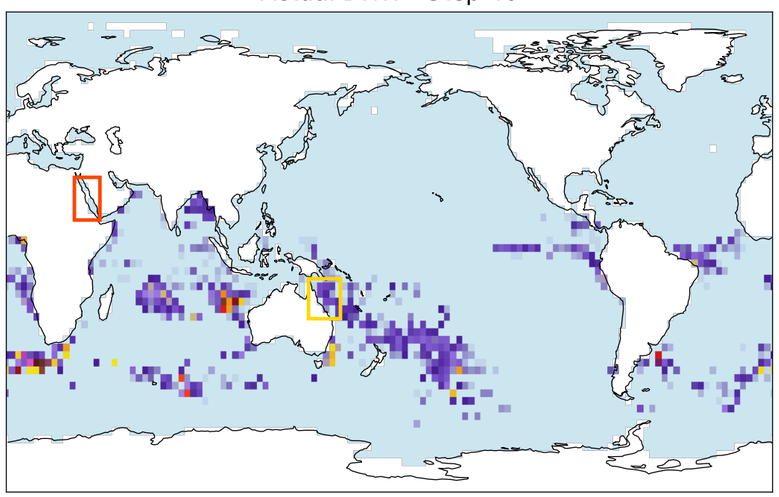
Predicted DHW - Step 1



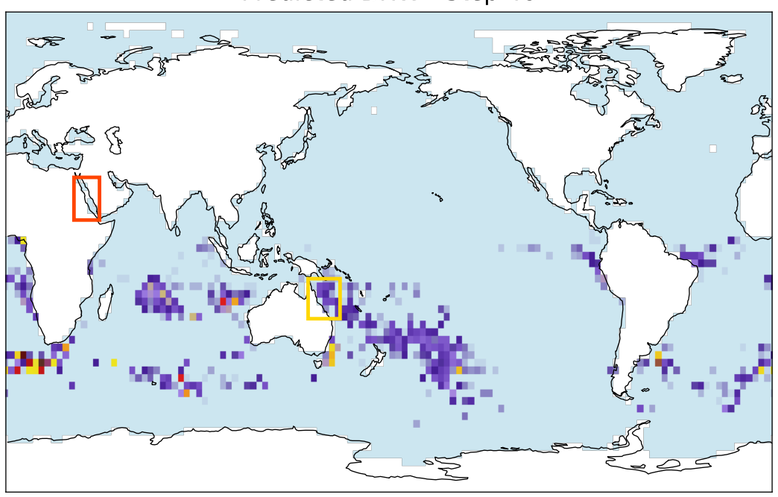
Error (Pred - Actual) - Step 1 [scale ±0.24]



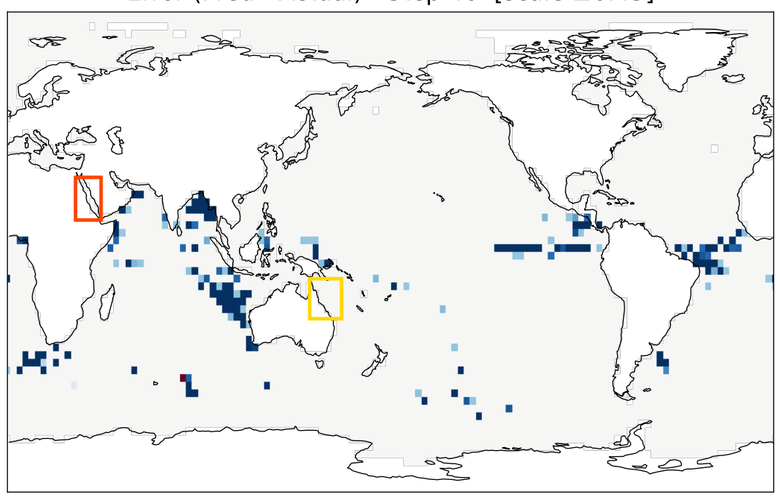
Actual DHW - Step 46



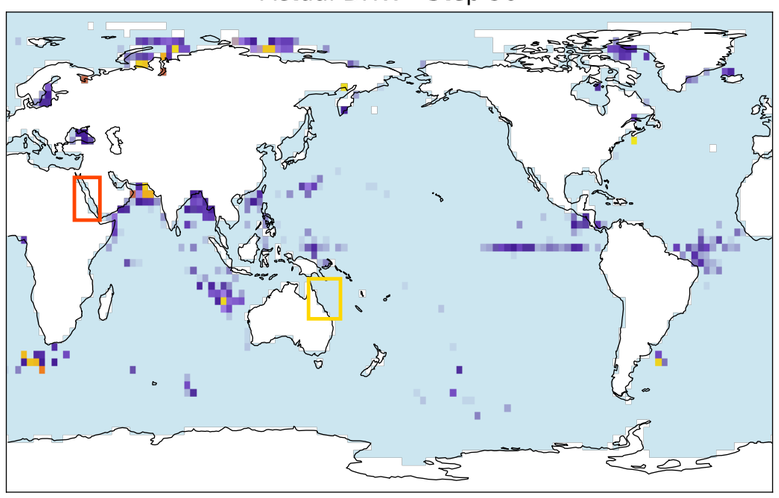
Predicted DHW - Step 46



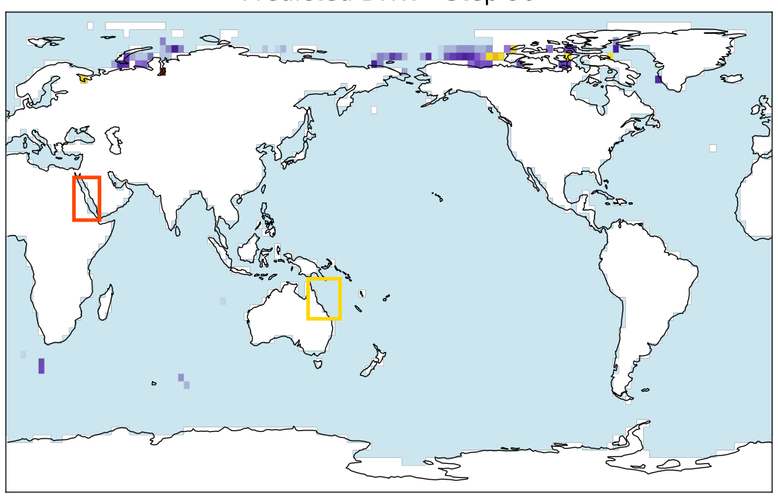
Error (Pred - Actual) - Step 46 [scale ±0.45]



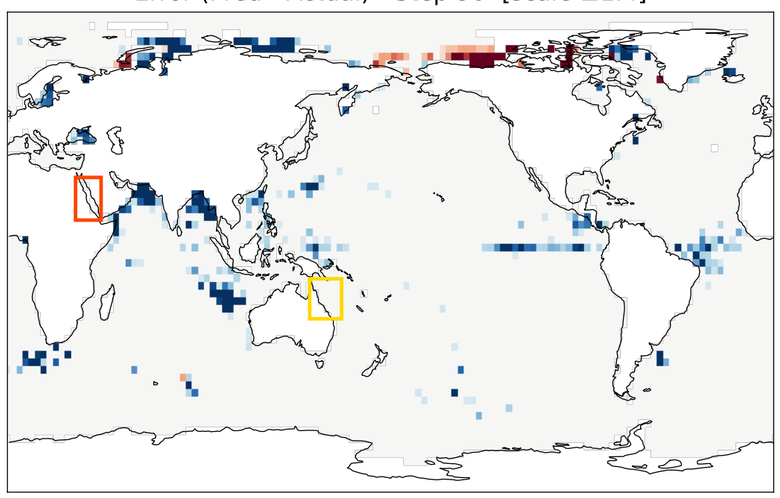
Actual DHW - Step 90

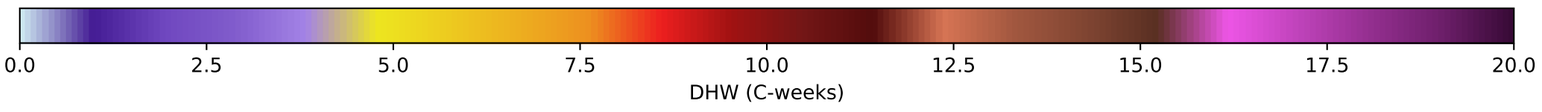


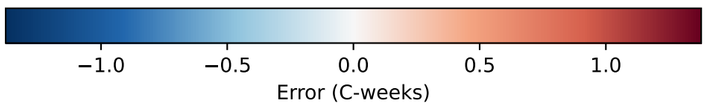
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.4]

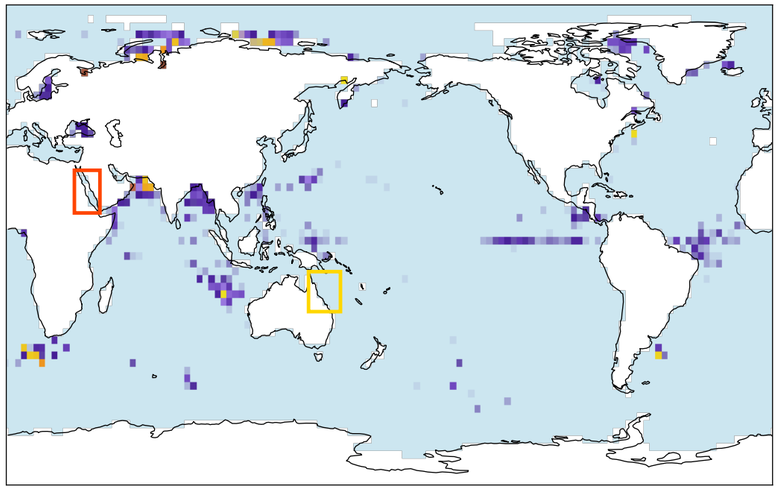




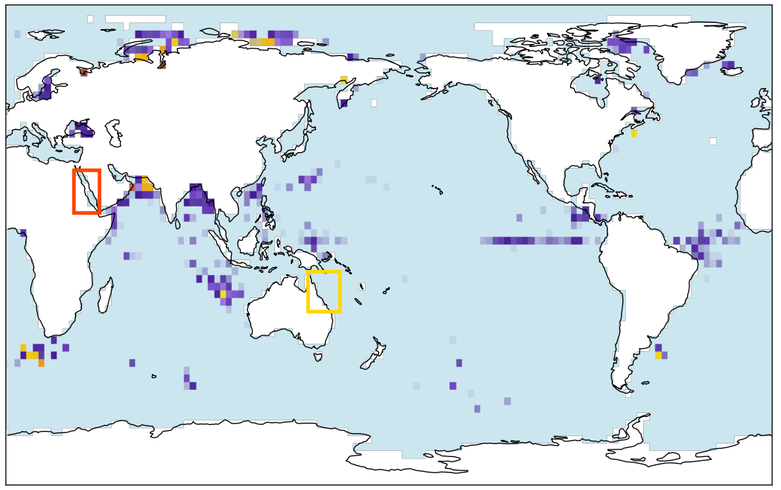


DHW Chain init: 2020-07-09 - Test (C-weeks)

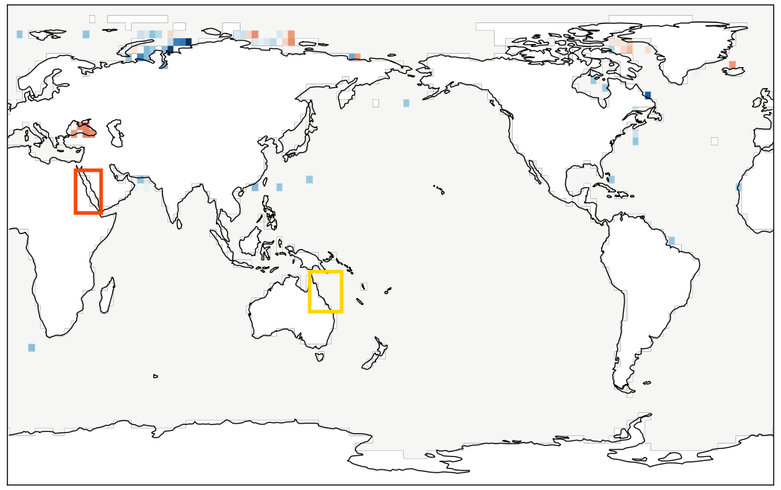
Actual DHW - Step 1



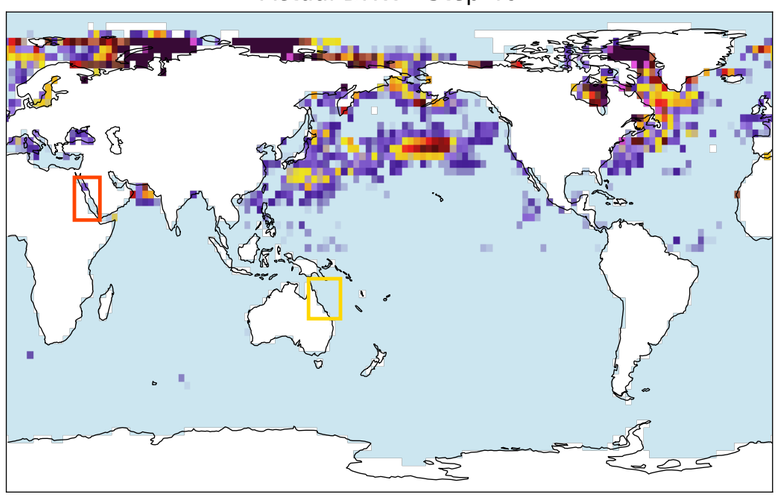
Predicted DHW - Step 1



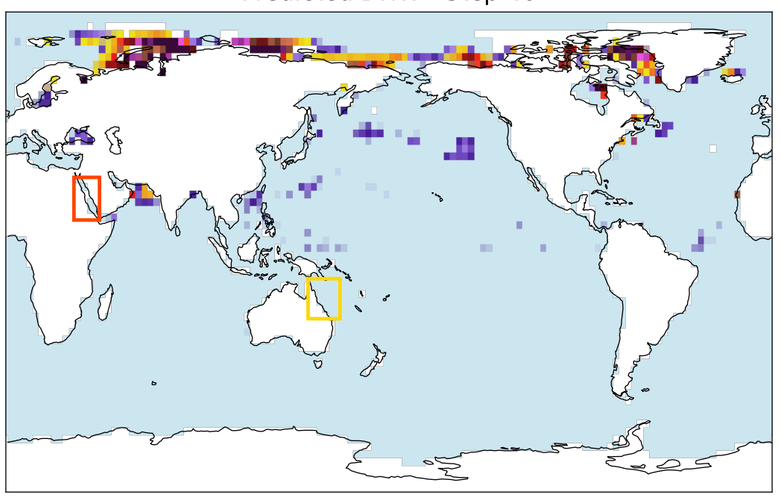
Error (Pred - Actual) - Step 1 [scale ±0.51]



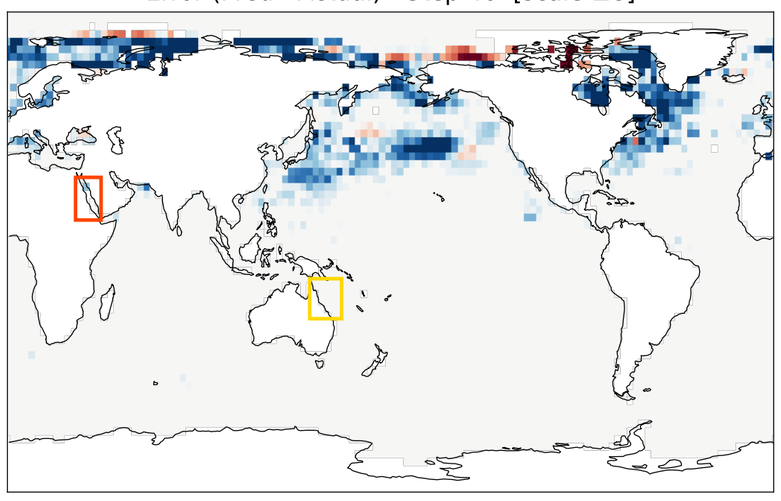
Actual DHW - Step 46



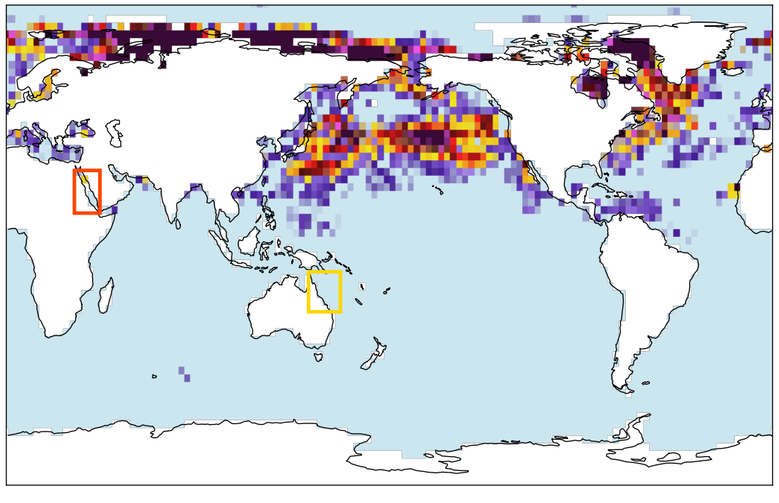
Predicted DHW - Step 46



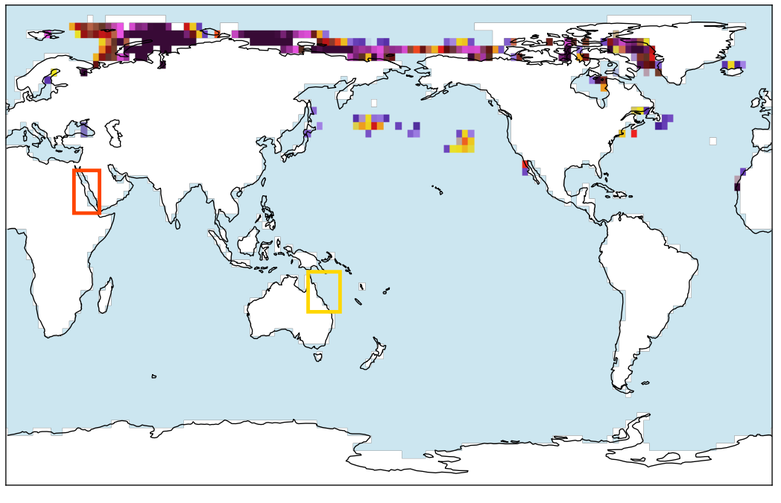
Error (Pred - Actual) - Step 46 [scale ±8]



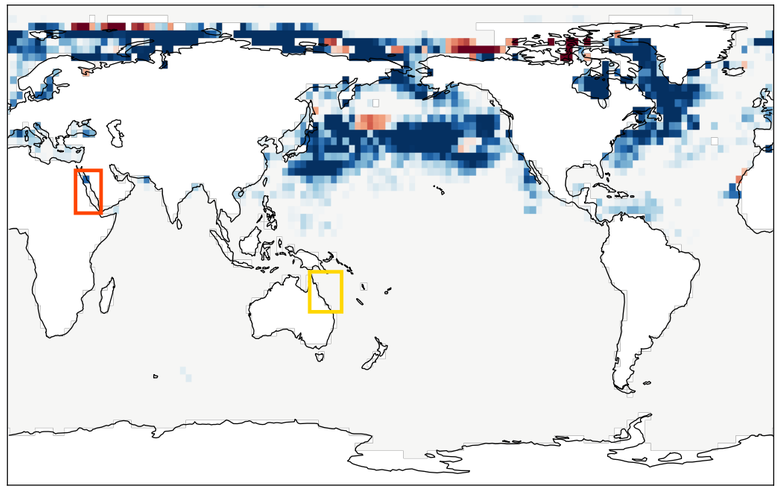
Actual DHW - Step 90

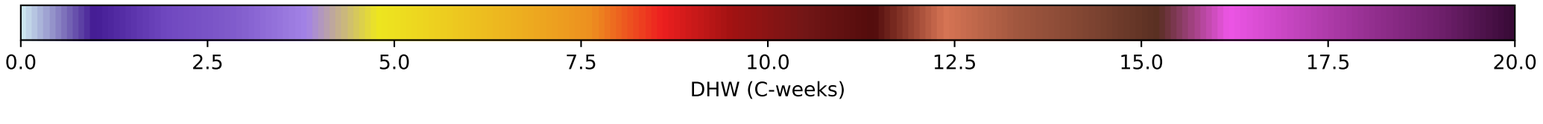


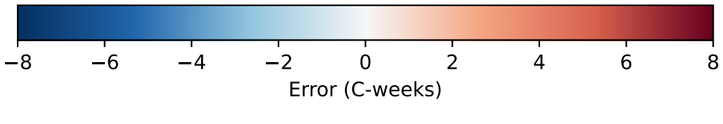
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

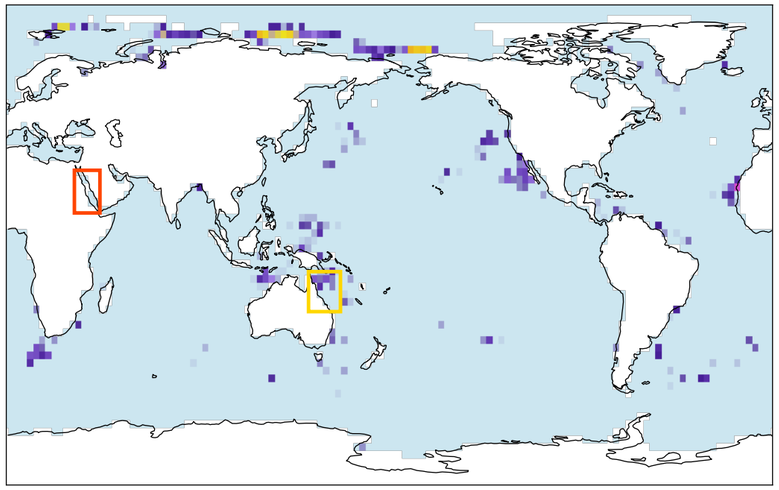




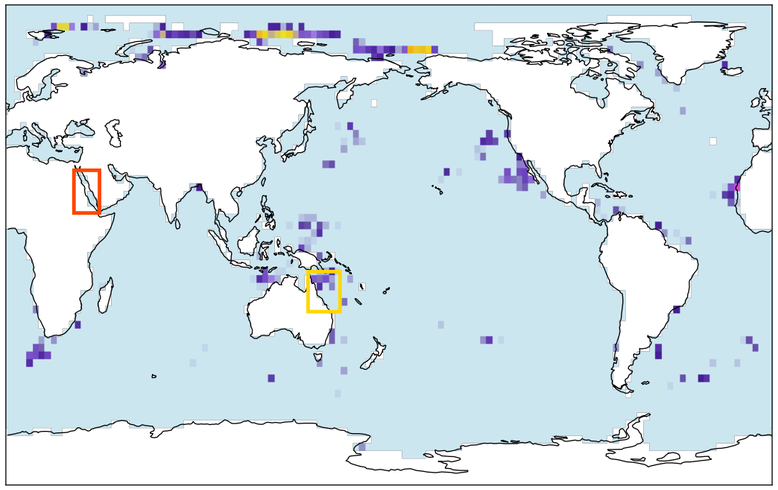


DHW Chain init: 2021-01-06 - Test (C-weeks)

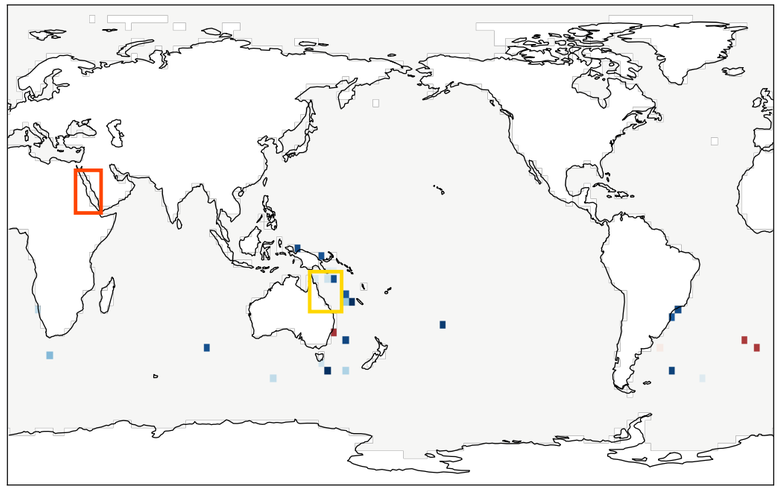
Actual DHW - Step 1



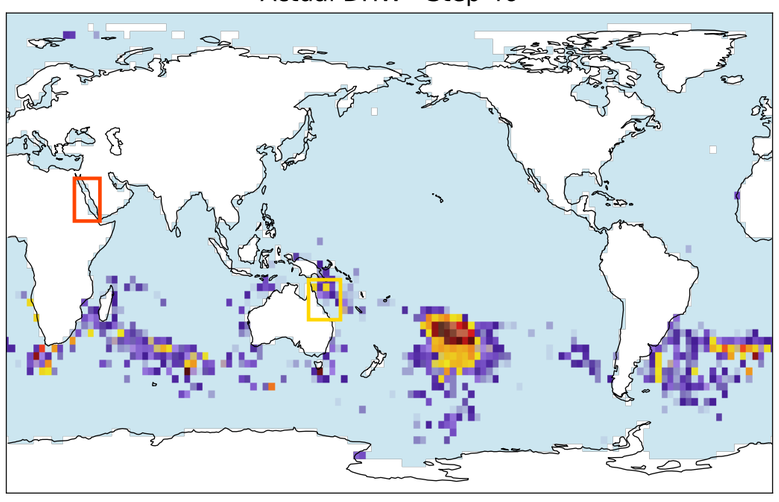
Predicted DHW - Step 1



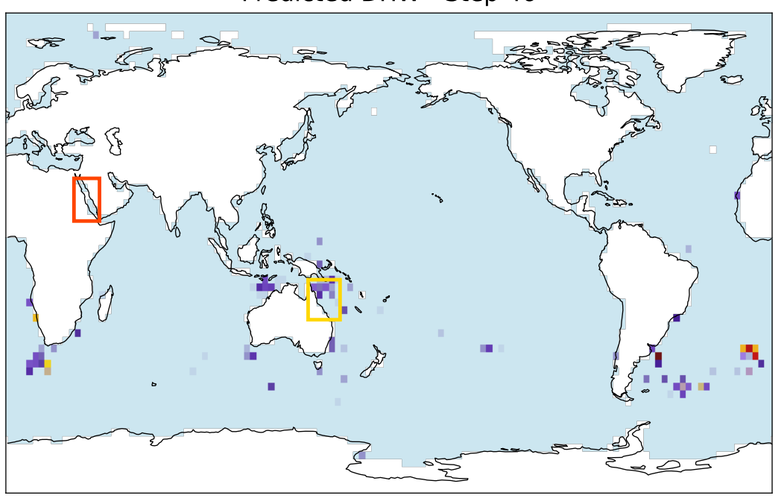
Error (Pred - Actual) - Step 1 [scale ±0.19]



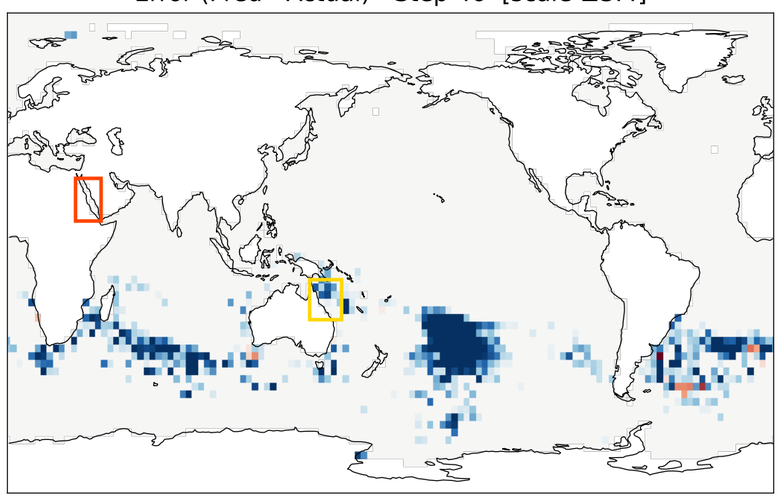
Actual DHW - Step 46



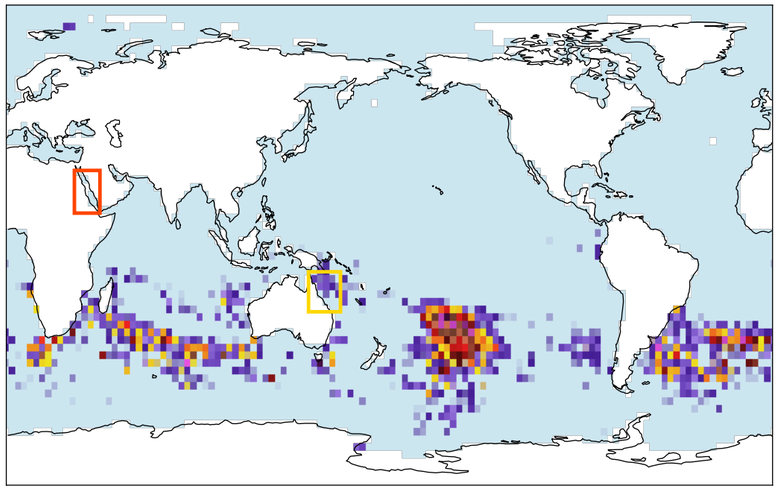
Predicted DHW - Step 46



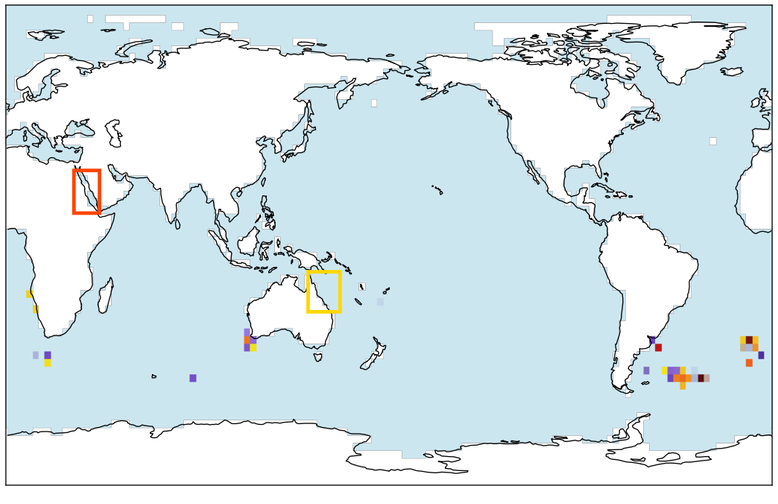
Error (Pred - Actual) - Step 46 [scale ±3.4]



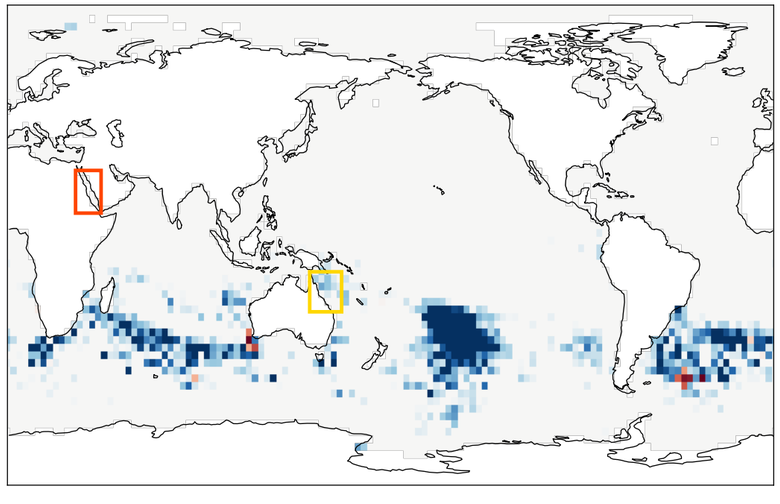
Actual DHW - Step 90

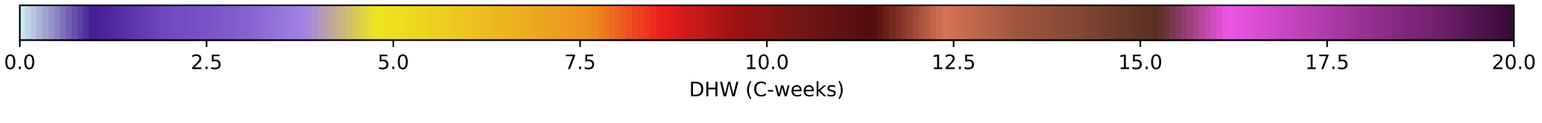


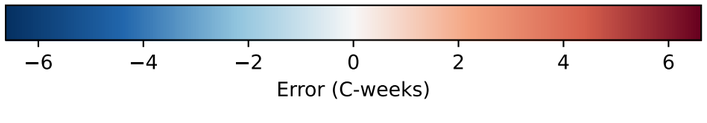
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

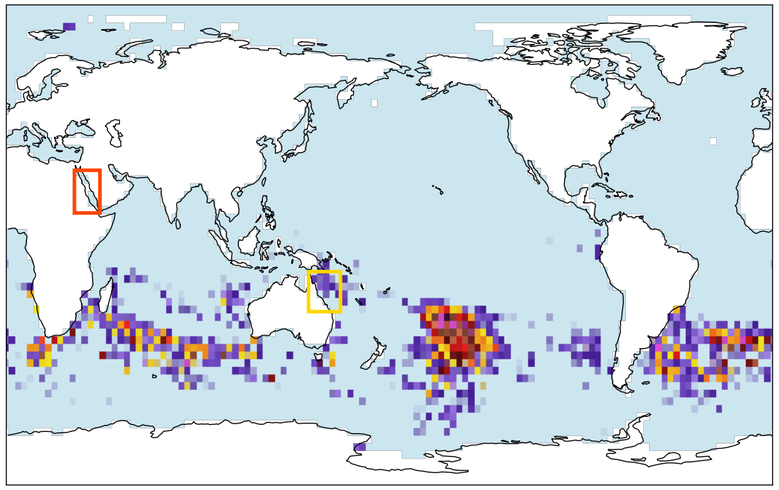




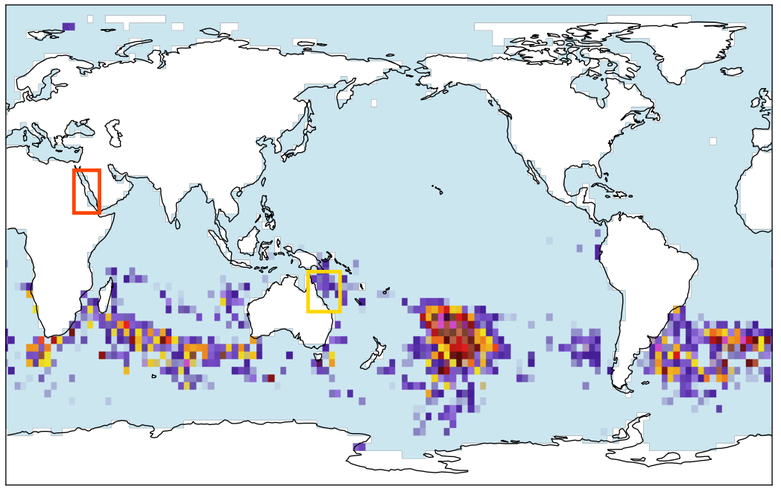


DHW Chain init: 2021-04-06 - Test (C-weeks)

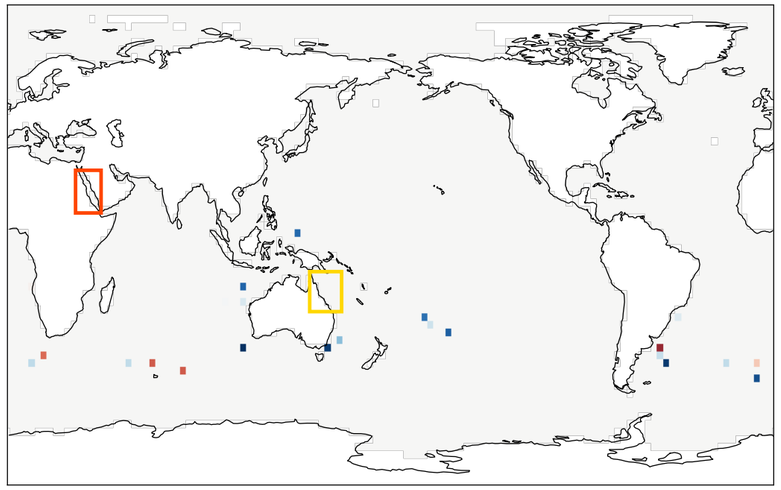
Actual DHW - Step 1



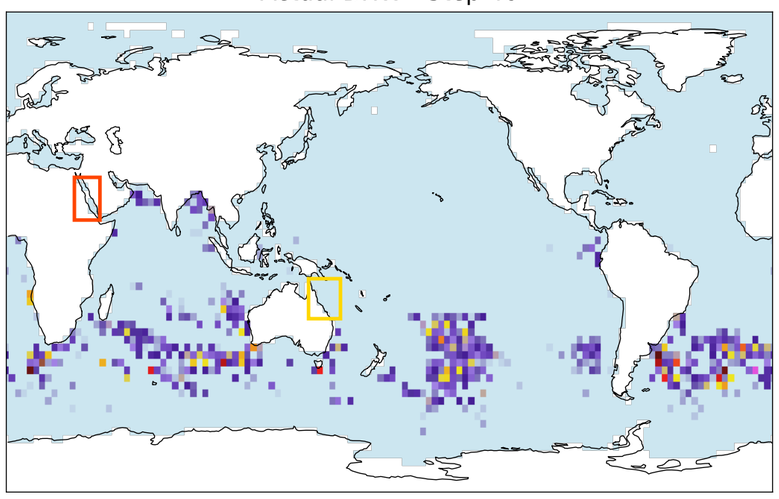
Predicted DHW - Step 1



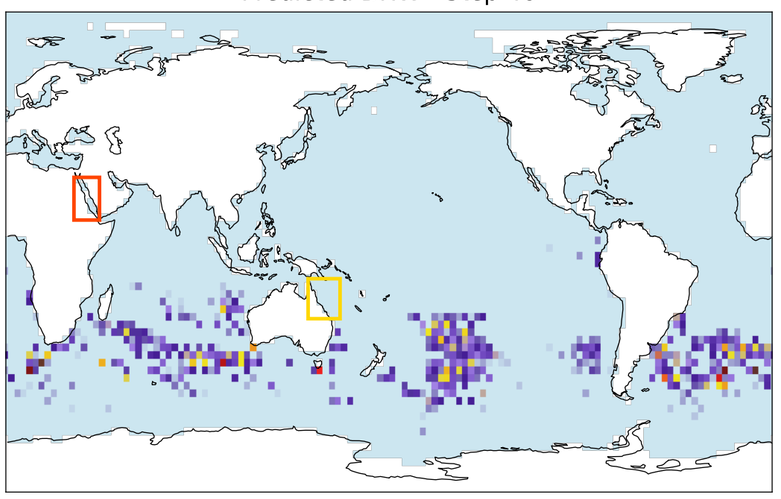
Error (Pred - Actual) - Step 1 [scale ±0.23]



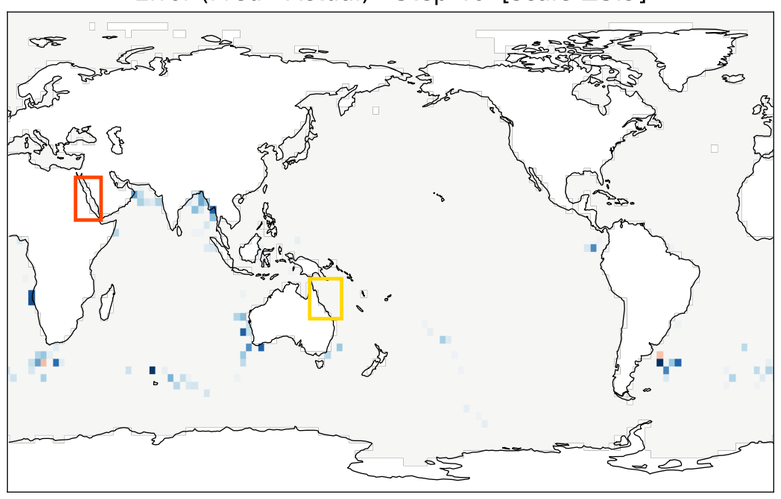
Actual DHW - Step 46



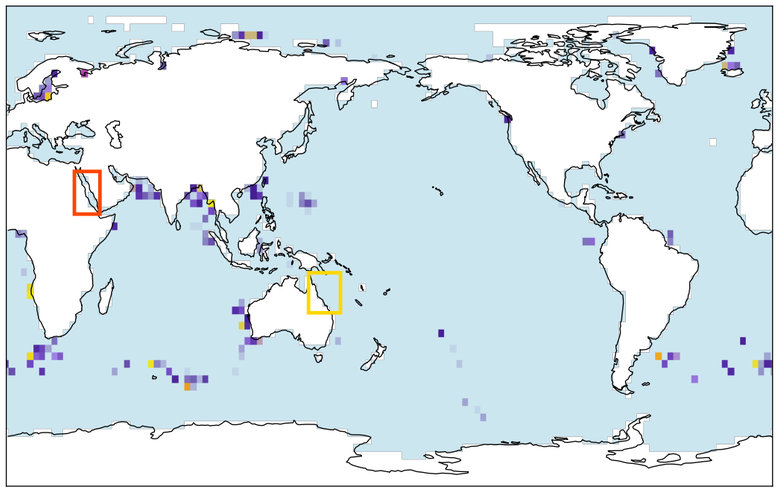
Predicted DHW - Step 46



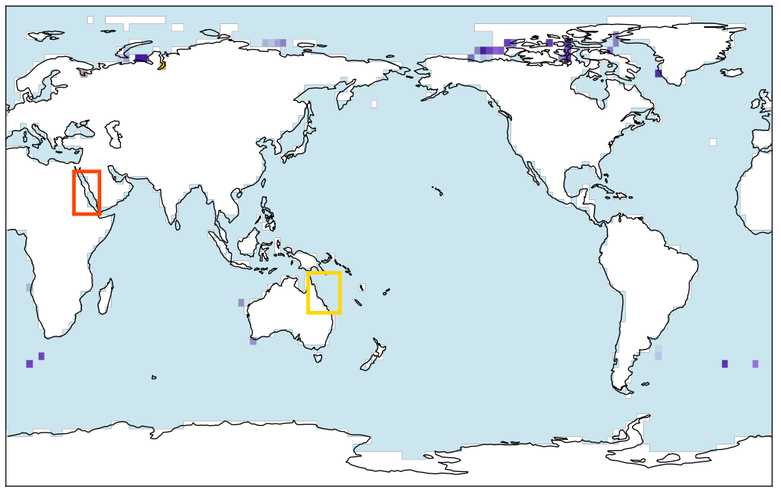
Error (Pred - Actual) - Step 46 [scale ±5.9]



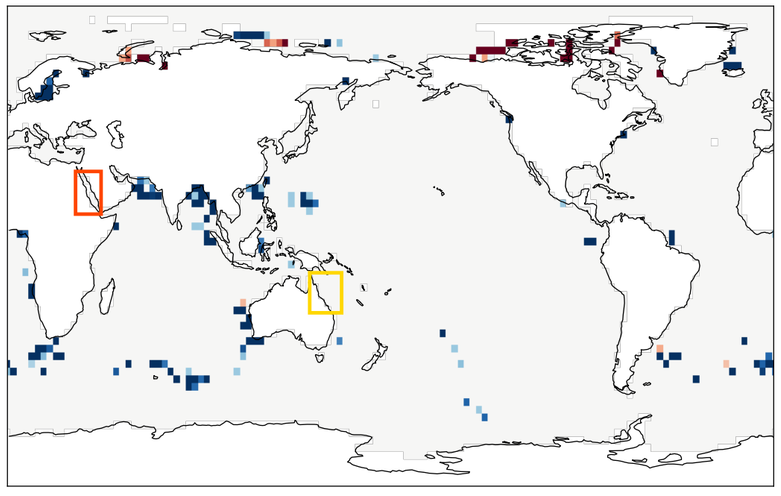
Actual DHW - Step 90

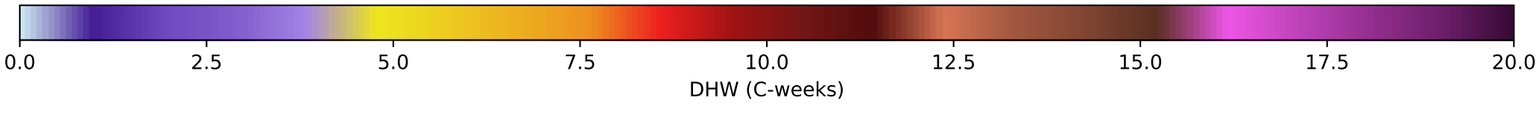


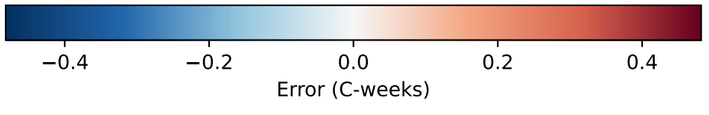
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.48]

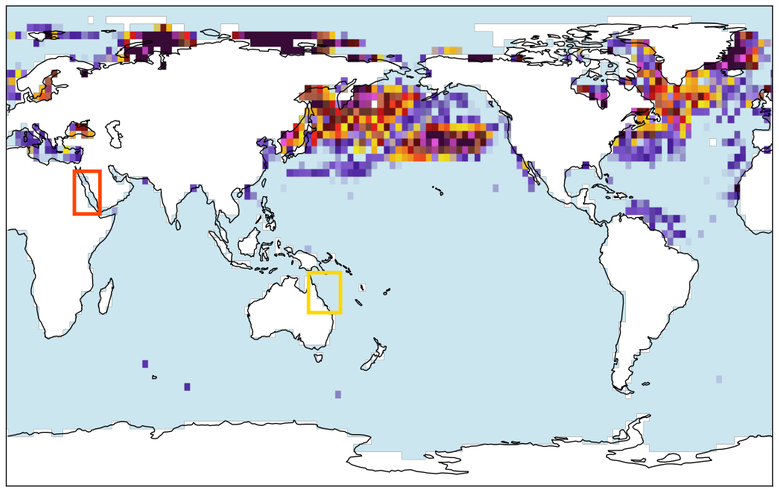




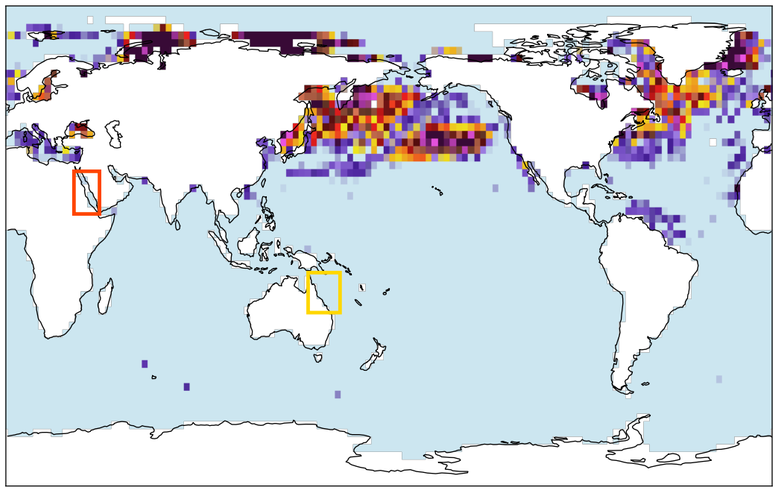


DHW Chain init: 2021-10-03 - Test (C-weeks)

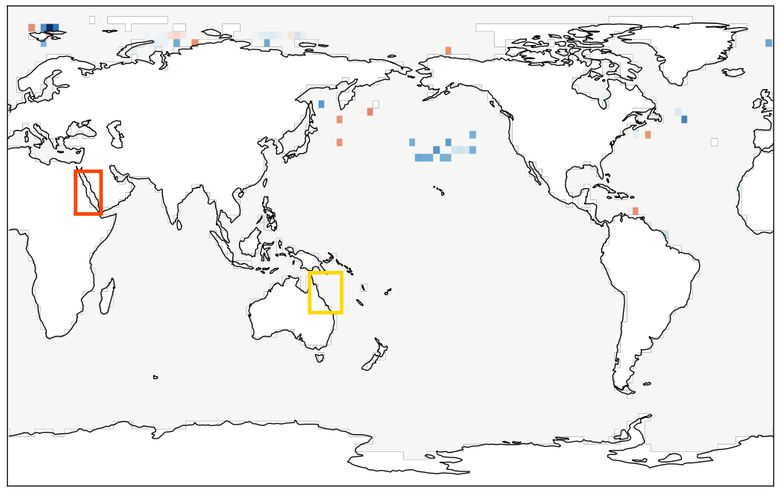
Actual DHW - Step 1



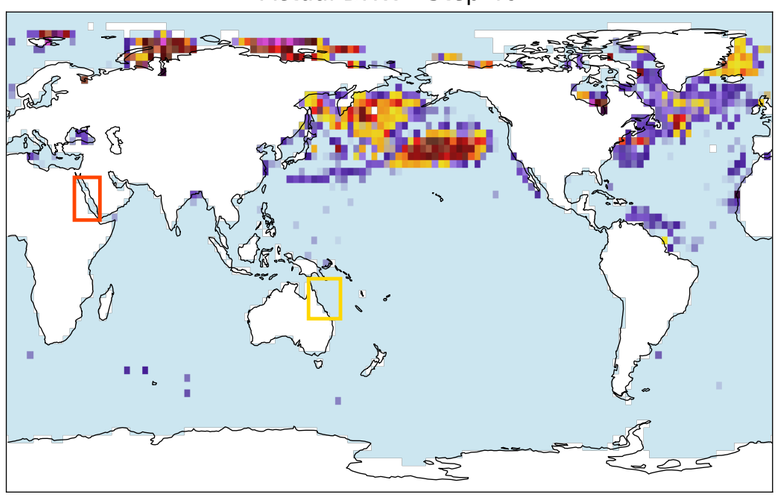
Predicted DHW - Step 1



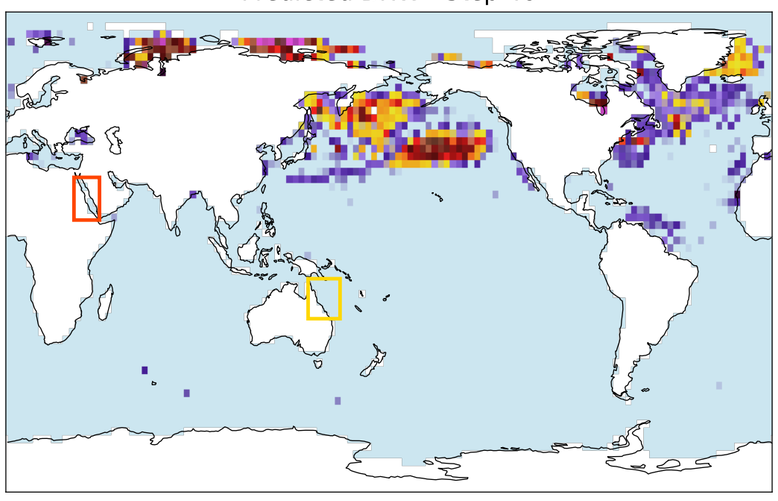
Error (Pred - Actual) - Step 1 [scale ±0.36]



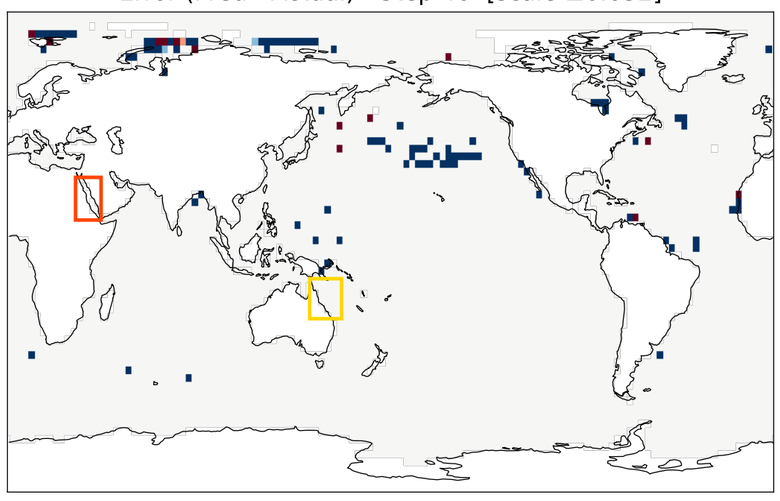
Actual DHW - Step 46



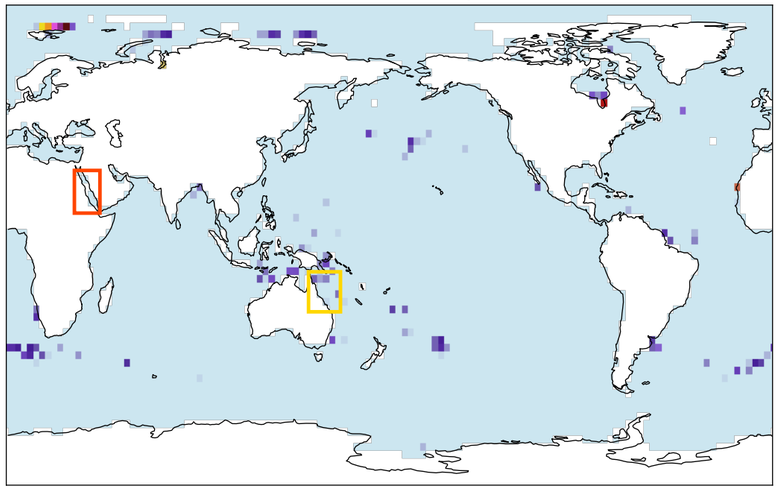
Predicted DHW - Step 46



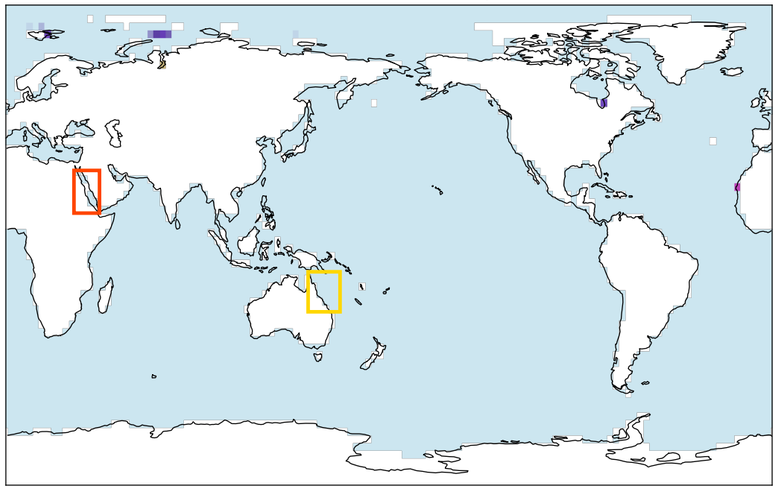
Error (Pred - Actual) - Step 46 [scale ±0.032]



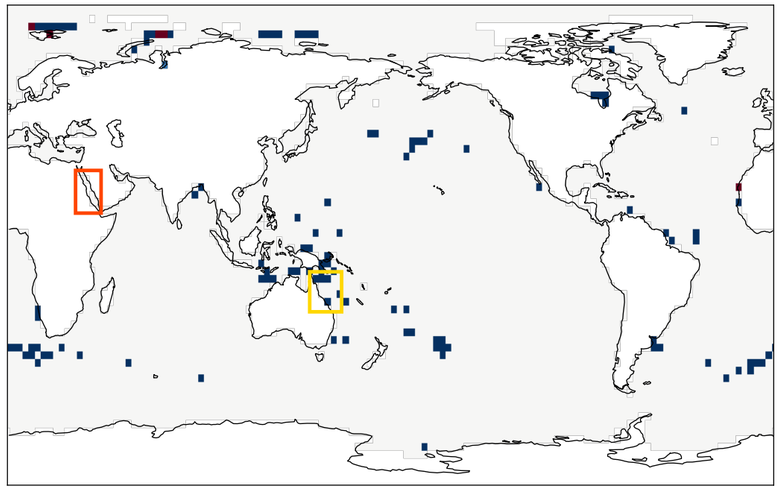
Actual DHW - Step 90

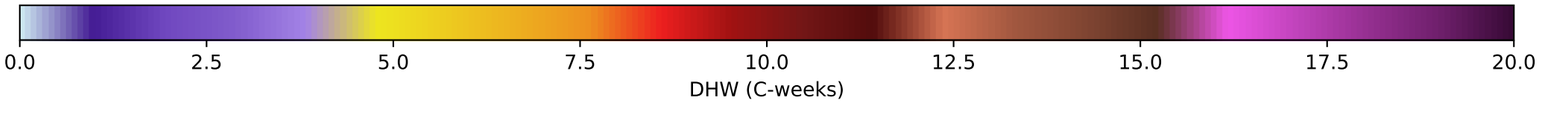


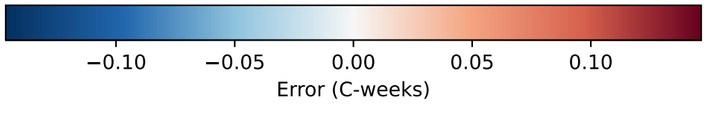
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

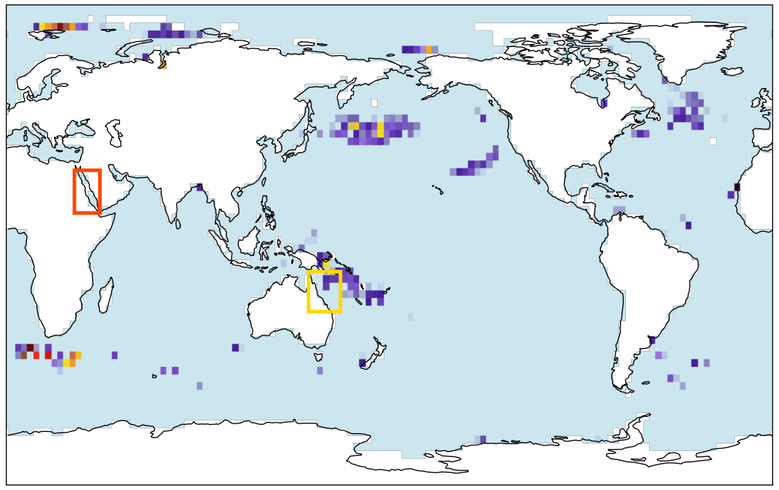




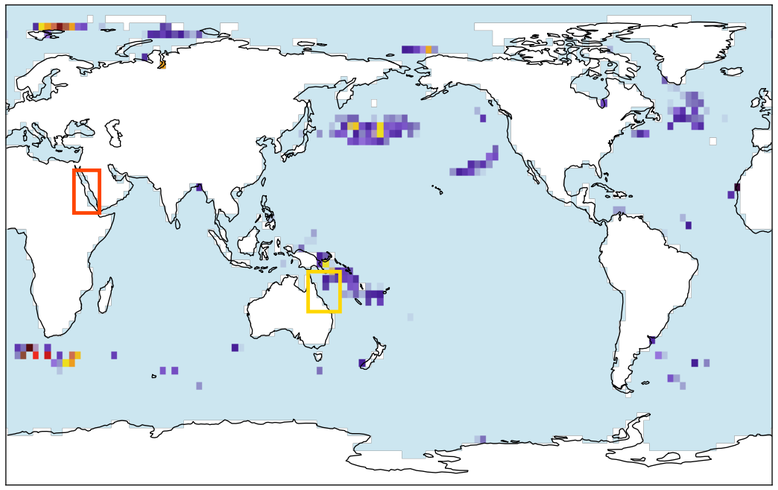


DHW Chain init: 2022-12-27 - Test (C-weeks)

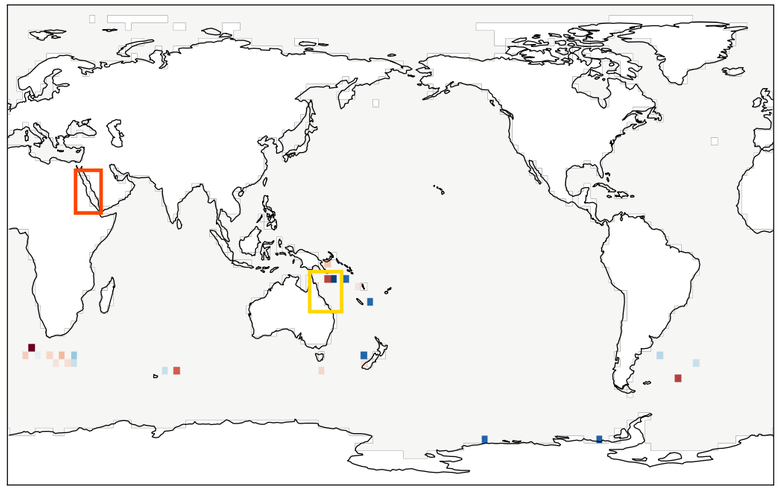
Actual DHW - Step 1



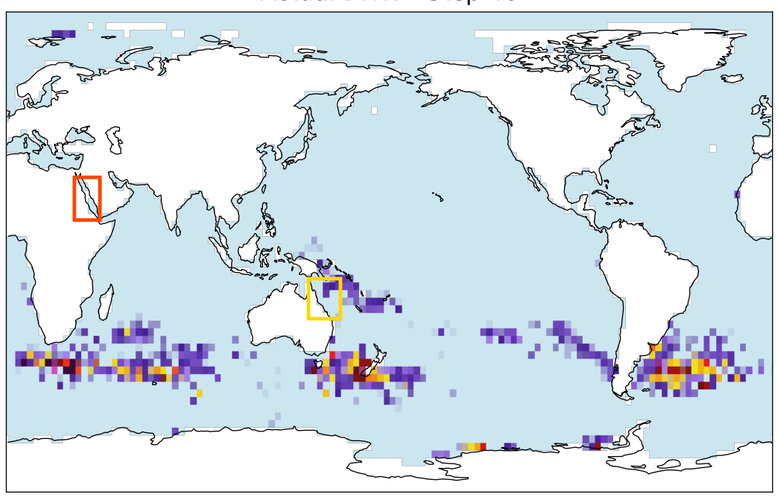
Predicted DHW - Step 1



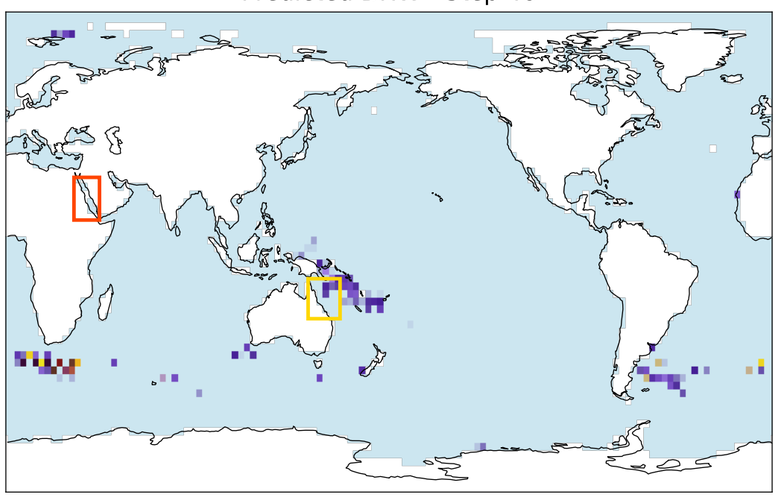
Error (Pred - Actual) - Step 1 [scale ±0.22]



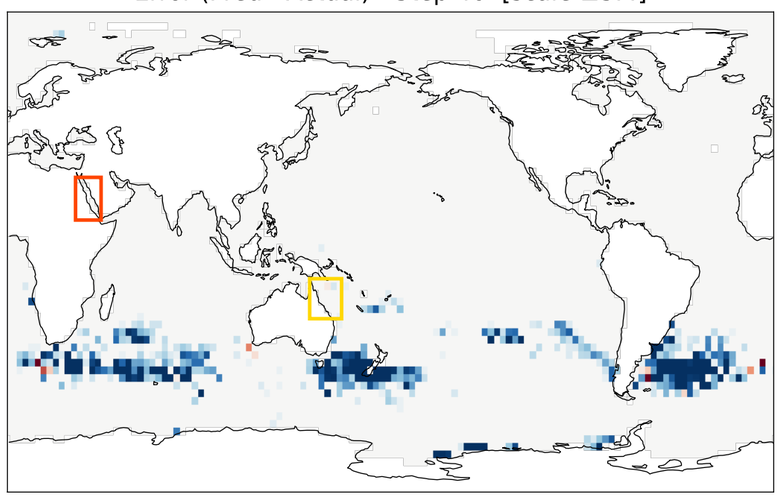
Actual DHW - Step 46



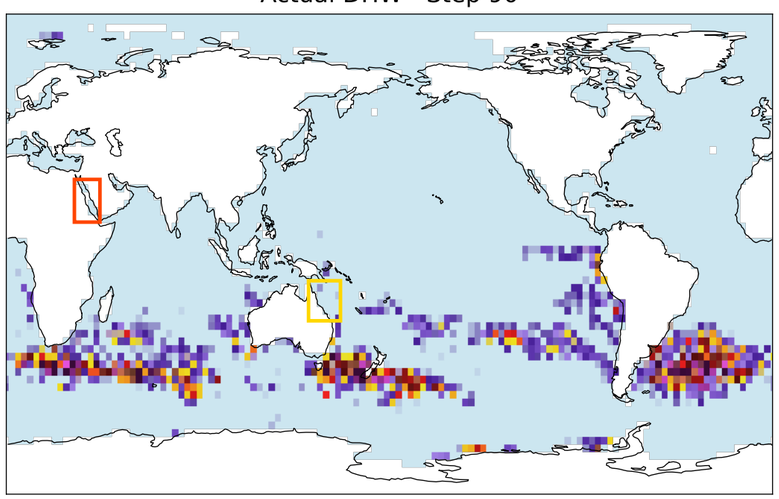
Predicted DHW - Step 46



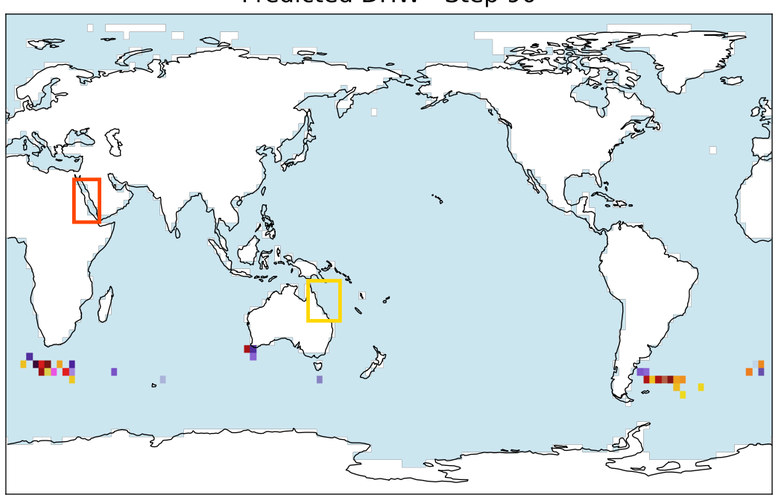
Error (Pred - Actual) - Step 46 [scale ±3.4]



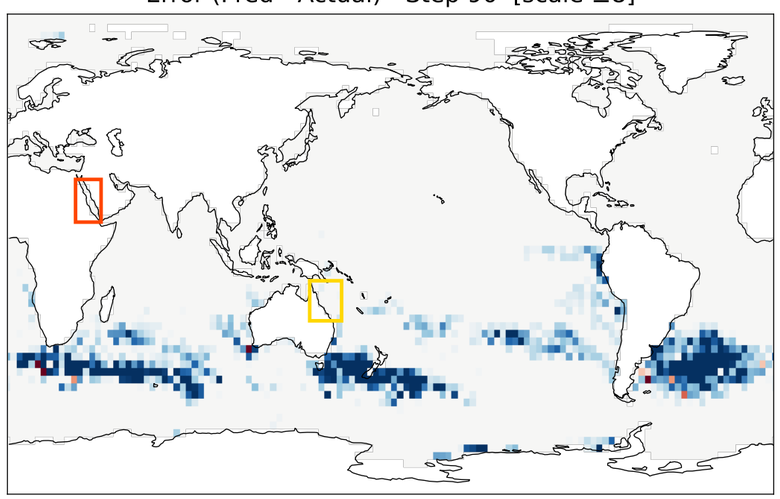
Actual DHW - Step 90

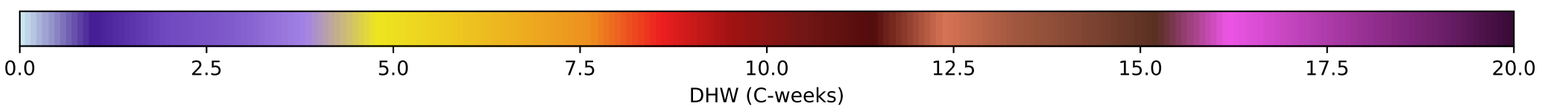


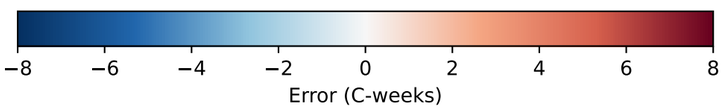
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

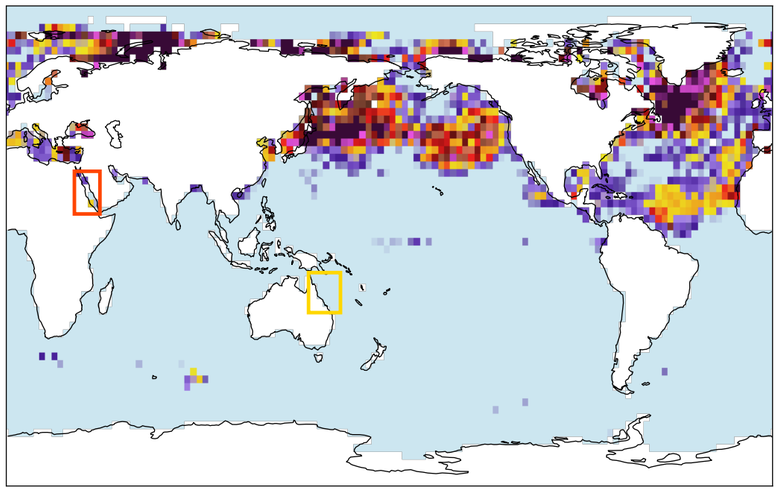




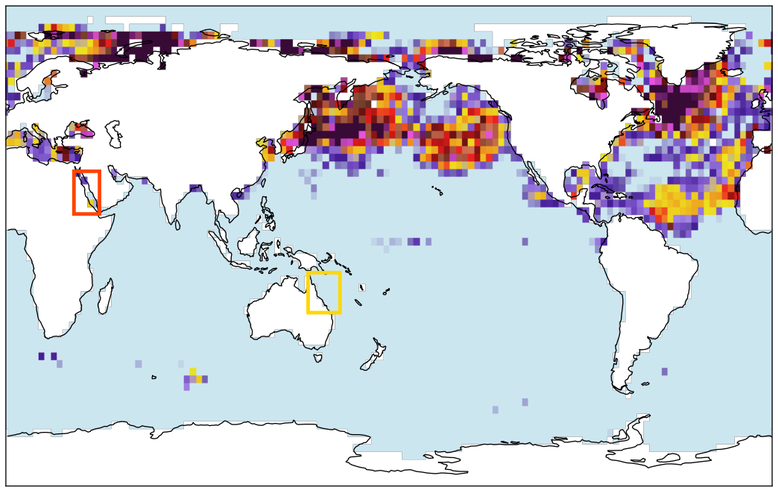


DHW Chain init: 2023-09-23 - Test (C-weeks)

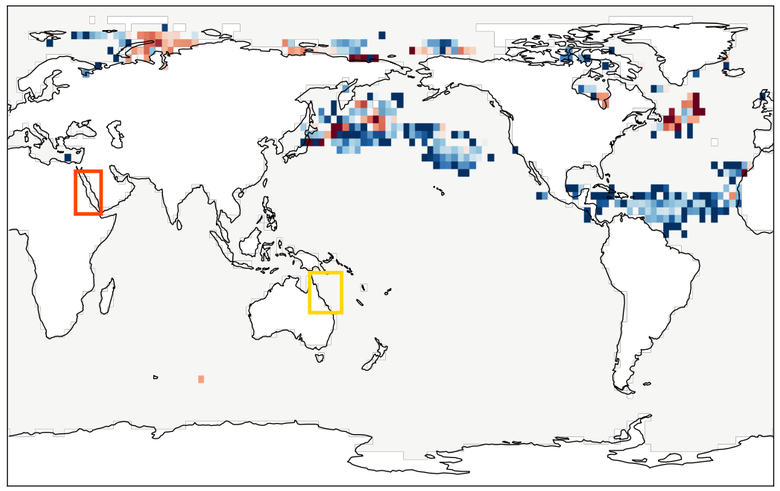
Actual DHW - Step 1



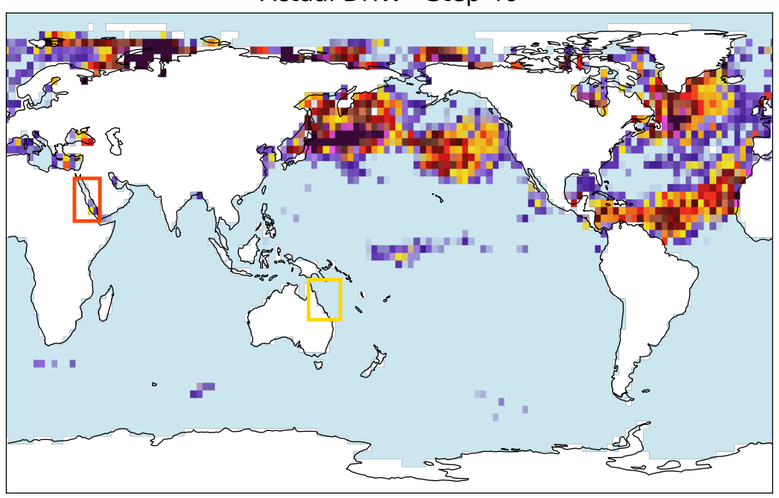
Predicted DHW - Step 1



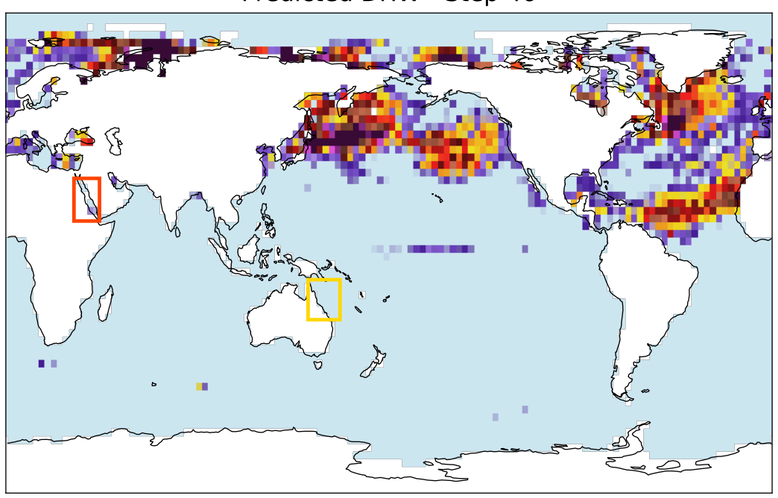
Error (Pred - Actual) - Step 1 [scale ±0.05]



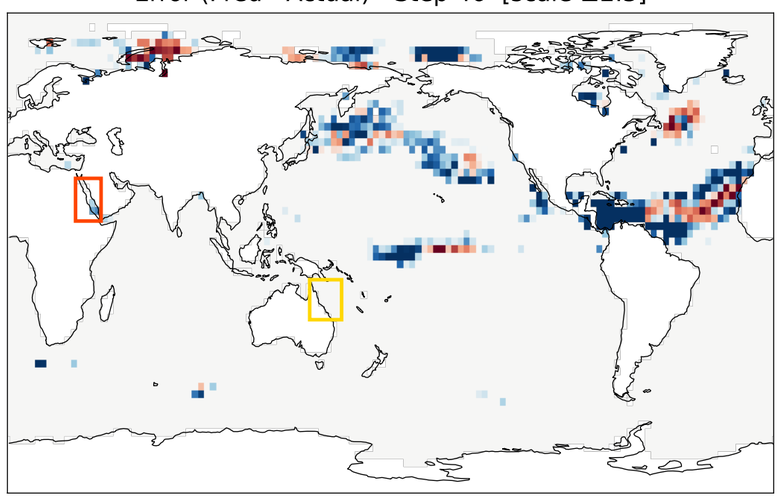
Actual DHW - Step 46



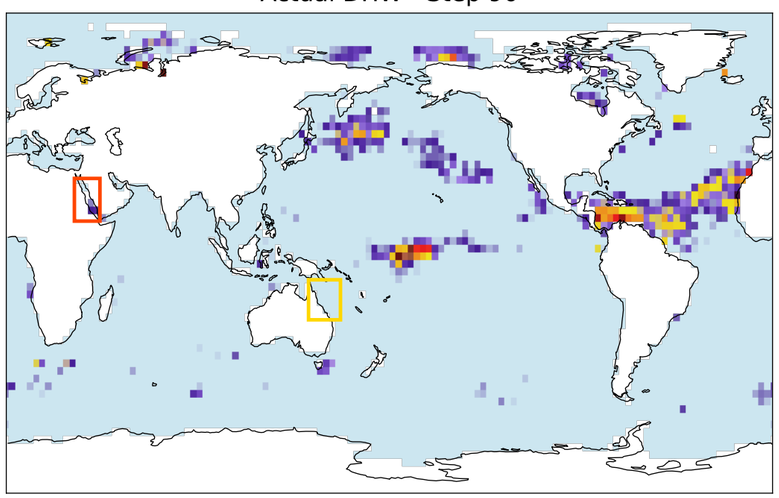
Predicted DHW - Step 46



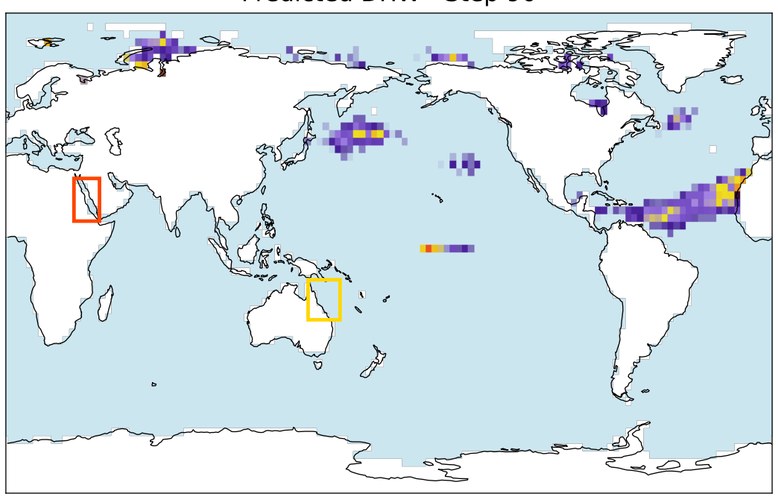
Error (Pred - Actual) - Step 46 [scale ±2.3]



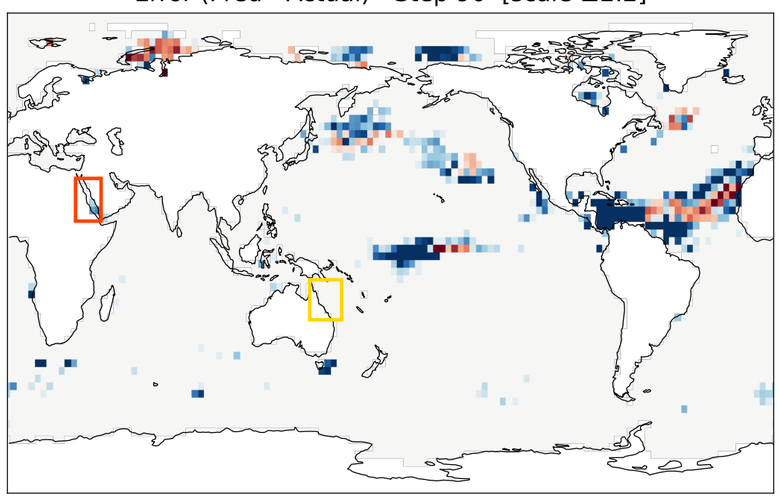
Actual DHW - Step 90

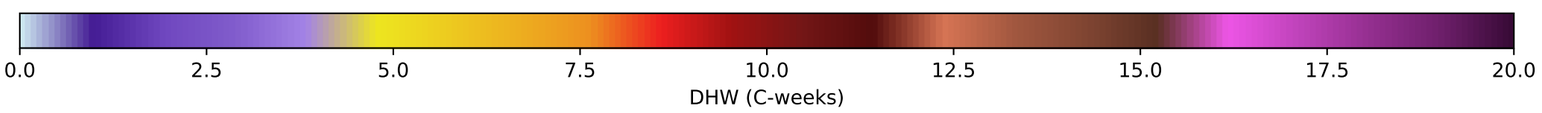


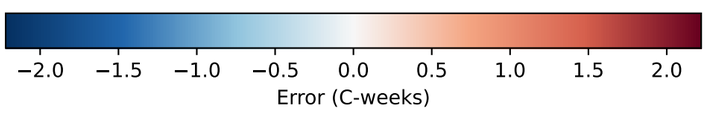
Predicted DHW - Step 90



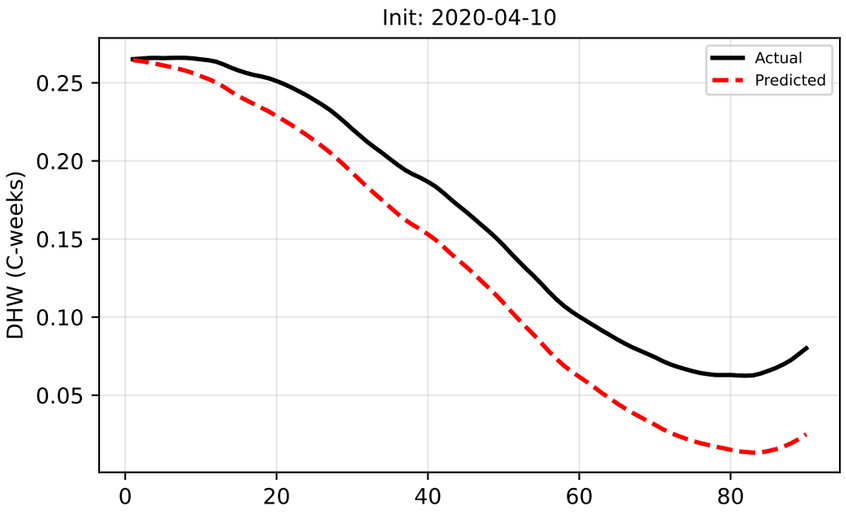
Error (Pred - Actual) - Step 90 [scale ±2.2]

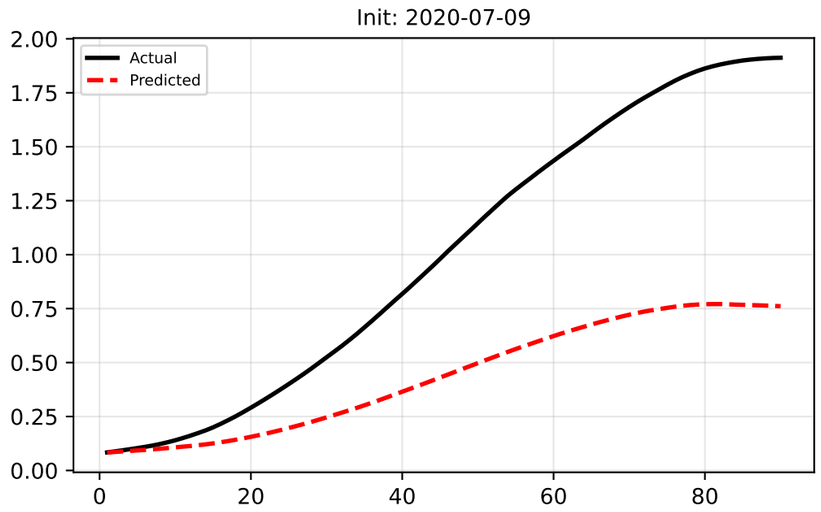


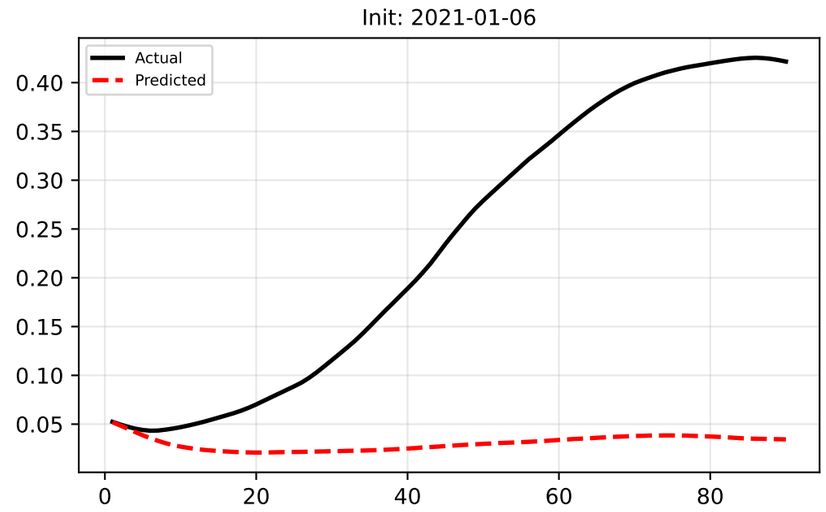


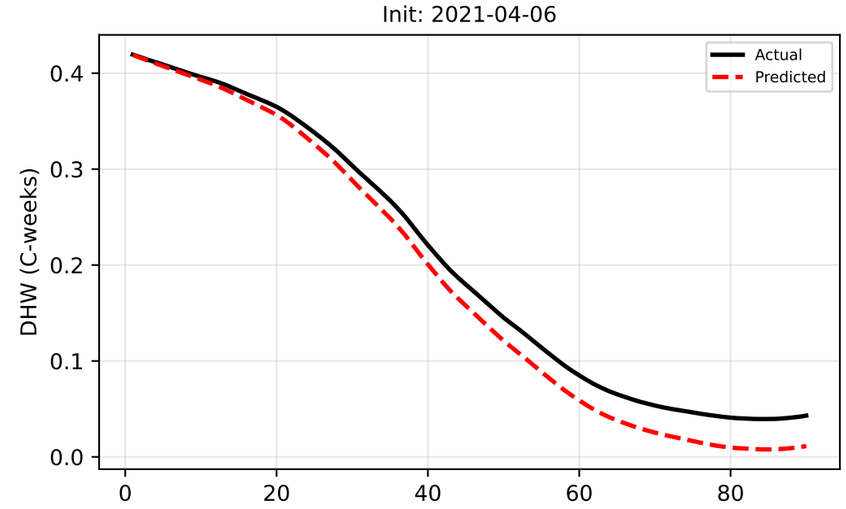


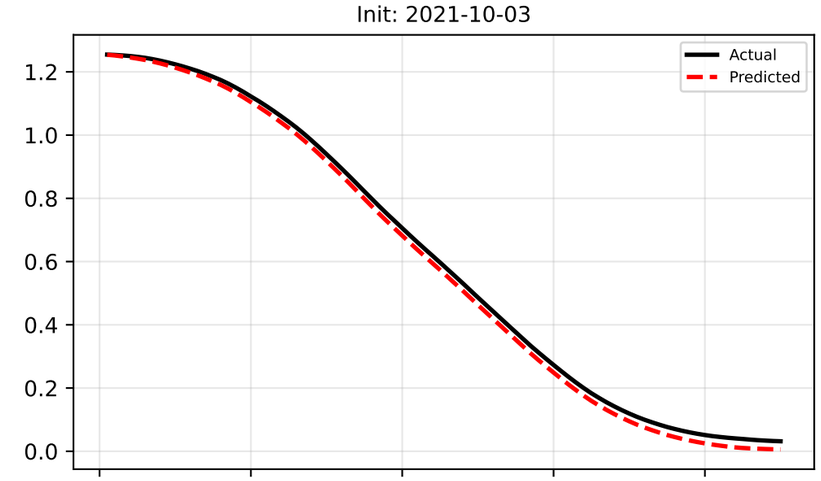
Global-Mean DHW - Test (C-weeks)

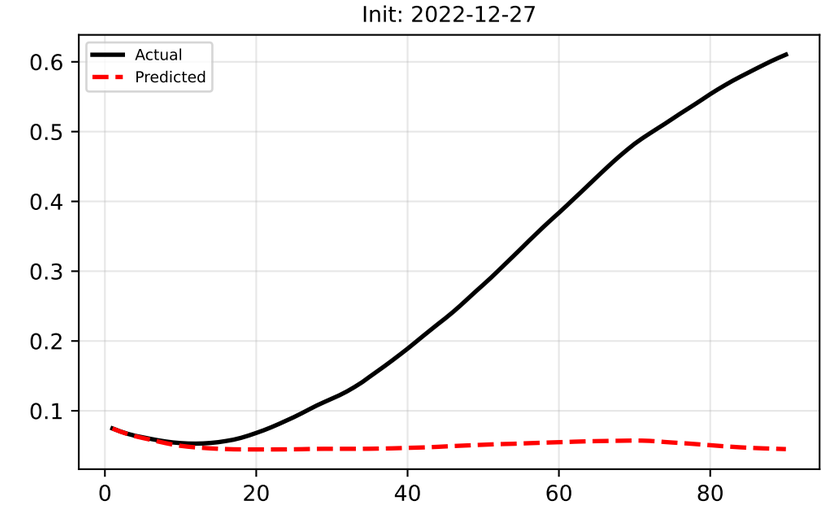


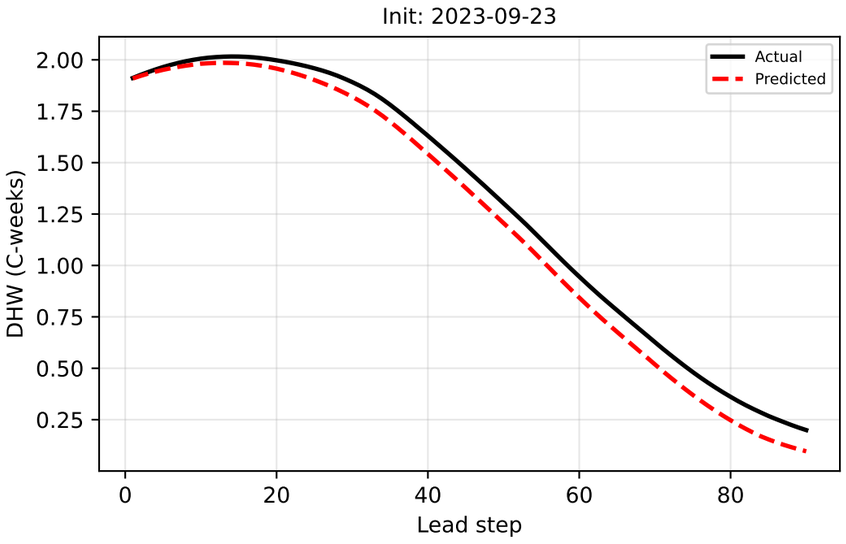




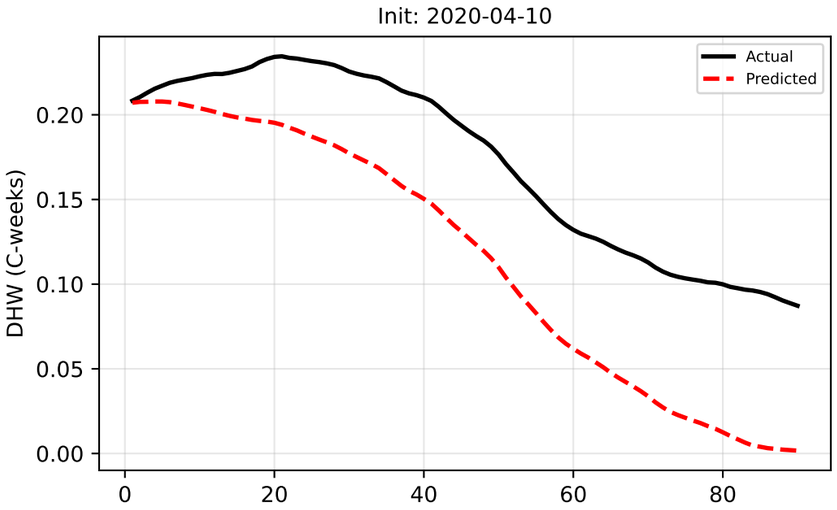


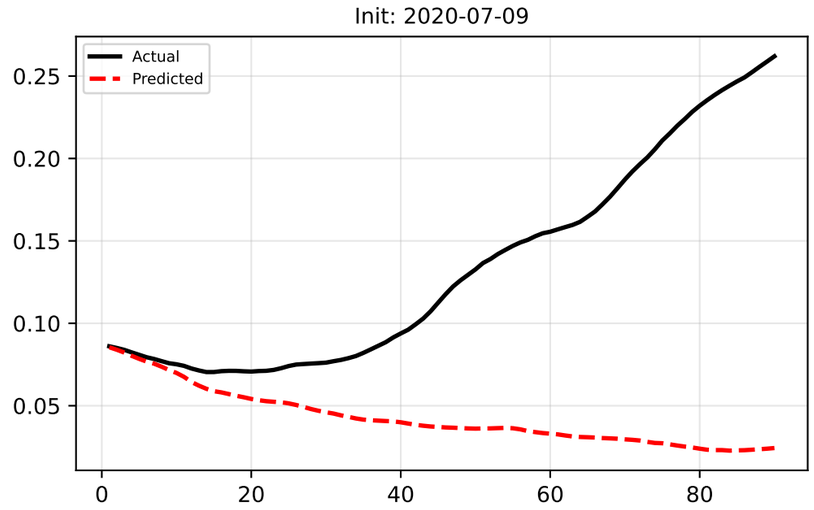


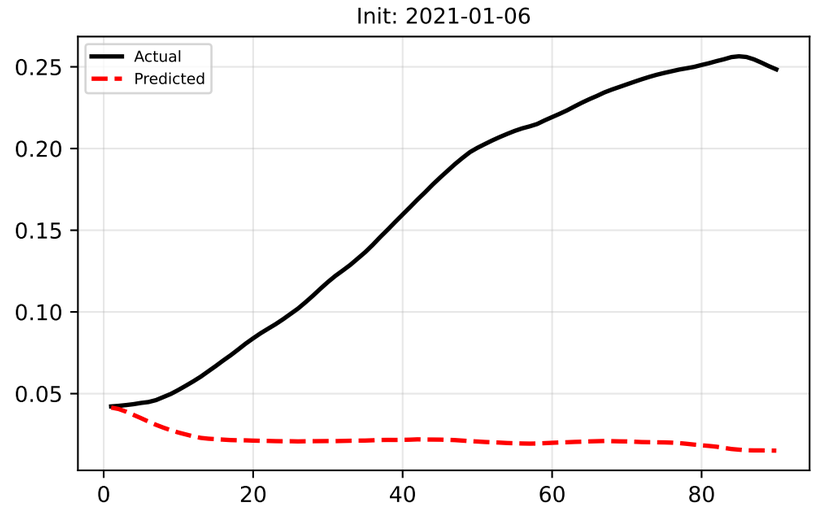


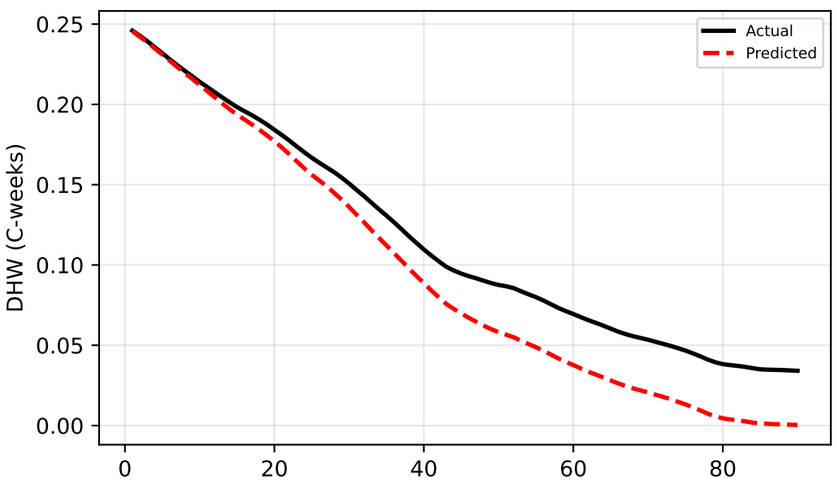


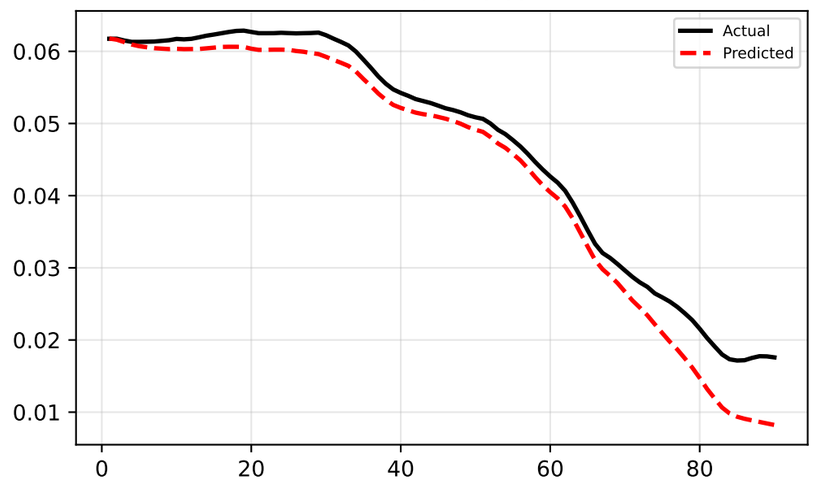
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

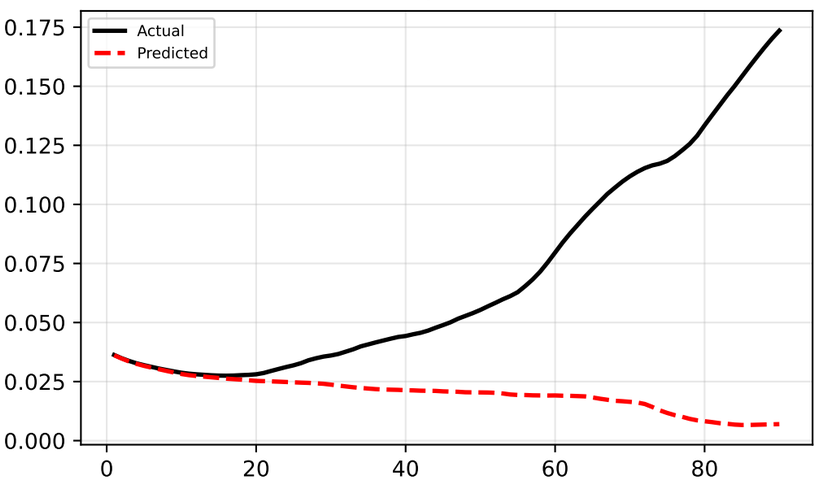


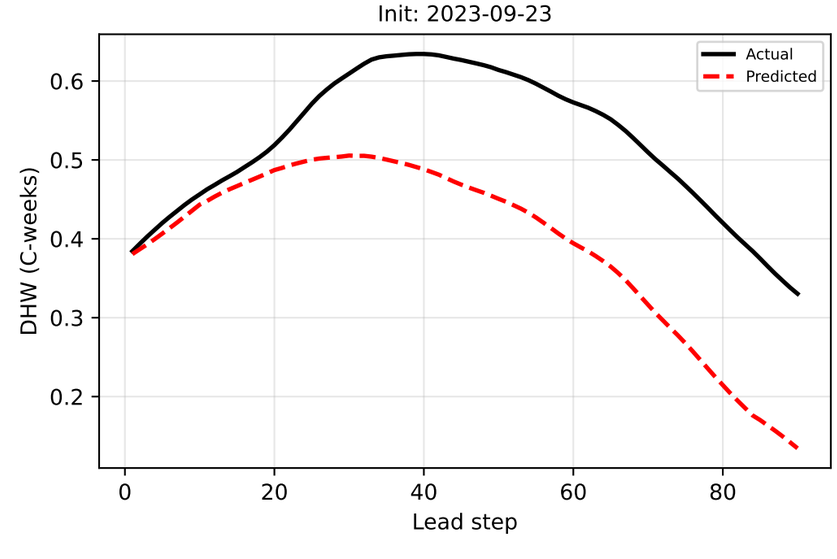


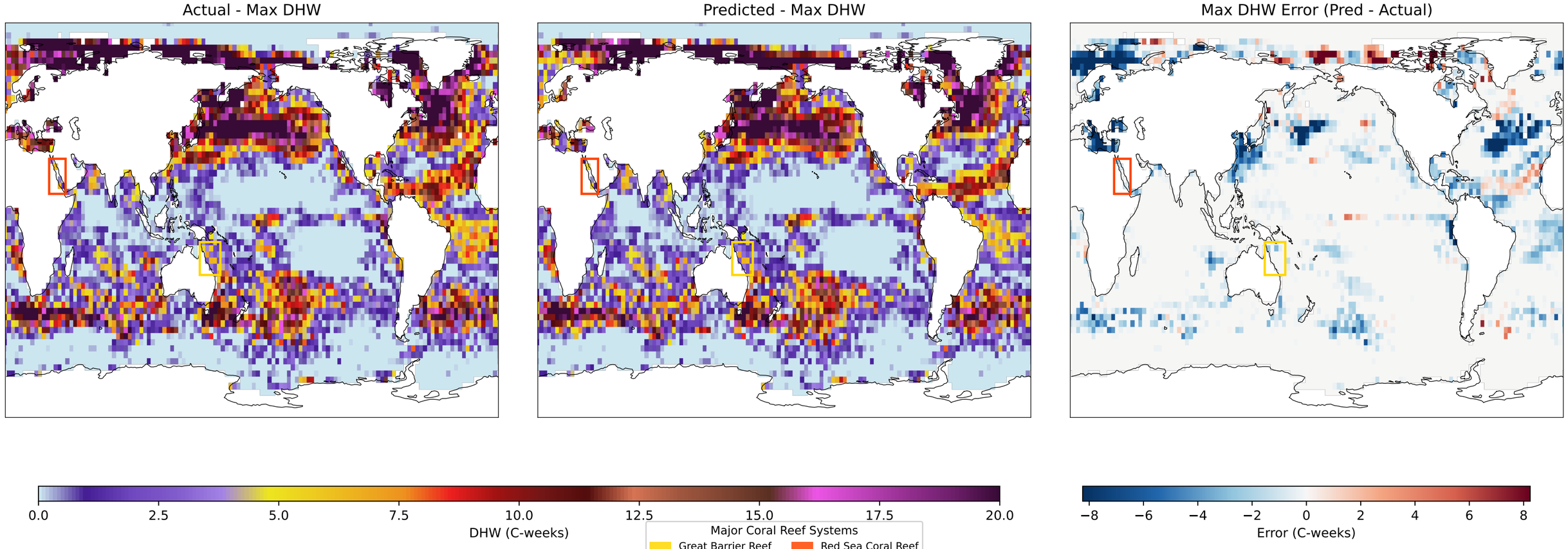


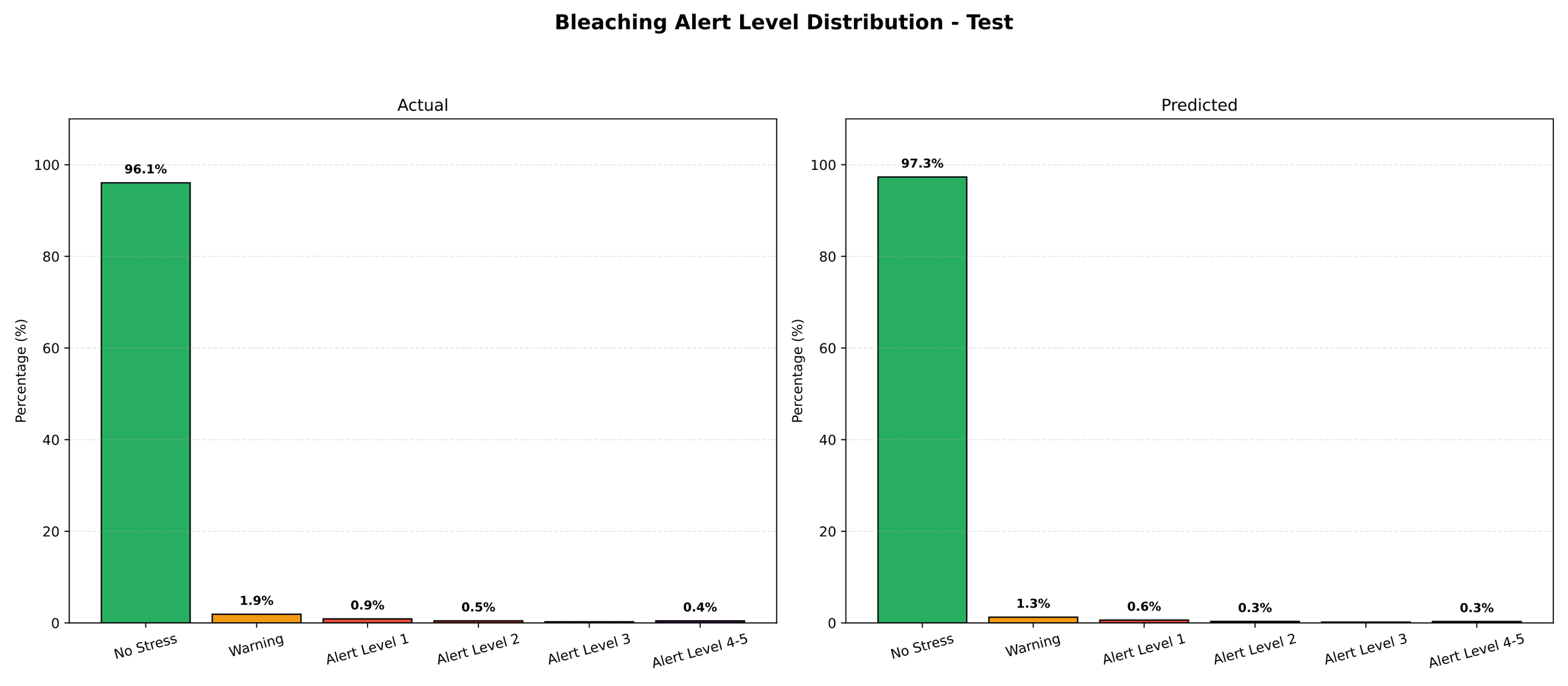




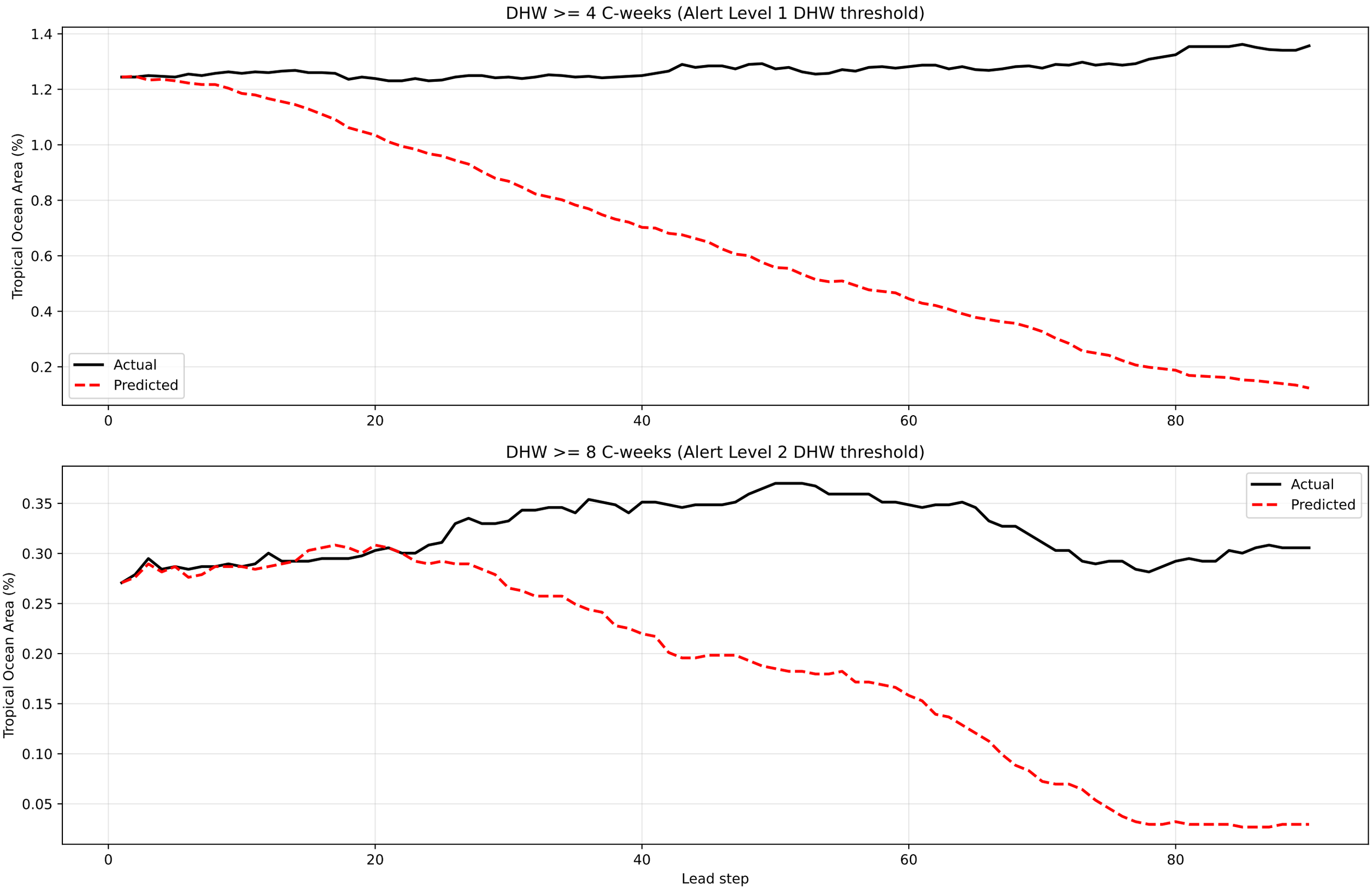





Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.1%
1.9%
0.9%
0.5%
0.4%
97.3%
1.3%
0.6%
0.3%
0.3%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test



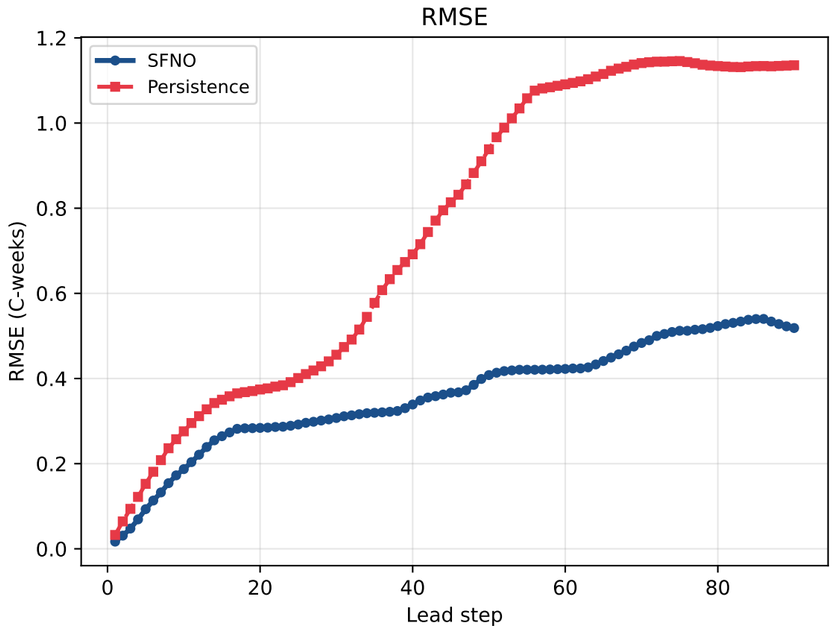
Great Barrier Reef Analysis - Test

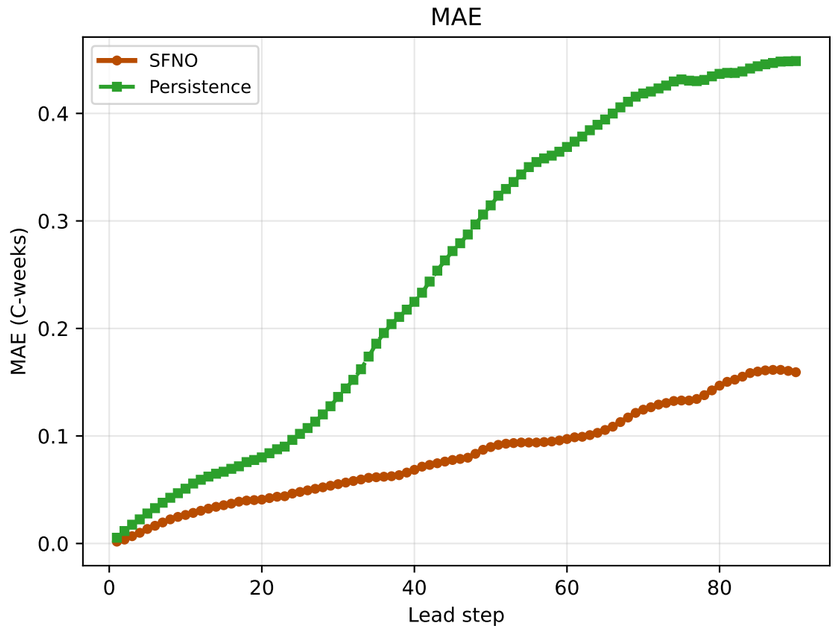
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                      Grid: 5 lat x 5 lon points
```

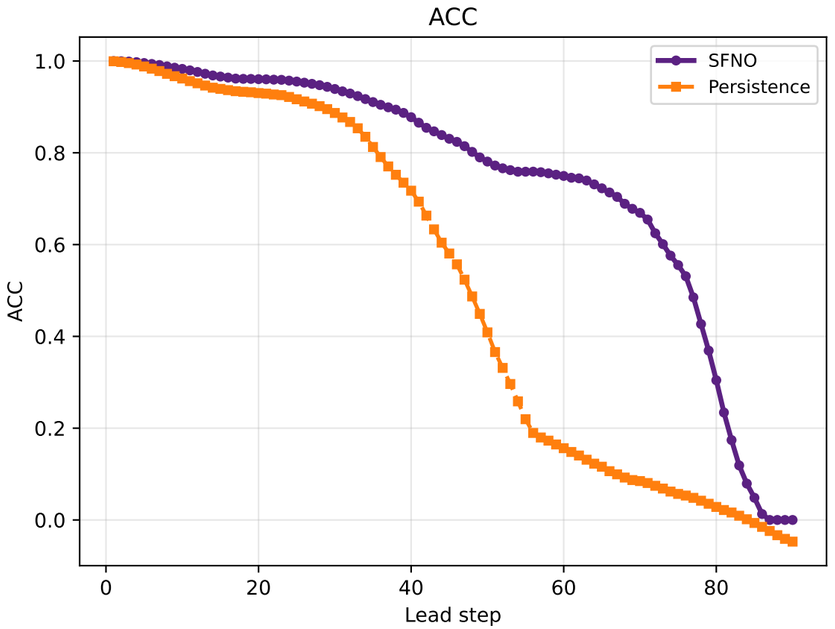
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

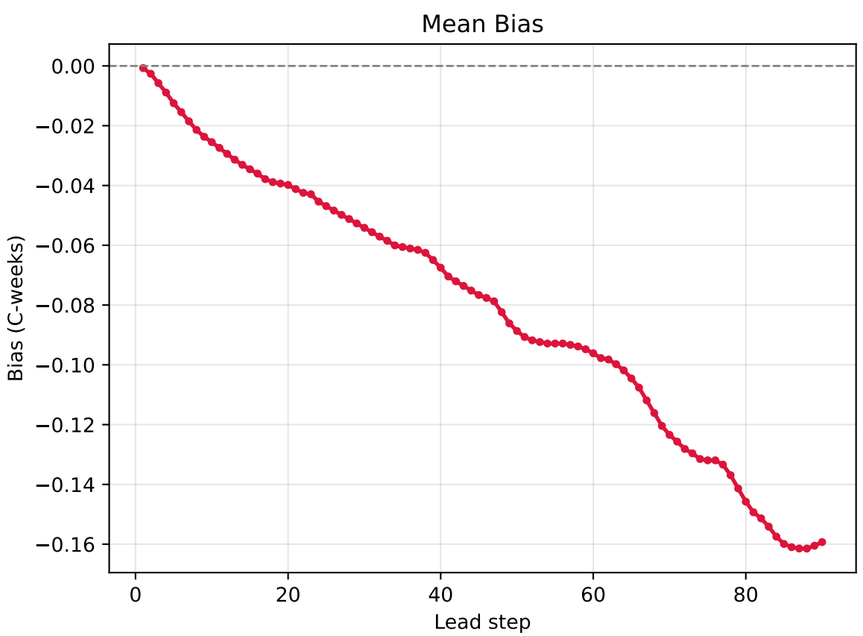
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0169	0.0325	48.1%	0.0017	0.0054	-0.0007	0.9999	0.9995	0.9997
2	0.0310	0.0643	51.9%	0.0037	0.0116	-0.0026	0.9995	0.9979	0.9990
3	0.0476	0.0939	49.3%	0.0068	0.0175	-0.0058	0.9988	0.9955	0.9978
4	0.0692	0.1219	43.3%	0.0100	0.0224	-0.0089	0.9976	0.9924	0.9953
5	0.0931	0.1525	38.9%	0.0135	0.0279	-0.0125	0.9956	0.9881	0.9915
6	0.1134	0.1810	37.4%	0.0165	0.0329	-0.0155	0.9936	0.9834	0.9875
7	0.1324	0.2083	36.4%	0.0196	0.0379	-0.0185	0.9913	0.9781	0.9831
8	0.1542	0.2362	34.7%	0.0225	0.0425	-0.0214	0.9883	0.9721	0.9773
9	0.1726	0.2574	33.0%	0.0247	0.0468	-0.0237	0.9854	0.9669	0.9717
10	0.1872	0.2759	32.2%	0.0265	0.0510	-0.0255	0.9828	0.9619	0.9667
11	0.2037	0.2955	31.1%	0.0285	0.0558	-0.0274	0.9797	0.9562	0.9605
12	0.2211	0.3116	29.1%	0.0304	0.0593	-0.0294	0.9761	0.9515	0.9537
13	0.2390	0.3277	27.1%	0.0324	0.0622	-0.0314	0.9722	0.9466	0.9463
14	0.2548	0.3424	25.6%	0.0341	0.0649	-0.0331	0.9686	0.9419	0.9393
15	0.2645	0.3502	24.5%	0.0356	0.0668	-0.0346	0.9661	0.9393	0.9347
16	0.2734	0.3582	23.7%	0.0370	0.0694	-0.0360	0.9639	0.9366	0.9304
17	0.2818	0.3652	22.8%	0.0389	0.0718	-0.0379	0.9617	0.9343	0.9263
18	0.2830	0.3680	23.1%	0.0399	0.0757	-0.0389	0.9611	0.9329	0.9252
19	0.2834	0.3705	23.5%	0.0404	0.0777	-0.0393	0.9610	0.9319	0.9249
20	0.2840	0.3742	24.1%	0.0408	0.0801	-0.0398	0.9605	0.9302	0.9239
21	0.2846	0.3768	24.5%	0.0422	0.0840	-0.0412	0.9601	0.9289	0.9233
22	0.2861	0.3810	24.9%	0.0435	0.0876	-0.0424	0.9594	0.9271	0.9220
23	0.2868	0.3841	25.3%	0.0440	0.0901	-0.0429	0.9590	0.9256	0.9213
24	0.2891	0.3914	26.2%	0.0464	0.0963	-0.0454	0.9573	0.9217	0.9181
25	0.2919	0.4011	27.2%	0.0479	0.1020	-0.0469	0.9552	0.9167	0.9143
26	0.2959	0.4104	27.9%	0.0494	0.1074	-0.0484	0.9528	0.9118	0.9099
27	0.2984	0.4189	28.8%	0.0509	0.1133	-0.0498	0.9503	0.9070	0.9053
28	0.3013	0.4287	29.7%	0.0522	0.1201	-0.0512	0.9474	0.9017	0.9001
29	0.3039	0.4402	31.0%	0.0537	0.1277	-0.0527	0.9437	0.8952	0.8935
30	0.3073	0.4558	32.6%	0.0552	0.1363	-0.0542	0.9390	0.8868	0.8851
31	0.3113	0.4739	34.3%	0.0567	0.1442	-0.0556	0.9339	0.8770	0.8760
32	0.3135	0.4915	36.2%	0.0581	0.1522	-0.0571	0.9292	0.8672	0.8676
33	0.3162	0.5150	38.6%	0.0595	0.1621	-0.0585	0.9236	0.8535	0.8577
34	0.3185	0.5448	41.5%	0.0611	0.1738	-0.0601	0.9171	0.8349	0.8462
35	0.3195	0.5777	44.7%	0.0616	0.1858	-0.0606	0.9105	0.8127	0.8347
36	0.3206	0.6076	47.2%	0.0621	0.1958	-0.0611	0.9046	0.7908	0.8243
37	0.3218	0.6333	49.2%	0.0626	0.2040	-0.0615	0.8991	0.7703	0.8151
38	0.3237	0.6548	50.6%	0.0636	0.2107	-0.0625	0.8940	0.7521	0.8064
39	0.3303	0.6735	51.0%	0.0660	0.2175	-0.0649	0.8869	0.7349	0.7946
40	0.3389	0.6918	51.0%	0.0685	0.2248	-0.0675	0.8776	0.7175	0.7789
41	0.3485	0.7155	51.3%	0.0715	0.2332	-0.0704	0.8656	0.6935	0.7588
42	0.3553	0.7441	52.3%	0.0731	0.2436	-0.0721	0.8545	0.6632	0.7399
43	0.3585	0.7709	53.5%	0.0746	0.2537	-0.0736	0.8468	0.6331	0.7261
44	0.3624	0.7952	54.4%	0.0762	0.2632	-0.0752	0.8387	0.6042	0.7116
45	0.3666	0.8139	55.0%	0.0777	0.2720	-0.0766	0.8306	0.5806	0.6964
46	0.3675	0.8315	55.8%	0.0786	0.2793	-0.0776	0.8238	0.5570	0.6838
47	0.3724	0.8560	56.5%	0.0798	0.2874	-0.0788	0.8144	0.5236	0.6660
48	0.3850	0.8827	56.4%	0.0834	0.2967	-0.0824	0.8020	0.4870	0.6431
49	0.3990	0.9103	56.2%	0.0872	0.3059	-0.0861	0.7899	0.4490	0.6208
50	0.4081	0.9384	56.5%	0.0897	0.3145	-0.0887	0.7809	0.4087	0.6033
51	0.4136	0.9668	57.2%	0.0917	0.3234	-0.0907	0.7723	0.3658	0.5879
52	0.4175	0.9892	57.8%	0.0928	0.3297	-0.0918	0.7662	0.3313	0.5773
53	0.4190	1.0111	58.6%	0.0934	0.3362	-0.0924	0.7622	0.2963	0.5692
54	0.4206	1.0344	59.3%	0.0939	0.3432	-0.0929	0.7588	0.2584	0.5617
55	0.4206	1.0582	60.3%	0.0939	0.3501	-0.0929	0.7588	0.2195	0.5591
56	0.4206	1.0761	60.9%	0.0939	0.3548	-0.0929	0.7590	0.1896	0.5572
57	0.4208	1.0812	61.1%	0.0943	0.3581	-0.0933	0.7577	0.1797	0.5526
58	0.4213	1.0841	61.1%	0.0949	0.3607	-0.0938	0.7552	0.1728	0.5467
59	0.4220	1.0875	61.2%	0.0958	0.3643	-0.0948	0.7523	0.1645	0.5396
60	0.4225	1.0909	61.3%	0.0972	0.3688	-0.0961	0.7496	0.1564	0.5329
61	0.4236	1.0943	61.3%	0.0988	0.3738	-0.0977	0.7457	0.1481	0.5231
62	0.4237	1.0980	61.4%	0.0992	0.3785	-0.0982	0.7444	0.1404	0.5168
63	0.4261	1.1030	61.4%	0.1008	0.3843	-0.0998	0.7396	0.1313	0.5089
64	0.4331	1.1092	61.0%	0.1029	0.3894	-0.1019	0.7312	0.1227	0.4980
65	0.4411	1.1152	60.4%	0.1056	0.3943	-0.1045	0.7226	0.1160	0.4883
66	0.4491	1.1228	60.0%	0.1087	0.3999	-0.1076	0.7136	0.1063	0.4780
67	0.4568	1.1280	59.5%	0.1129	0.4056	-0.1119	0.7040	0.0994	0.4677
68	0.4654	1.1321	58.9%	0.1172	0.4108	-0.1161	0.6889	0.0924	0.4508
69	0.4751	1.1374	58.2%	0.1215	0.4156	-0.1204	0.6780	0.0870	0.4400
70	0.4835	1.1409	57.6%	0.1245	0.4186	-0.1234	0.6692	0.0847	0.4312
71	0.4899	1.1428	57.1%	0.1267	0.4203	-0.1257	0.6546	0.0805	0.4154
72	0.5000	1.1442	56.3%	0.1292	0.4232	-0.1281	0.6246	0.0744	0.3838
73	0.5046	1.1443	55.9%	0.1306	0.4259	-0.1296	0.6010	0.0684	0.3572
74	0.5092	1.1451	55.5%	0.1325	0.4297	-0.1315	0.5760	0.0619	0.3323
75	0.5121	1.1456	55.3%	0.1330	0.4316	-0.1319	0.5554	0.0567	0.3119
76	0.5121	1.1433	55.2%	0.1330	0.4305	-0.1319	0.5310	0.0532	0.2909
77	0.5146	1.1405	54.9%	0.1344	0.4300	-0.1334	0.4850	0.0482	0.2544
78	0.5159	1.1370	54.6%	0.1379	0.4312	-0.1369	0.4268	0.0419	0.2120
79	0.5188	1.1351	54.3%	0.1424	0.4344	-0.1414	0.3690	0.0353	0.1726
80	0.5232	1.1339	53.9%	0.1468	0.4368	-0.1458	0.3044	0.0284	0.1288
81	0.5279	1.1328	53.4%	0.1504	0.4377	-0.1493	0.2341	0.0216	0.0811
82	0.5307	1.1317	53.1%	0.1524	0.4376	-0.1514	0.1741	0.0162	0.0394
83	0.5338	1.1313	52.8%	0.1552	0.4390	-0.1541	0.1188	0.0092	-0.0006
84	0.5379	1.1328	52.5%	0.1585	0.4417	-0.1575	0.0792	0.0015	-0.0307
85	0.5397	1.1337	52.4%	0.1599	0.4438	-0.1599	0.0483	-0.0065	-0.0564
86	0.5400	1.1340	52.4%	0.1610	0.4457	-0.1610	0.0129	-0.0150	-0.0864
87	0.5338	1.1333	52.9%	0.1615	0.4469	-0.1615	0.0000	-0.0240	-0.1007
88	0.5279	1.1341	53.5%	0.1615	0.4482	-0.1615	0.0000	-0.0332	-0.1032
89	0.5225	1.1349	54.0%	0.1605	0.4484	-0.1605	-0.0000	-0.0412	-0.1042
90	0.5186	1.1357	54.3%	0.1593	0.4487	-0.1593	0.0000	-0.0472	-0.1042

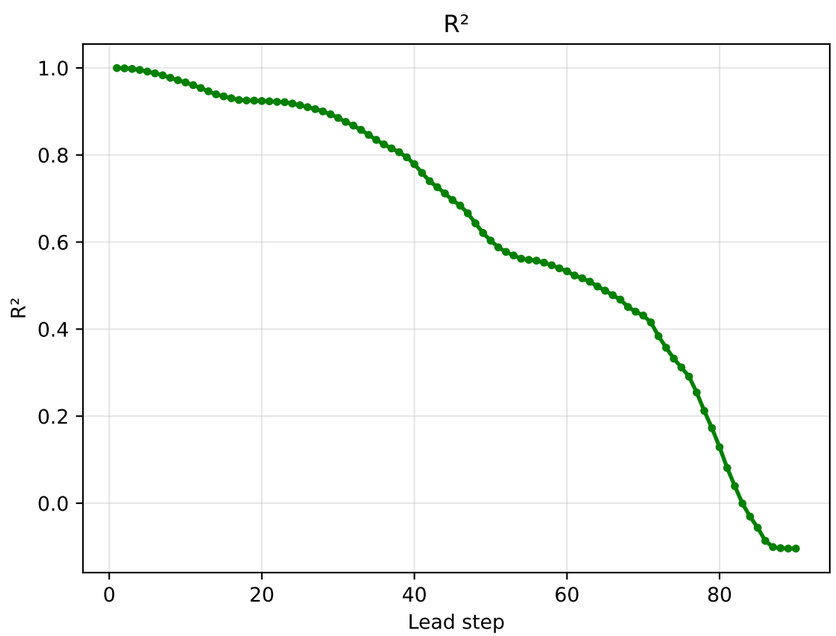
DHW Forecast Skill vs Lead Step - Test - GBR

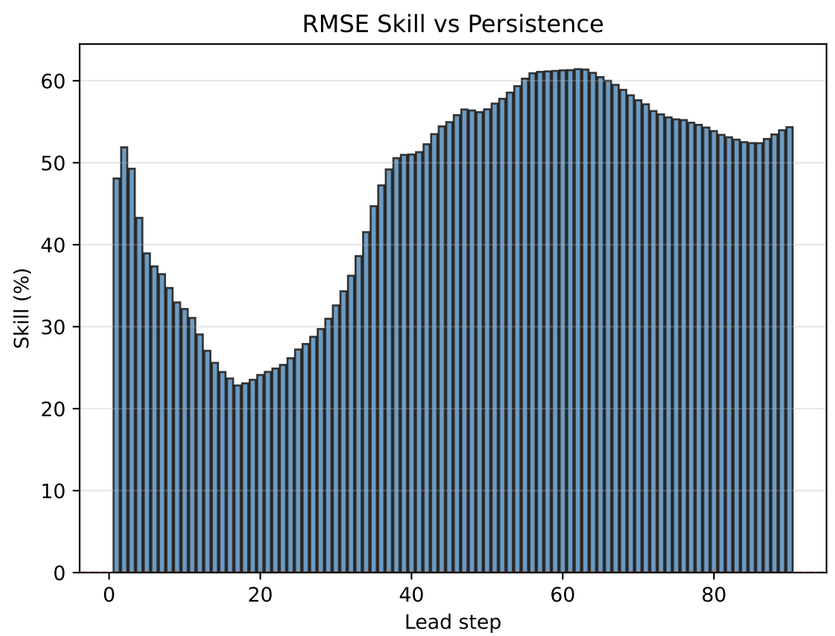


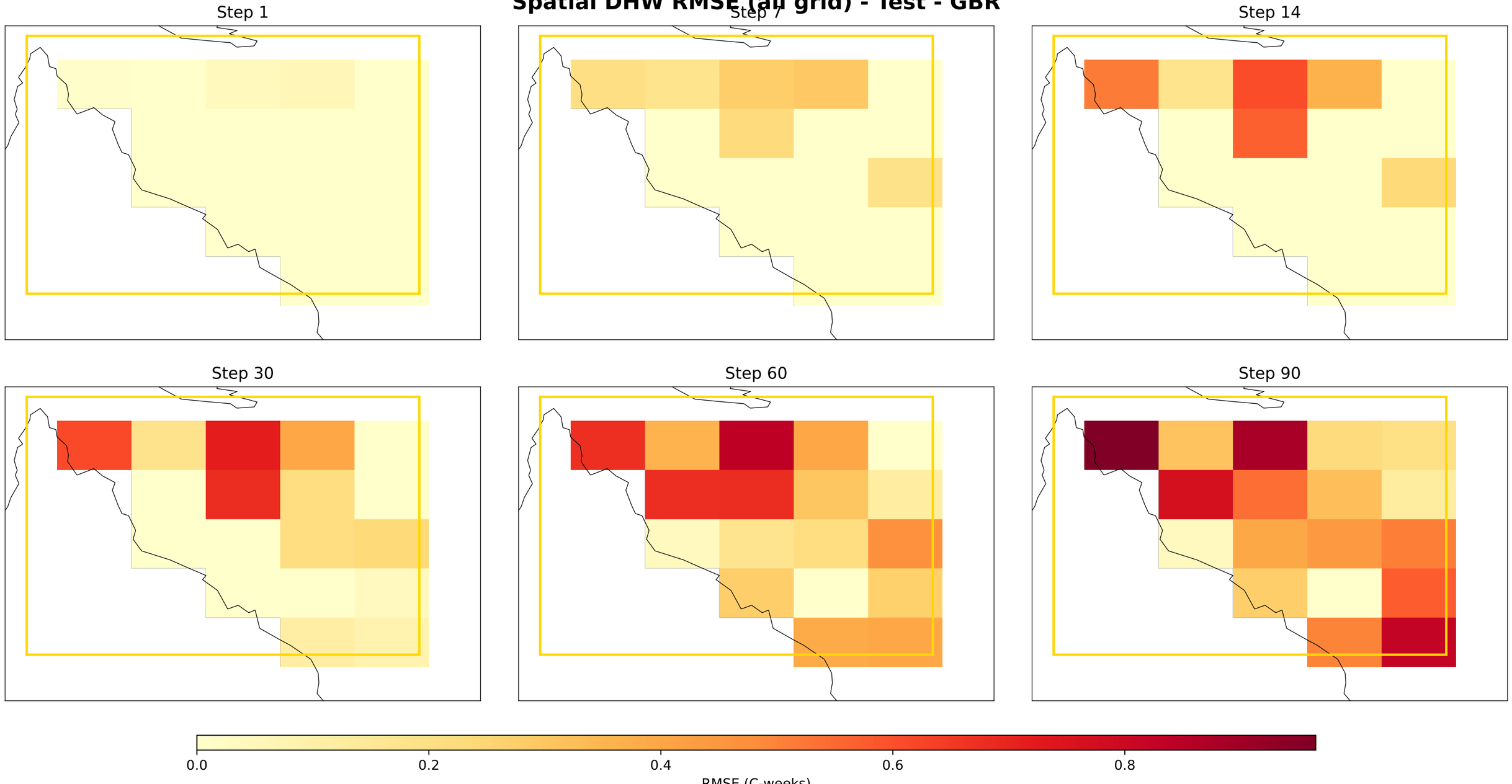


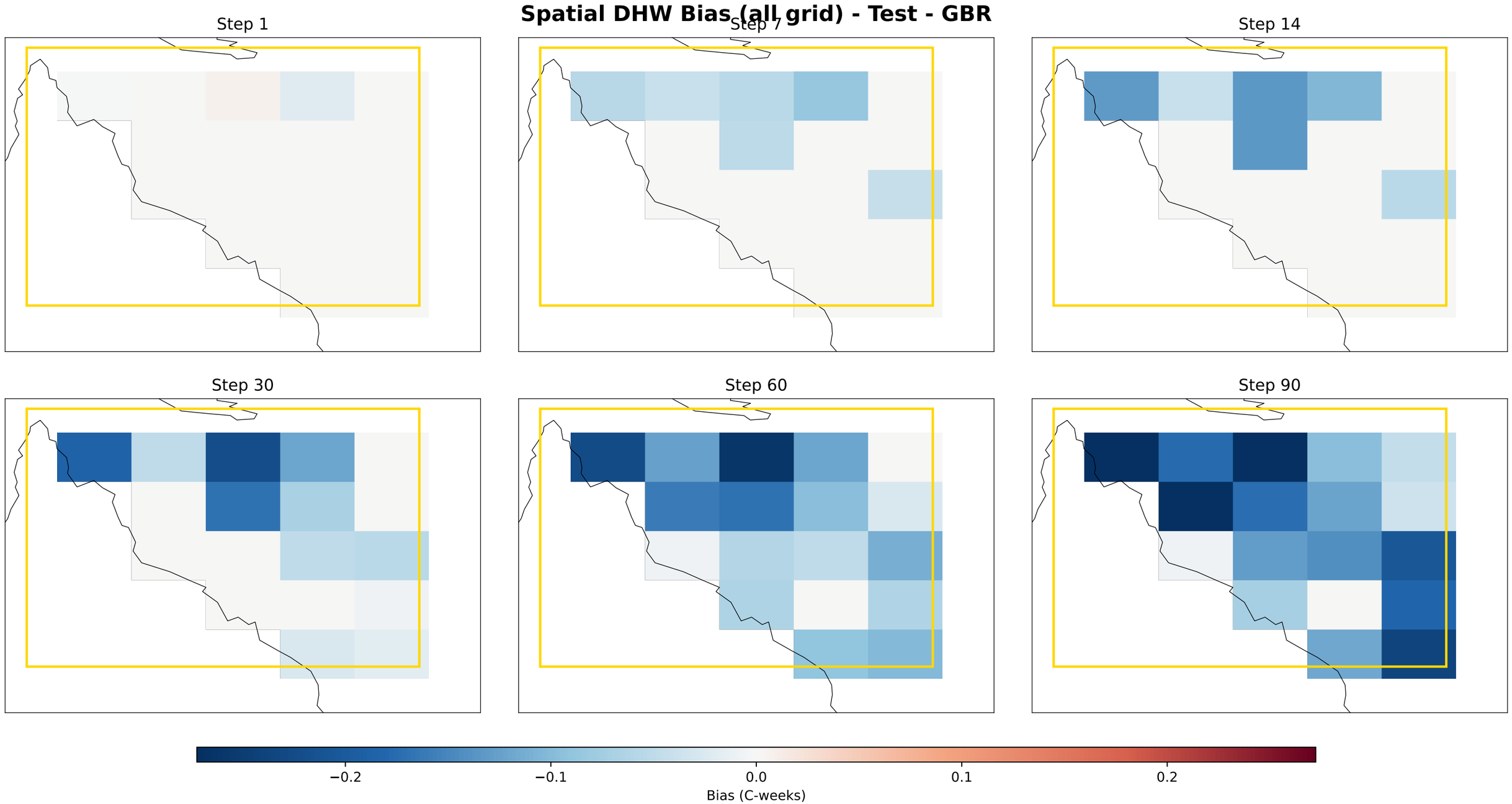


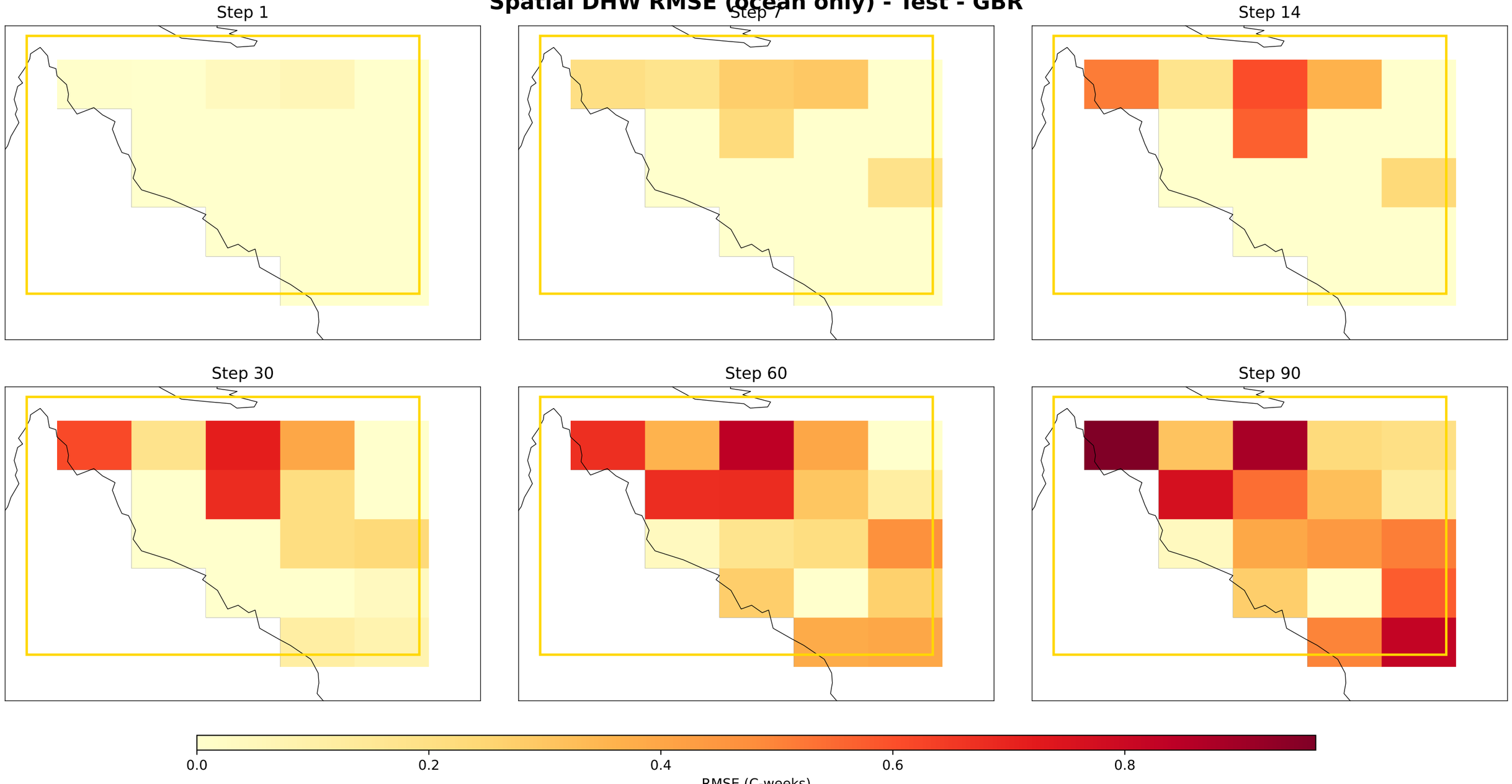


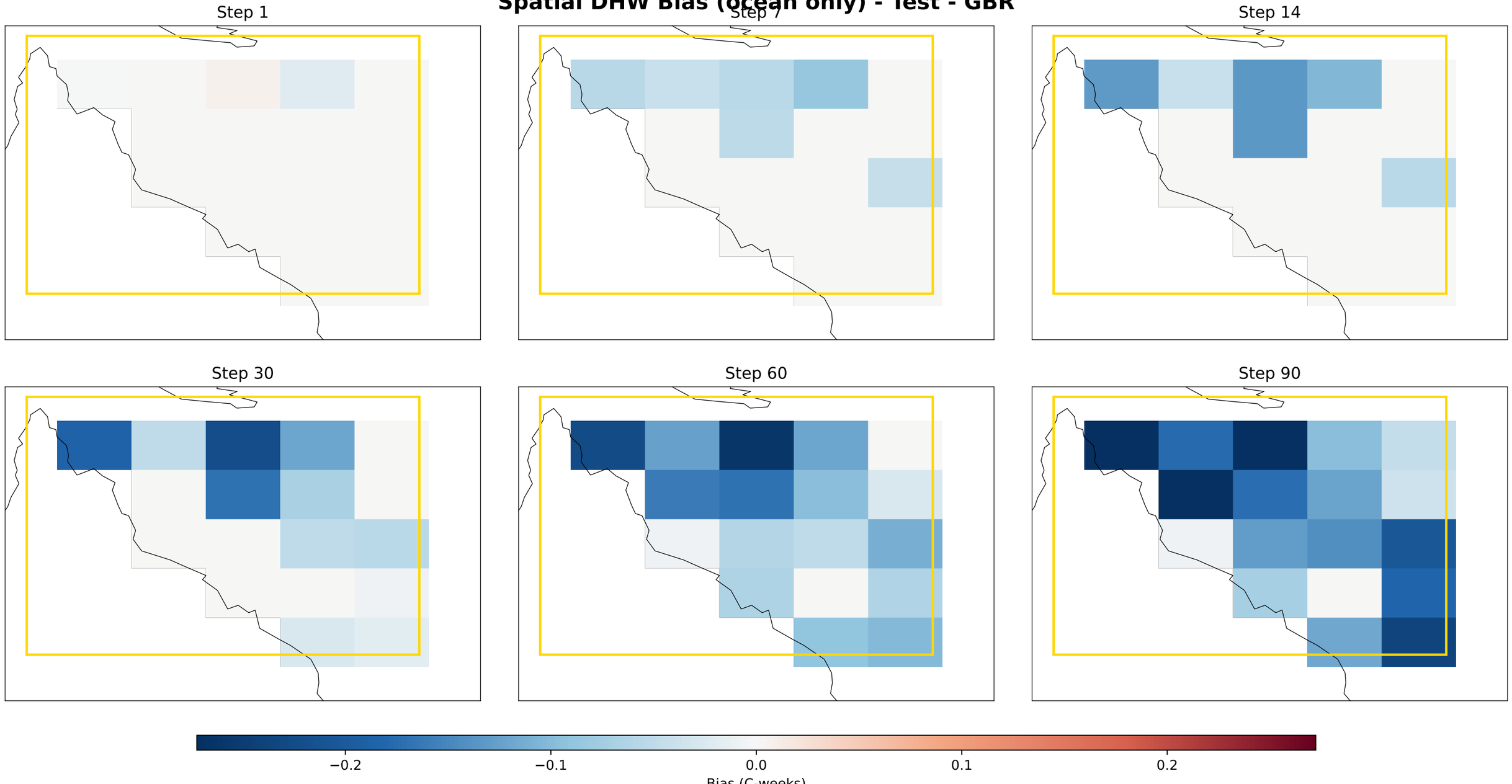





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

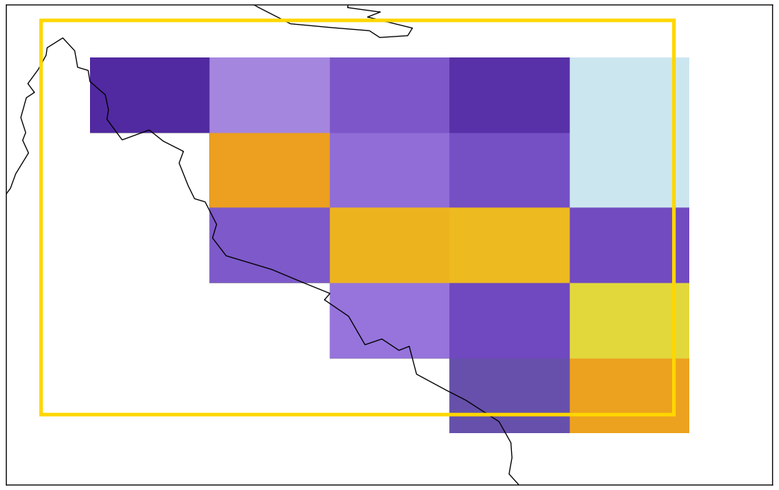

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

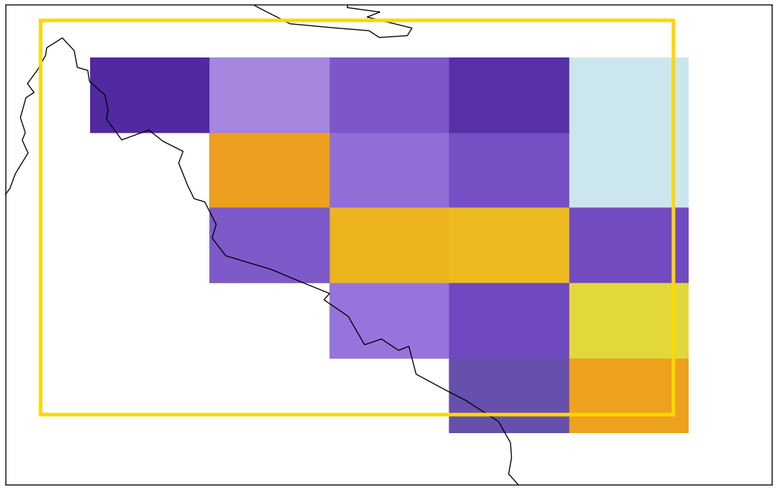

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

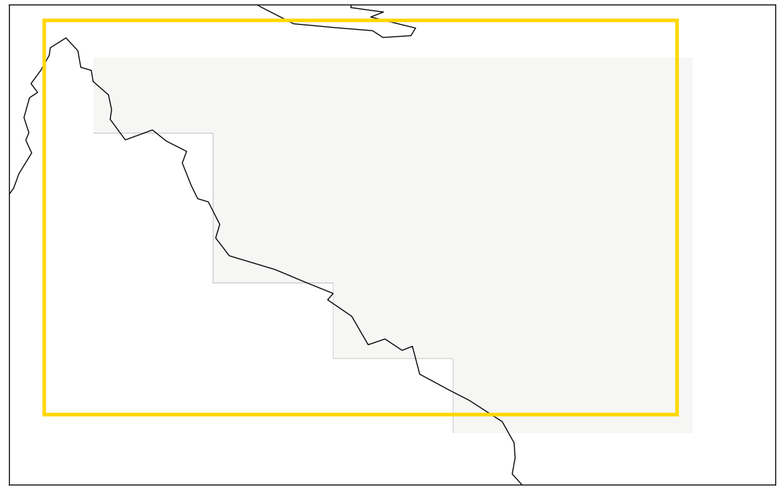
Actual DHW - Step 1



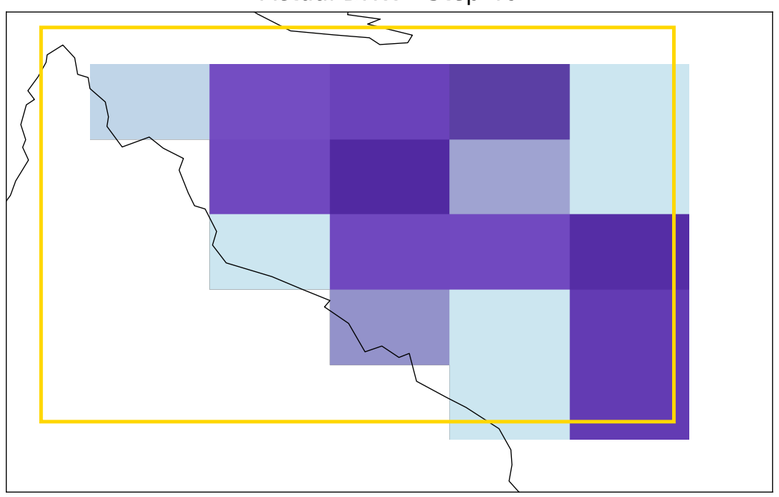
Predicted DHW - Step 1



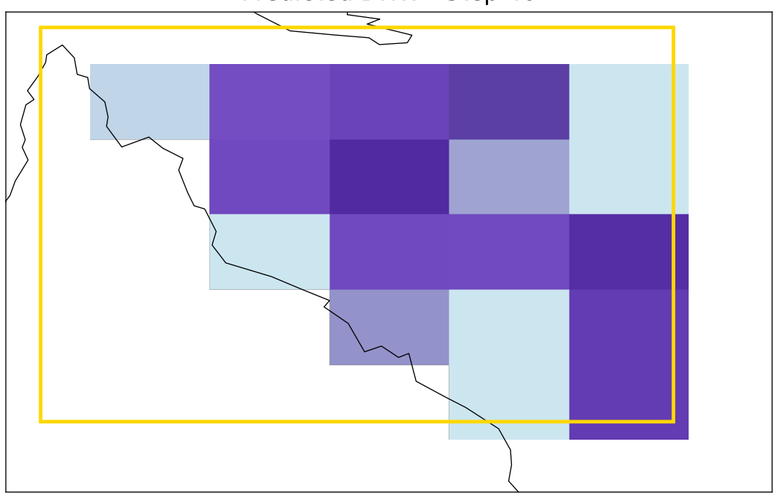
Error (Pred - Actual) - Step 1 [scale ±8]



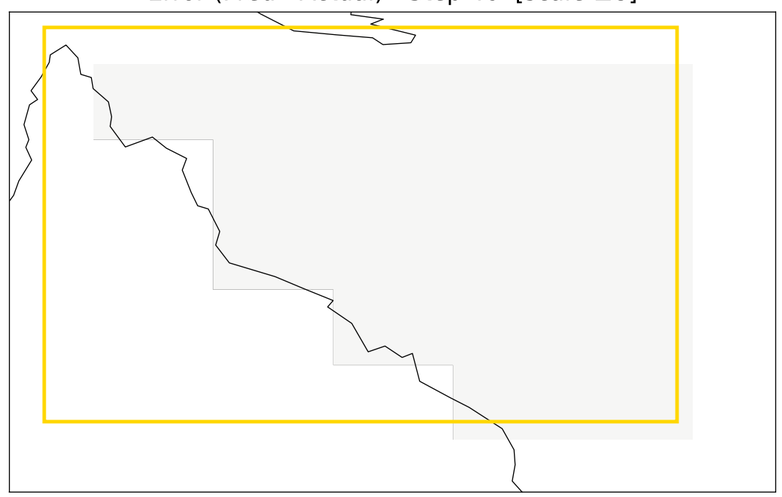
Actual DHW - Step 46



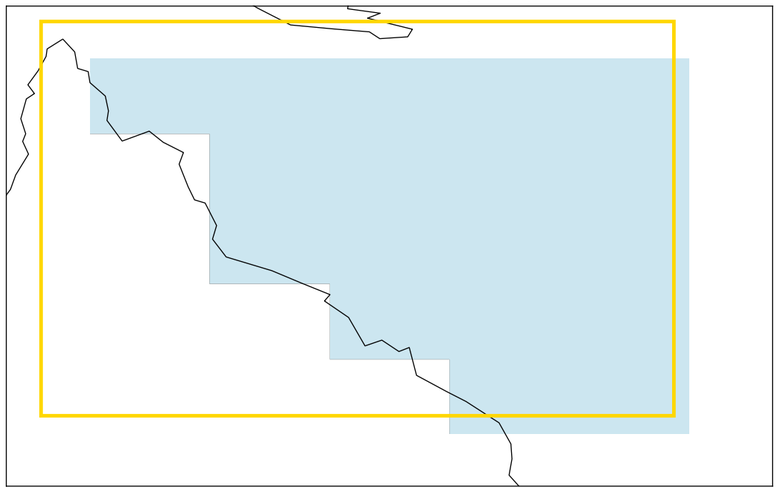
Predicted DHW - Step 46



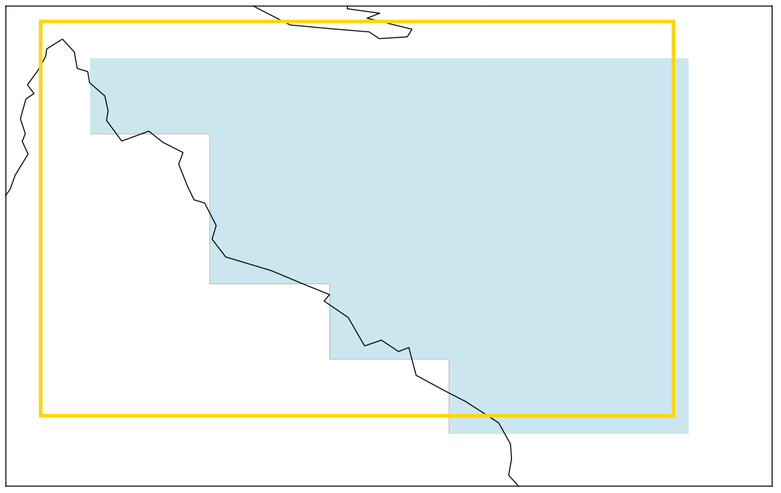
Error (Pred - Actual) - Step 46 [scale ±8]



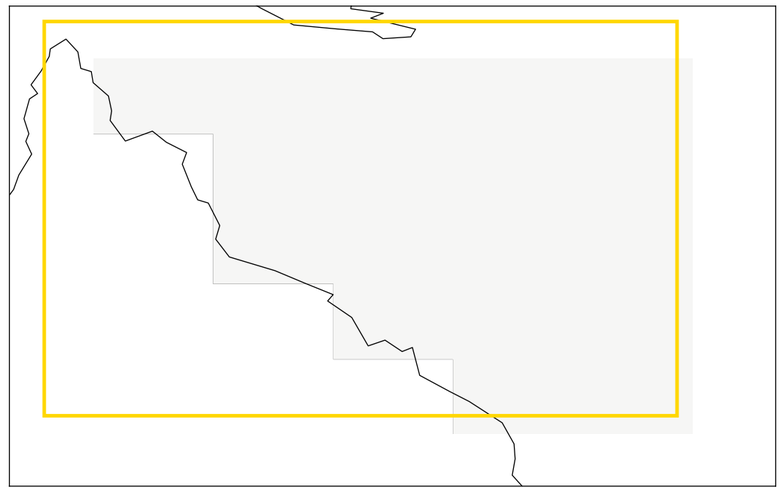
Actual DHW - Step 90

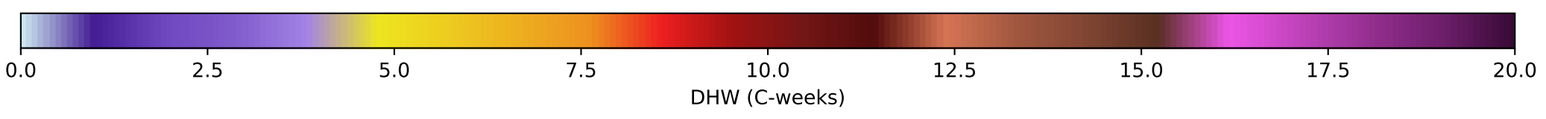


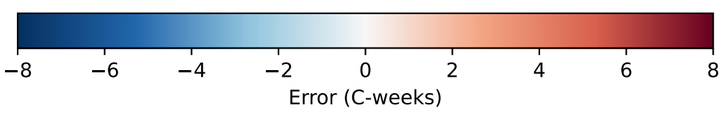
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

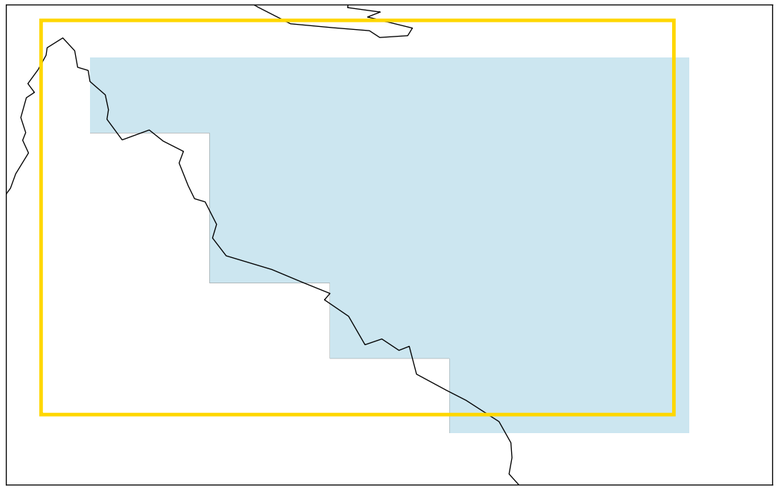




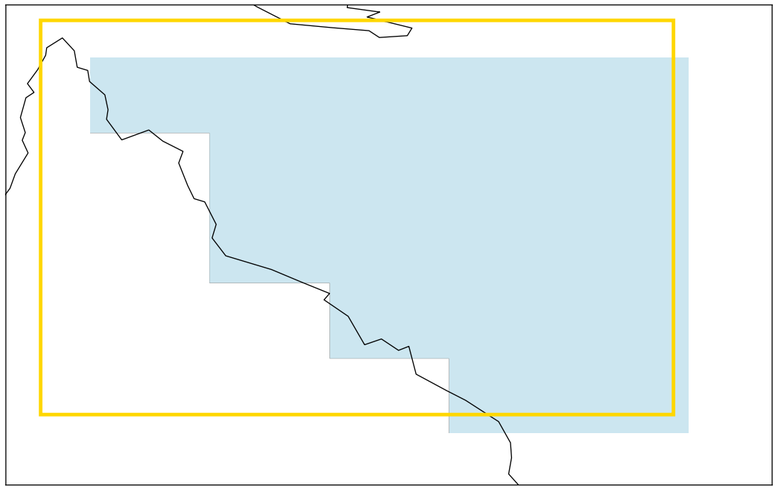


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

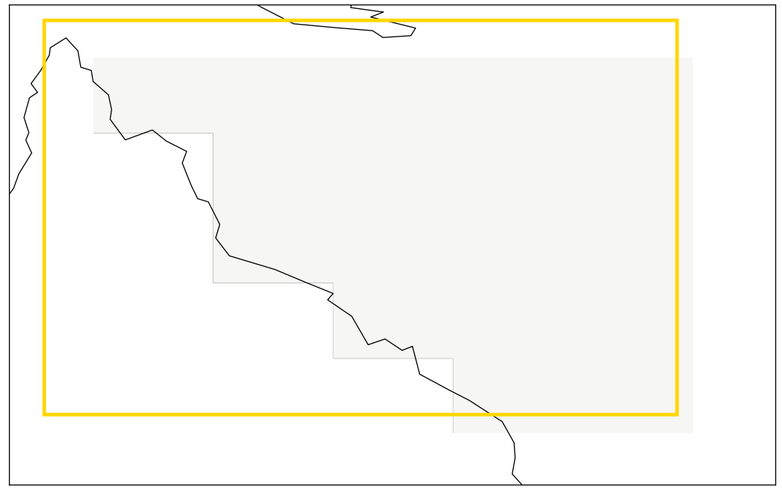
Actual DHW - Step 1



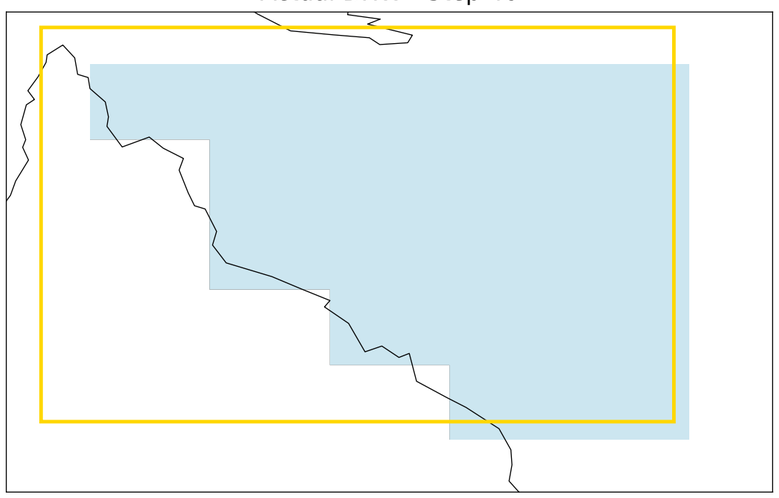
Predicted DHW - Step 1



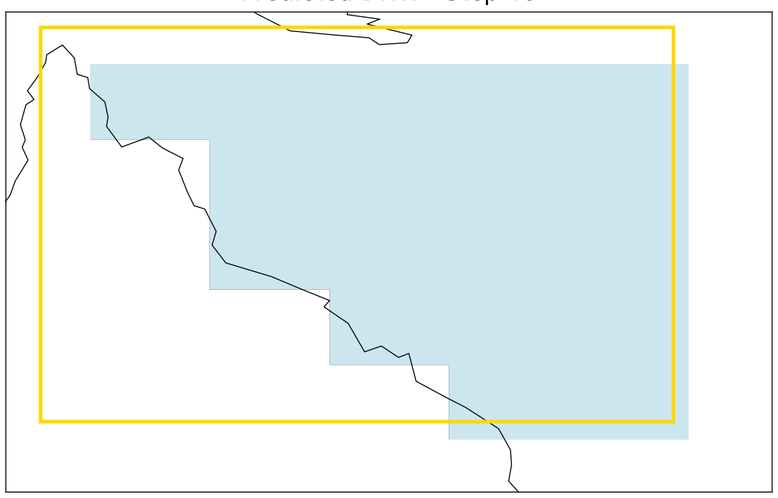
Error (Pred - Actual) - Step 1 [scale ±8]



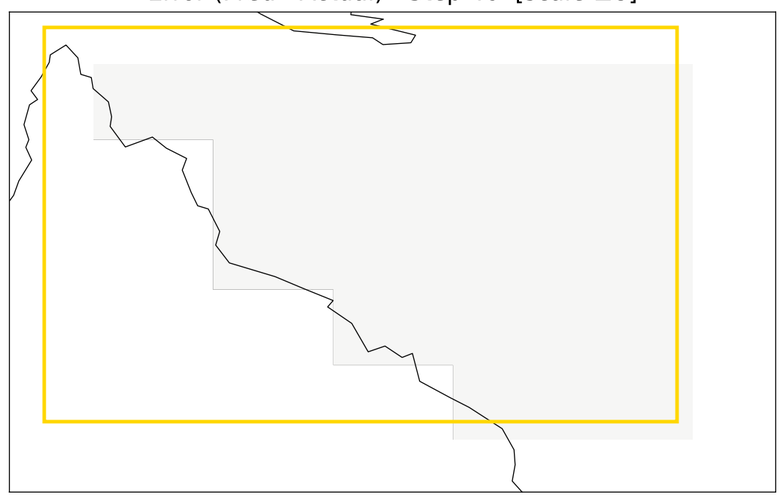
Actual DHW - Step 46



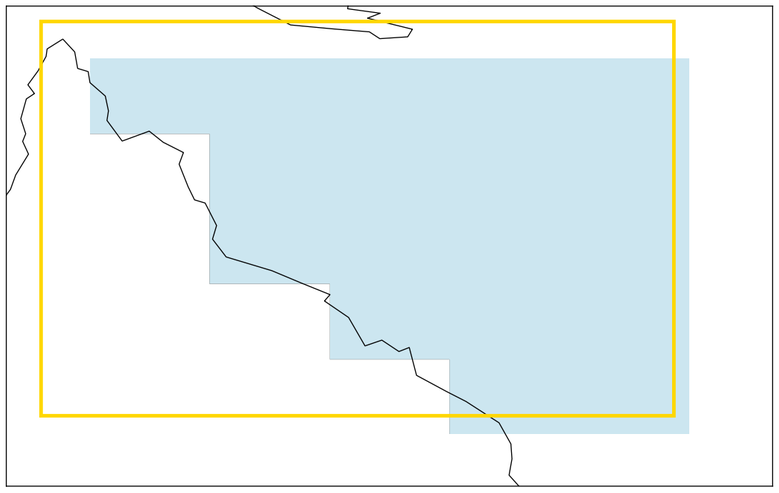
Predicted DHW - Step 46



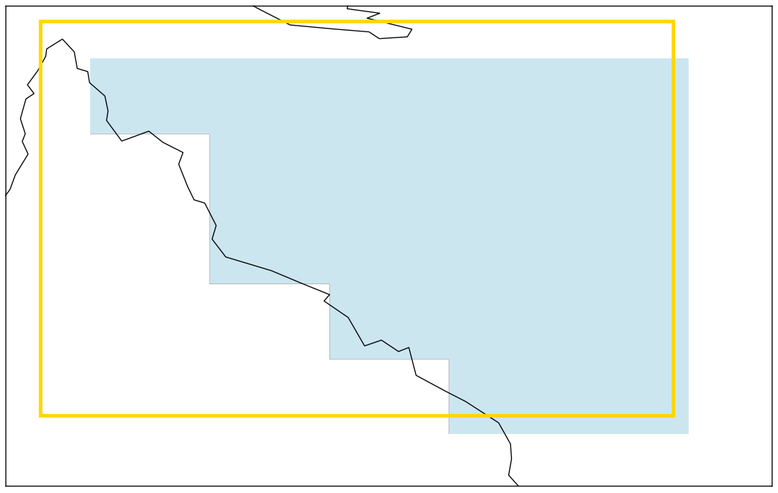
Error (Pred - Actual) - Step 46 [scale ±8]



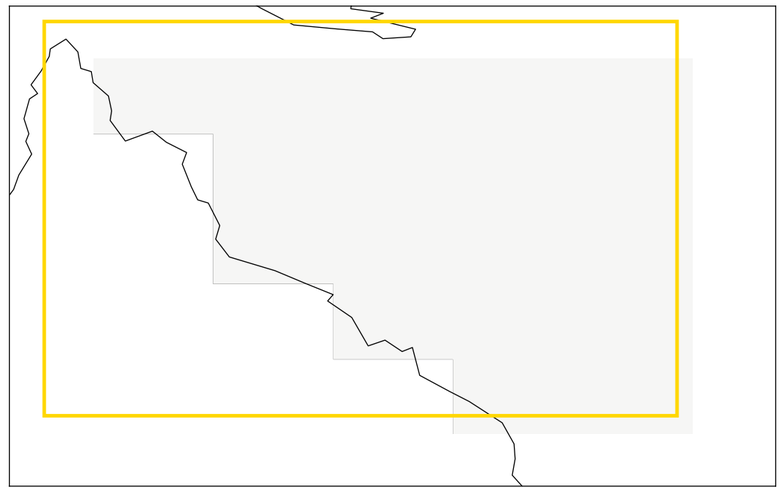
Actual DHW - Step 90

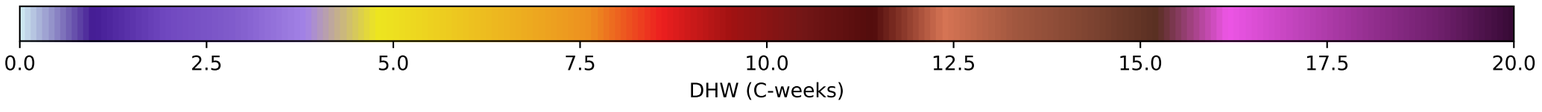


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

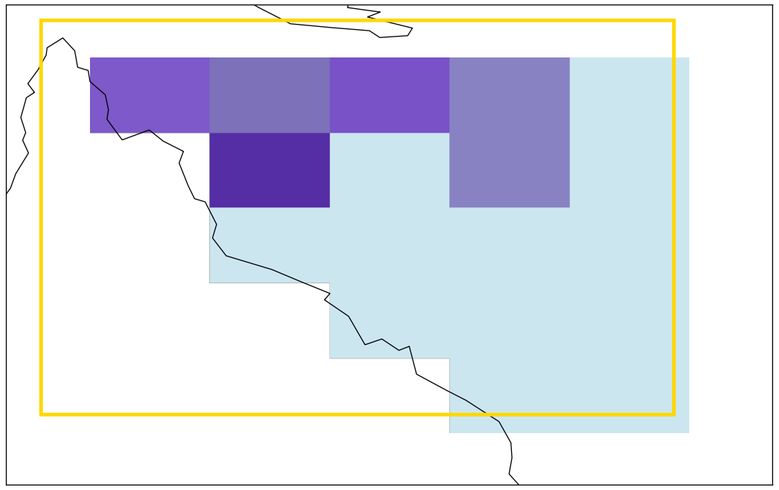




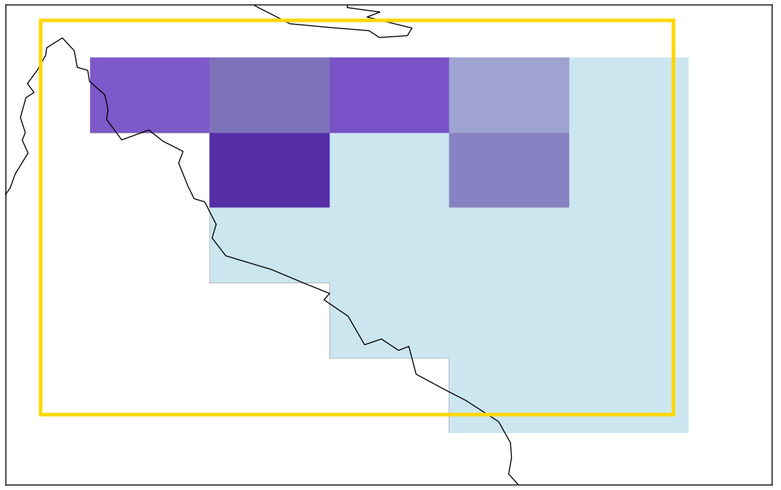


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

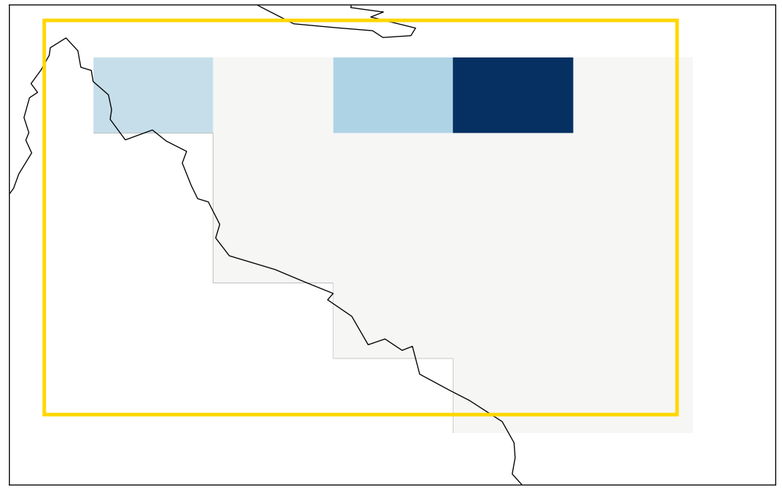
Actual DHW - Step 1



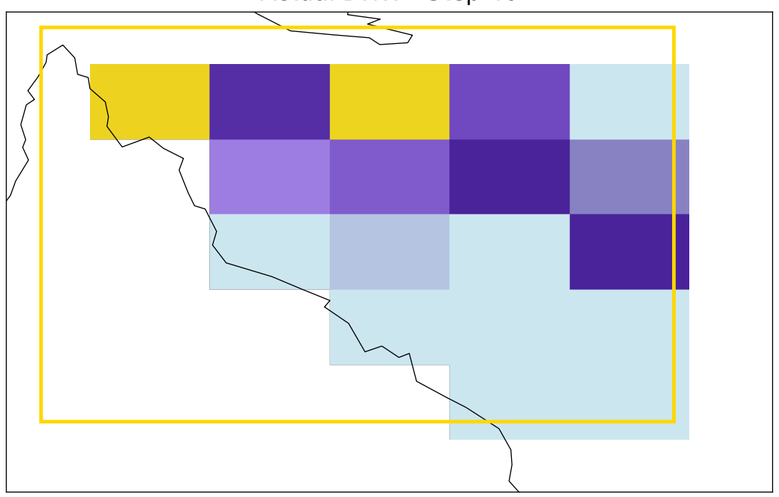
Predicted DHW - Step 1



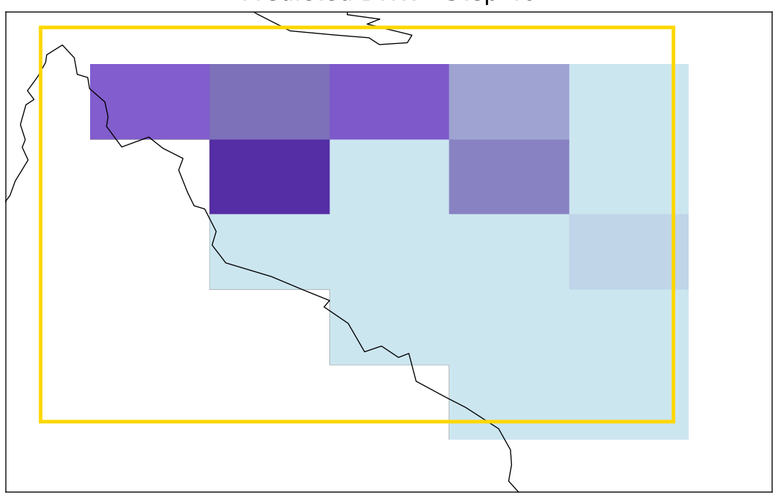
Error (Pred - Actual) - Step 1 [scale ±0.11]



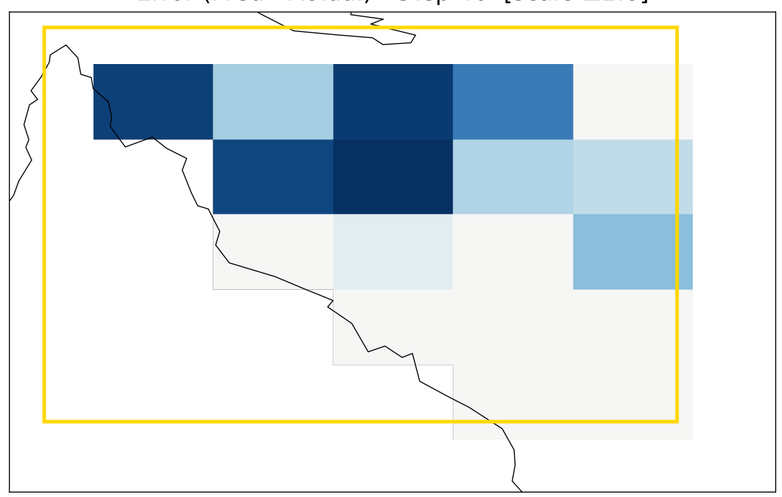
Actual DHW - Step 46



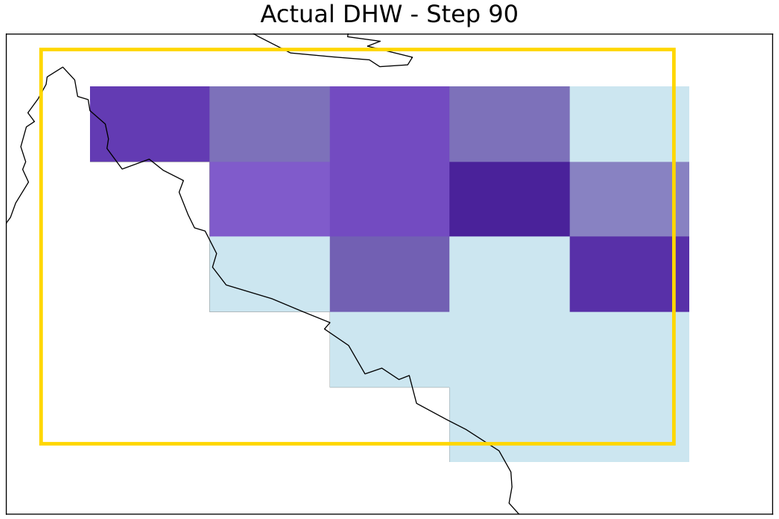
Predicted DHW - Step 46



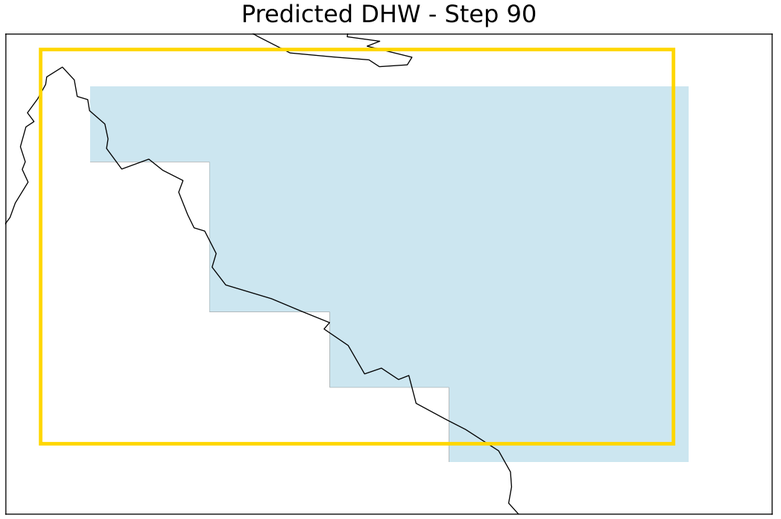
Error (Pred - Actual) - Step 46 [scale ±2.8]



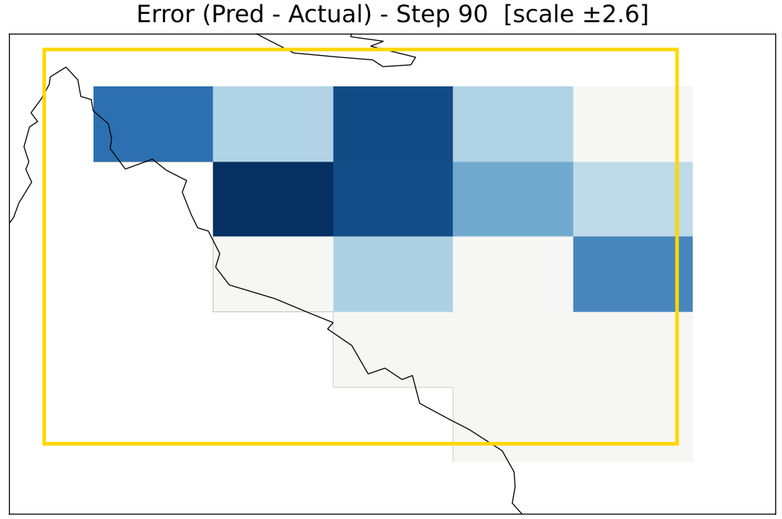
Actual DHW - Step 90

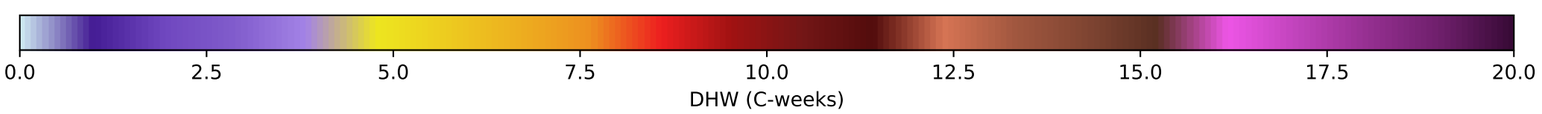


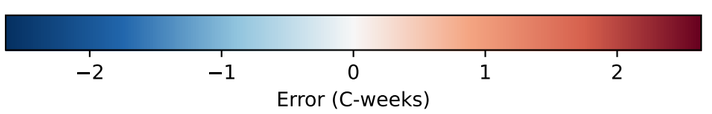
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

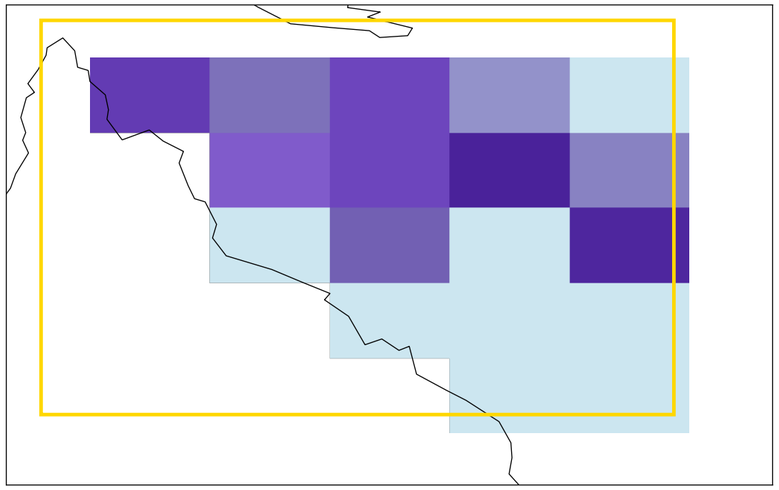




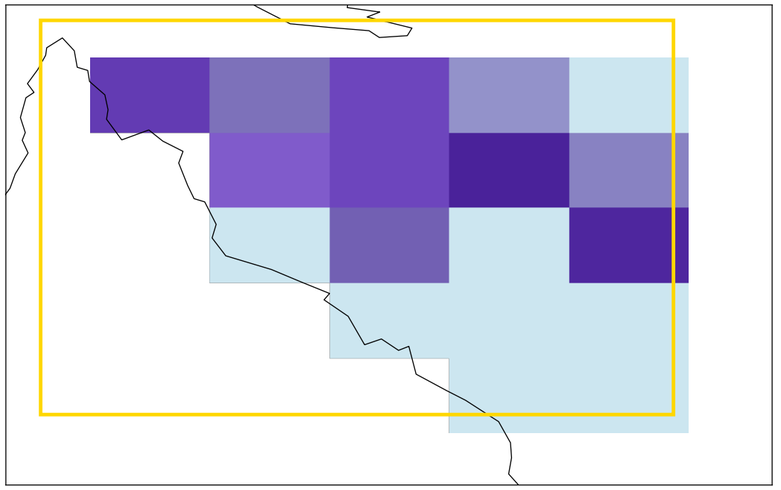


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

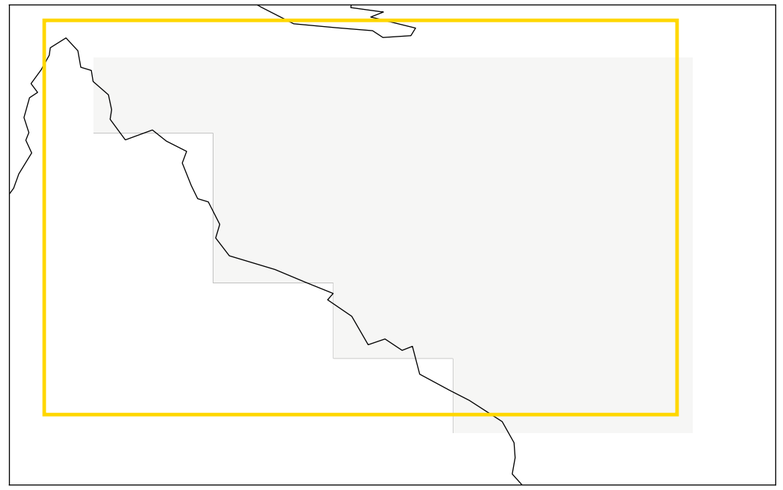
Actual DHW - Step 1



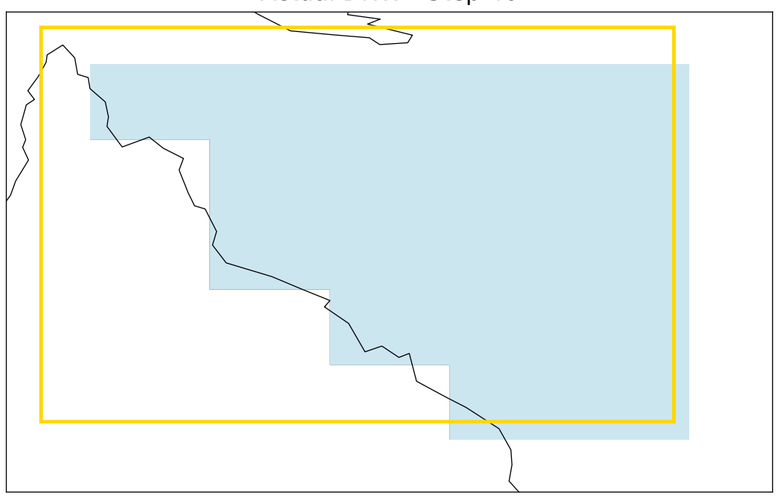
Predicted DHW - Step 1



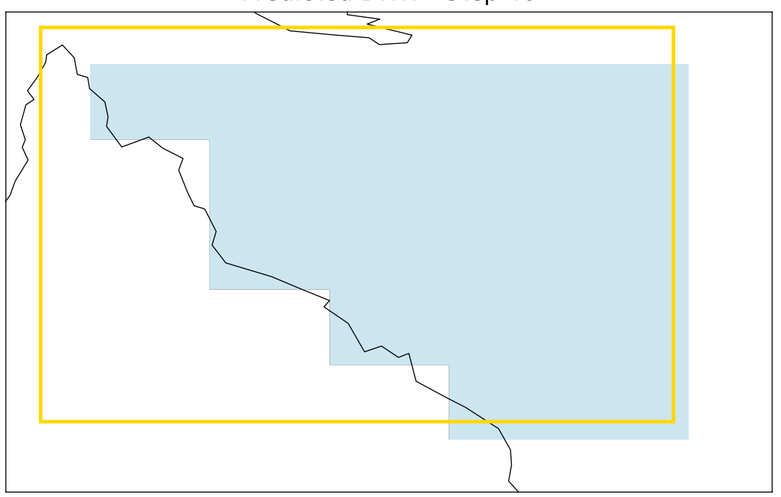
Error (Pred - Actual) - Step 1 [scale ±8]



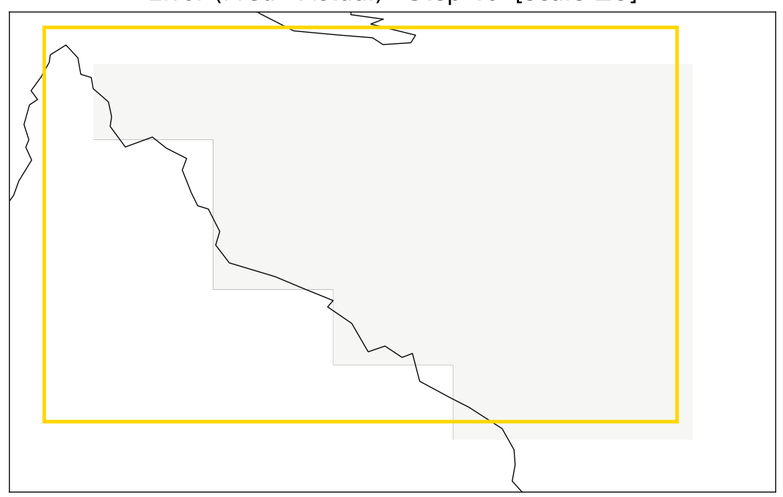
Actual DHW - Step 46



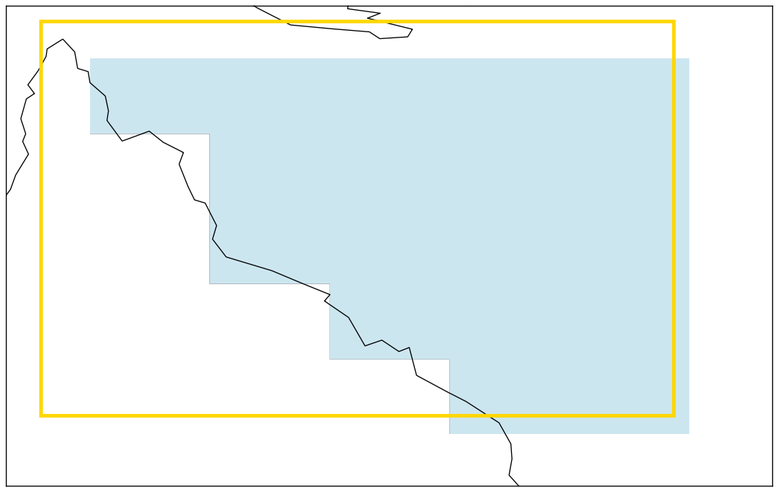
Predicted DHW - Step 46



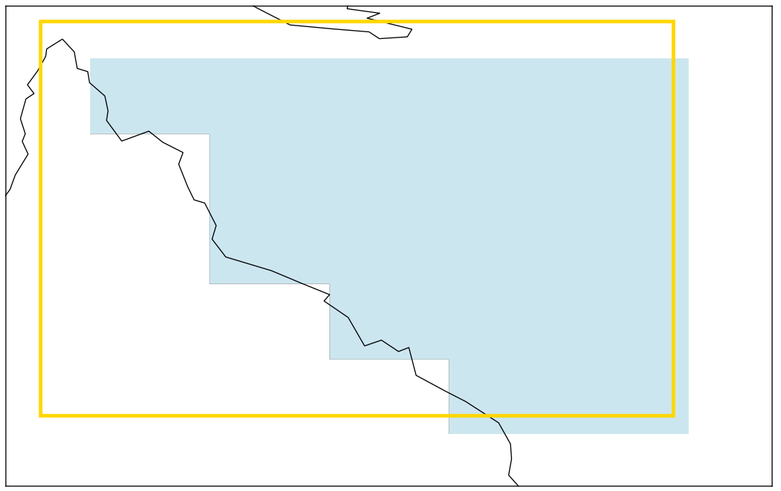
Error (Pred - Actual) - Step 46 [scale ±8]



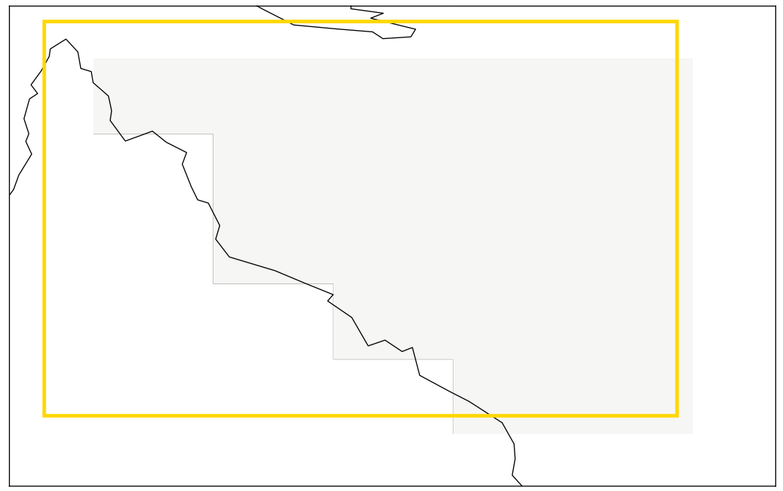
Actual DHW - Step 90

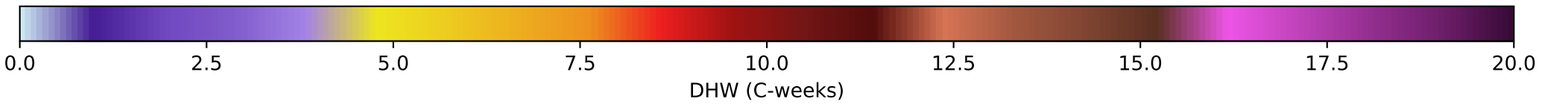


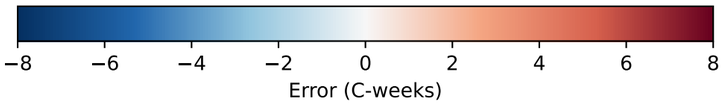
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

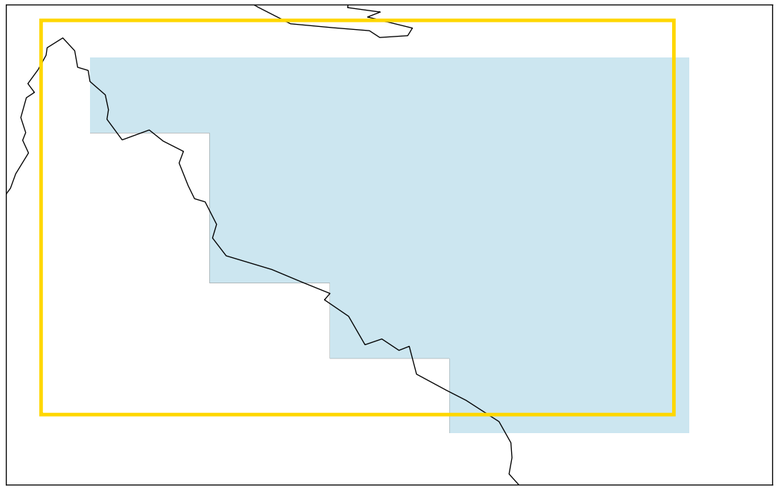




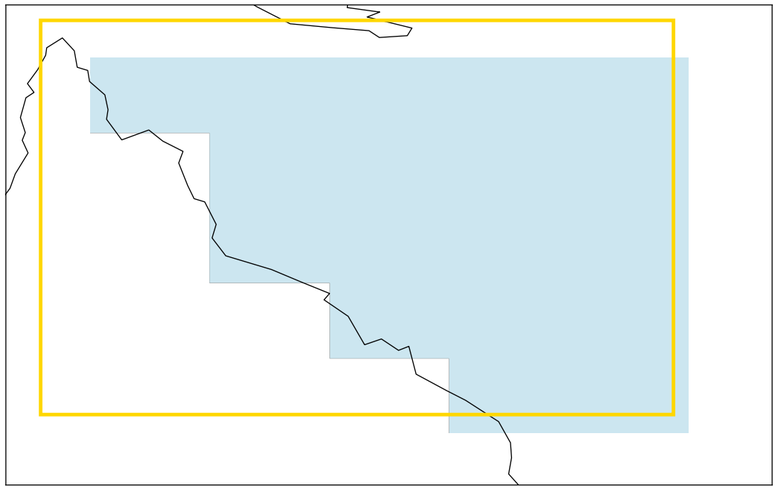


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

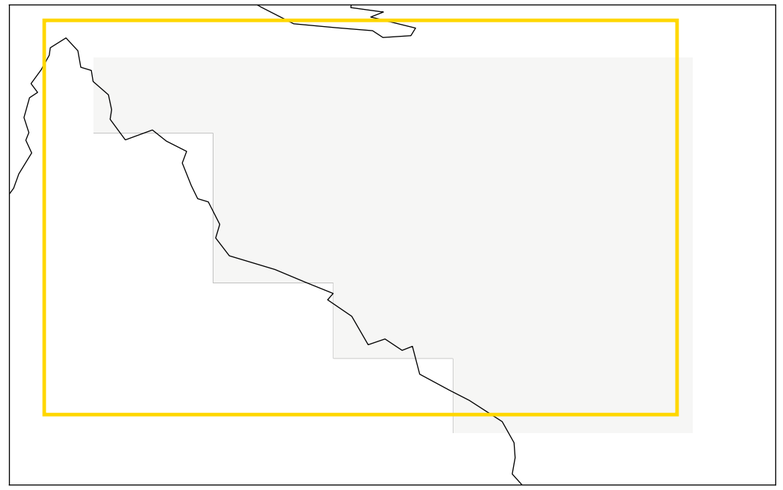
Actual DHW - Step 1



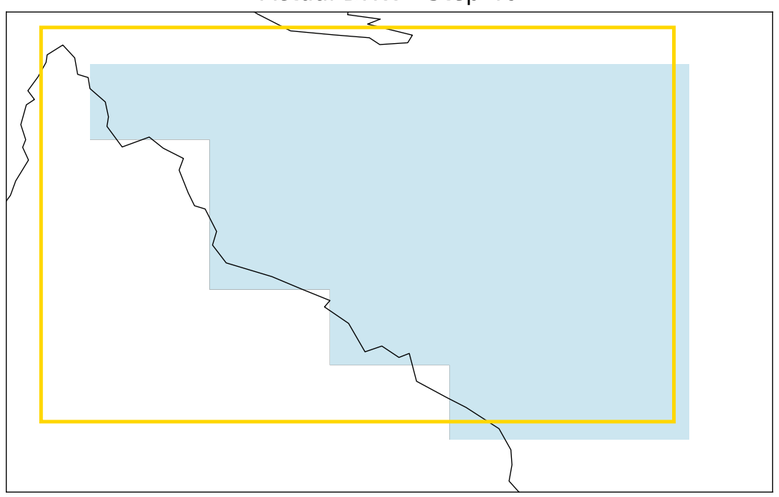
Predicted DHW - Step 1



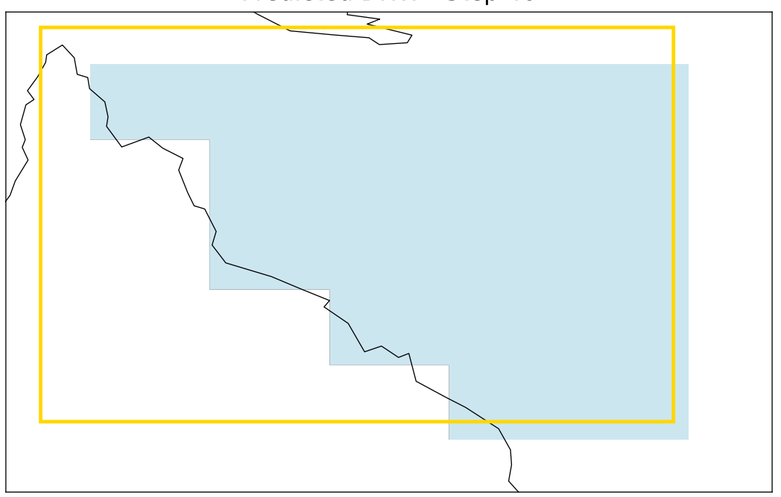
Error (Pred - Actual) - Step 1 [scale ±8]



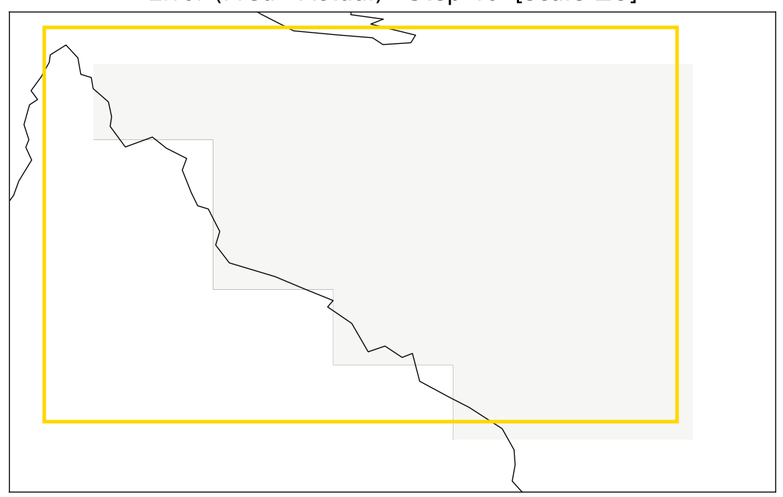
Actual DHW - Step 46



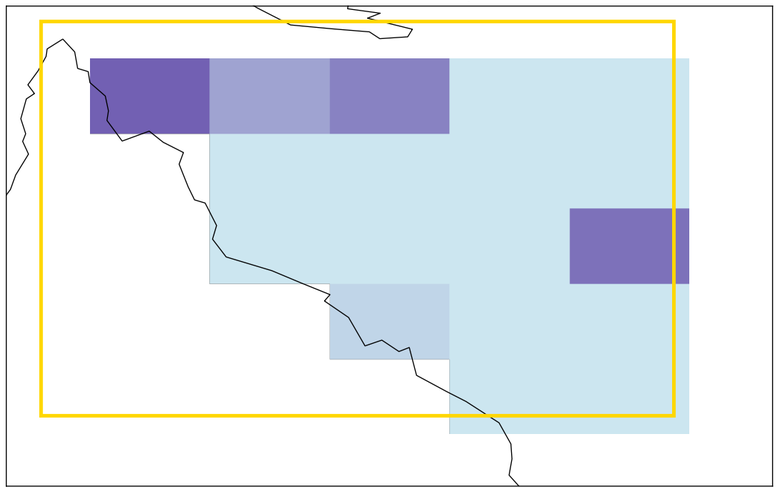
Predicted DHW - Step 46



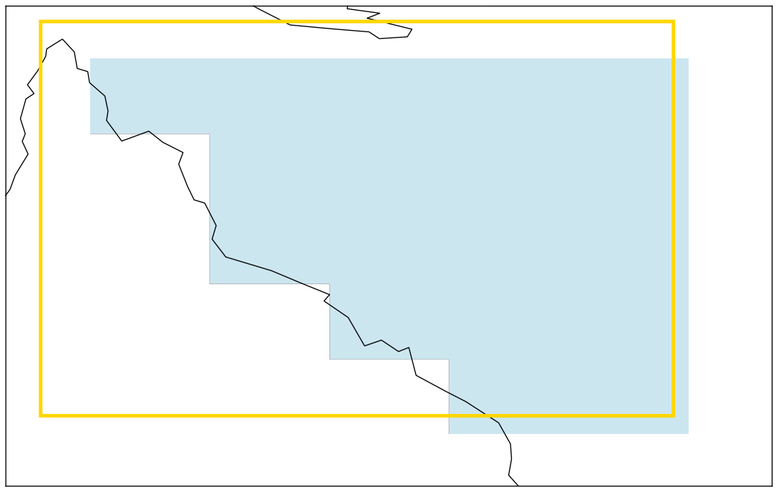
Error (Pred - Actual) - Step 46 [scale ±8]



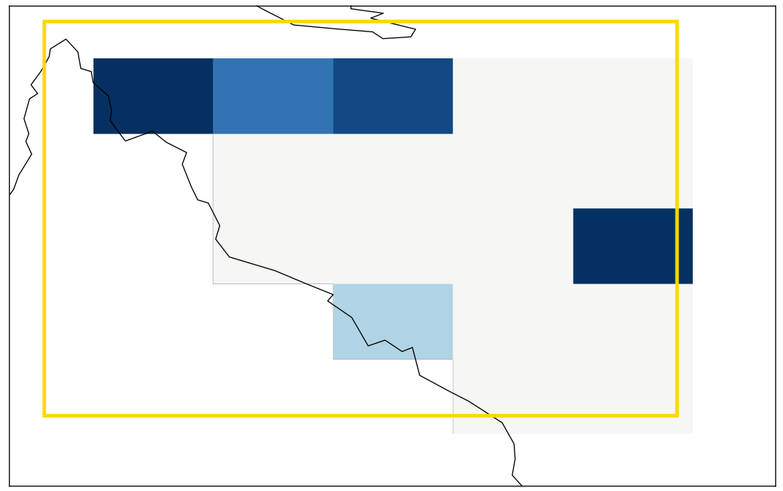
Actual DHW - Step 90

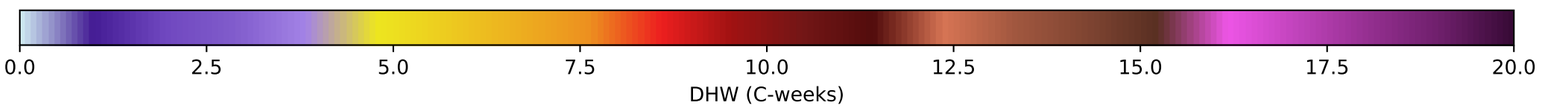


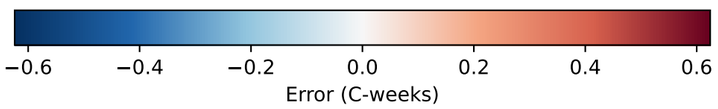
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

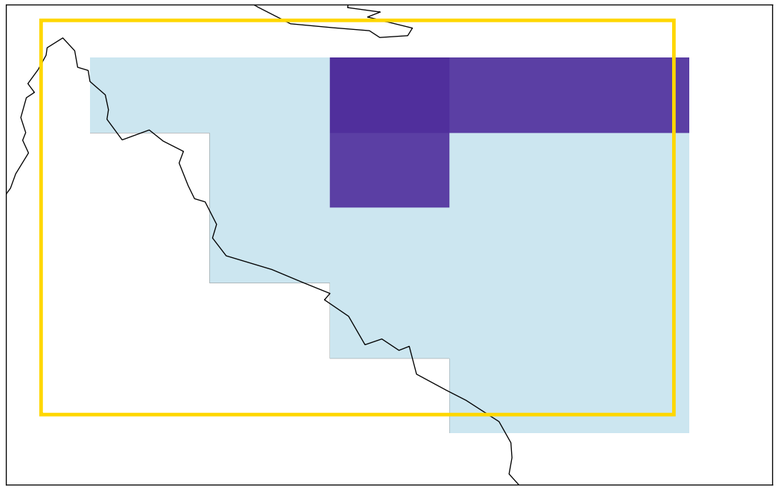




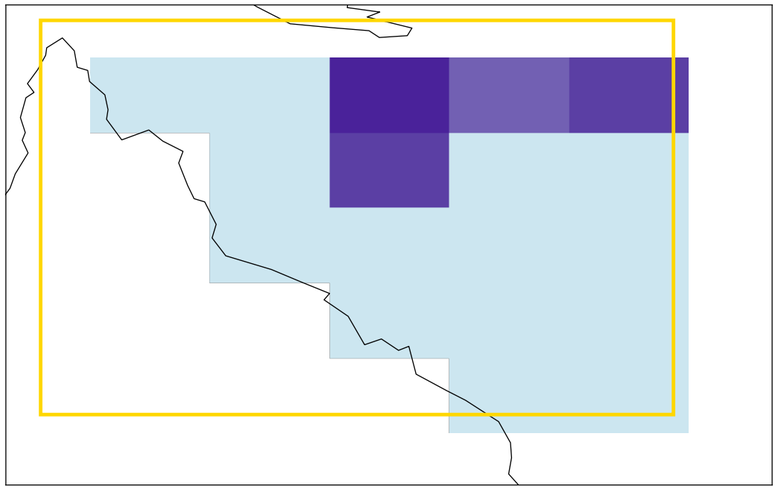


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

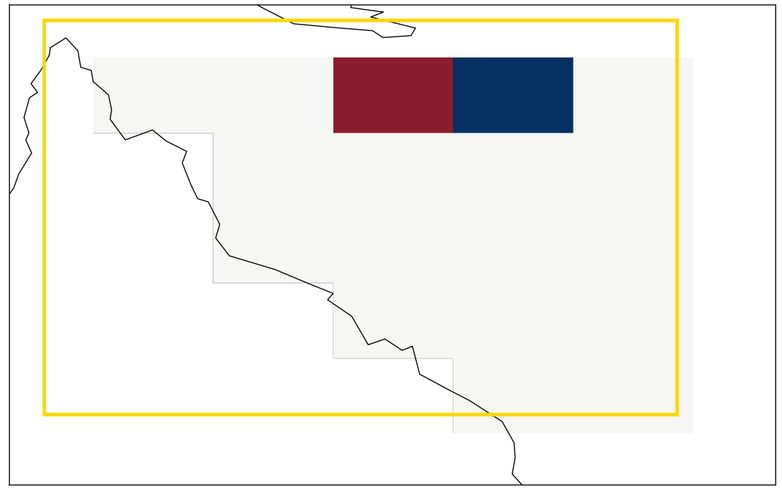
Actual DHW - Step 1



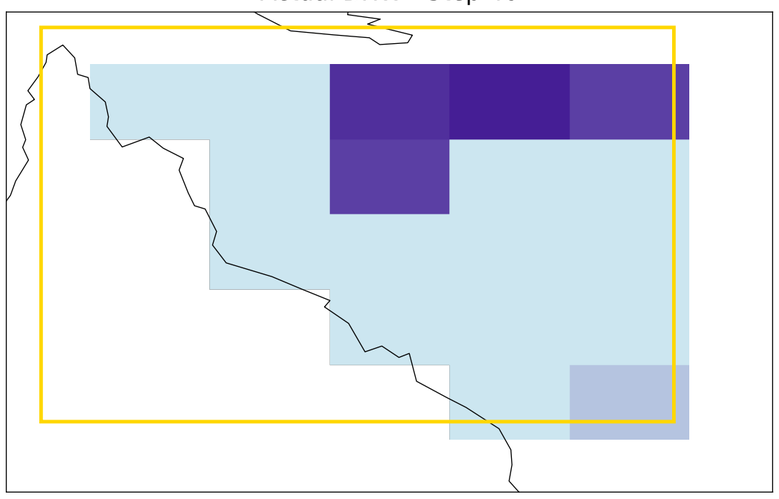
Predicted DHW - Step 1



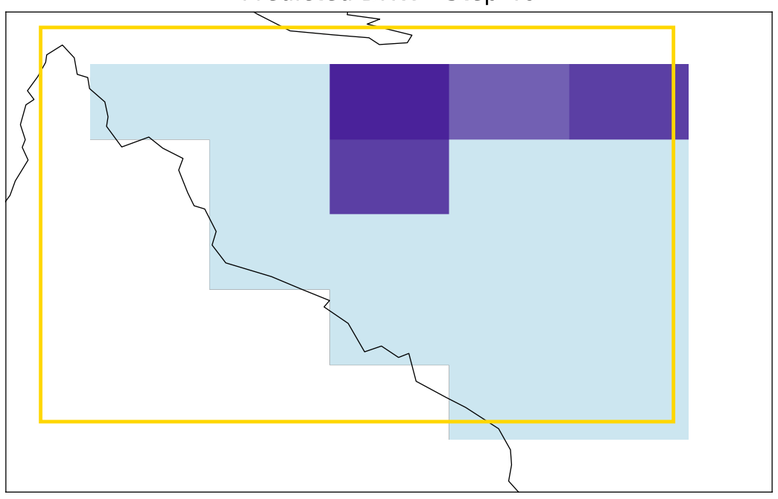
Error (Pred - Actual) - Step 1 [scale ±0.19]



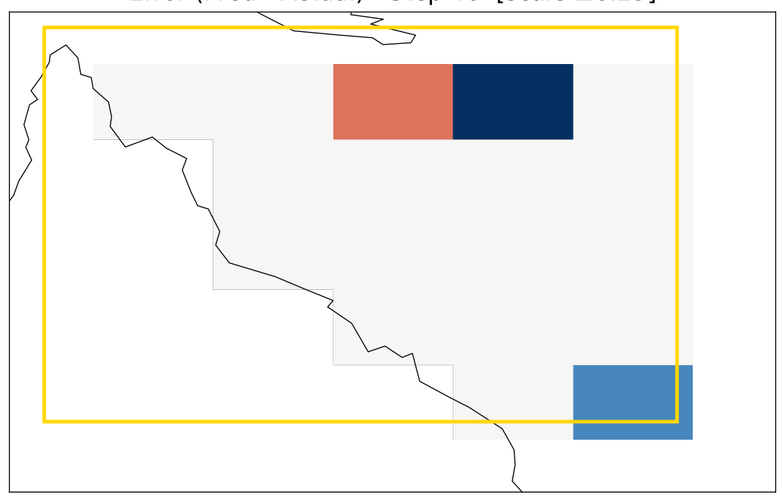
Actual DHW - Step 46



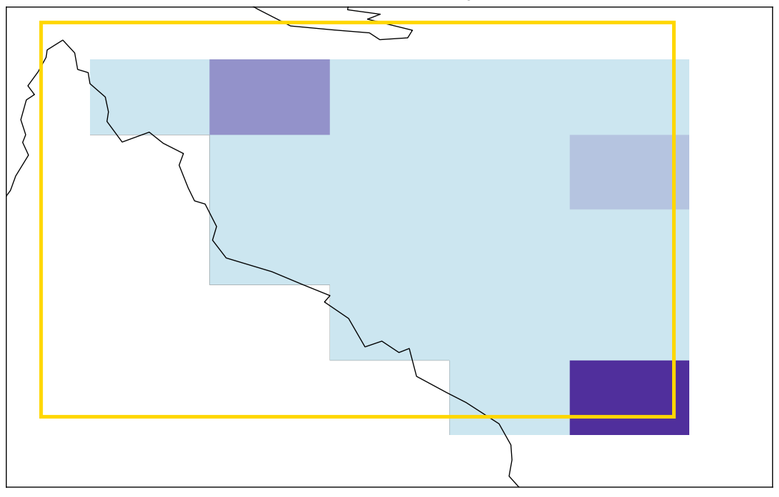
Predicted DHW - Step 46



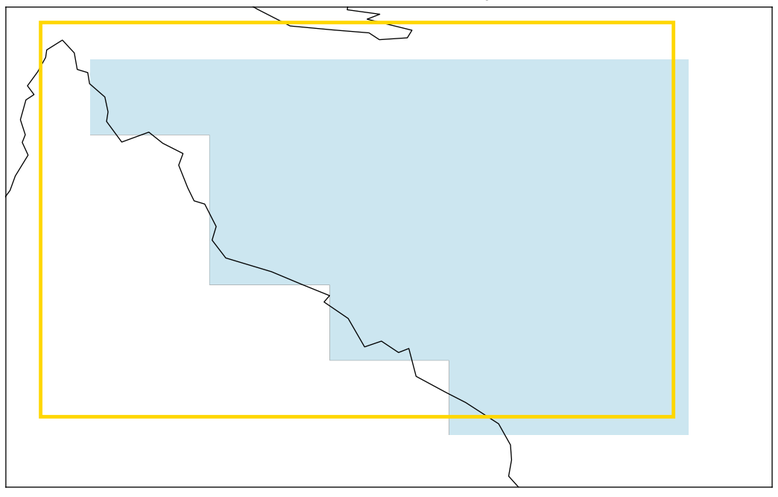
Error (Pred - Actual) - Step 46 [scale ±0.29]



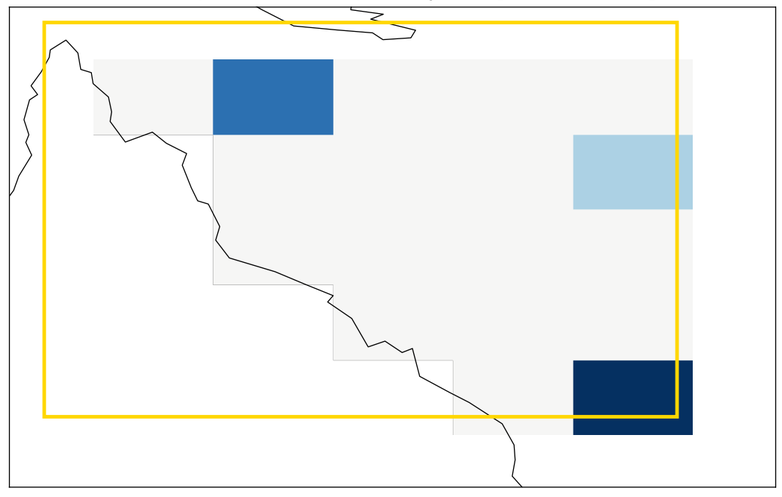
Actual DHW - Step 90

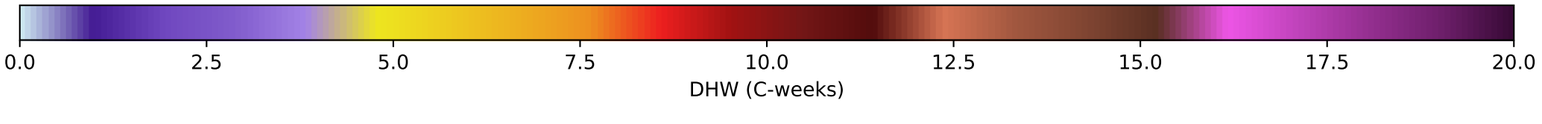


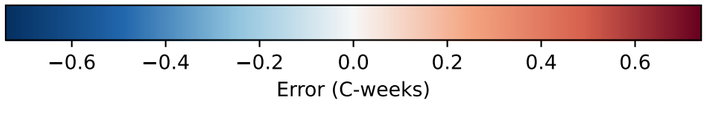
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.74]

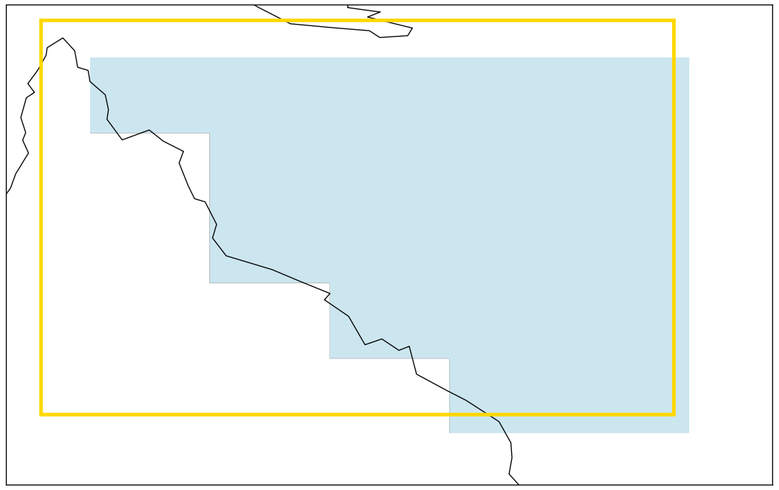




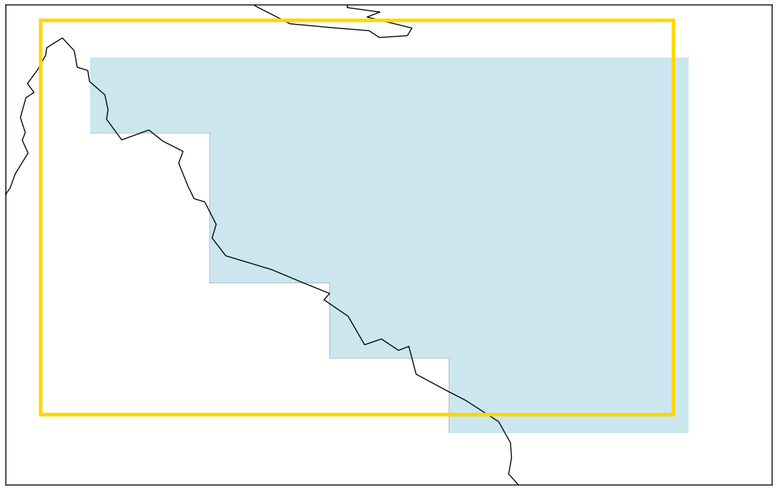


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

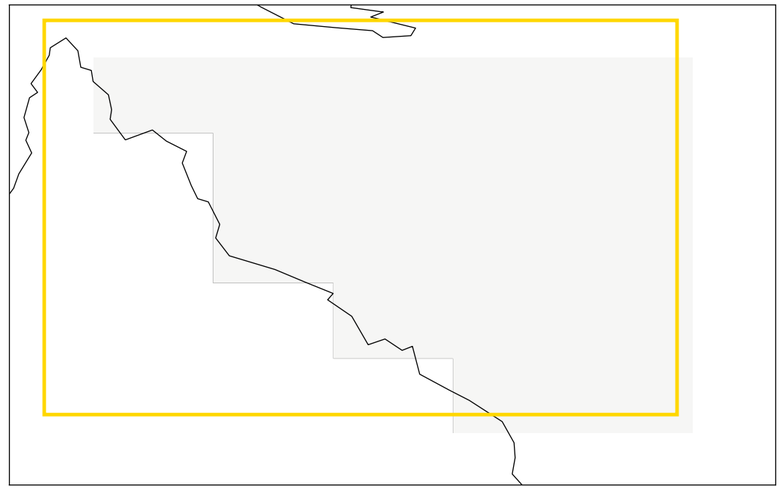
Actual DHW - Step 1



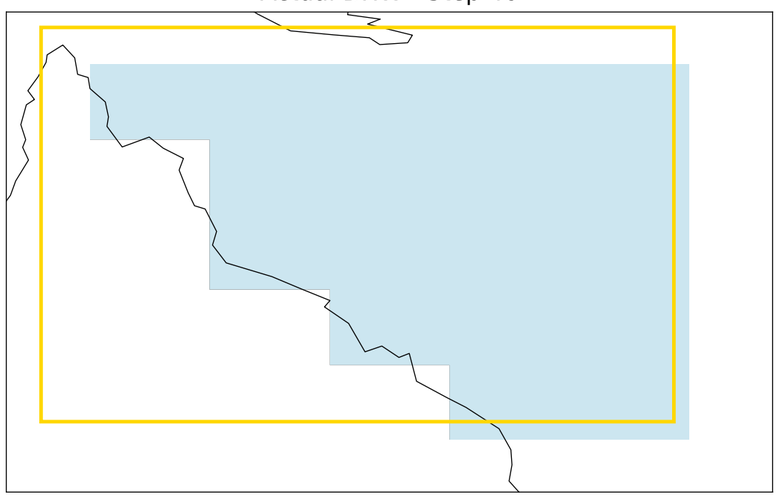
Predicted DHW - Step 1



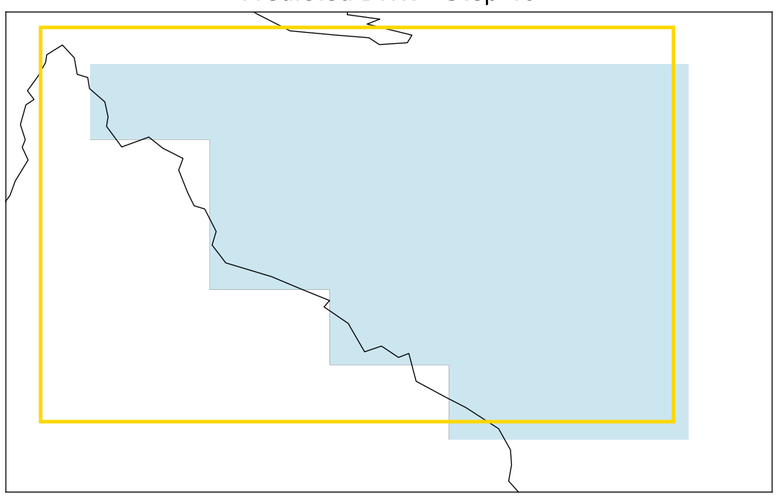
Error (Pred - Actual) - Step 1 [scale ±8]



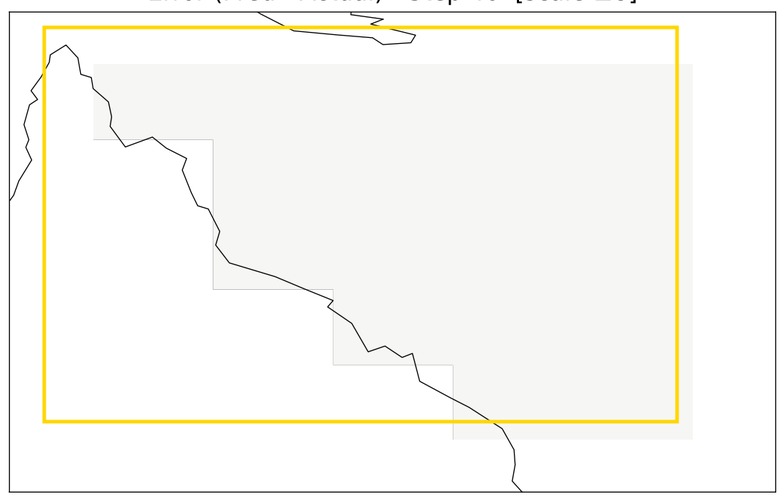
Actual DHW - Step 46



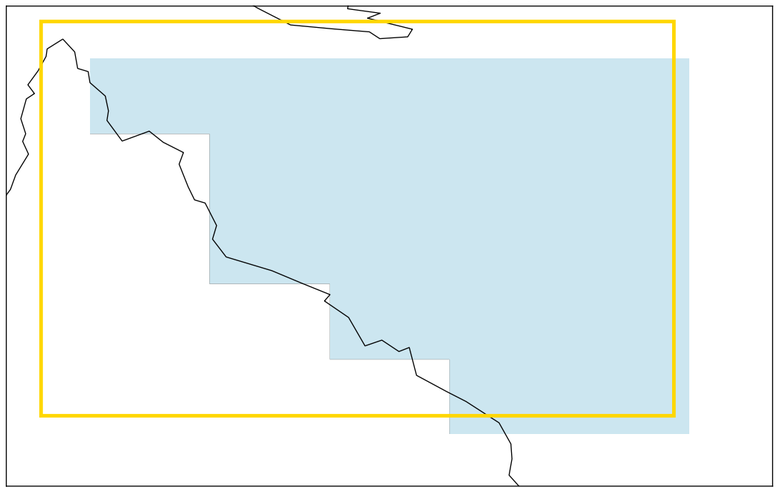
Predicted DHW - Step 46



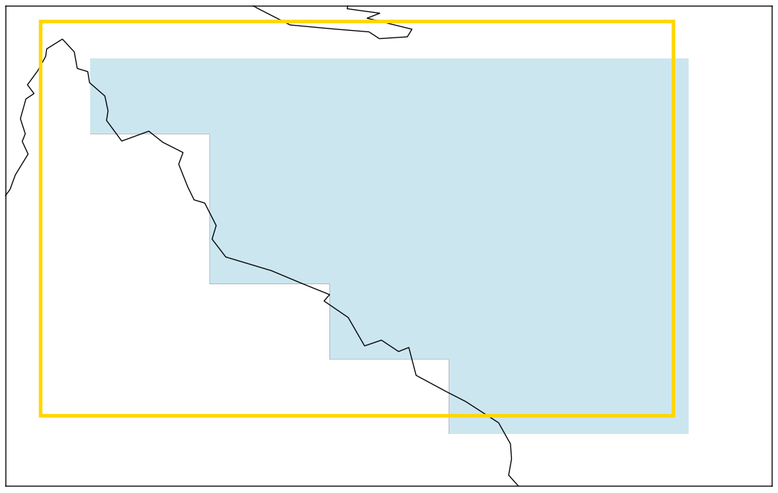
Error (Pred - Actual) - Step 46 [scale ±8]



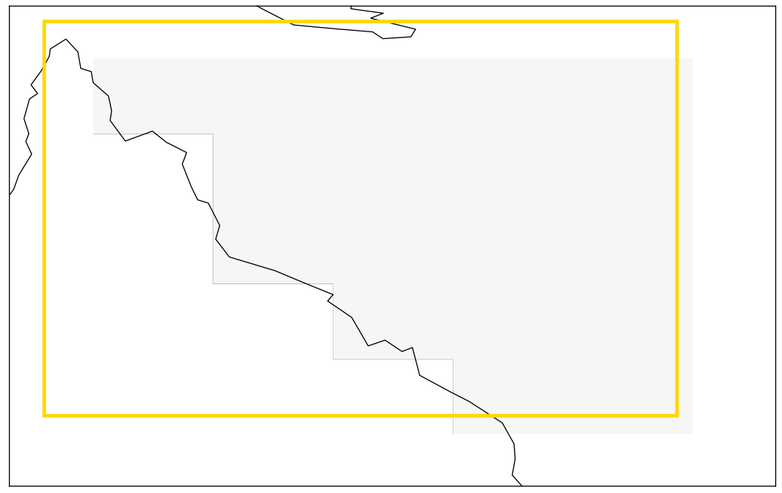
Actual DHW - Step 90

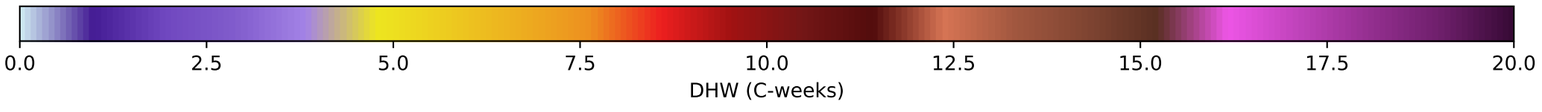


Predicted DHW - Step 90



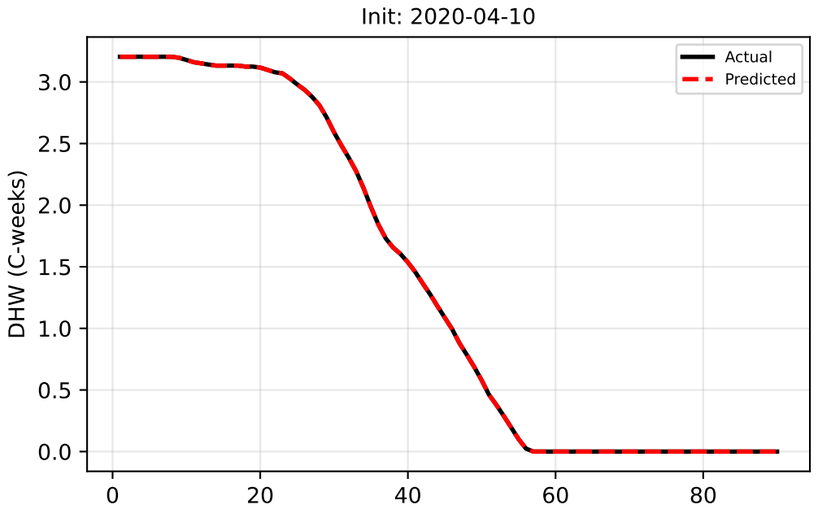
Error (Pred - Actual) - Step 90 [scale ±8]

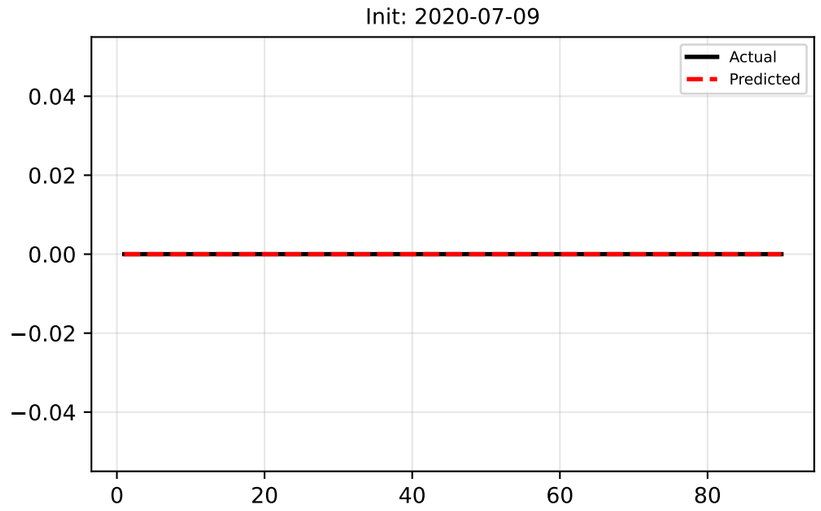


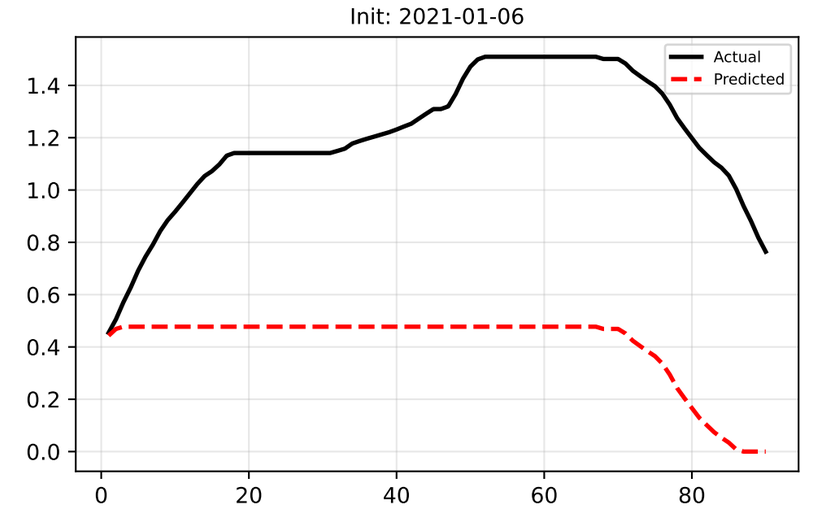


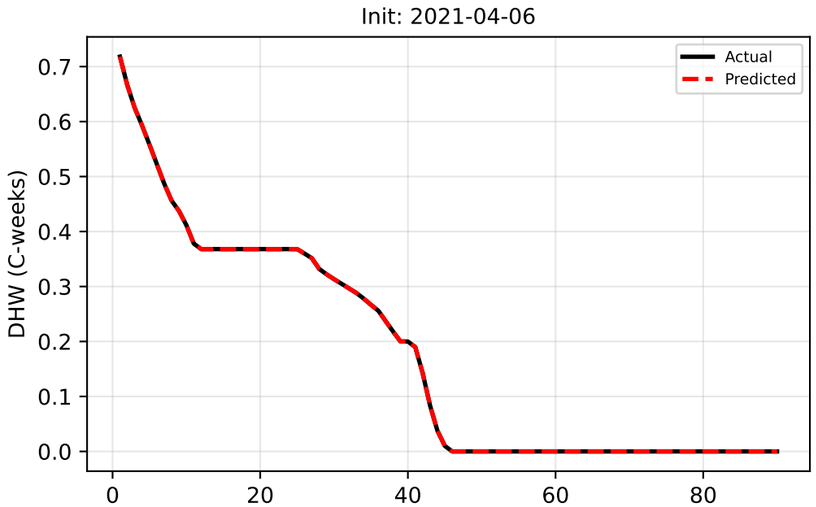


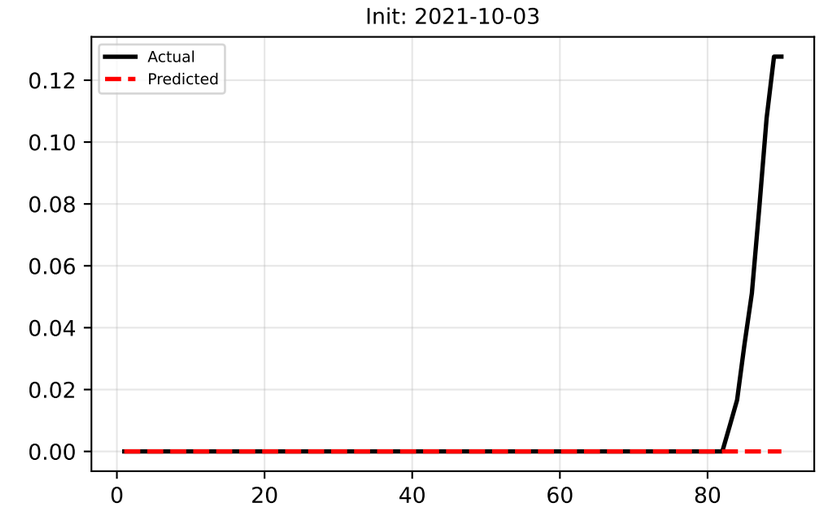
Global-Mean DHW - Test - GBR (C-weeks)

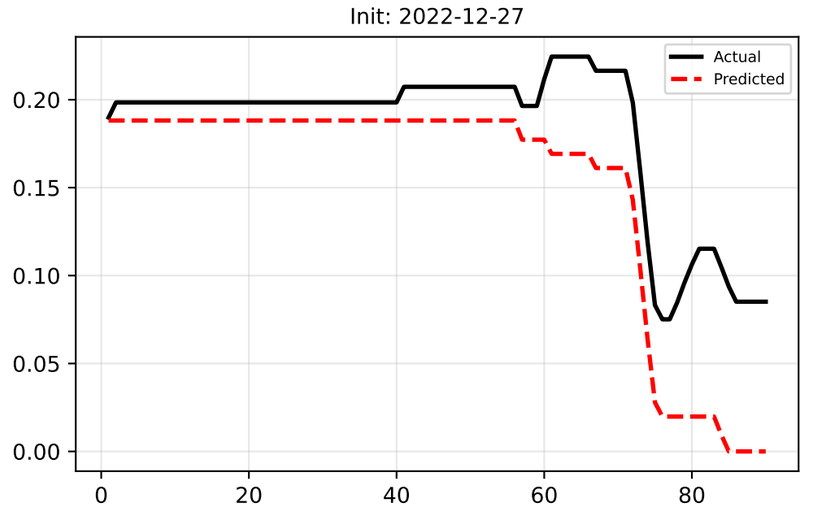


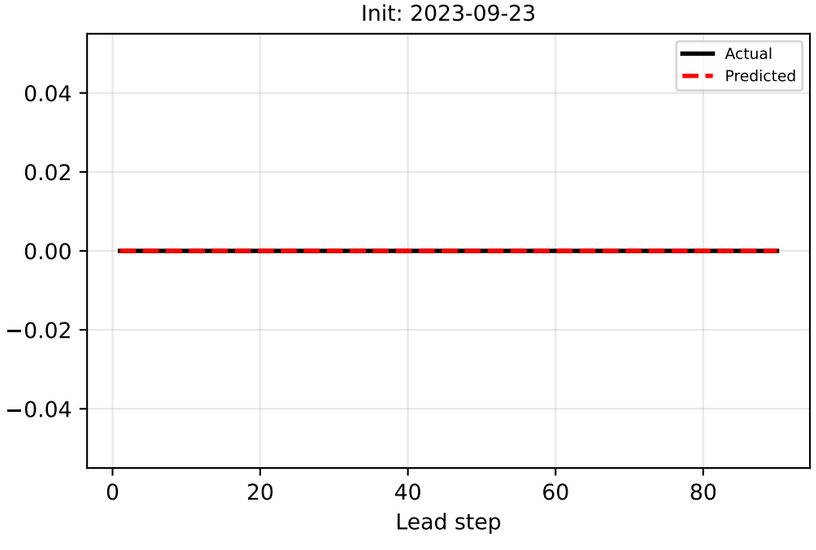


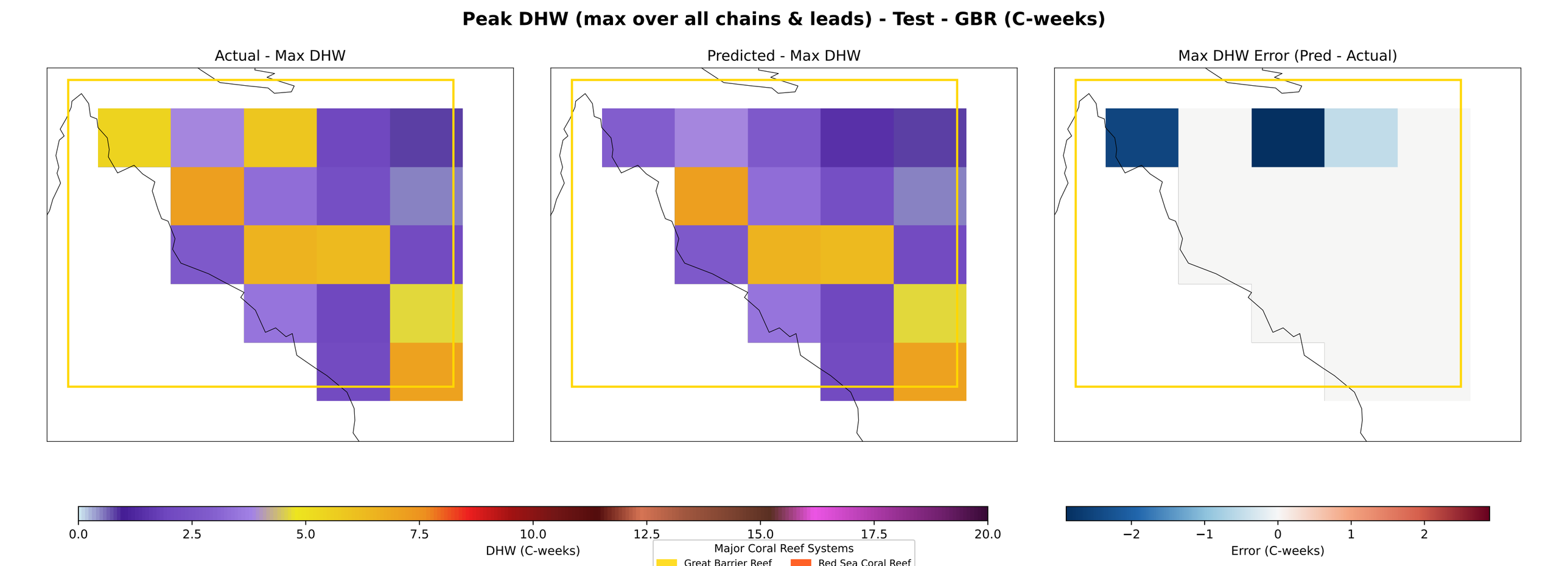


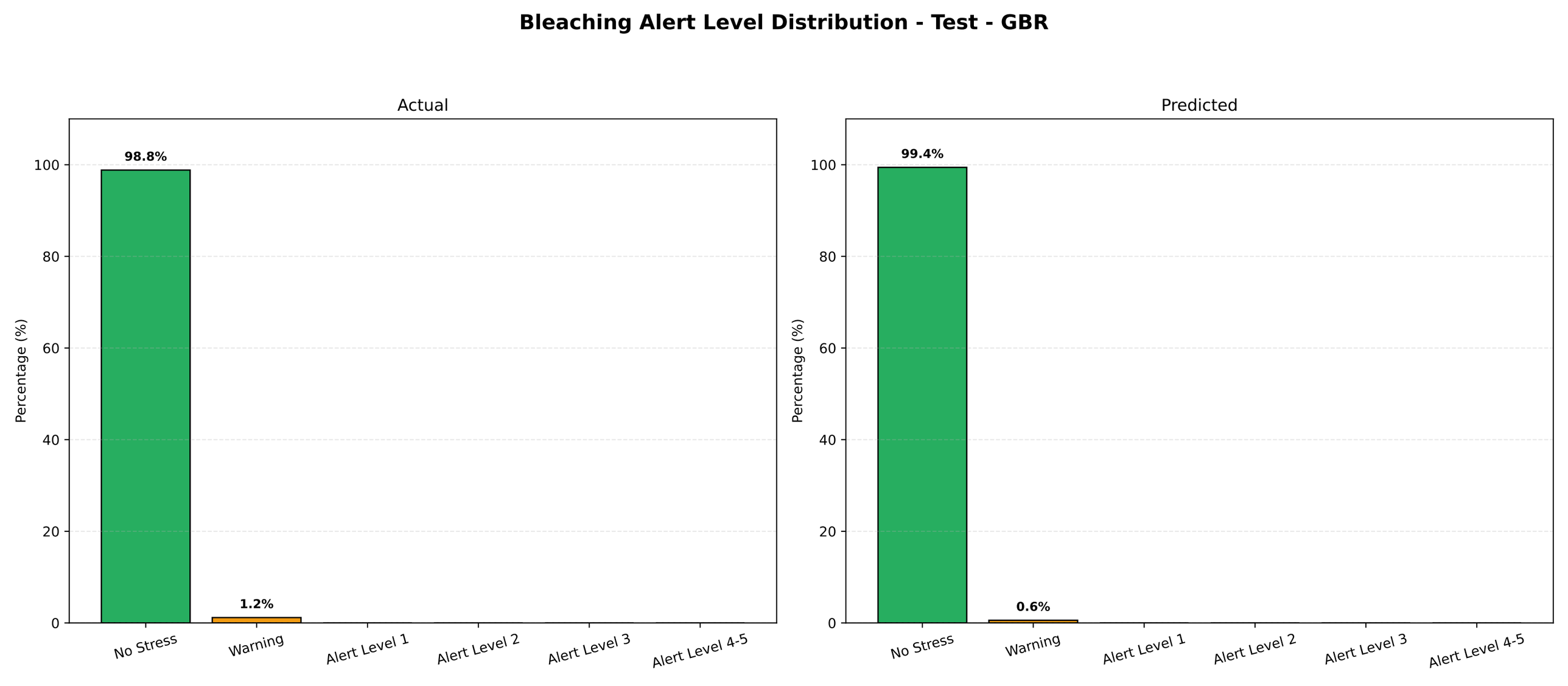




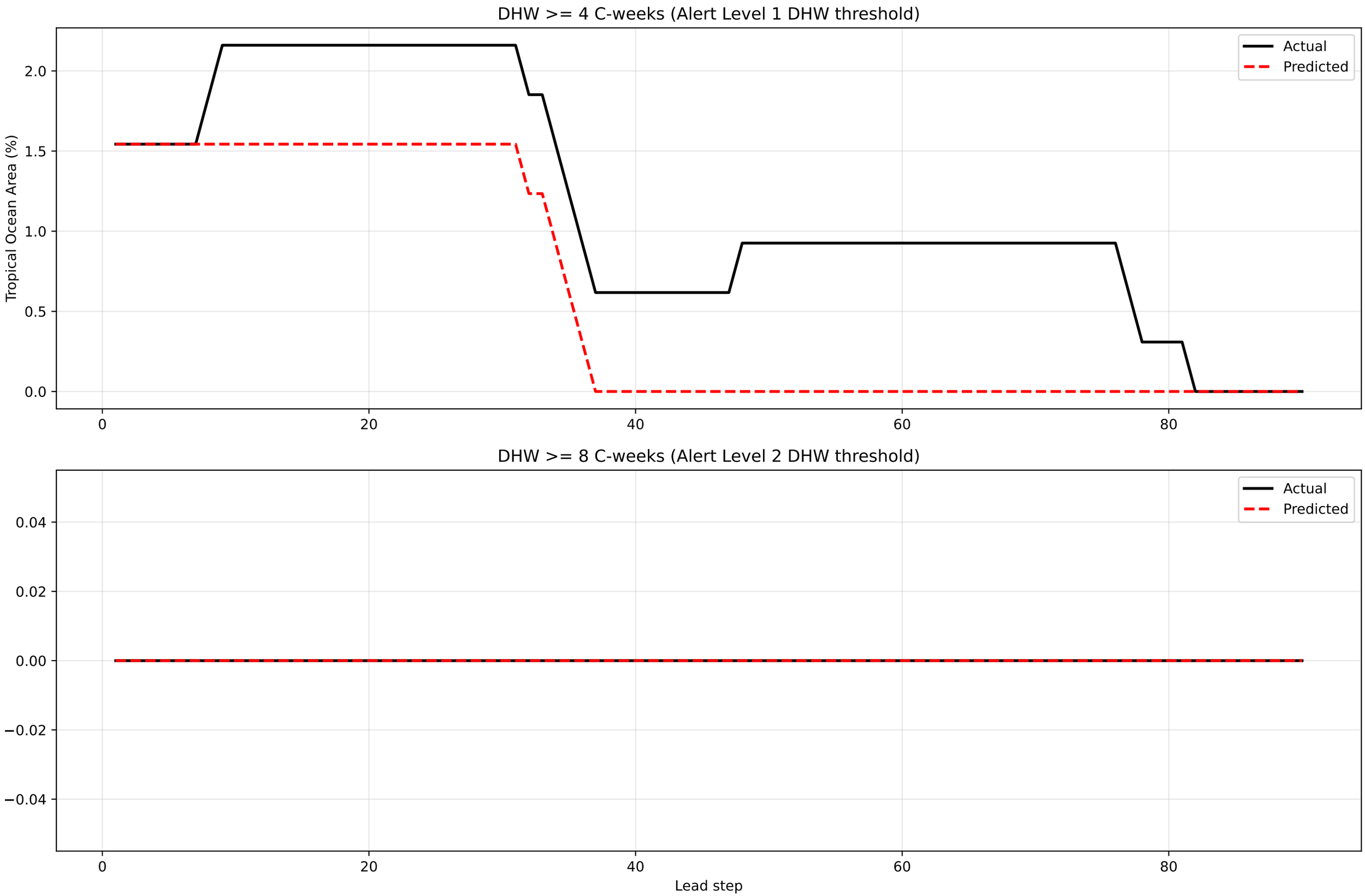





Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−2
−1
0
1
2
Error (C-weeks)


Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.8%
1.2%
99.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

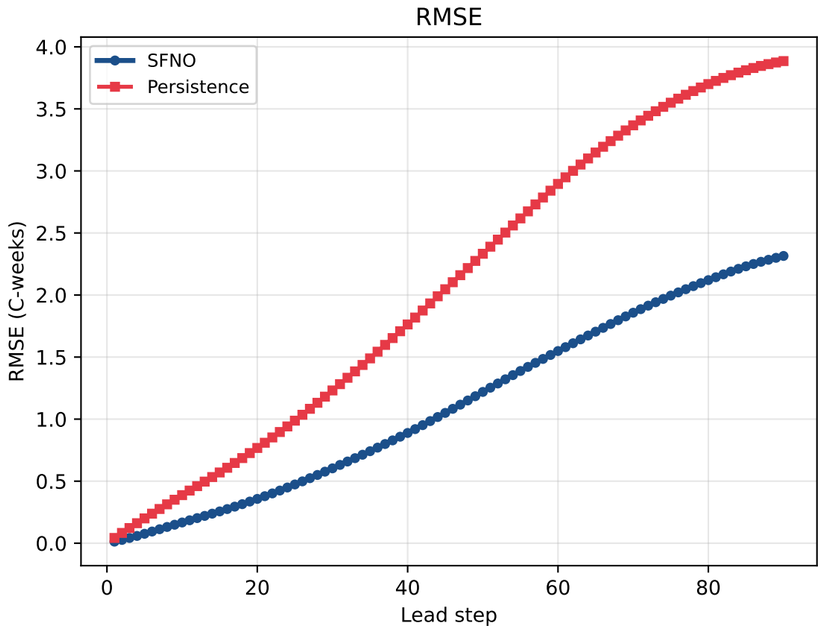
Tropical Ocean Area Under Bleaching Stress - Test - GBR

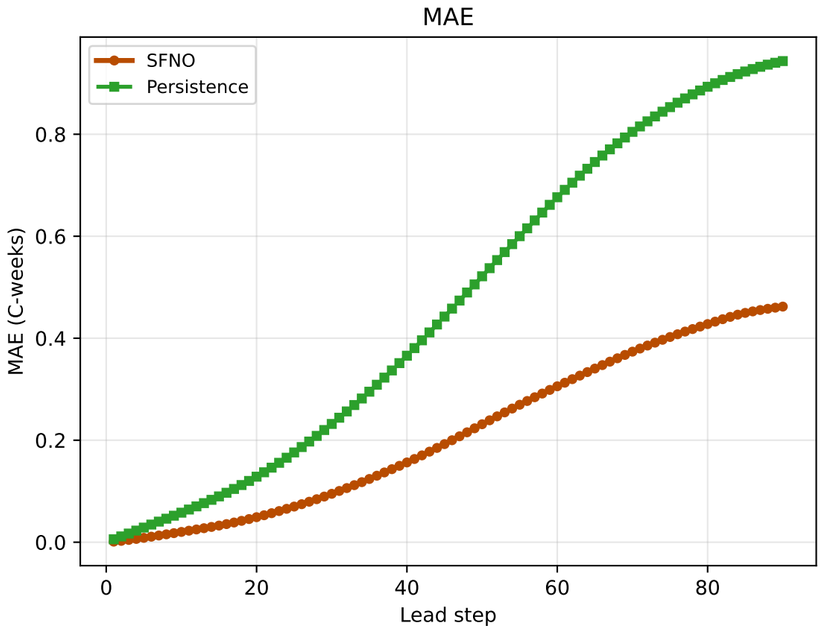


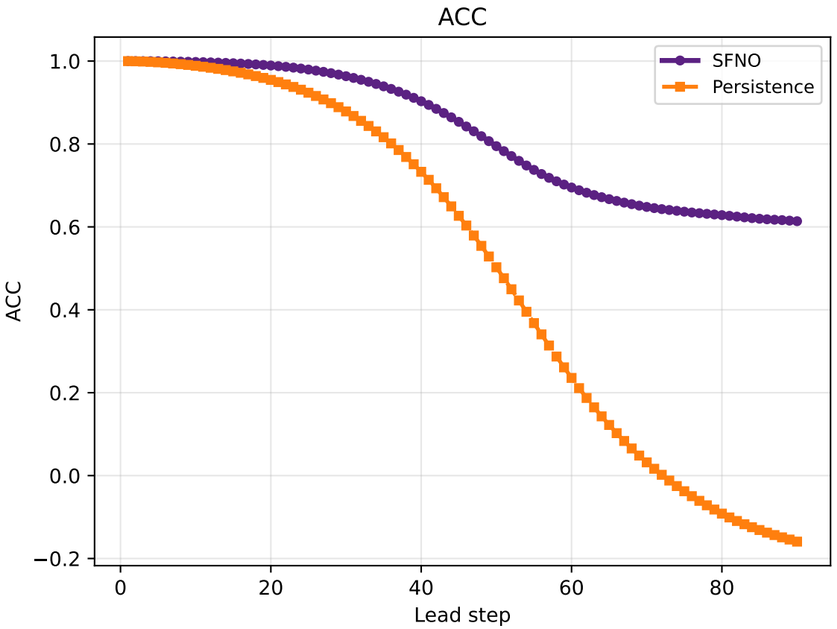
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

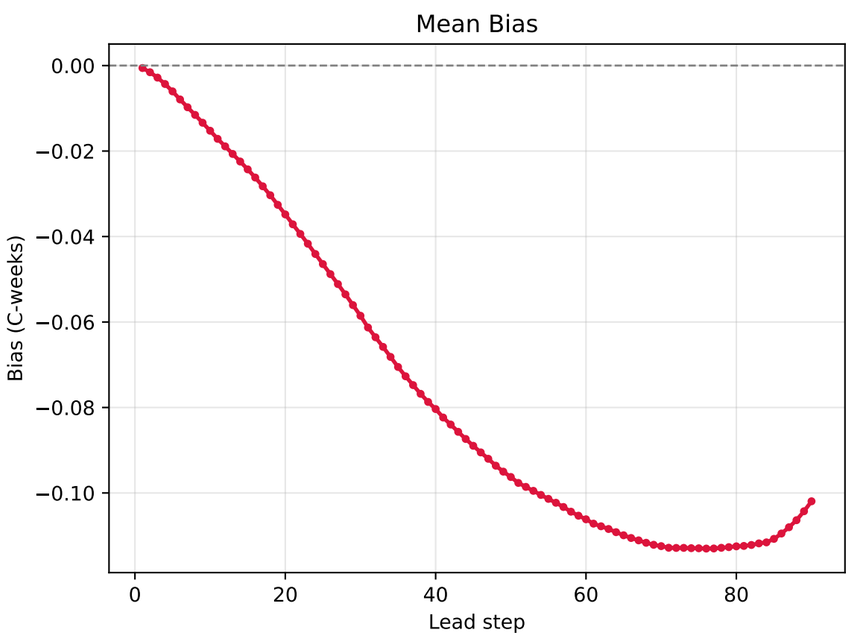
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0128	0.0425	69.9%	0.0011	0.0057	-0.0006	1.0000	0.9999	1.0000
2	0.0266	0.0829	67.9%	0.0026	0.0114	-0.0016	0.9999	0.9995	0.9999
3	0.0423	0.1222	65.4%	0.0045	0.0170	-0.0028	0.9999	0.9989	0.9998
4	0.0586	0.1613	63.6%	0.0064	0.0227	-0.0043	0.9997	0.9980	0.9996
5	0.0760	0.2000	62.0%	0.0086	0.0286	-0.0060	0.9996	0.9970	0.9993
6	0.0943	0.2381	60.4%	0.0108	0.0346	-0.0079	0.9993	0.9957	0.9989
7	0.1127	0.2758	59.1%	0.0131	0.0404	-0.0098	0.9991	0.9943	0.9984
8	0.1309	0.3133	58.2%	0.0154	0.0461	-0.0115	0.9987	0.9926	0.9979
9	0.1489	0.3504	57.5%	0.0178	0.0520	-0.0134	0.9983	0.9907	0.9972
10	0.1670	0.3874	56.9%	0.0201	0.0579	-0.0152	0.9979	0.9886	0.9965
11	0.1848	0.4241	56.4%	0.0226	0.0641	-0.0171	0.9974	0.9863	0.9957
12	0.2024	0.4601	56.0%	0.0250	0.0702	-0.0189	0.9969	0.9839	0.9948
13	0.2201	0.4965	55.7%	0.0274	0.0766	-0.0207	0.9963	0.9812	0.9938
14	0.2382	0.5332	55.3%	0.0299	0.0831	-0.0225	0.9956	0.9783	0.9927
15	0.2563	0.5701	55.0%	0.0326	0.0898	-0.0243	0.9948	0.9752	0.9915
16	0.2752	0.6080	54.7%	0.0355	0.0970	-0.0262	0.9940	0.9717	0.9902
17	0.2948	0.6467	54.4%	0.0386	0.1044	-0.0282	0.9930	0.9680	0.9886
18	0.3147	0.6861	54.1%	0.0419	0.1121	-0.0303	0.9919	0.9639	0.9869
19	0.3357	0.7264	53.8%	0.0454	0.1201	-0.0326	0.9907	0.9594	0.9849
20	0.3571	0.7676	53.5%	0.0491	0.1285	-0.0348	0.9893	0.9546	0.9828
21	0.3790	0.8096	53.2%	0.0528	0.1371	-0.0371	0.9878	0.9494	0.9804
22	0.4013	0.8524	52.9%	0.0569	0.1462	-0.0394	0.9861	0.9437	0.9777
23	0.4244	0.8961	52.6%	0.0611	0.1557	-0.0417	0.9842	0.9376	0.9748
24	0.4489	0.9412	52.3%	0.0654	0.1657	-0.0441	0.9820	0.9309	0.9714
25	0.4734	0.9875	52.1%	0.0699	0.1760	-0.0464	0.9796	0.9237	0.9677
26	0.4985	1.0350	51.8%	0.0745	0.1866	-0.0488	0.9770	0.9158	0.9637
27	0.5241	1.0830	51.6%	0.0794	0.1974	-0.0511	0.9740	0.9074	0.9594
28	0.5501	1.1316	51.4%	0.0843	0.2084	-0.0535	0.9709	0.8984	0.9546
29	0.5765	1.1811	51.2%	0.0895	0.2200	-0.0560	0.9674	0.8888	0.9494
30	0.6033	1.2311	51.0%	0.0949	0.2319	-0.0585	0.9635	0.8785	0.9438
31	0.6310	1.2818	50.8%	0.1005	0.2441	-0.0613	0.9593	0.8675	0.9377
32	0.6583	1.3332	50.6%	0.1061	0.2564	-0.0636	0.9548	0.8558	0.9312
33	0.6854	1.3846	50.5%	0.1119	0.2690	-0.0658	0.9501	0.8434	0.9244
34	0.7130	1.4364	50.4%	0.1178	0.2818	-0.0682	0.9448	0.8304	0.9170
35	0.7415	1.4894	50.2%	0.1240	0.2951	-0.0705	0.9391	0.8164	0.9089
36	0.7703	1.5434	50.1%	0.1304	0.3088	-0.0727	0.9329	0.8014	0.9003
37	0.7994	1.5981	50.0%	0.1368	0.3227	-0.0748	0.9262	0.7854	0.8911
38	0.8287	1.6531	49.9%	0.1432	0.3368	-0.0768	0.9190	0.7687	0.8813
39	0.8583	1.7081	49.8%	0.1497	0.3512	-0.0787	0.9113	0.7511	0.8710
40	0.8885	1.7629	49.6%	0.1564	0.3657	-0.0804	0.9031	0.7329	0.8600
41	0.9195	1.8185	49.4%	0.1632	0.3806	-0.0824	0.8942	0.7136	0.8482
42	0.9515	1.8750	49.3%	0.1703	0.3959	-0.0840	0.8848	0.6932	0.8358
43	0.9843	1.9323	49.1%	0.1776	0.4113	-0.0857	0.8747	0.6717	0.8226
44	1.0170	1.9898	48.9%	0.1849	0.4269	-0.0874	0.8643	0.6495	0.8090
45	1.0501	2.0466	48.7%	0.1924	0.4424	-0.0890	0.8534	0.6267	0.7950
46	1.0831	2.1032	48.5%	0.1999	0.4581	-0.0905	0.8422	0.6033	0.7805
47	1.1165	2.1599	48.3%	0.2077	0.4738	-0.0920	0.8307	0.5792	0.7655
48	1.1503	2.2173	48.1%	0.2155	0.4897	-0.0936	0.8188	0.5542	0.7501
49	1.1846	2.2749	47.9%	0.2233	0.5055	-0.0950	0.8068	0.5286	0.7343
50	1.2192	2.3325	47.7%	0.2313	0.5215	-0.0963	0.7948	0.5026	0.7183
51	1.2533	2.3896	47.6%	0.2391	0.5374	-0.0976	0.7827	0.4762	0.7021
52	1.2871	2.4464	47.4%	0.2468	0.5532	-0.0986	0.7709	0.4495	0.6859
53	1.3209	2.5032	47.2%	0.2545	0.5689	-0.0995	0.7593	0.4226	0.6698
54	1.3546	2.5604	47.1%	0.2621	0.5845	-0.1005	0.7481	0.3952	0.6539
55	1.3879	2.6175	47.0%	0.2696	0.6001	-0.1014	0.7375	0.3679	0.6385
56	1.4209	2.6741	46.9%	0.2769	0.6155	-0.1023	0.7275	0.3407	0.6236
57	1.4534	2.7302	46.8%	0.2843	0.6310	-0.1033	0.7184	0.3138	0.6095
58	1.4852	2.7859	46.7%	0.2915	0.6464	-0.1044	0.7100	0.2872	0.5961
59	1.5169	2.8409	46.6%	0.2986	0.6616	-0.1053	0.7021	0.2611	0.5831
60	1.5486	2.8953	46.5%	0.3058	0.6764	-0.1062	0.6950	0.2357	0.5707
61	1.5800	2.9487	46.4%	0.3128	0.6909	-0.1072	0.6884	0.2110	0.5589
62	1.6114	3.0007	46.3%	0.3197	0.7050	-0.1078	0.6824	0.1873	0.5474
63	1.6427	3.0513	46.2%	0.3267	0.7188	-0.1084	0.6768	0.1647	0.5363
64	1.6738	3.1006	46.0%	0.3336	0.7323	-0.1092	0.6717	0.1429	0.5256
65	1.7047	3.1484	45.9%	0.3405	0.7456	-0.1099	0.6669	0.1222	0.5152
66	1.7355	3.1948	45.7%	0.3473	0.7582	-0.1105	0.6626	0.1024	0.5051
67	1.7665	3.2402	45.5%	0.3540	0.7704	-0.1111	0.6585	0.0835	0.4951
68	1.7971	3.2842	45.3%	0.3607	0.7822	-0.1117	0.6548	0.0655	0.4853
69	1.8276	3.3266	45.1%	0.3672	0.7936	-0.1121	0.6513	0.0483	0.4756
70	1.8575	3.3677	44.8%	0.3737	0.8047	-0.1124	0.6481	0.0319	0.4662
71	1.8861	3.4069	44.6%	0.3797	0.8153	-0.1128	0.6453	0.0165	0.4573
72	1.9145	3.4445	44.4%	0.3856	0.8253	-0.1129	0.6429	0.0020	0.4487
73	1.9419	3.4811	44.2%	0.3913	0.8349	-0.1128	0.6408	-0.0119	0.4406
74	1.9686	3.5164	44.0%	0.3969	0.8443	-0.1129	0.6388	-0.0252	0.4327
75	1.9951	3.5499	43.8%	0.4024	0.8532	-0.1129	0.6367	-0.0377	0.4248
76	2.0213	3.5823	43.6%	0.4078	0.8619	-0.1130	0.6347	-0.0496	0.4170
77	2.0470	3.6139	43.4%	0.4130	0.8703	-0.1130	0.6331	-0.0609	0.4097
78	2.0717	3.6437	43.1%	0.4180	0.8782	-0.1128	0.6315	-0.0716	0.4024
79	2.0960	3.6724	42.9%	0.4229	0.8858	-0.1127	0.6299	-0.0819	0.3951
80	2.1200	3.6998	42.7%	0.4278	0.8931	-0.1125	0.6284	-0.0917	0.3876
81	2.1434	3.7255	42.5%	0.4325	0.9000	-0.1124	0.6267	-0.1009	0.3801
82	2.1667	3.7494	42.2%	0.4373	0.9064	-0.1122	0.6248	-0.1095	0.3722
83	2.1894	3.7716	42.0%	0.4418	0.9123	-0.1118	0.6230	-0.1173	0.3644
84	2.2110	3.7922	41.7%	0.4462	0.9179	-0.1116	0.6212	-0.1245	0.3569
85	2.2313	3.8114	41.5%	0.4497	0.9231	-0.1107	0.6196	-0.1313	0.3499
86	2.2502	3.8295	41.2%	0.4526	0.9278	-0.1095	0.6184	-0.1376	0.3434
87	2.2678	3.8461	41.0%	0.4553	0.9323	-0.1080	0.6173	-0.1434	0.3373
88	2.2839	3.8609	40.8%	0.4577	0.9365	-0.1064	0.6164	-0.1489	0.3312
89	2.2998	3.8738	40.6%	0.4599	0.9402	-0.1043	0.6151	-0.1542	0.3242
90	2.3149	3.8852	40.4%	0.4620	0.9435	-0.1019	0.6138	-0.1593	0.3170

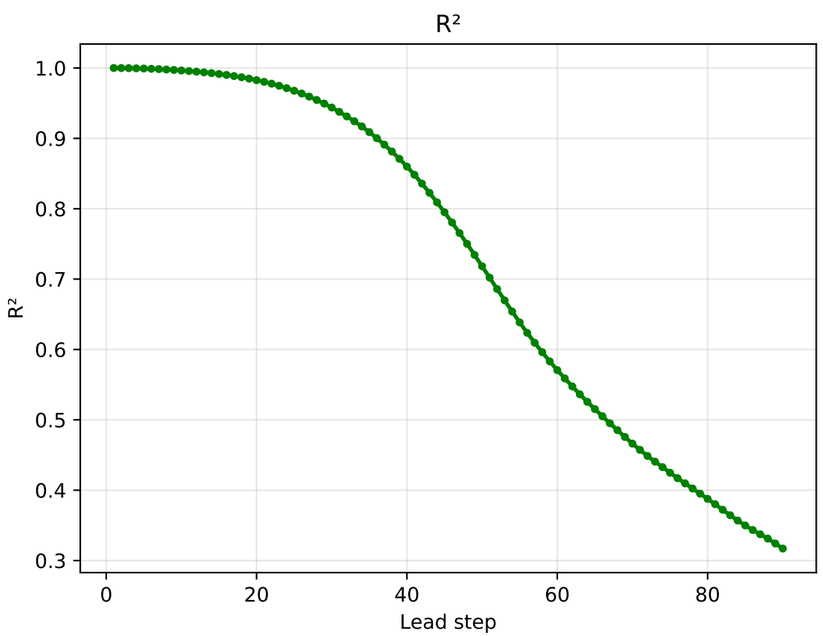
DHW Forecast Skill vs Lead Step - Val

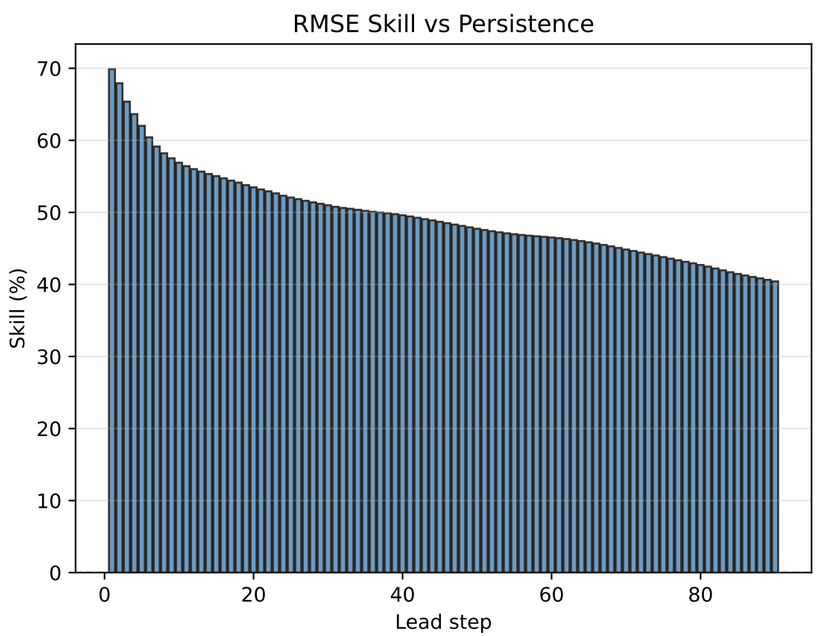


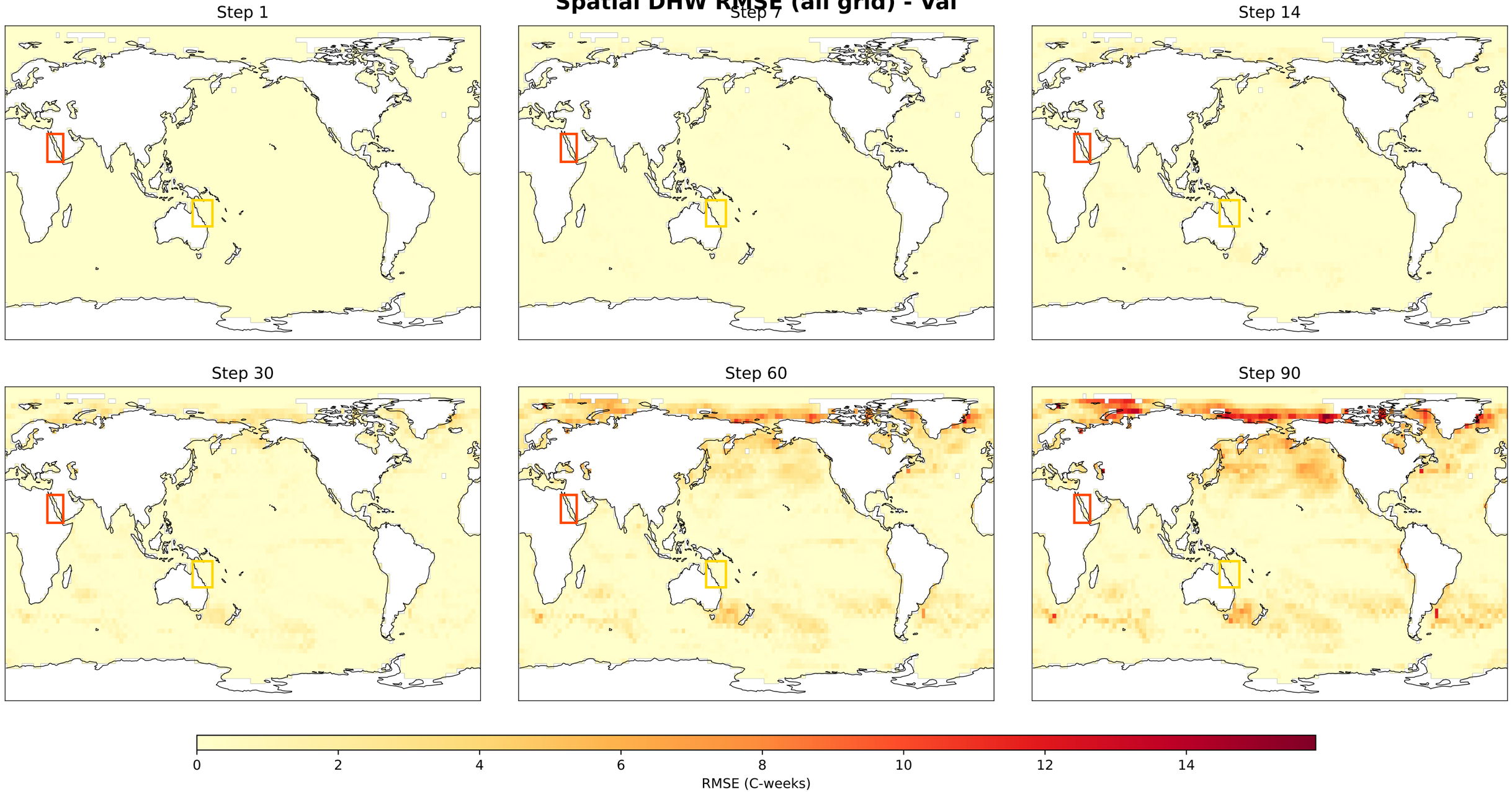


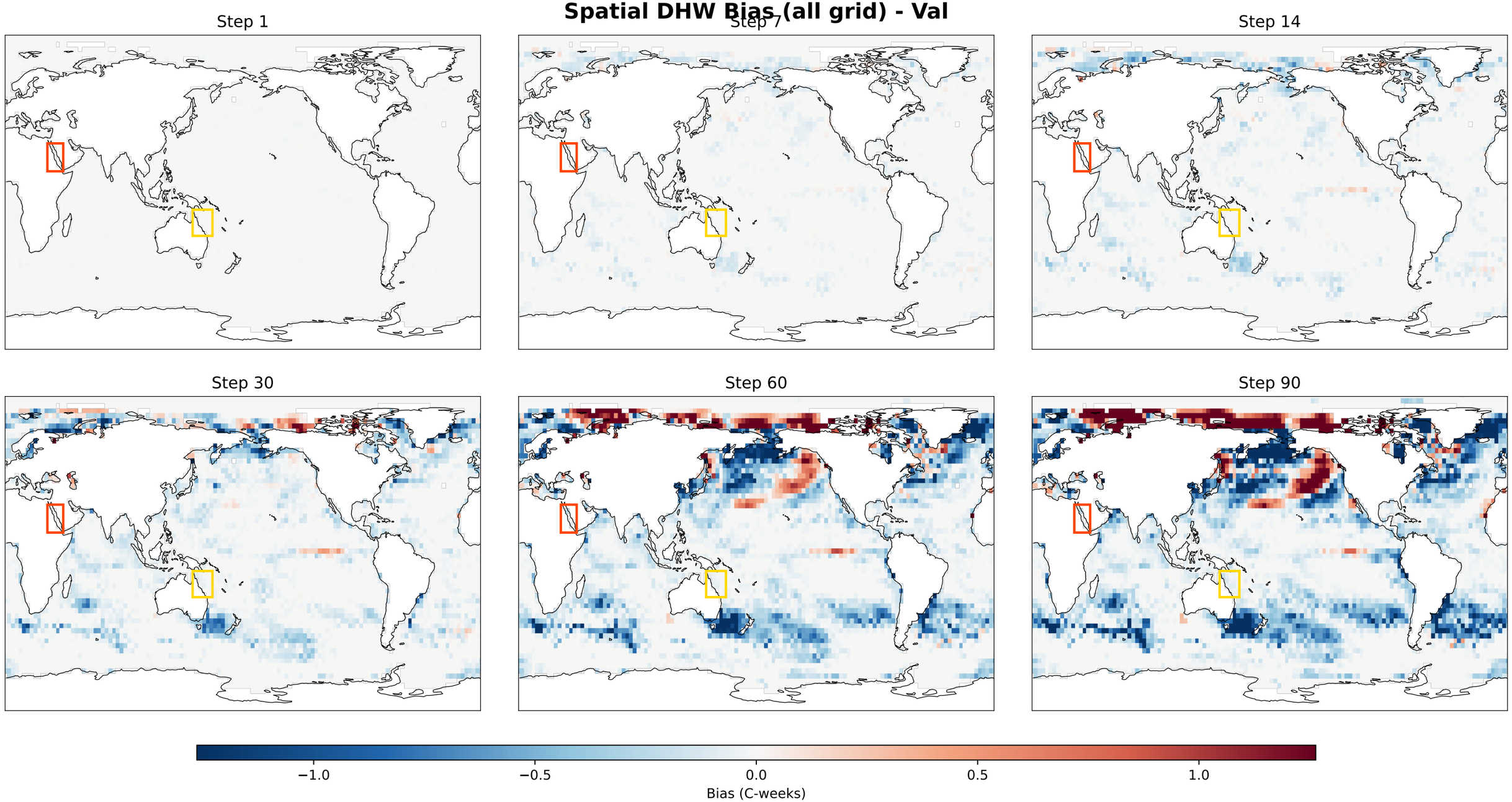


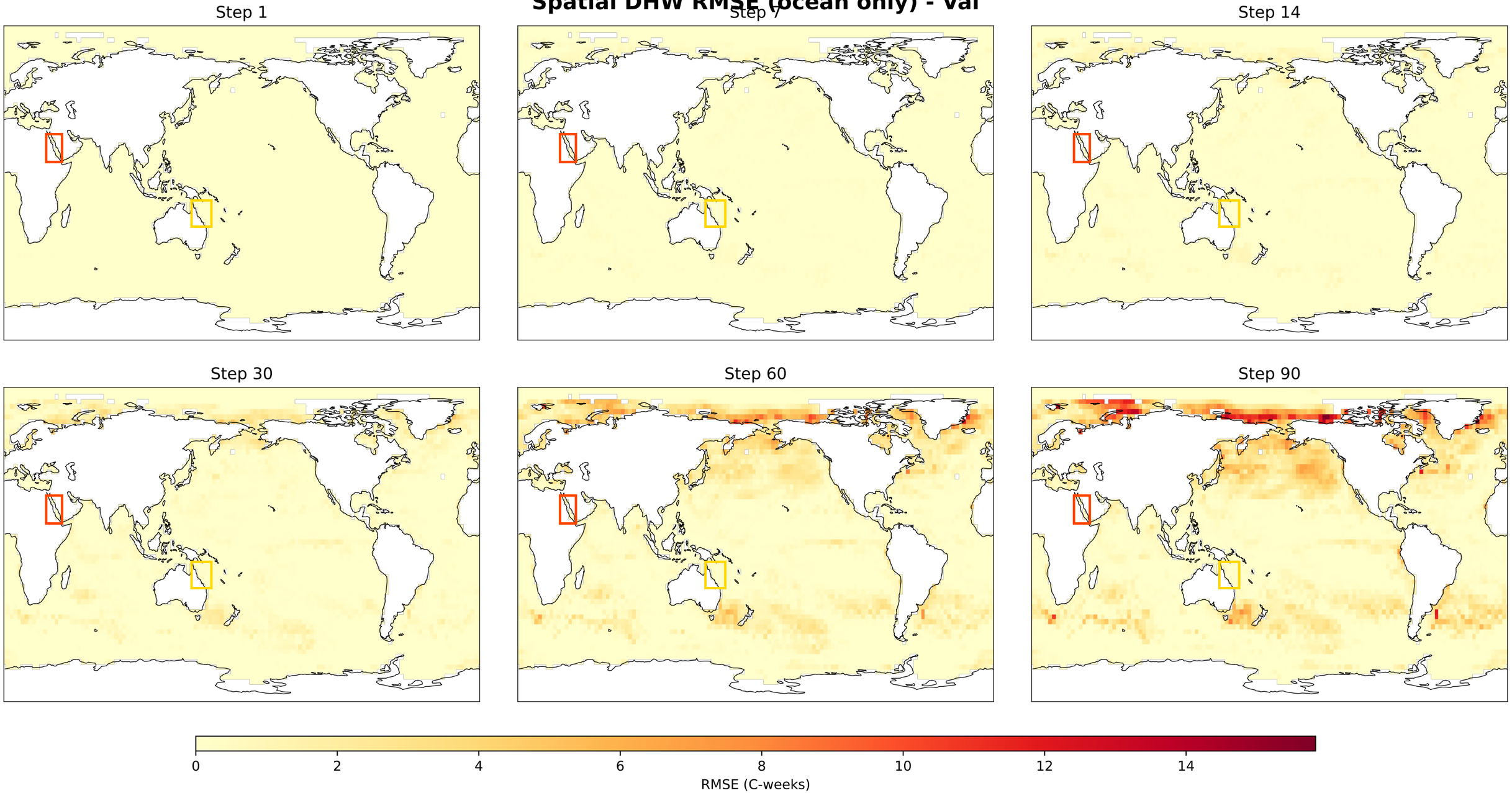


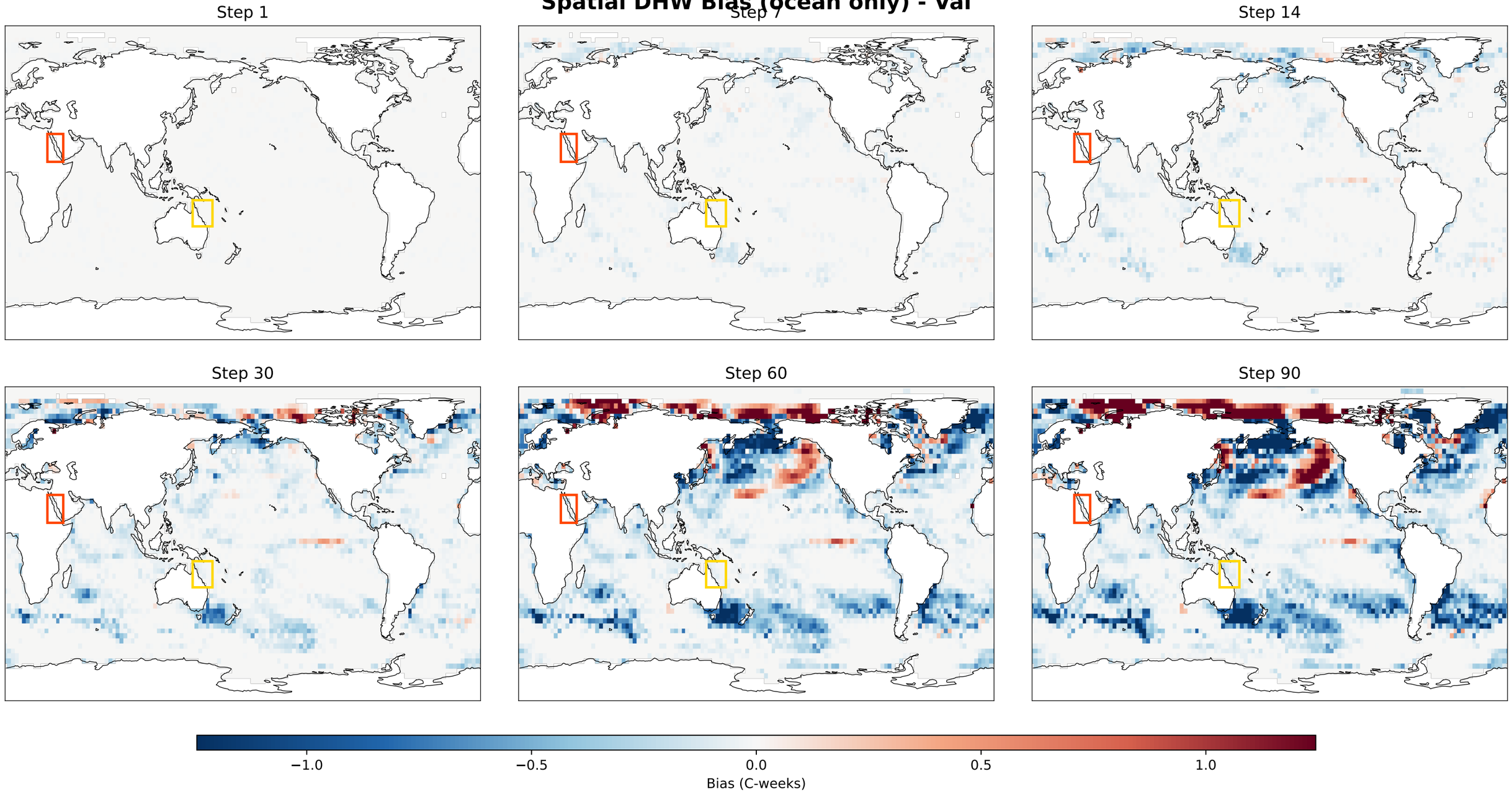





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
RMSE (C-weeks)

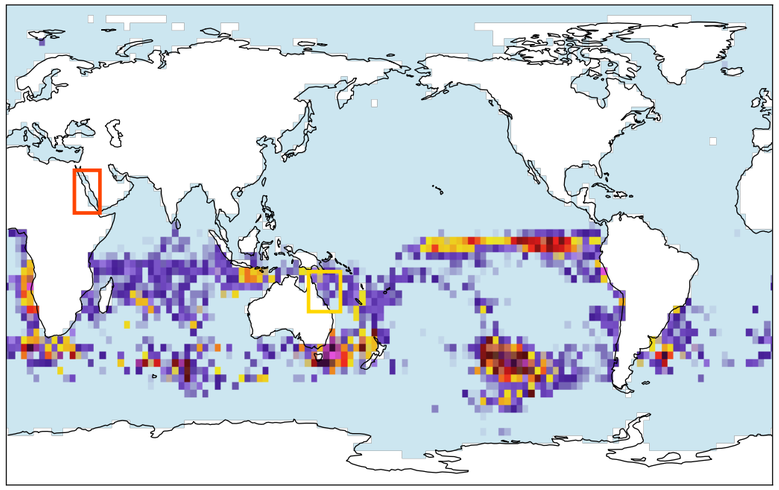

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
RMSE (C-weeks)

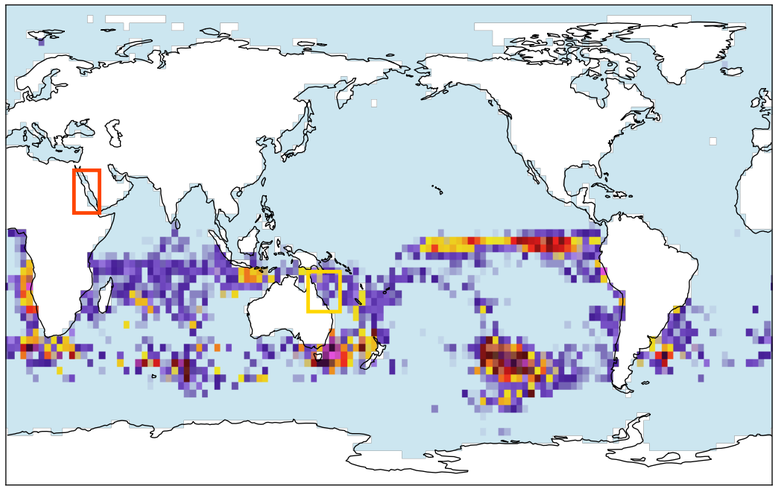

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

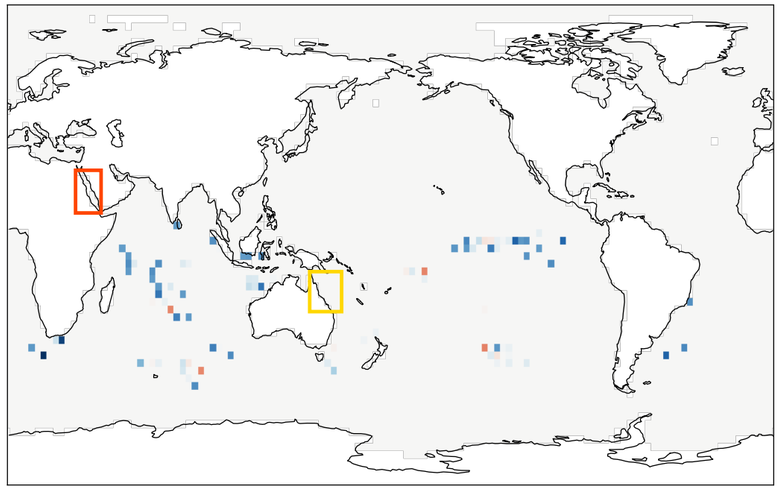
Actual DHW - Step 1



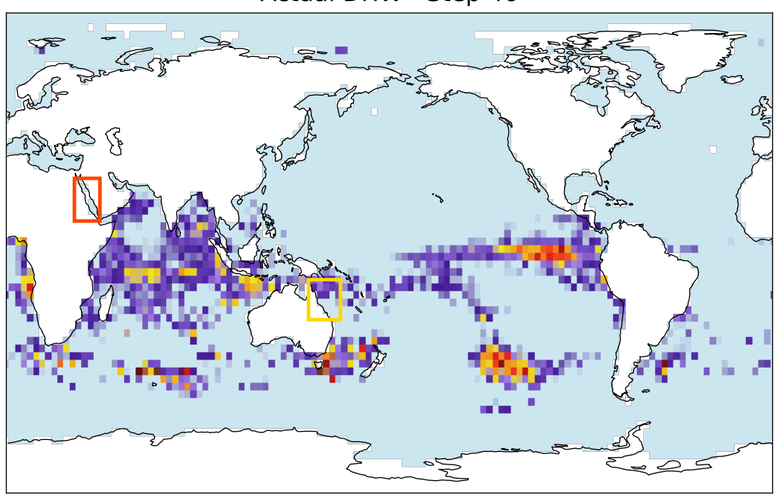
Predicted DHW - Step 1



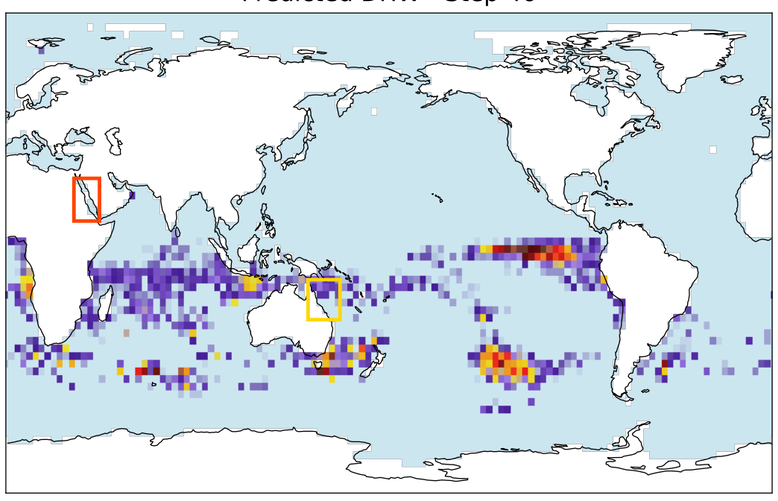
Error (Pred - Actual) - Step 1 [scale ±0.31]



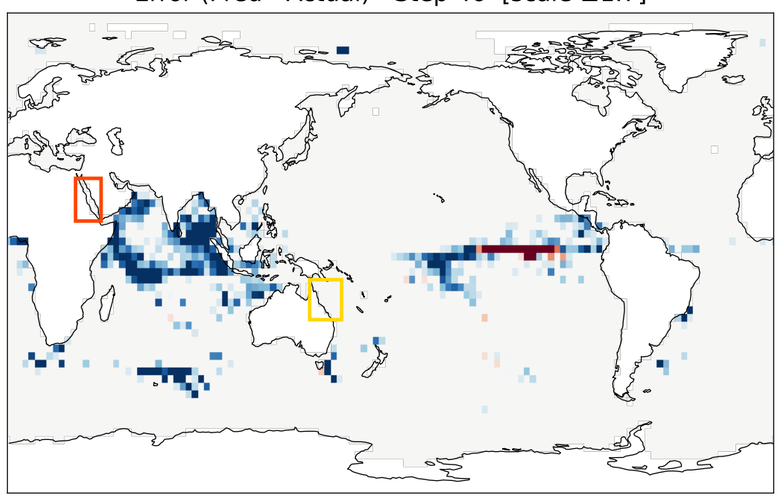
Actual DHW - Step 46



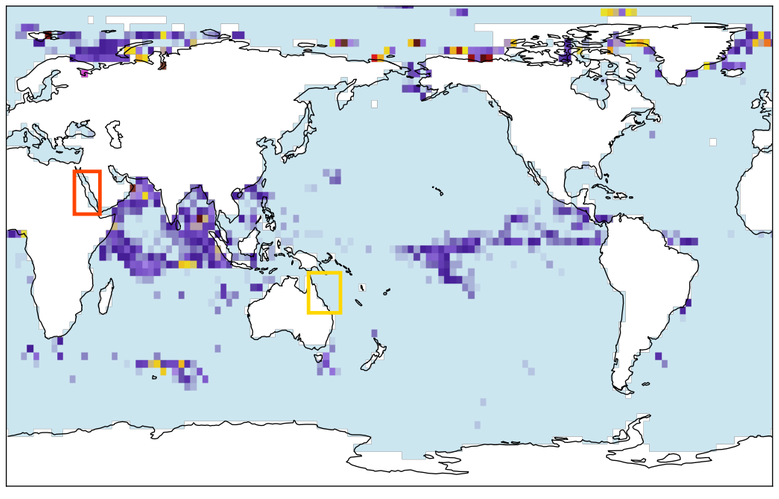
Predicted DHW - Step 46



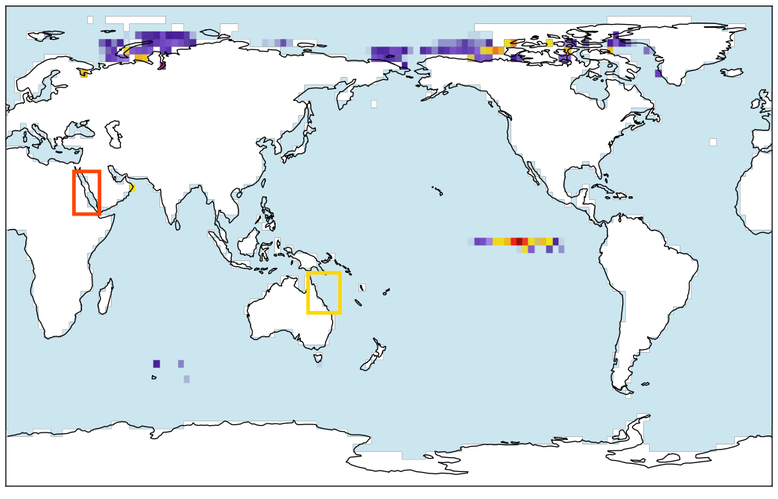
Error (Pred - Actual) - Step 46 [scale ±1.7]



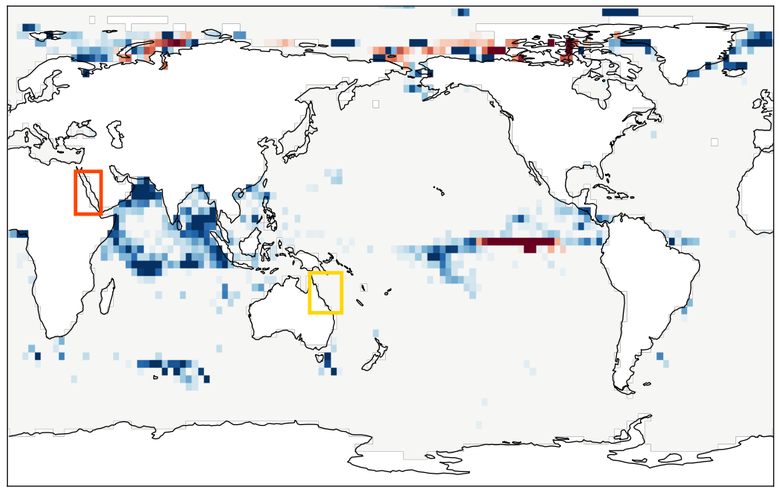
Actual DHW - Step 90

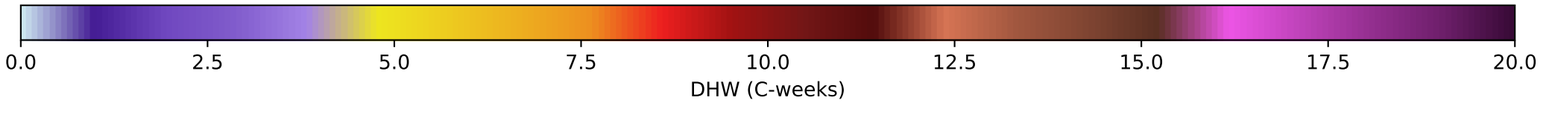


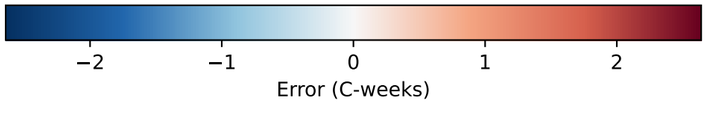
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

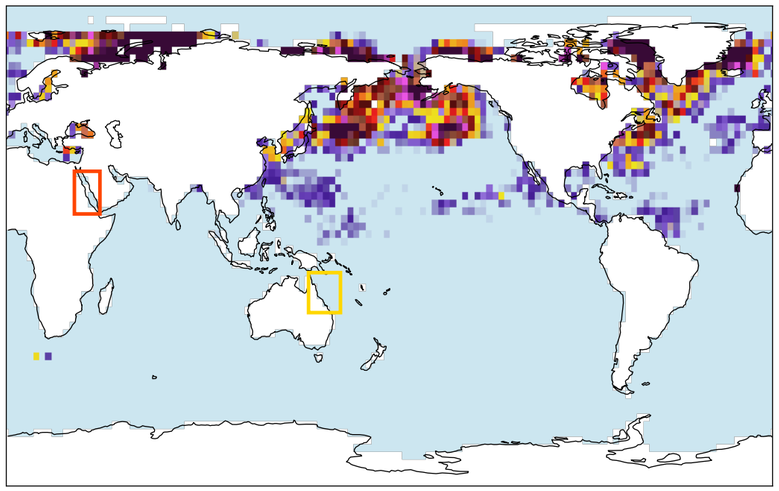




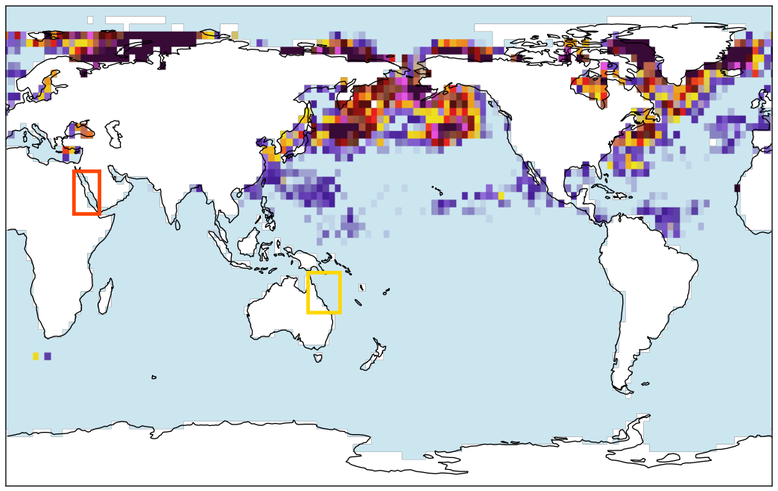


DHW Chain init: 2016-10-07 - Val (C-weeks)

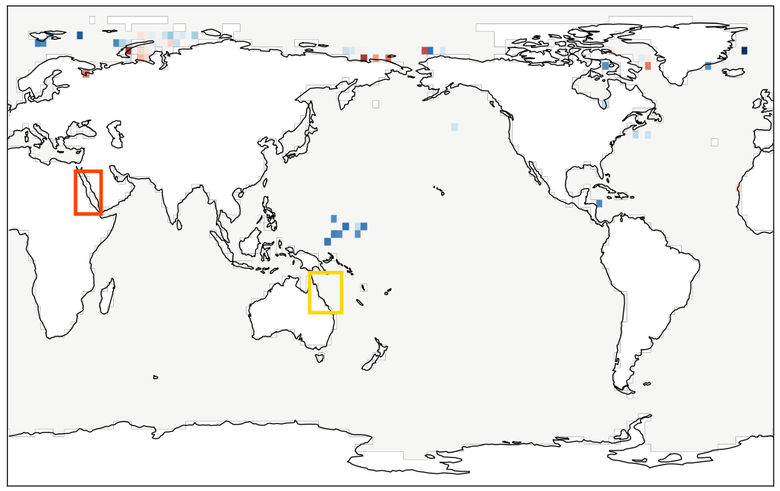
Actual DHW - Step 1



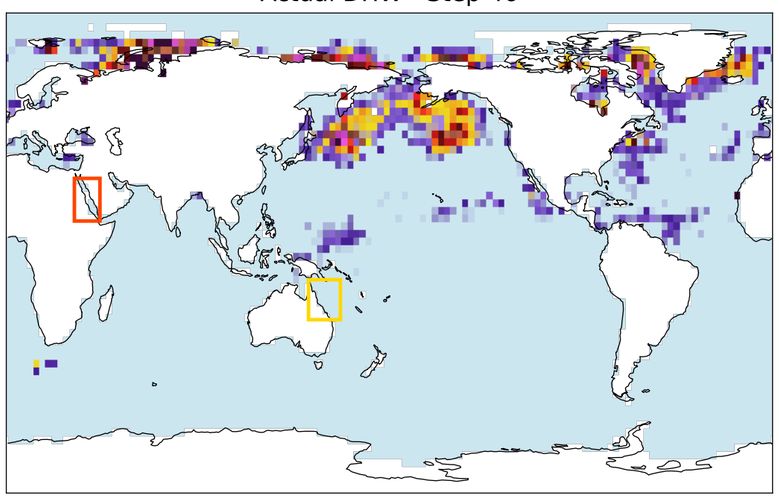
Predicted DHW - Step 1



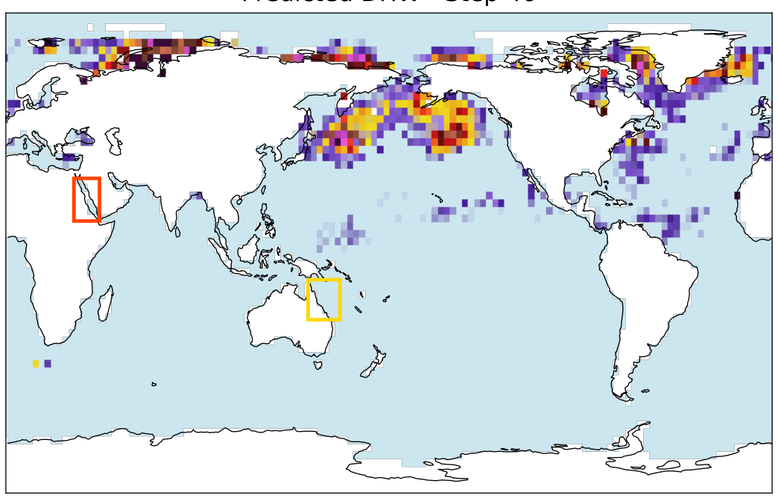
Error (Pred - Actual) - Step 1 [scale ±0.28]



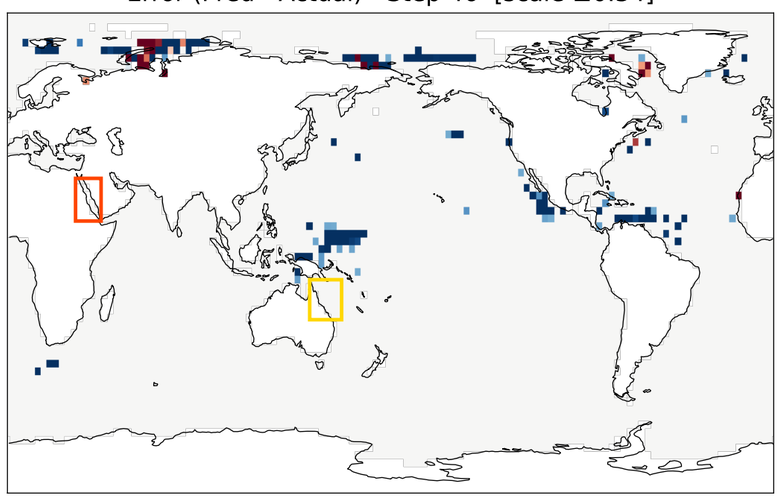
Actual DHW - Step 46



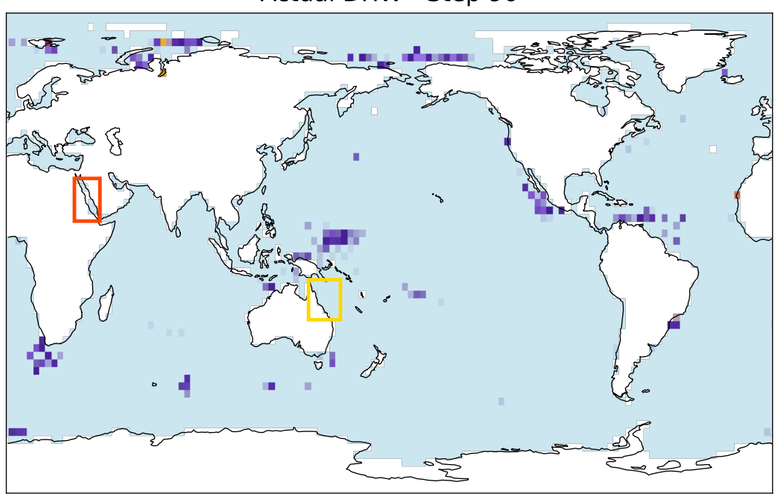
Predicted DHW - Step 46



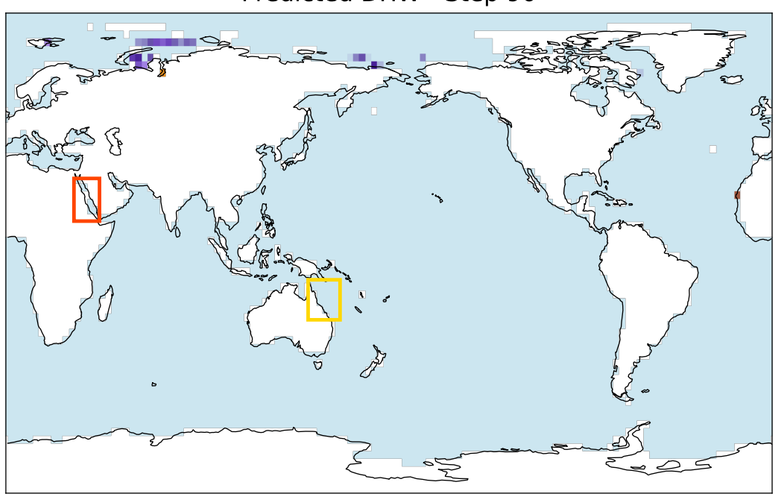
Error (Pred - Actual) - Step 46 [scale ±0.34]



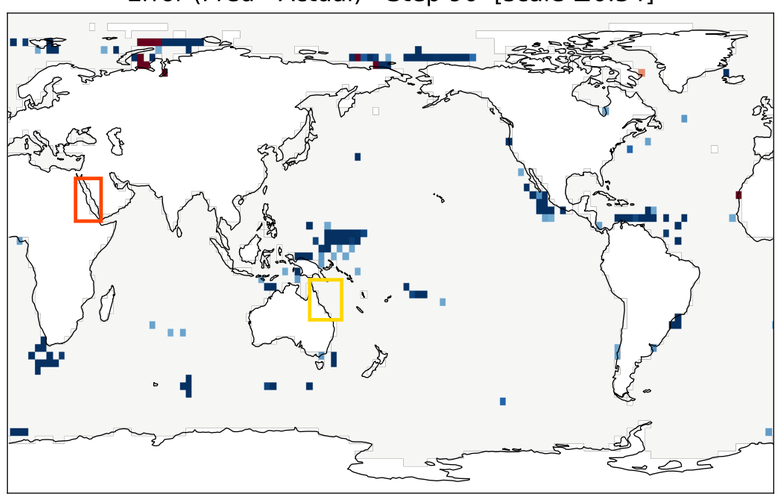
Actual DHW - Step 90

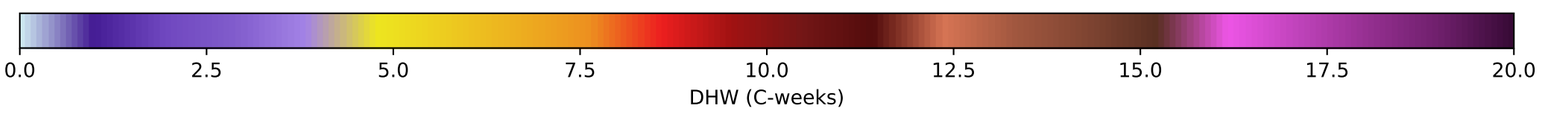


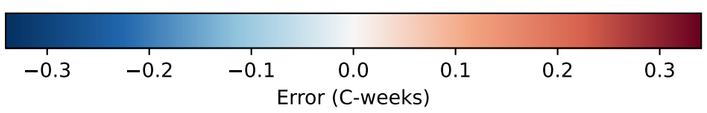
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.34]

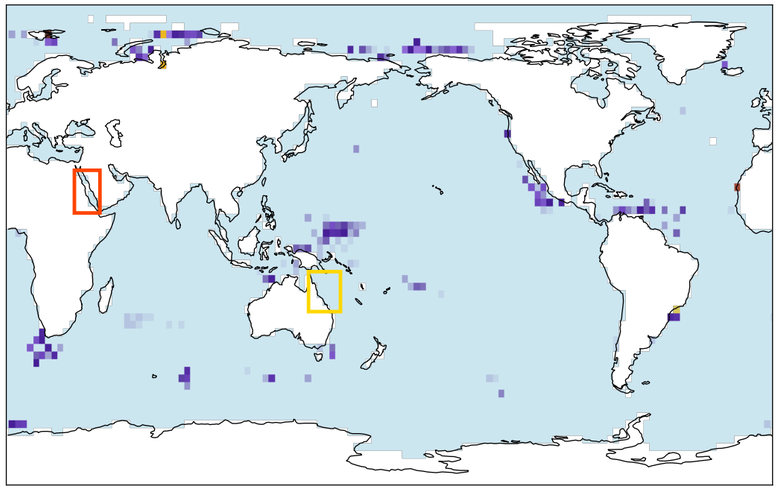




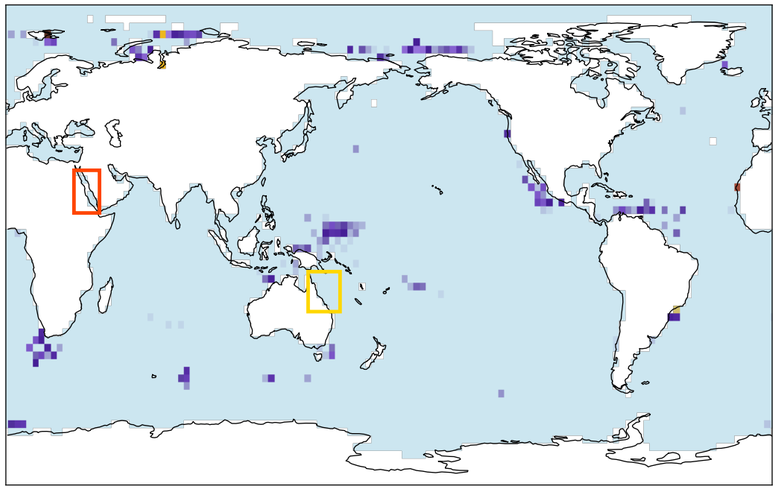


DHW Chain init: 2017-01-05 - Val (C-weeks)

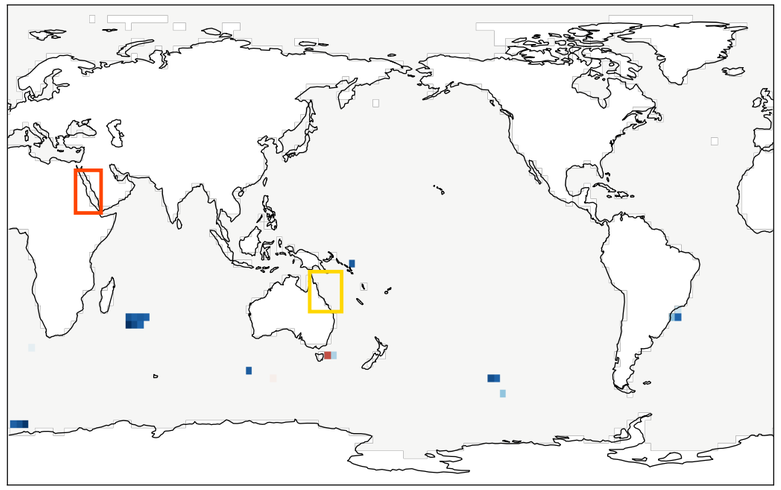
Actual DHW - Step 1



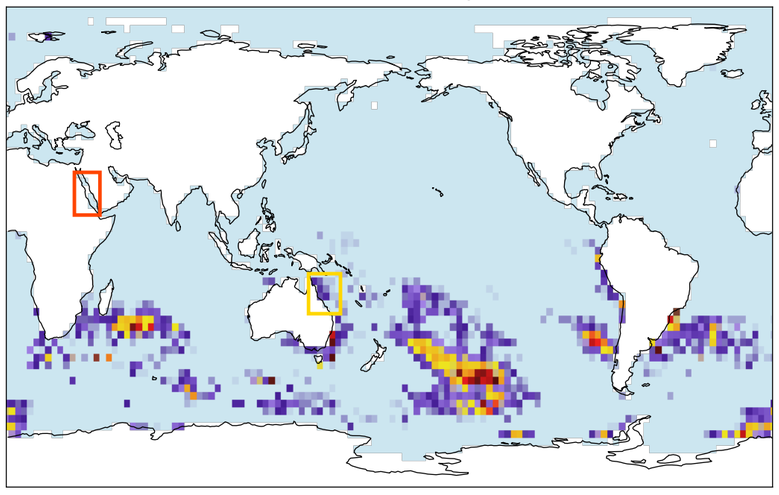
Predicted DHW - Step 1



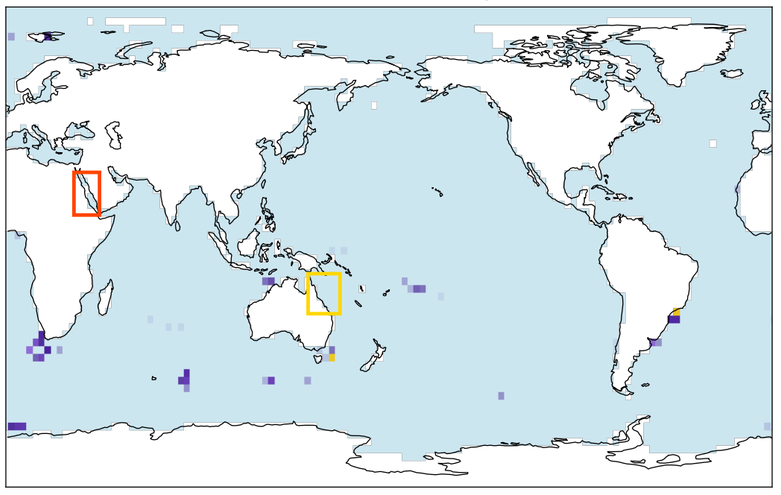
Error (Pred - Actual) - Step 1 [scale ±0.22]



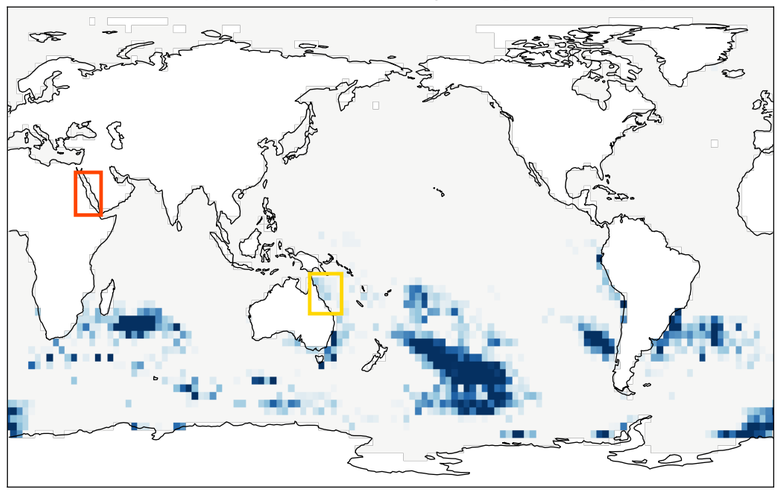
Actual DHW - Step 46



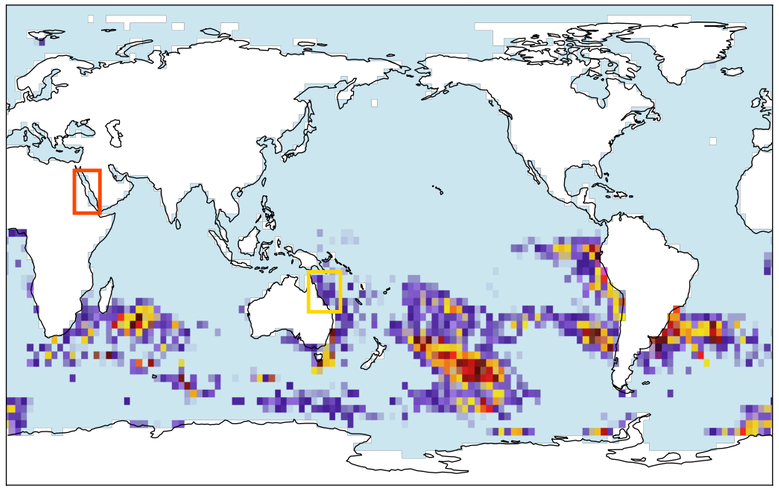
Predicted DHW - Step 46



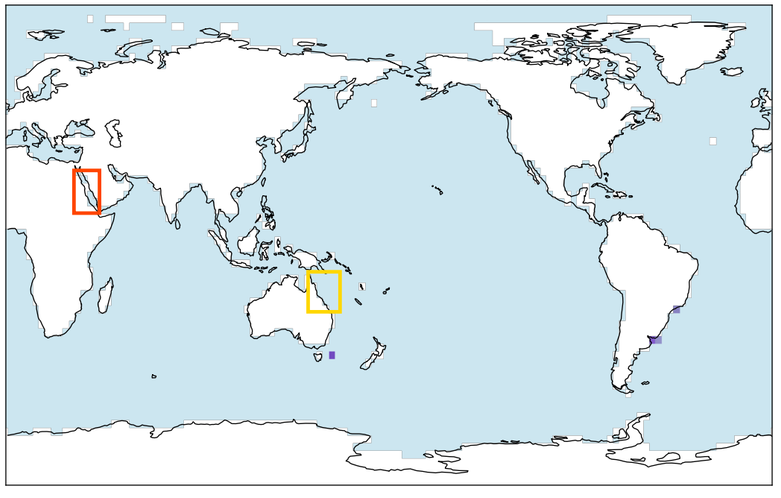
Error (Pred - Actual) - Step 46 [scale ±4.8]



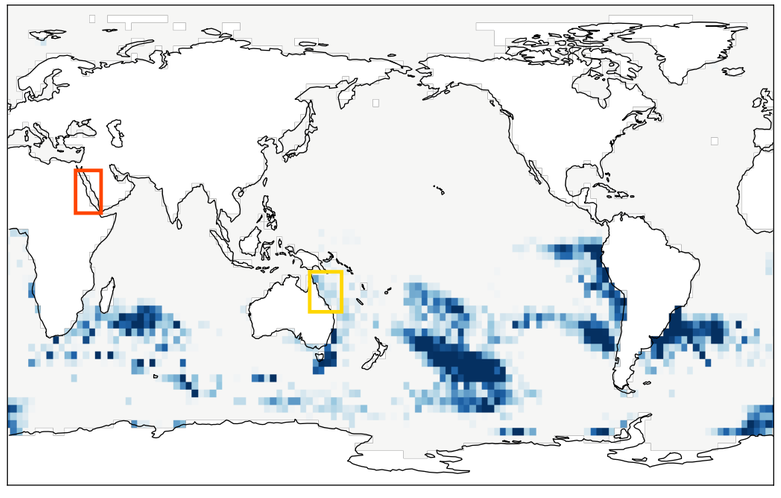
Actual DHW - Step 90

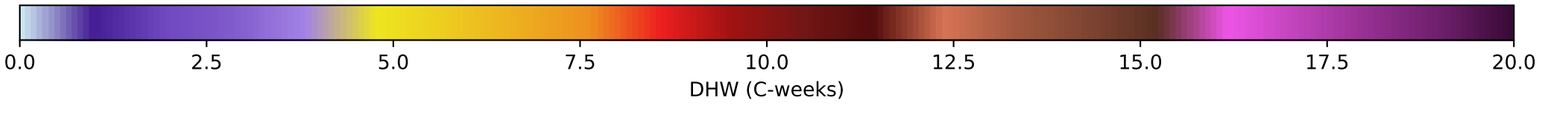


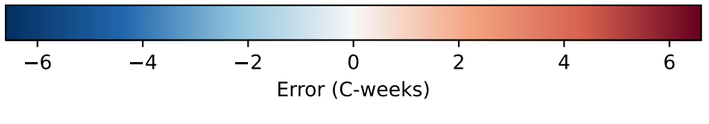
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

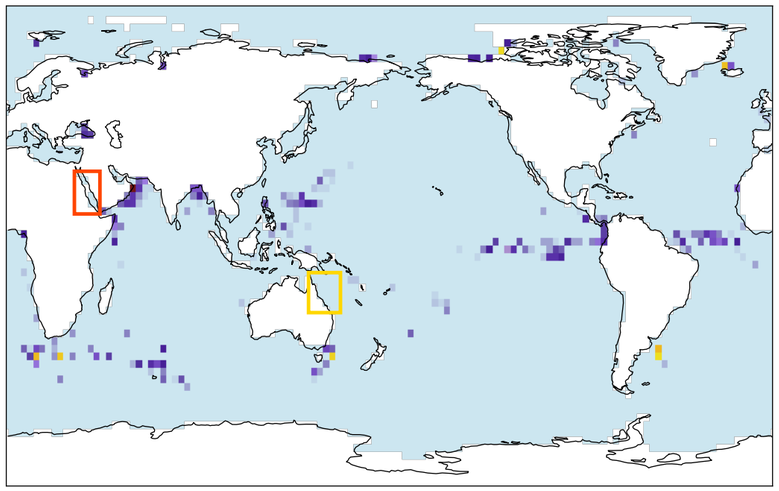




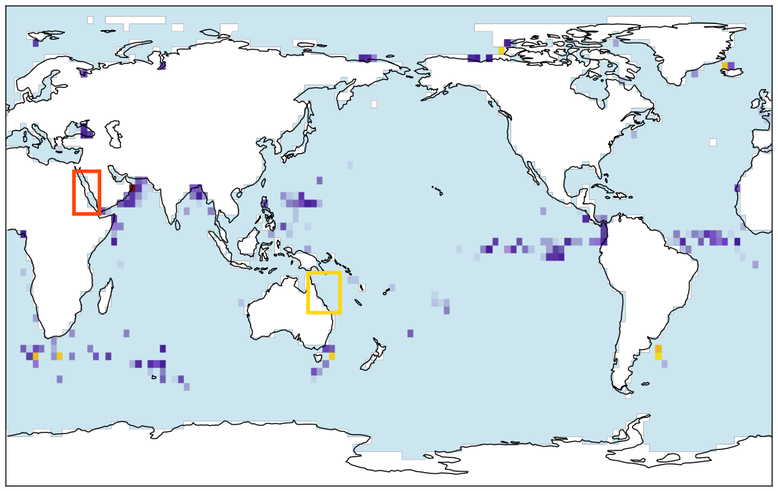


DHW Chain init: 2017-07-04 - Val (C-weeks)

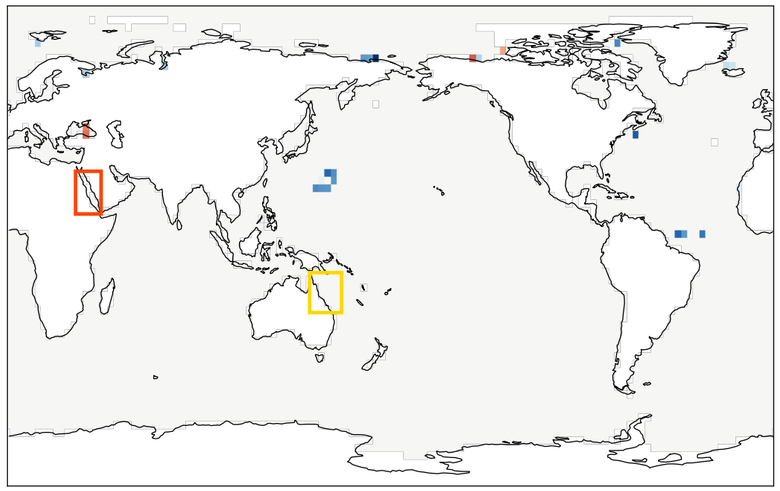
Actual DHW - Step 1



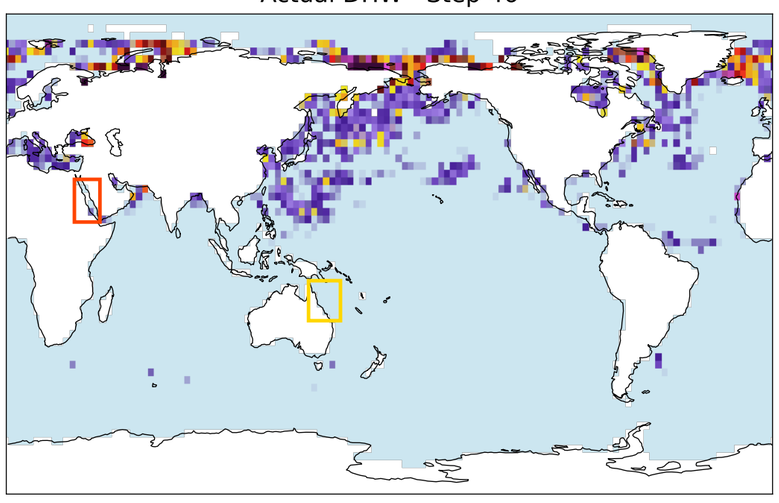
Predicted DHW - Step 1



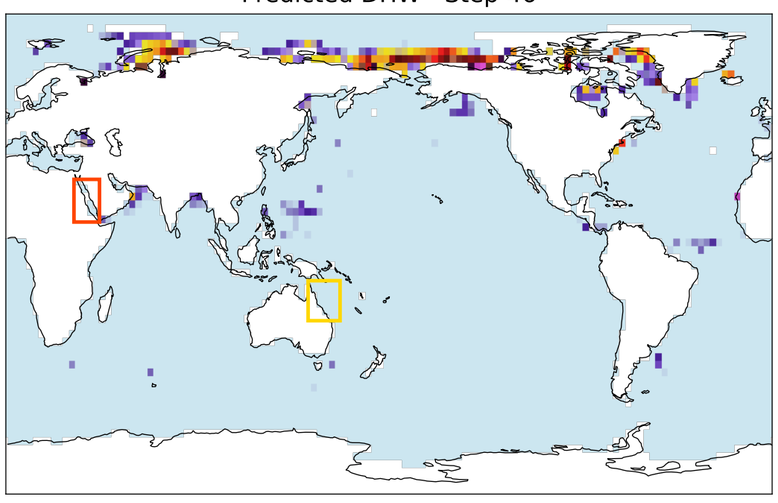
Error (Pred - Actual) - Step 1 [scale ±0.31]



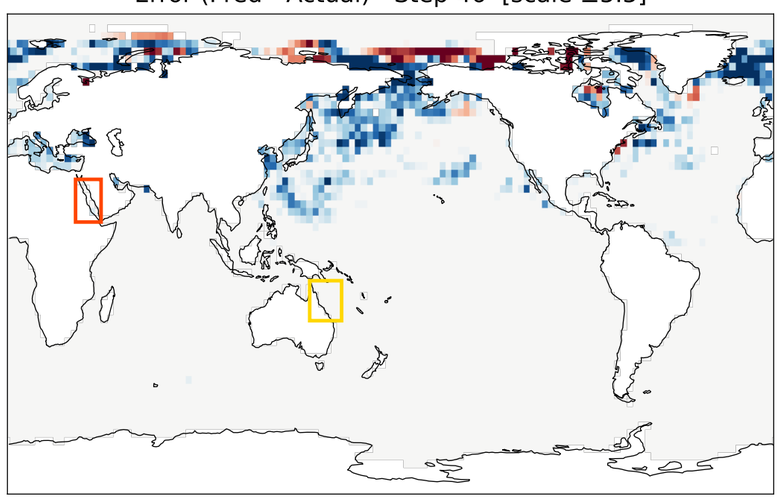
Actual DHW - Step 46



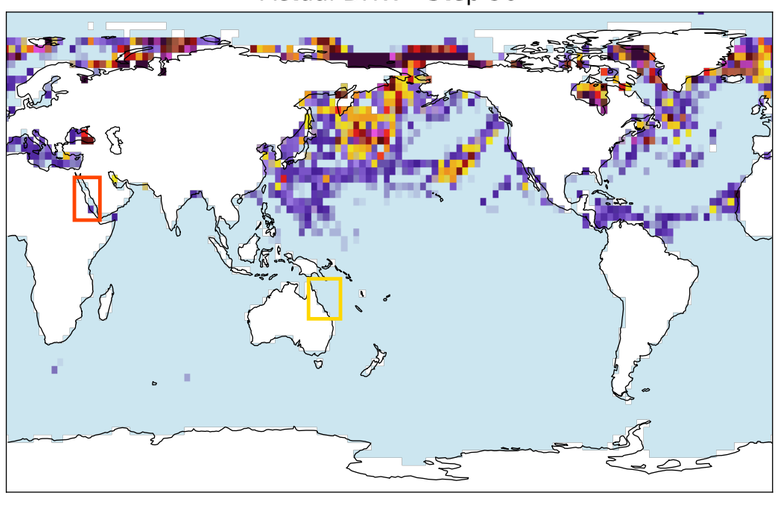
Predicted DHW - Step 46



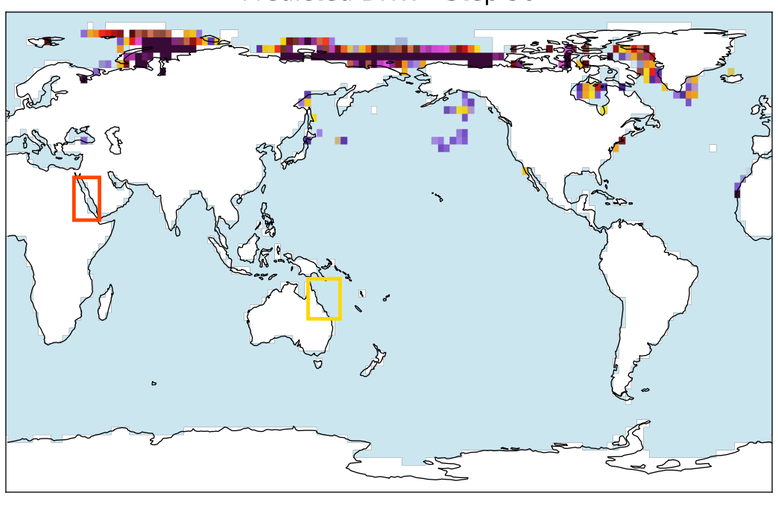
Error (Pred - Actual) - Step 46 [scale ±5.5]



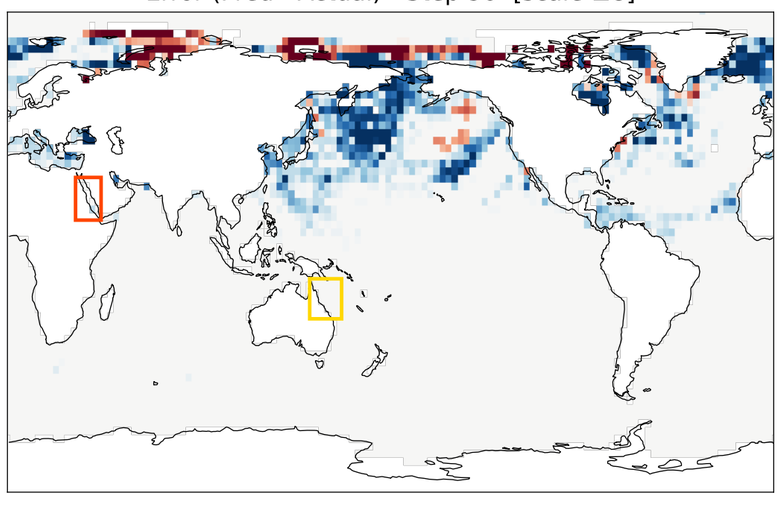
Actual DHW - Step 90

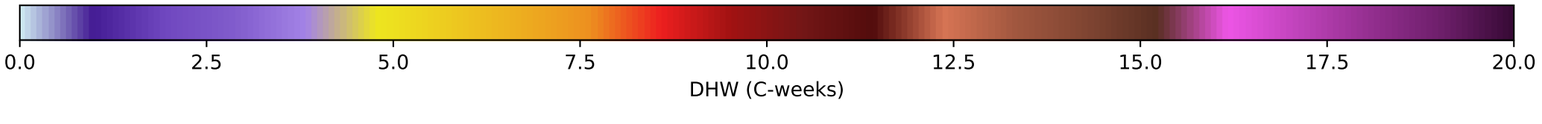


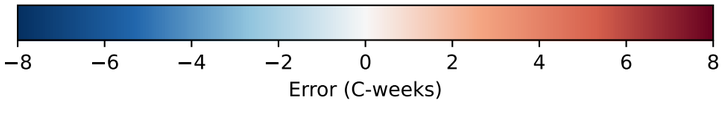
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

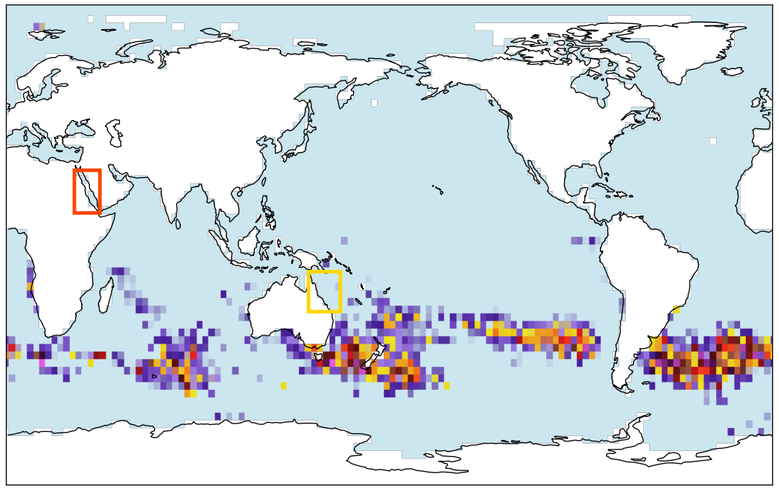




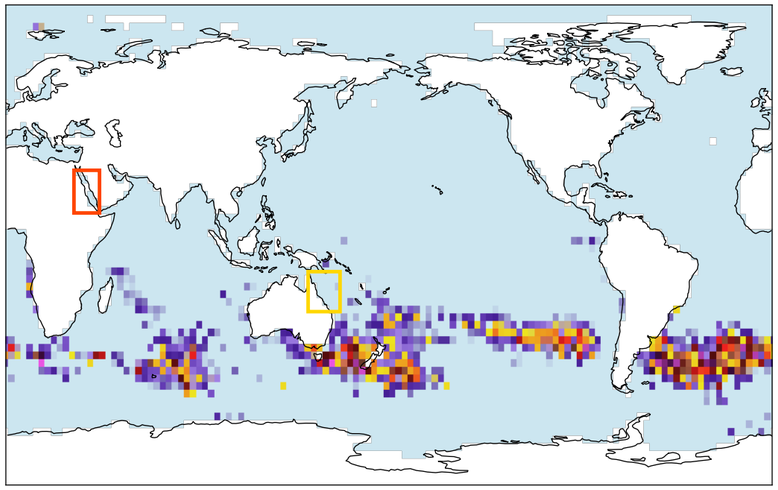


DHW Chain init: 2018-03-31 - Val (C-weeks)

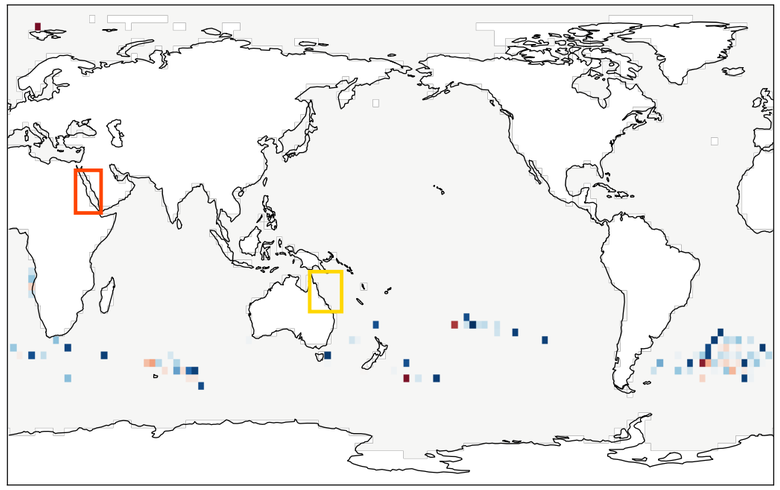
Actual DHW - Step 1



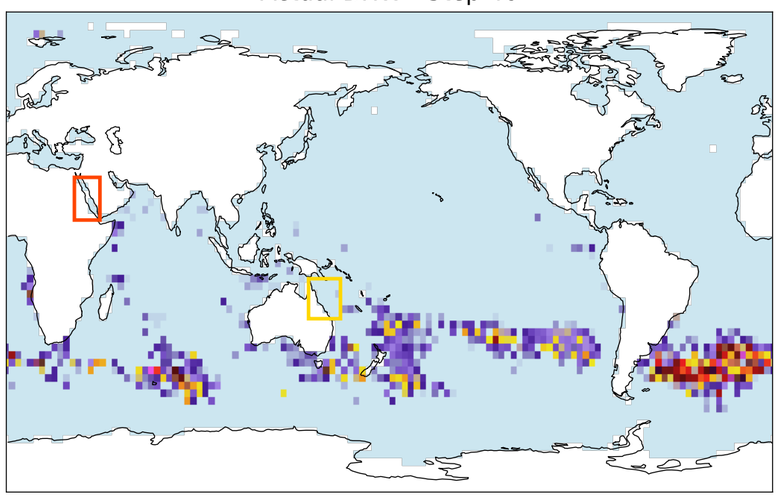
Predicted DHW - Step 1



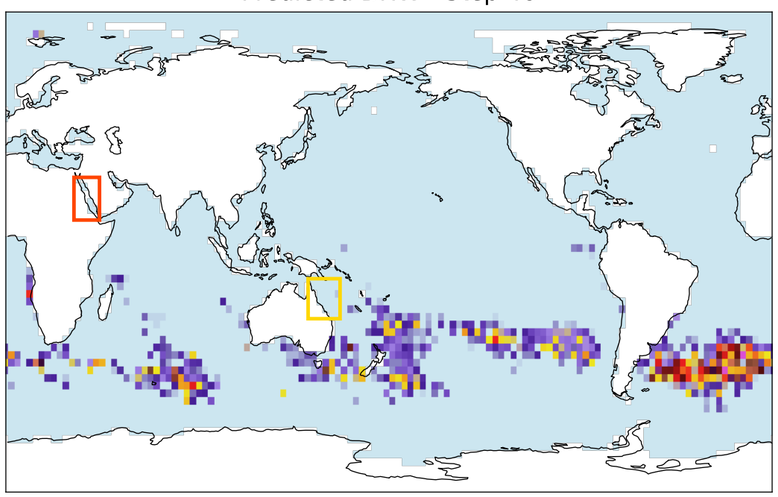
Error (Pred - Actual) - Step 1 [scale ±0.18]



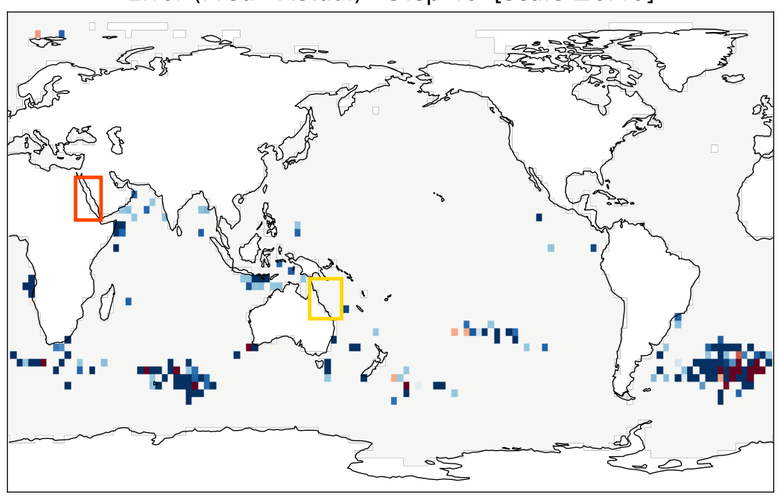
Actual DHW - Step 46



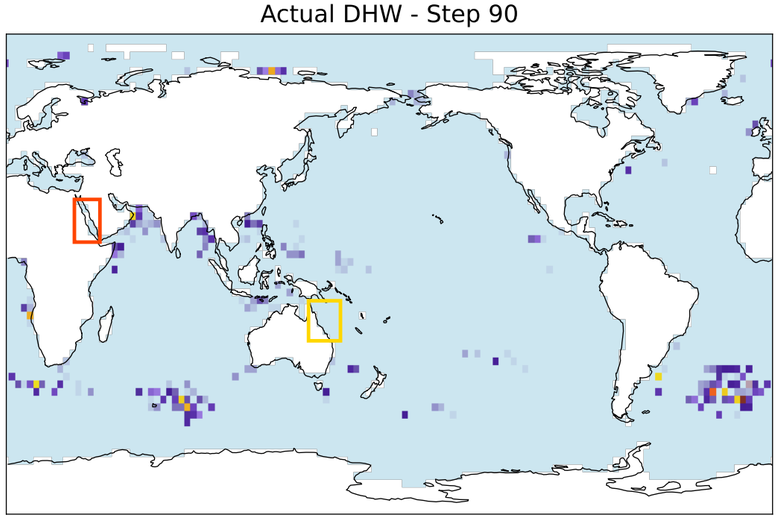
Predicted DHW - Step 46



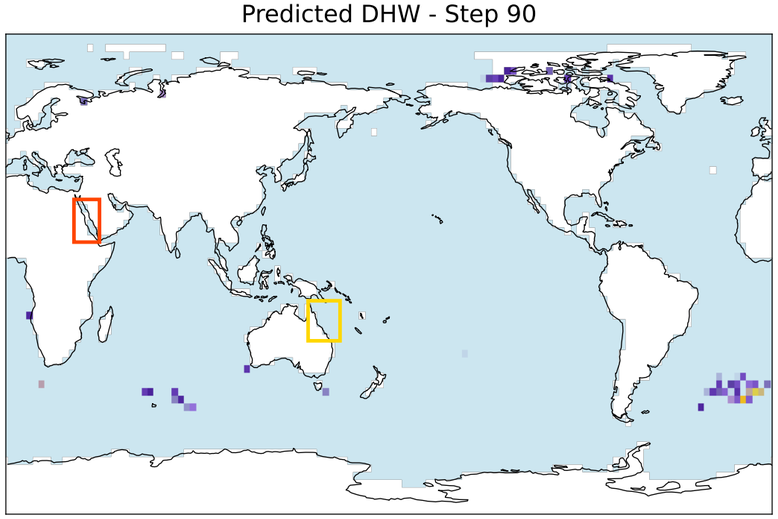
Error (Pred - Actual) - Step 46 [scale ±0.46]



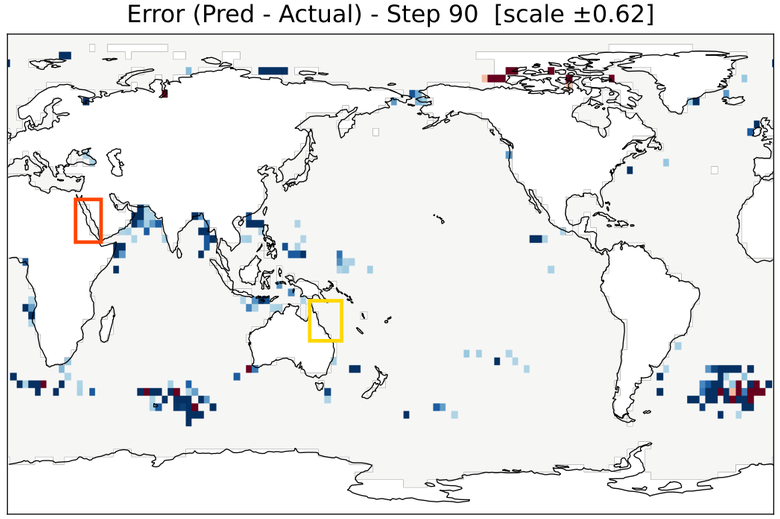
Actual DHW - Step 90

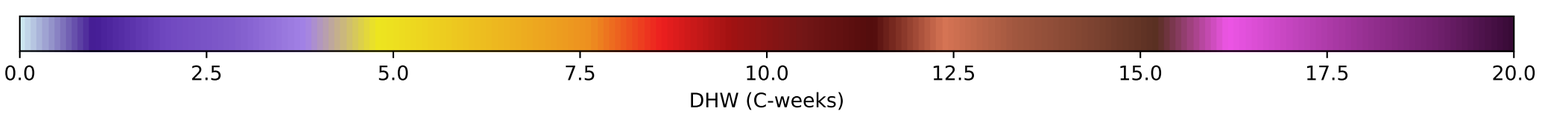


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

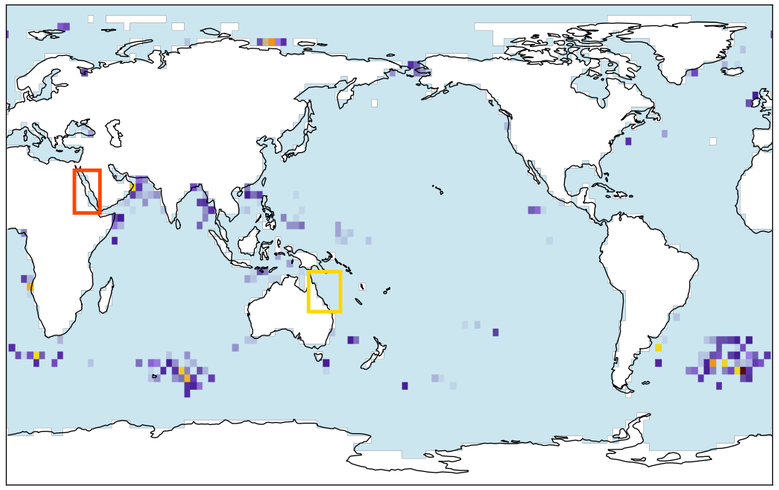




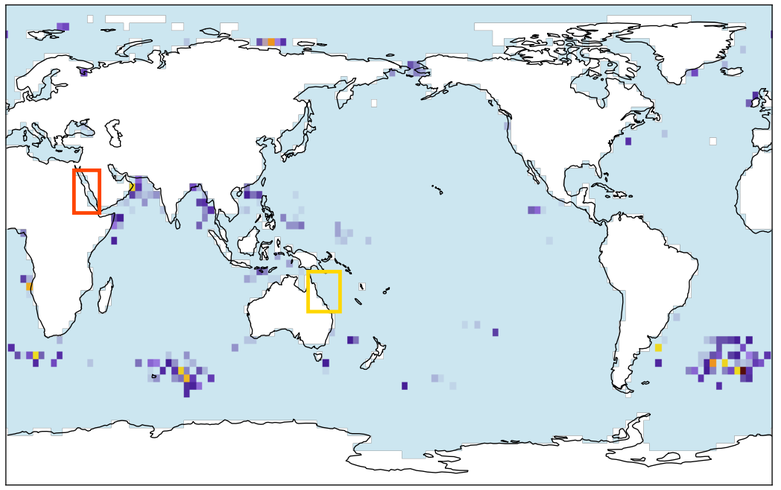


DHW Chain init: 2018-06-29 - Val (C-weeks)

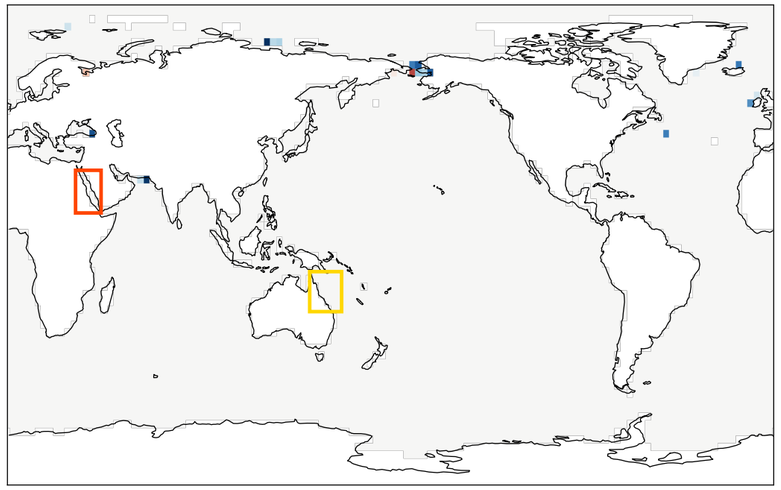
Actual DHW - Step 1



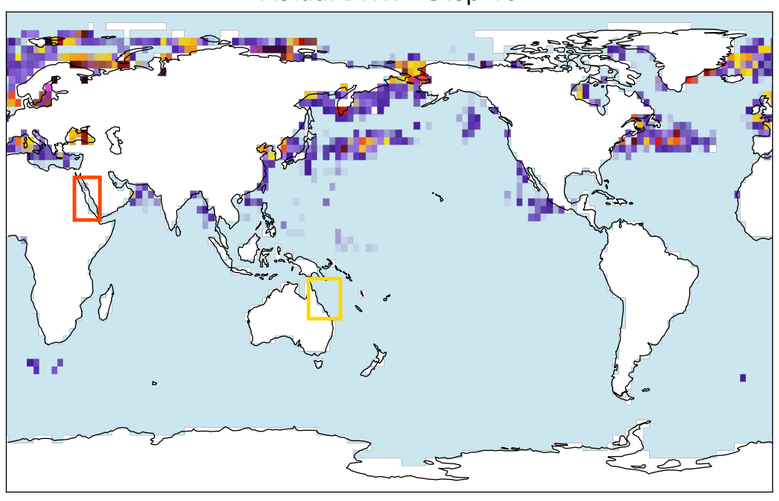
Predicted DHW - Step 1



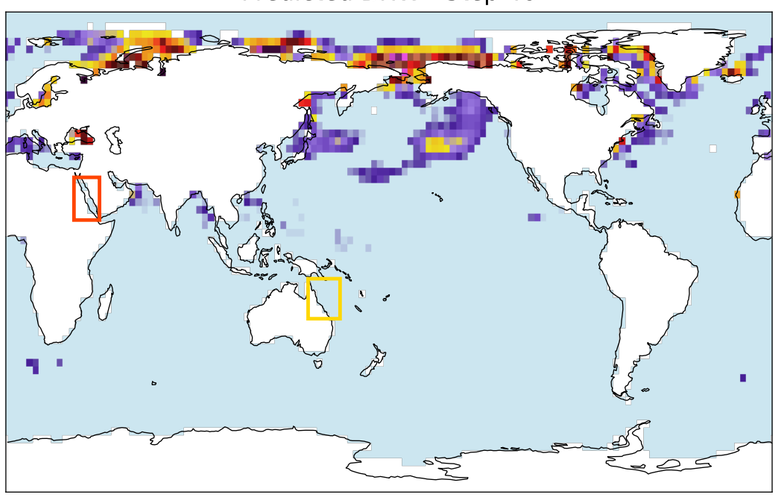
Error (Pred - Actual) - Step 1 [scale ±0.25]



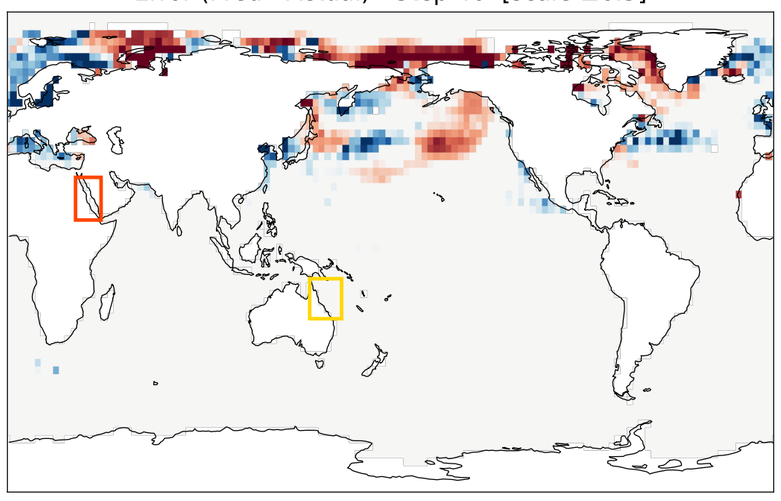
Actual DHW - Step 46



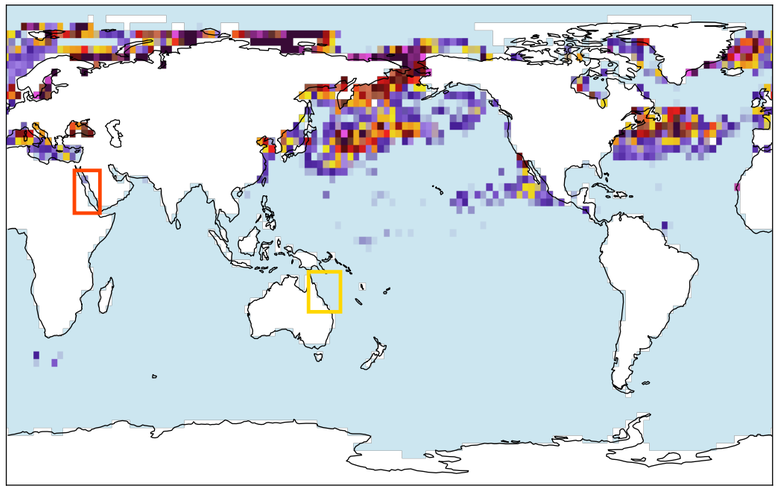
Predicted DHW - Step 46



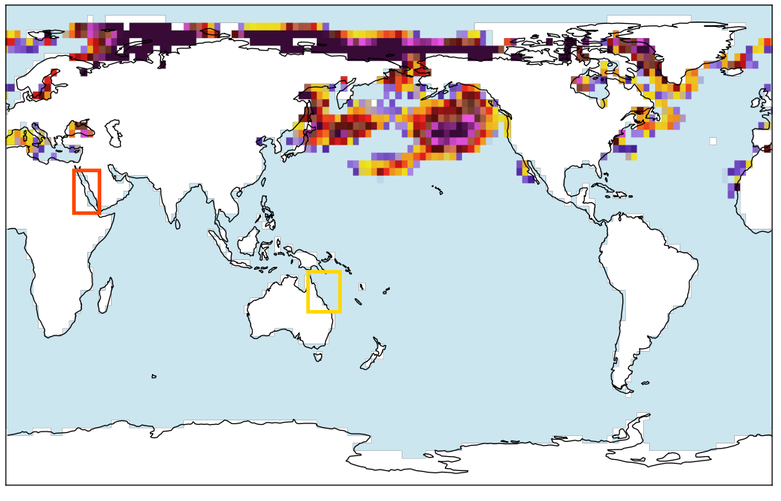
Error (Pred - Actual) - Step 46 [scale ±6.5]



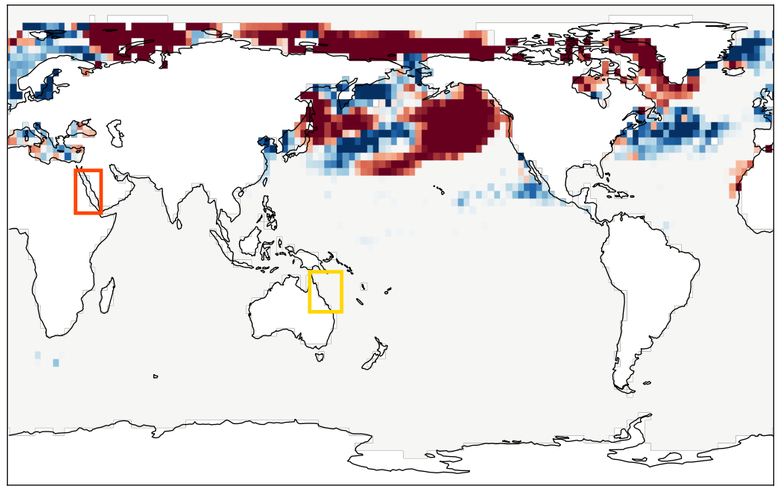
Actual DHW - Step 90

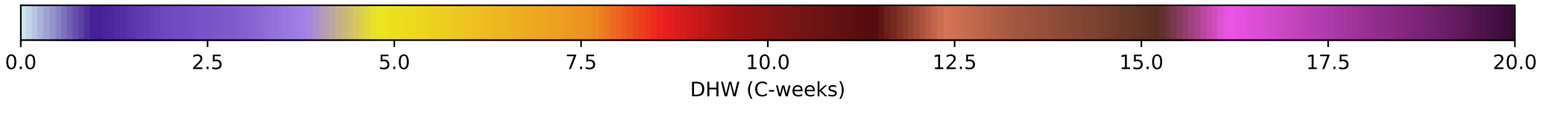


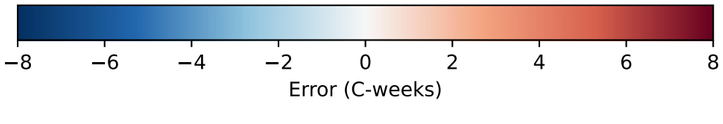
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

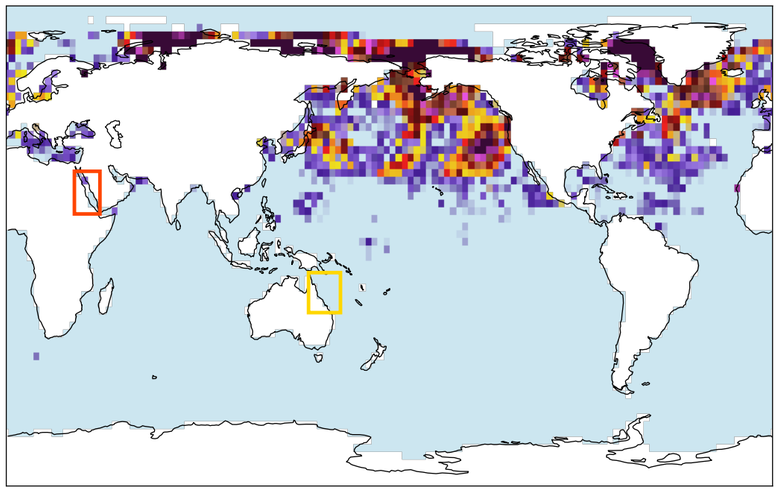




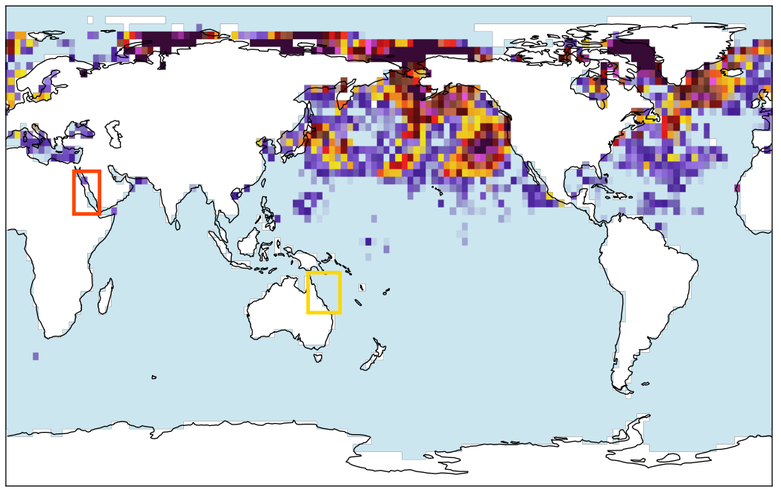


DHW Chain init: 2019-09-22 - Val (C-weeks)

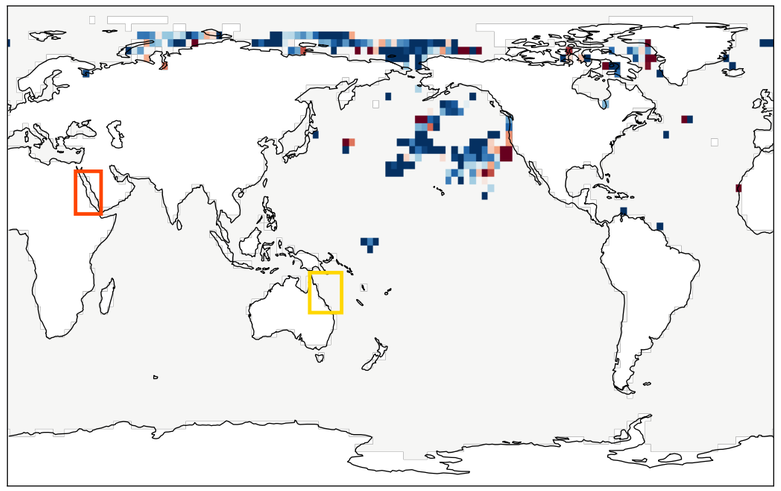
Actual DHW - Step 1



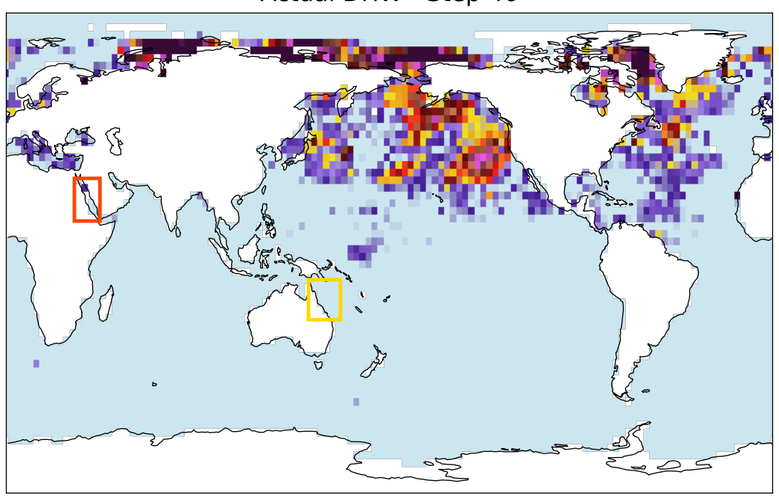
Predicted DHW - Step 1



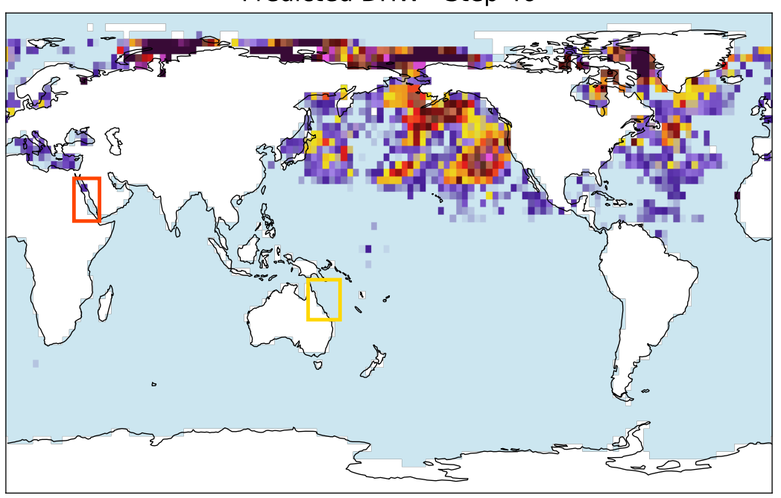
Error (Pred - Actual) - Step 1 [scale ±0.037]



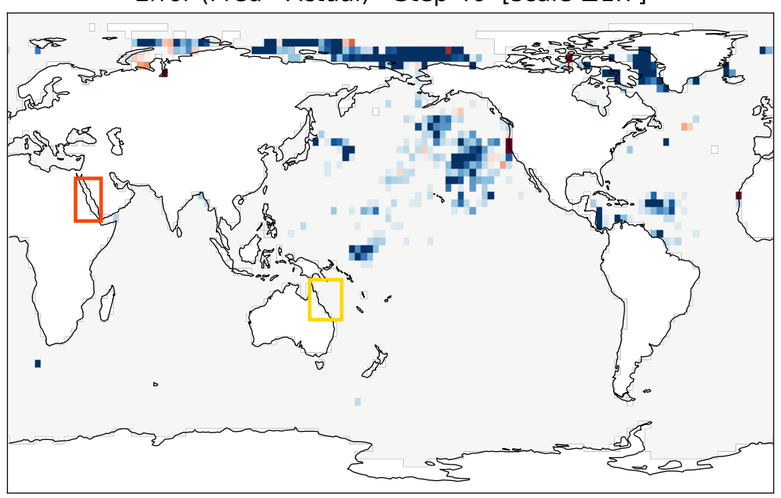
Actual DHW - Step 46



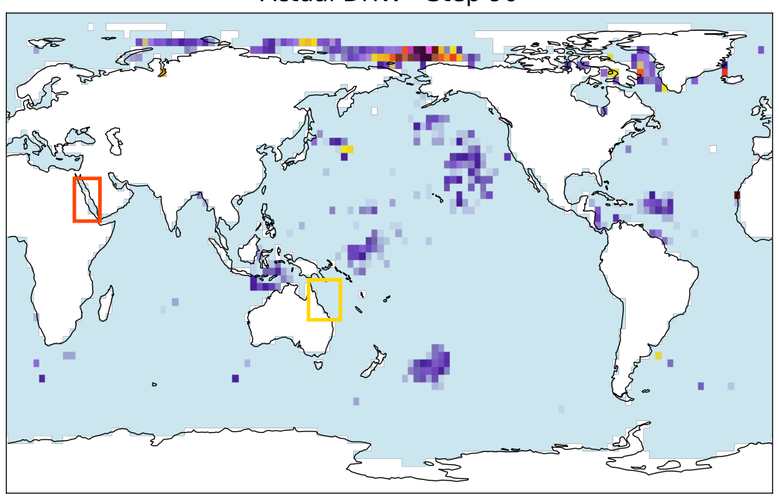
Predicted DHW - Step 46



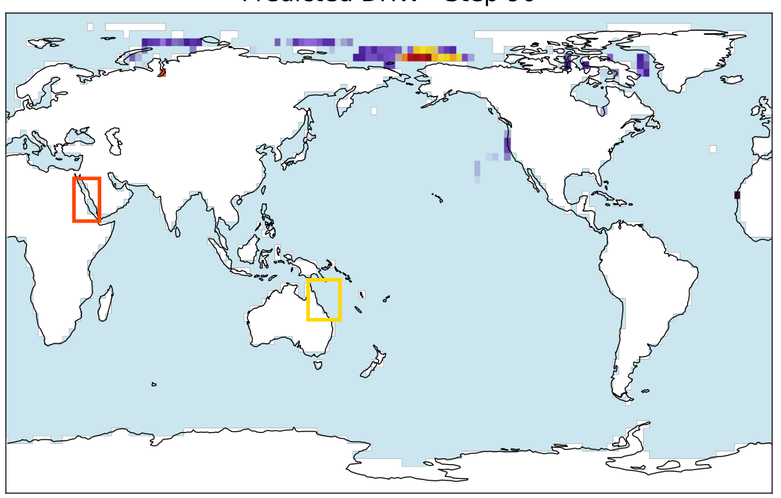
Error (Pred - Actual) - Step 46 [scale ±1.7]



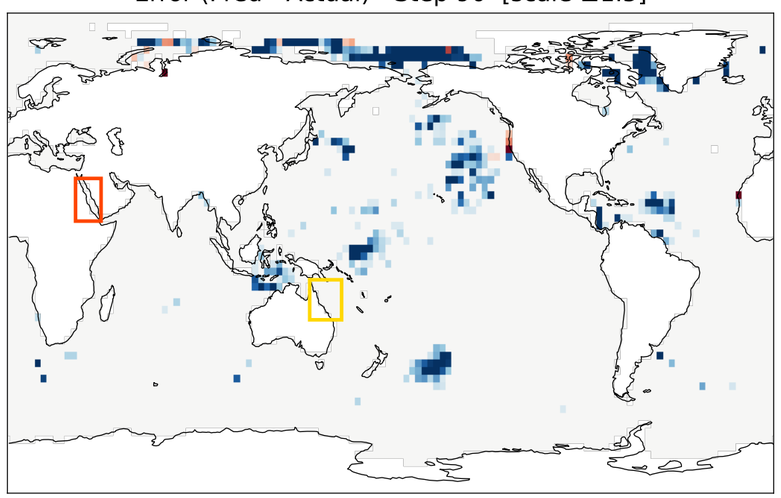
Actual DHW - Step 90

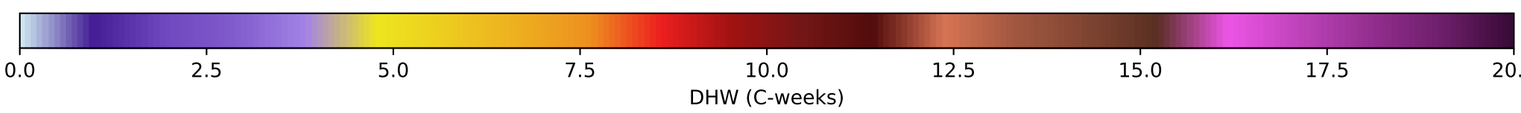


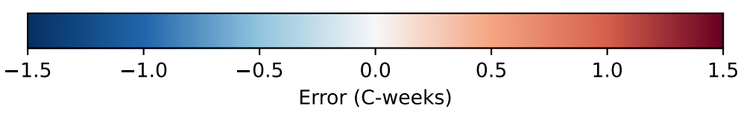
Predicted DHW - Step 90



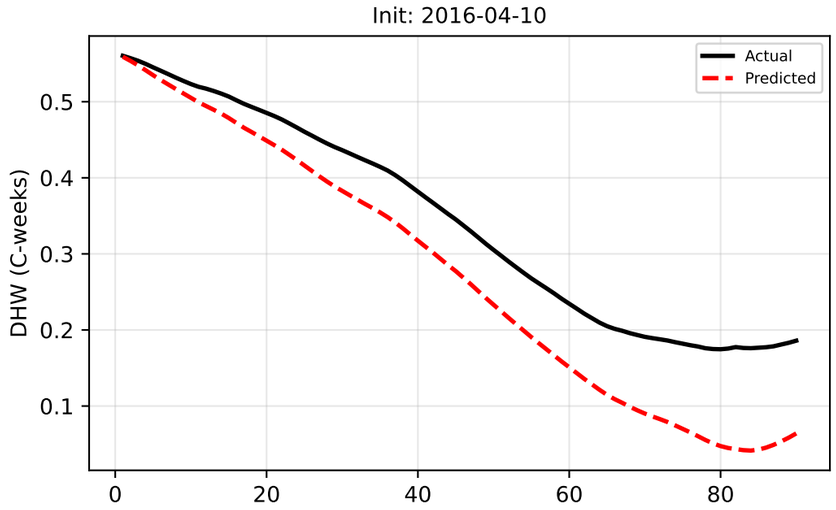
Error (Pred - Actual) - Step 90 [scale ±1.5]

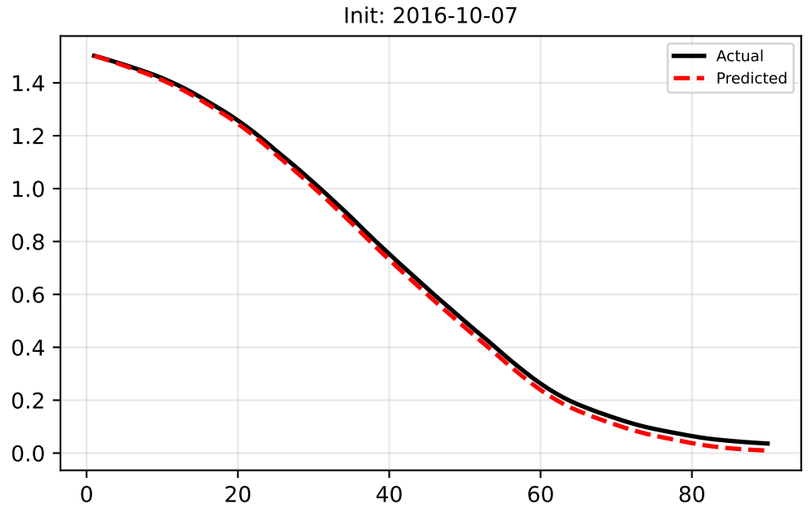


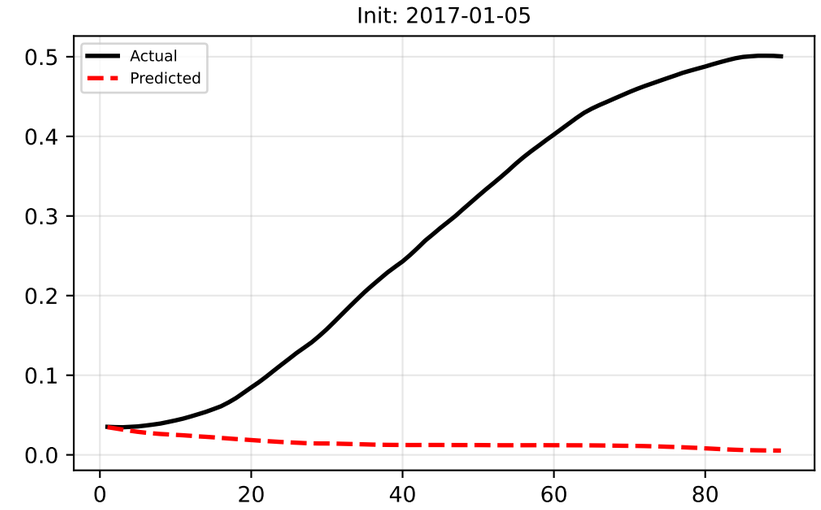


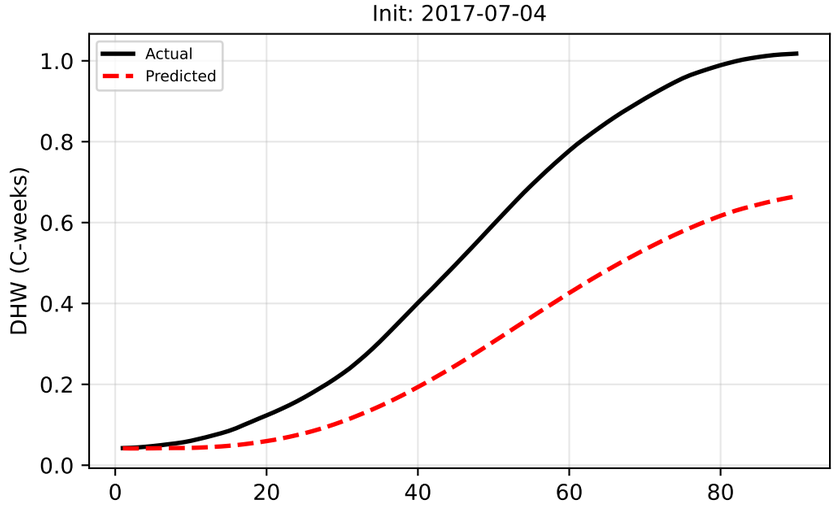


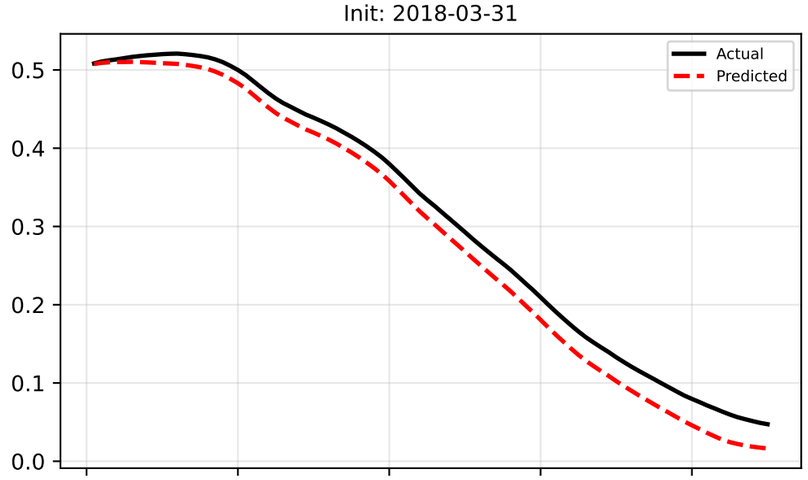
Global-Mean DHW - Val (C-weeks)

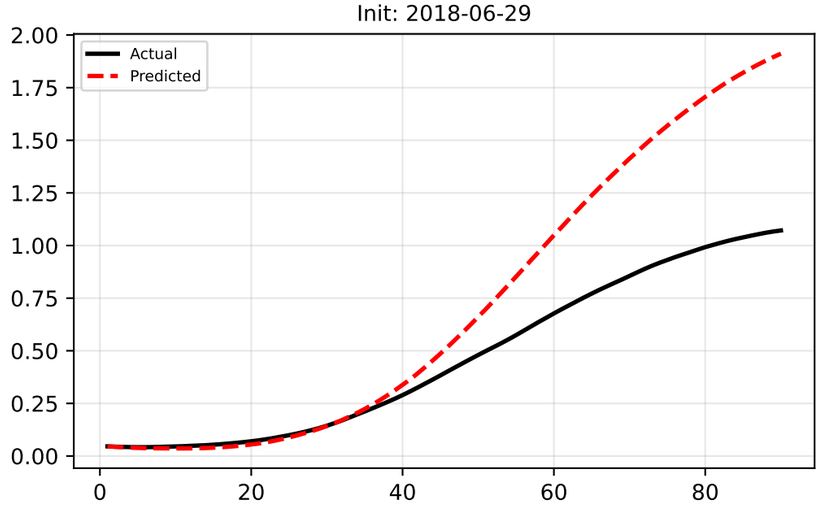


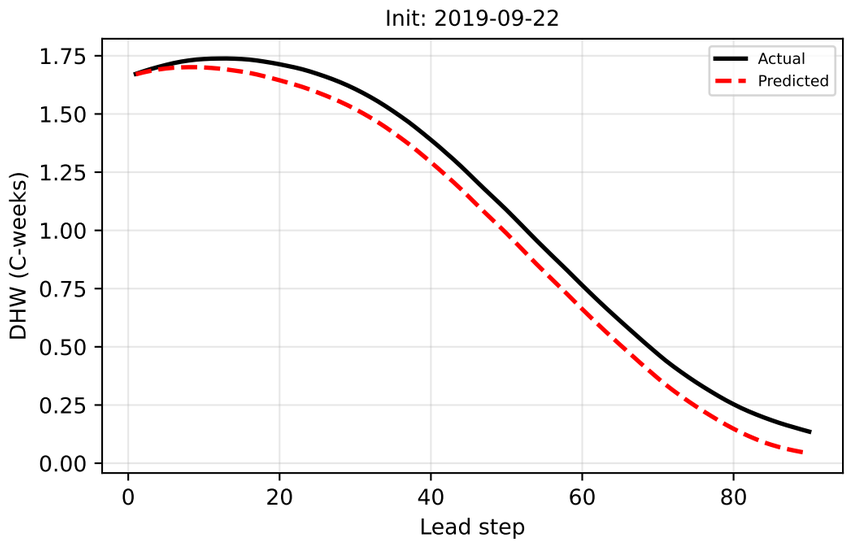




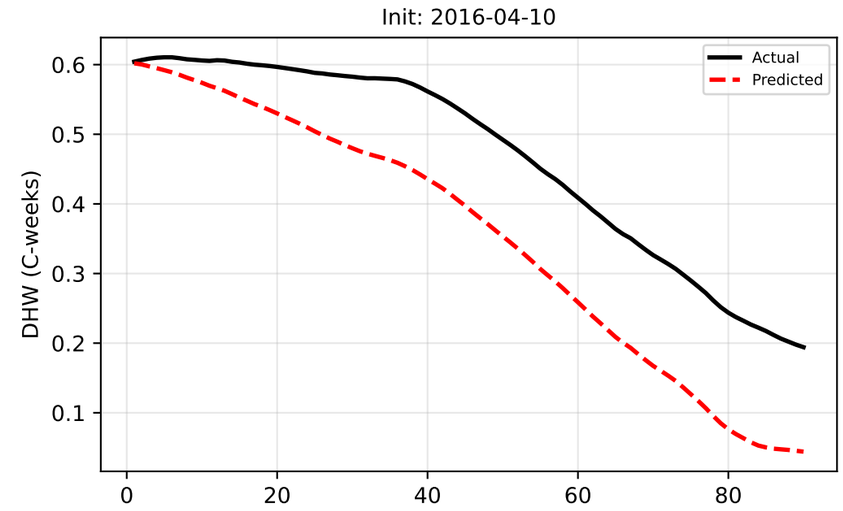


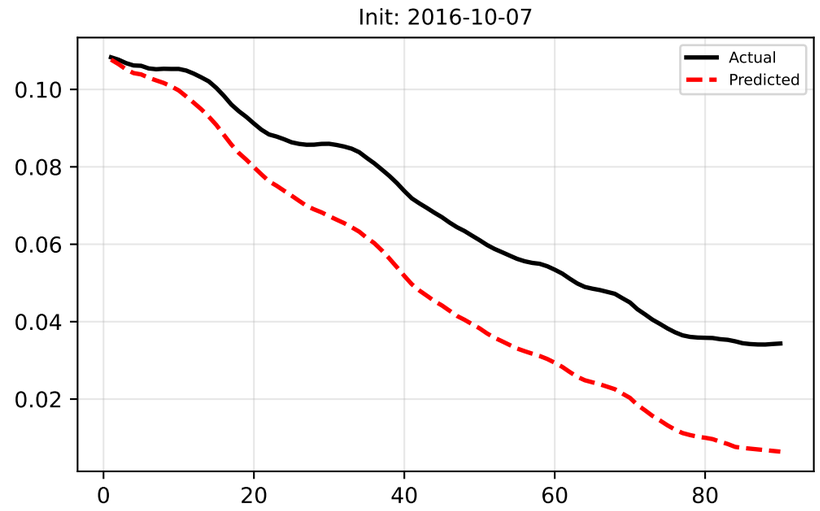


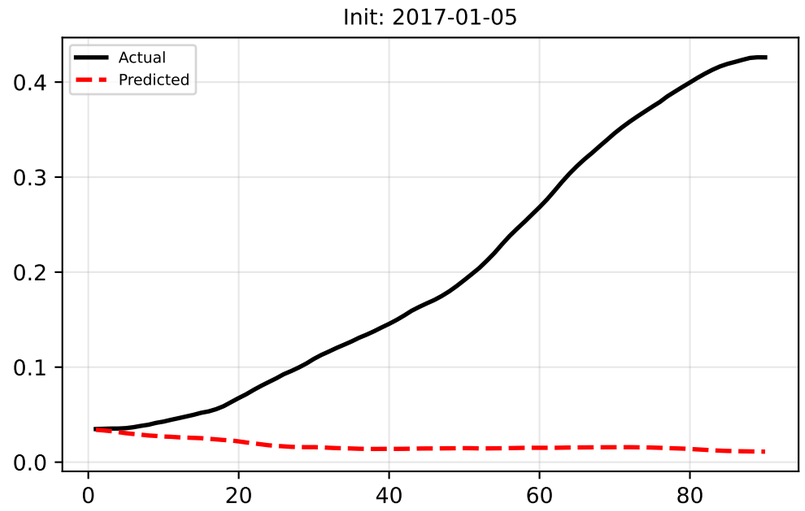


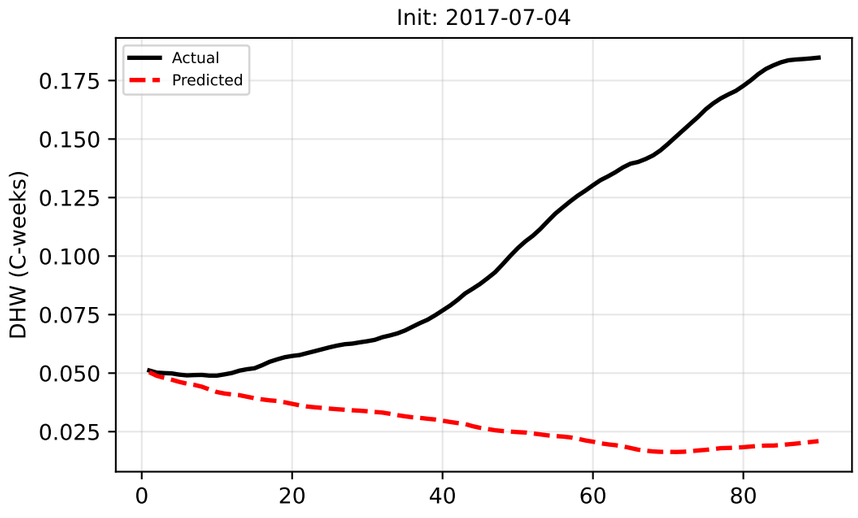


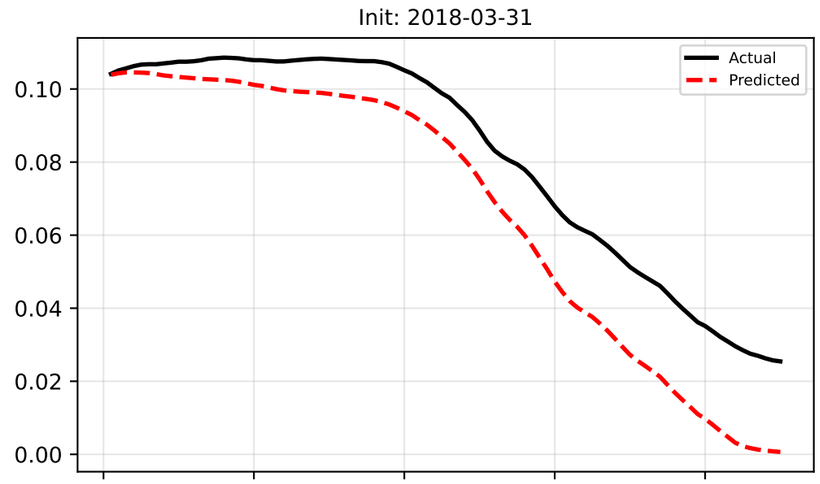
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

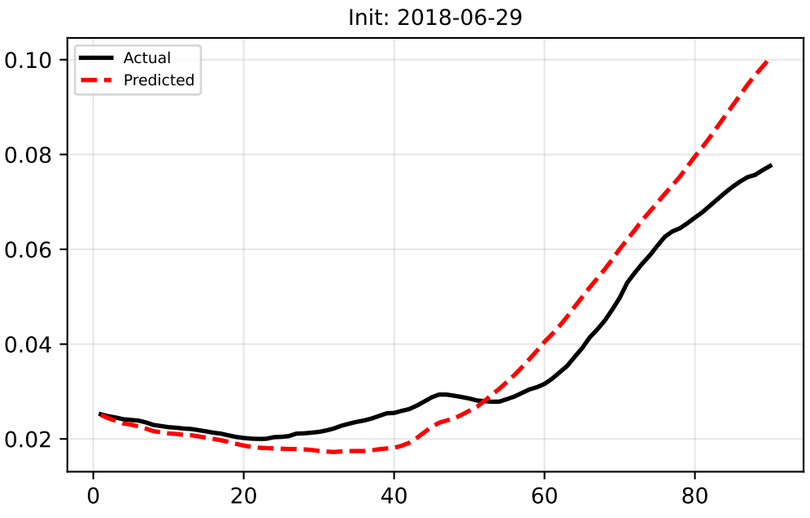


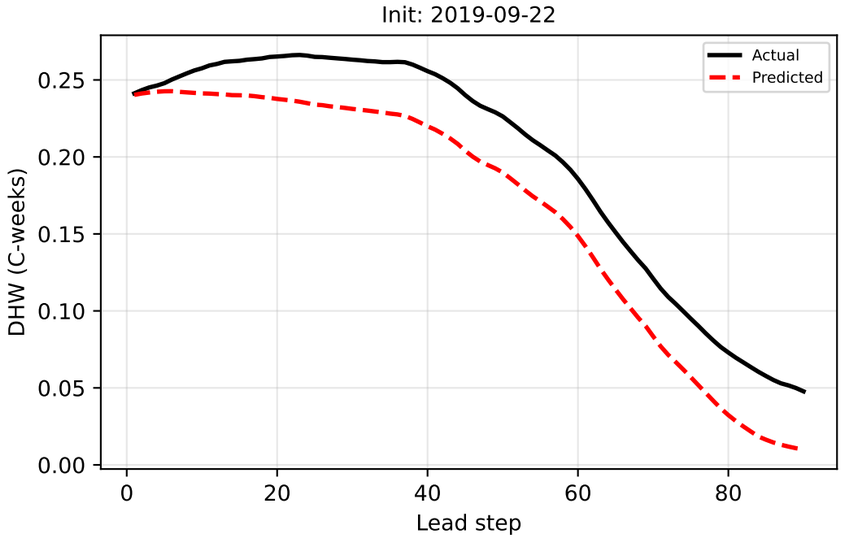




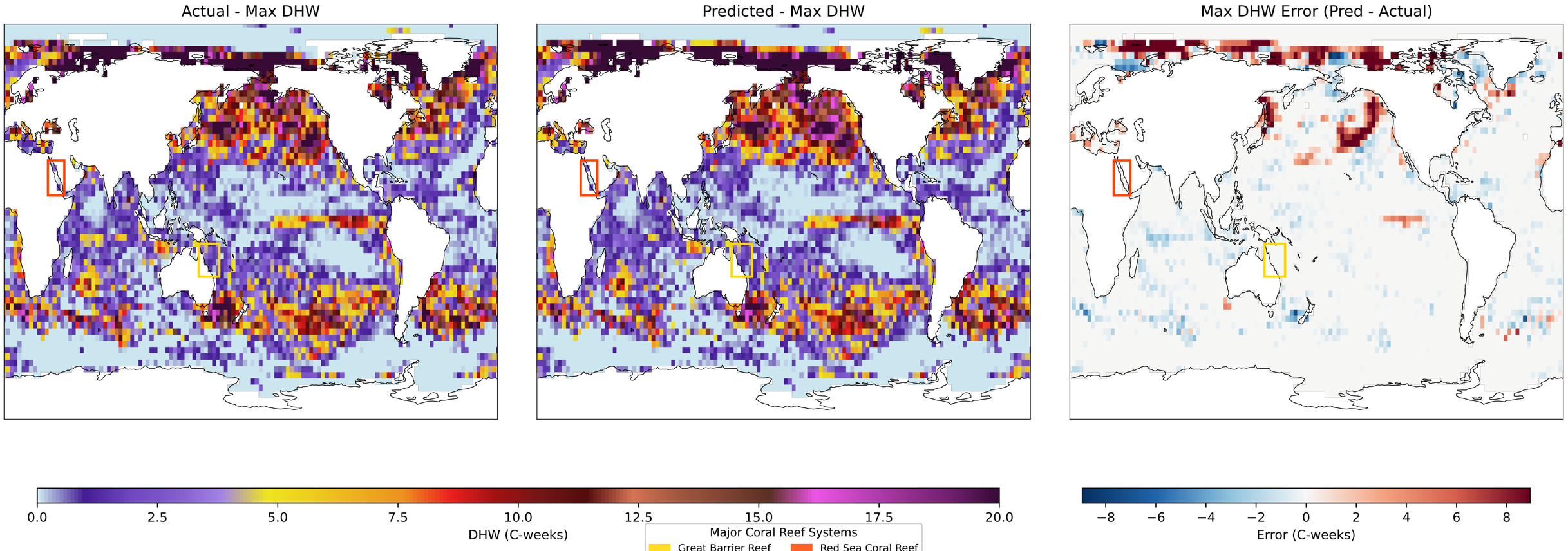


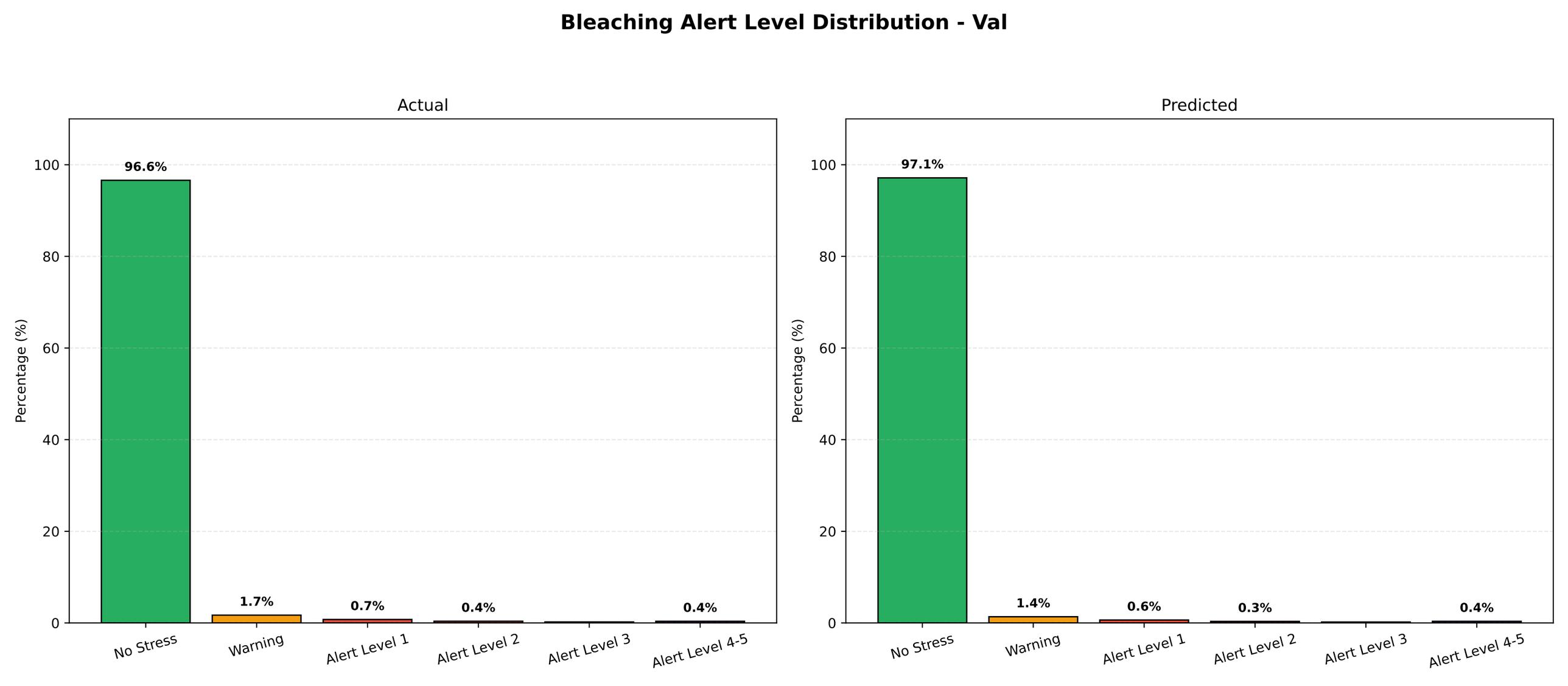






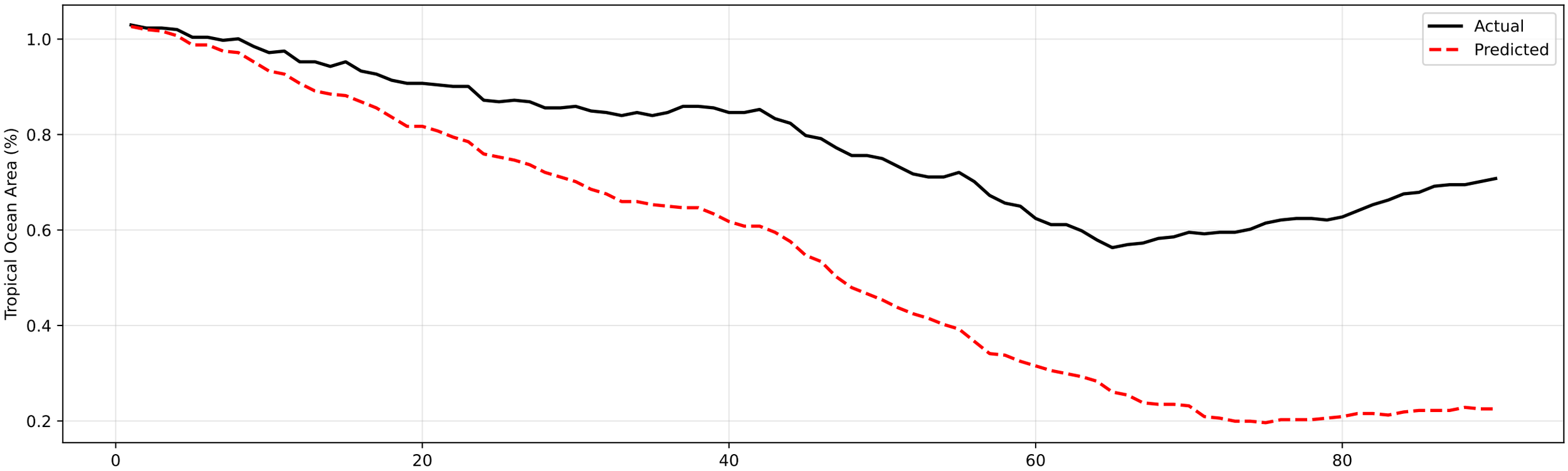
Peak DHW (max over all chains & leads) - Val (C-weeks)



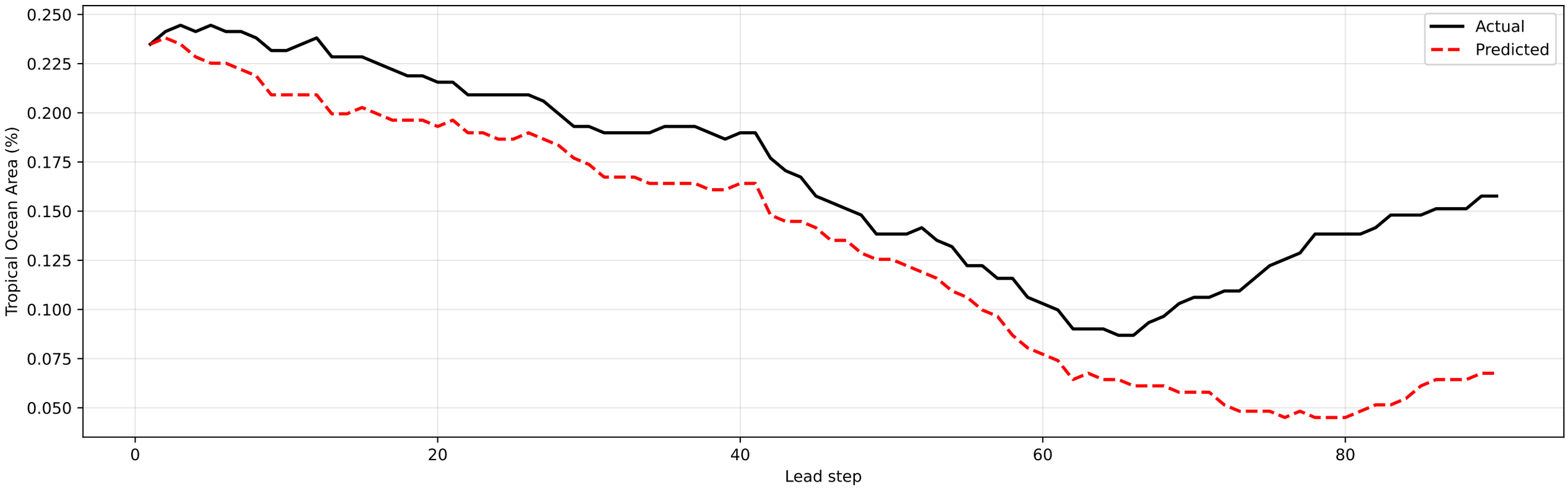

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.7%
0.4%
0.4%
97.1%
1.4%
0.6%
0.3%
0.4%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



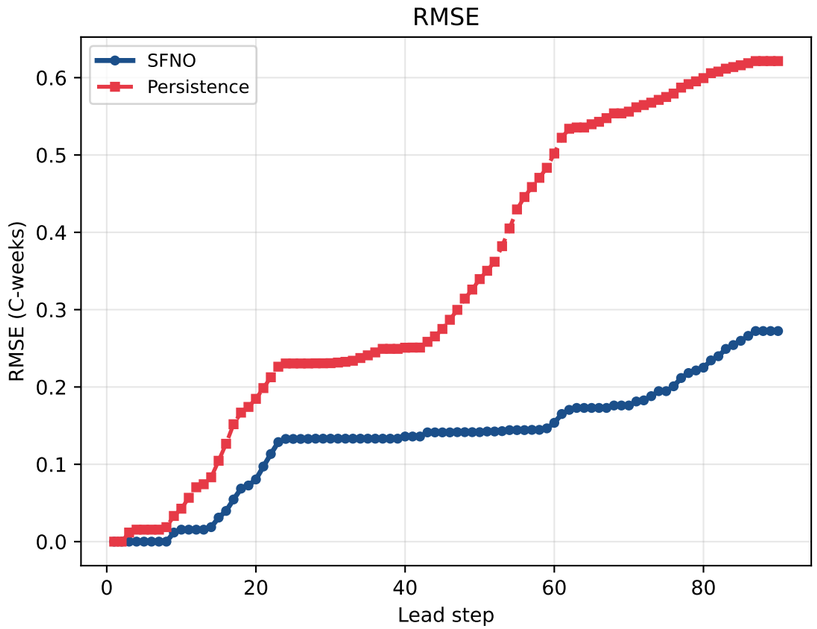
Great Barrier Reef Analysis - Val

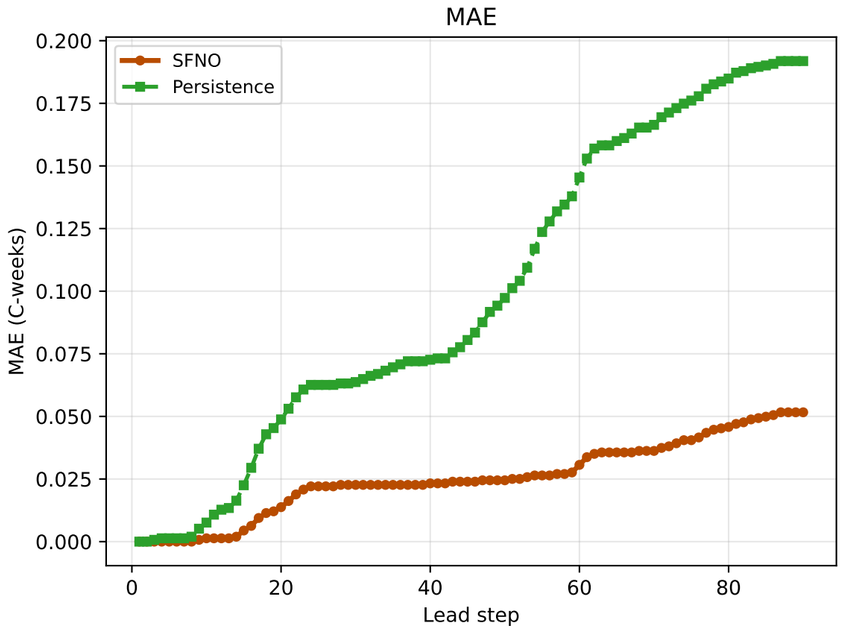
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 5 lat x 5 lon points
```

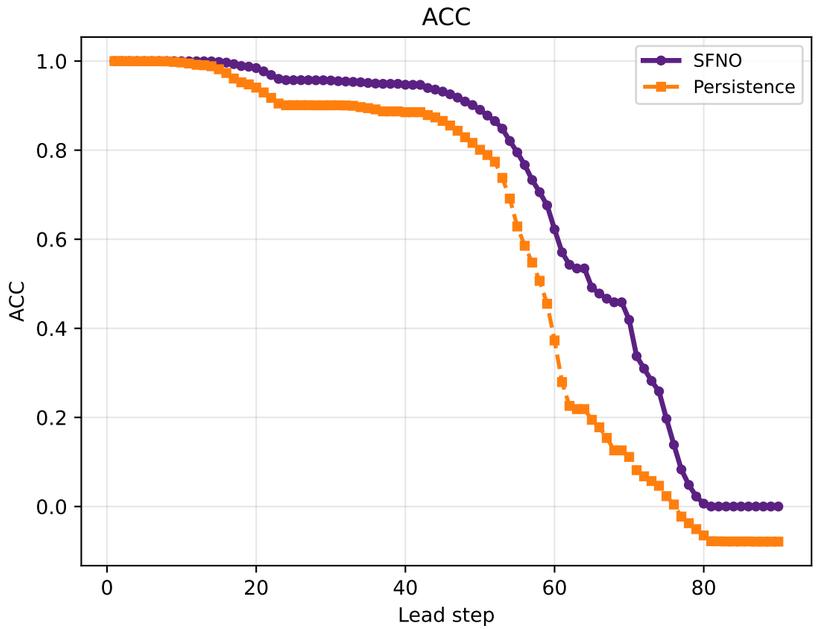
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

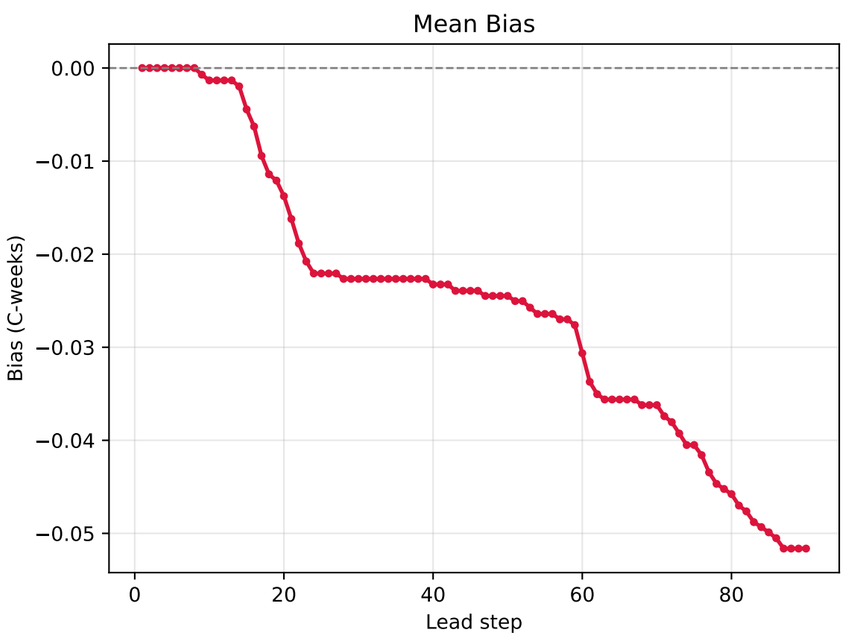
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
2	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
3	0.0000	0.0117	100.0%	0.0000	0.0007	0.0000	1.0000	0.9998	1.0000
4	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
5	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
6	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
7	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
8	0.0000	0.0187	100.0%	0.0000	0.0020	0.0000	1.0000	0.9994	1.0000
9	0.0117	0.0332	64.6%	0.0007	0.0052	-0.0007	0.9997	0.9981	0.9995
10	0.0155	0.0426	63.7%	0.0013	0.0076	-0.0013	0.9996	0.9969	0.9991
11	0.0155	0.0567	72.7%	0.0013	0.0108	-0.0013	0.9995	0.9947	0.9991
12	0.0155	0.0703	78.0%	0.0013	0.0127	-0.0013	0.9995	0.9917	0.9991
13	0.0155	0.0742	79.2%	0.0013	0.0134	-0.0013	0.9995	0.9907	0.9991
14	0.0187	0.0832	77.5%	0.0020	0.0163	-0.0020	0.9993	0.9884	0.9987
15	0.0311	0.1046	70.3%	0.0044	0.0225	-0.0044	0.9980	0.9817	0.9962
16	0.0397	0.1265	68.6%	0.0063	0.0295	-0.0063	0.9966	0.9732	0.9935
17	0.0545	0.1518	64.1%	0.0094	0.0371	-0.0094	0.9932	0.9608	0.9871
18	0.0686	0.1668	58.9%	0.0114	0.0429	-0.0114	0.9889	0.9523	0.9789
19	0.0726	0.1742	58.3%	0.0121	0.0453	-0.0121	0.9874	0.9478	0.9761
20	0.0804	0.1846	56.4%	0.0138	0.0488	-0.0138	0.9843	0.9404	0.9703
21	0.0971	0.1986	51.1%	0.0162	0.0531	-0.0162	0.9770	0.9296	0.9566
22	0.1133	0.2125	46.7%	0.0189	0.0576	-0.0189	0.9688	0.9174	0.9416
23	0.1287	0.2262	43.1%	0.0208	0.0607	-0.0208	0.9600	0.9047	0.9259
24	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
25	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
26	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
27	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
28	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
29	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
30	0.1332	0.2308	42.3%	0.0227	0.0637	-0.0227	0.9564	0.9008	0.9192
31	0.1332	0.2316	42.5%	0.0227	0.0649	-0.0227	0.9550	0.9008	0.9167
32	0.1332	0.2325	42.7%	0.0227	0.0662	-0.0227	0.9542	0.9004	0.9152
33	0.1332	0.2339	43.1%	0.0227	0.0669	-0.0227	0.9533	0.8996	0.9136
34	0.1332	0.2374	43.9%	0.0227	0.0683	-0.0227	0.9525	0.8967	0.9121
35	0.1332	0.2409	44.7%	0.0227	0.0696	-0.0227	0.9509	0.8944	0.9093
36	0.1332	0.2447	45.6%	0.0227	0.0708	-0.0227	0.9496	0.8915	0.9068
37	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
38	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
39	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
40	0.1359	0.2508	45.8%	0.0233	0.0726	-0.0233	0.9469	0.8852	0.9019
41	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
42	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
43	0.1413	0.2584	45.3%	0.0239	0.0756	-0.0239	0.9396	0.8788	0.8887
44	0.1413	0.2654	46.8%	0.0239	0.0776	-0.0239	0.9361	0.8735	0.8822
45	0.1413	0.2750	48.6%	0.0239	0.0805	-0.0239	0.9315	0.8656	0.8740
46	0.1413	0.2869	50.8%	0.0239	0.0834	-0.0239	0.9255	0.8554	0.8631
47	0.1415	0.2997	52.8%	0.0245	0.0876	-0.0245	0.9180	0.8435	0.8499
48	0.1415	0.3142	55.0%	0.0245	0.0917	-0.0245	0.9092	0.8291	0.8344
49	0.1415	0.3260	56.6%	0.0245	0.0942	-0.0245	0.9013	0.8163	0.8207
50	0.1415	0.3395	58.3%	0.0245	0.0973	-0.0245	0.8907	0.8009	0.8025
51	0.1424	0.3503	59.3%	0.0250	0.1012	-0.0250	0.8778	0.7892	0.7809
52	0.1424	0.3620	60.7%	0.0250	0.1041	-0.0250	0.8653	0.7741	0.7605
53	0.1429	0.3821	62.6%	0.0257	0.1093	-0.0257	0.8482	0.7377	0.7322
54	0.1442	0.4049	64.4%	0.0264	0.1169	-0.0264	0.8206	0.6912	0.6890
55	0.1442	0.4296	66.4%	0.0264	0.1236	-0.0264	0.7950	0.6287	0.6501
56	0.1442	0.4457	67.6%	0.0264	0.1278	-0.0264	0.7668	0.5854	0.6088
57	0.1445	0.4585	68.5%	0.0270	0.1318	-0.0270	0.7330	0.5477	0.5614
58	0.1445	0.4705	69.3%	0.0270	0.1346	-0.0270	0.7057	0.5066	0.5248
59	0.1465	0.4834	69.7%	0.0276	0.1379	-0.0276	0.6760	0.4551	0.4861
60	0.1537	0.5019	69.4%	0.0306	0.1454	-0.0306	0.6222	0.3728	0.4186
61	0.1651	0.5223	68.4%	0.0337	0.1529	-0.0337	0.5707	0.2794	0.3594
62	0.1704	0.5339	68.1%	0.0350	0.1569	-0.0350	0.5428	0.2259	0.3295
63	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
64	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
65	0.1729	0.5397	68.0%	0.0356	0.1599	-0.0356	0.4915	0.1947	0.2746
66	0.1729	0.5429	68.2%	0.0356	0.1611	-0.0356	0.4781	0.1779	0.2597
67	0.1729	0.5476	68.4%	0.0356	0.1629	-0.0356	0.4666	0.1539	0.2469
68	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
69	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
70	0.1760	0.5561	68.3%	0.0362	0.1664	-0.0362	0.4189	0.1113	0.2001
71	0.1812	0.5617	67.7%	0.0374	0.1695	-0.0374	0.3375	0.0816	0.1329
72	0.1827	0.5646	67.6%	0.0380	0.1713	-0.0380	0.3096	0.0678	0.1116
73	0.1881	0.5677	66.9%	0.0393	0.1731	-0.0393	0.2819	0.0571	0.0915
74	0.1946	0.5713	65.9%	0.0405	0.1749	-0.0405	0.2585	0.0465	0.0756
75	0.1946	0.5750	66.2%	0.0405	0.1761	-0.0405	0.1969	0.0233	0.0384
76	0.2009	0.5794	65.3%	0.0416	0.1778	-0.0416	0.1385	0.0044	0.0092
77	0.2116	0.5871	64.0%	0.0434	0.1808	-0.0434	0.0831	-0.0226	-0.0142
78	0.2180	0.5916	63.1%	0.0447	0.1826	-0.0447	0.0485	-0.0374	-0.0277
79	0.2214	0.5952	62.8%	0.0452	0.1837	-0.0452	0.0225	-0.0507	-0.0367
80	0.2250	0.5994	62.5%	0.0458	0.1849	-0.0458	0.0064	-0.0649	-0.0418
81	0.2344	0.6058	61.3%	0.0470	0.1872	-0.0470	0.0000	-0.0779	-0.0419
82	0.2398	0.6079	60.6%	0.0476	0.1879	-0.0476	0.0000	-0.0782	-0.0411
83	0.2492	0.6117	59.3%	0.0488	0.1890	-0.0488	0.0000	-0.0785	-0.0399
84	0.2542	0.6137	58.6%	0.0493	0.1896	-0.0493	0.0000	-0.0786	-0.0391
85	0.2597	0.6160	57.8%	0.0499	0.1901	-0.0499	0.0000	-0.0786	-0.0383
86	0.2661	0.6187	57.0%	0.0505	0.1907	-0.0505	0.0000	-0.0785	-0.0374
87	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
88	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
89	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
90	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373

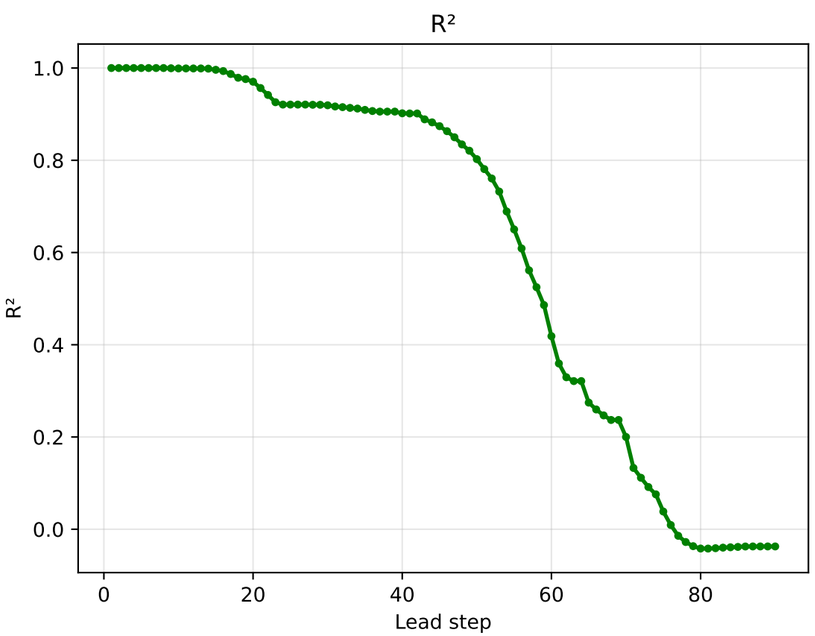
DHW Forecast Skill vs Lead Step - Val - GBR

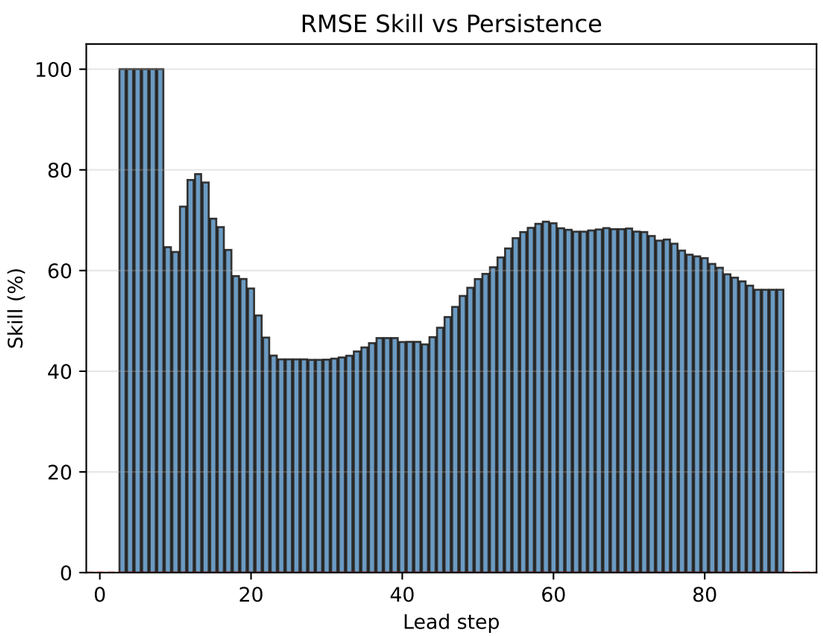


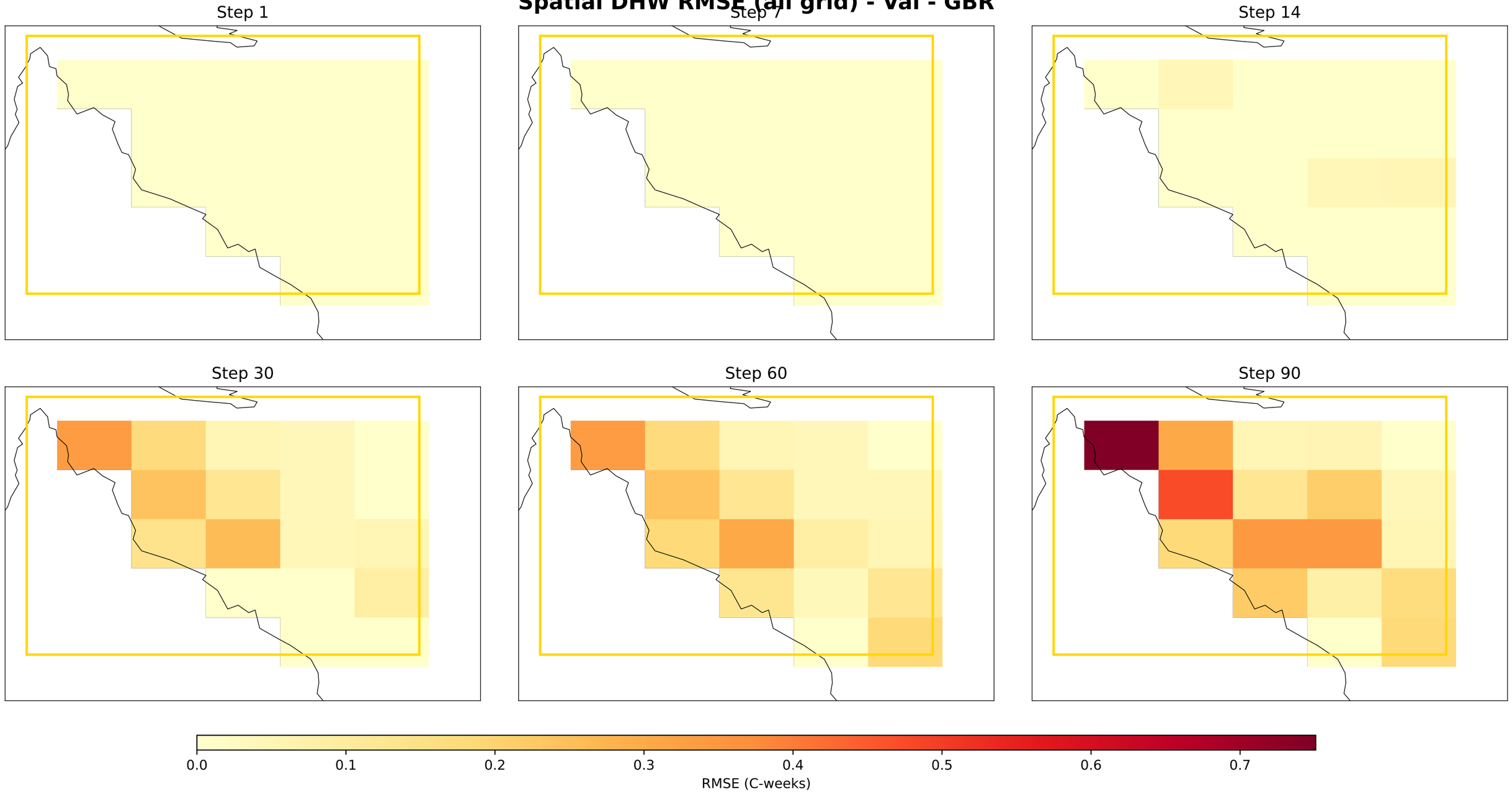


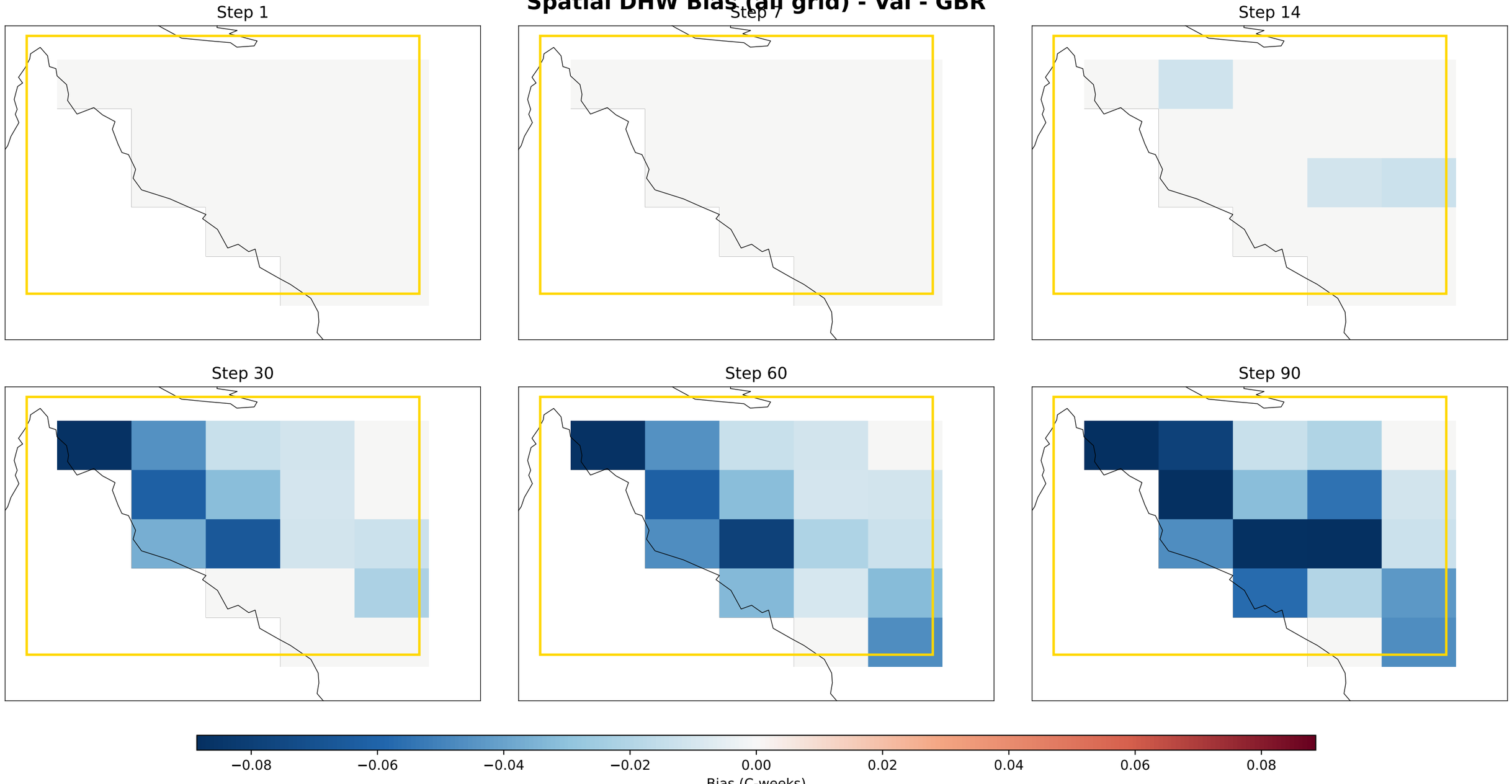


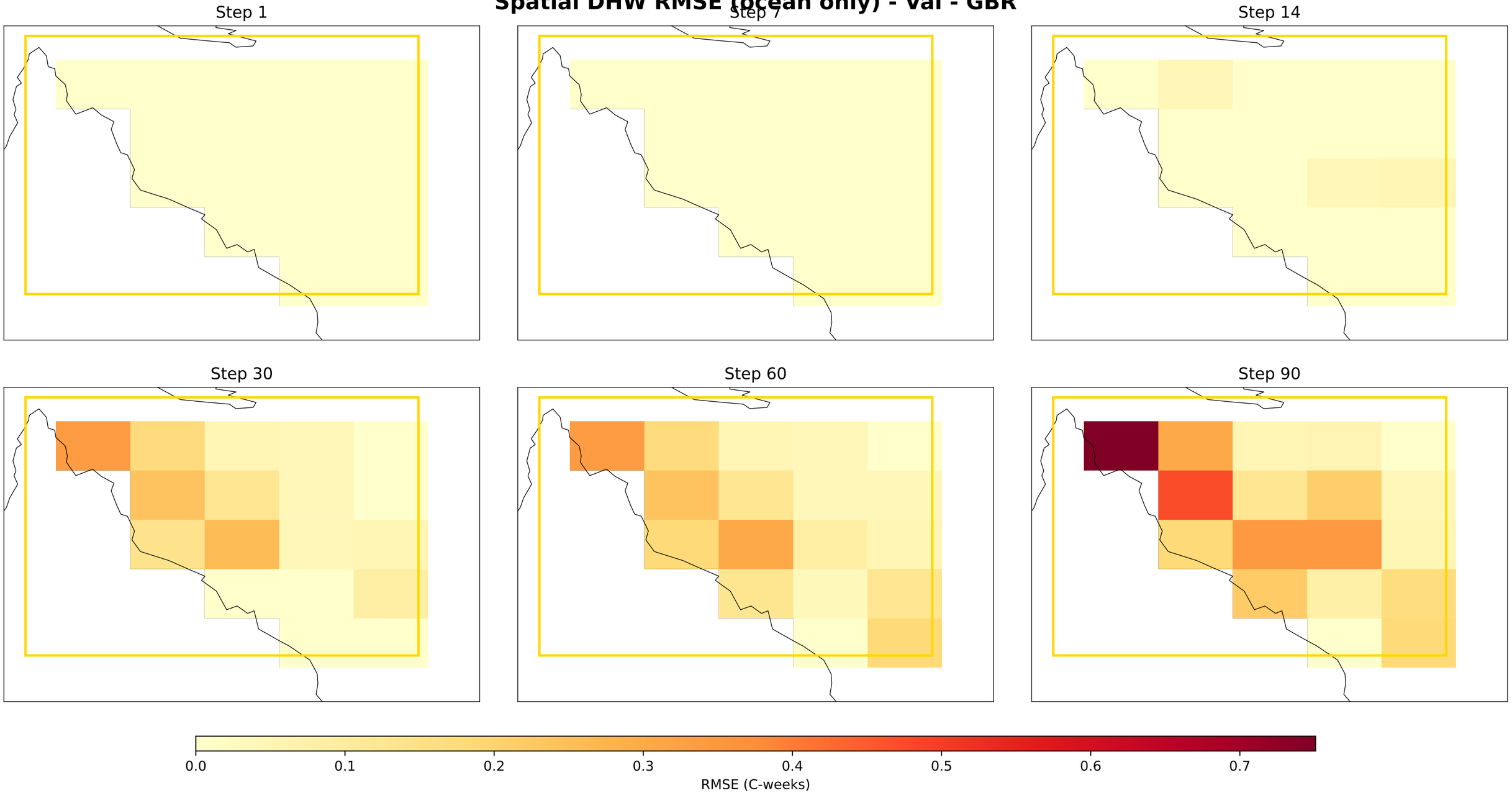


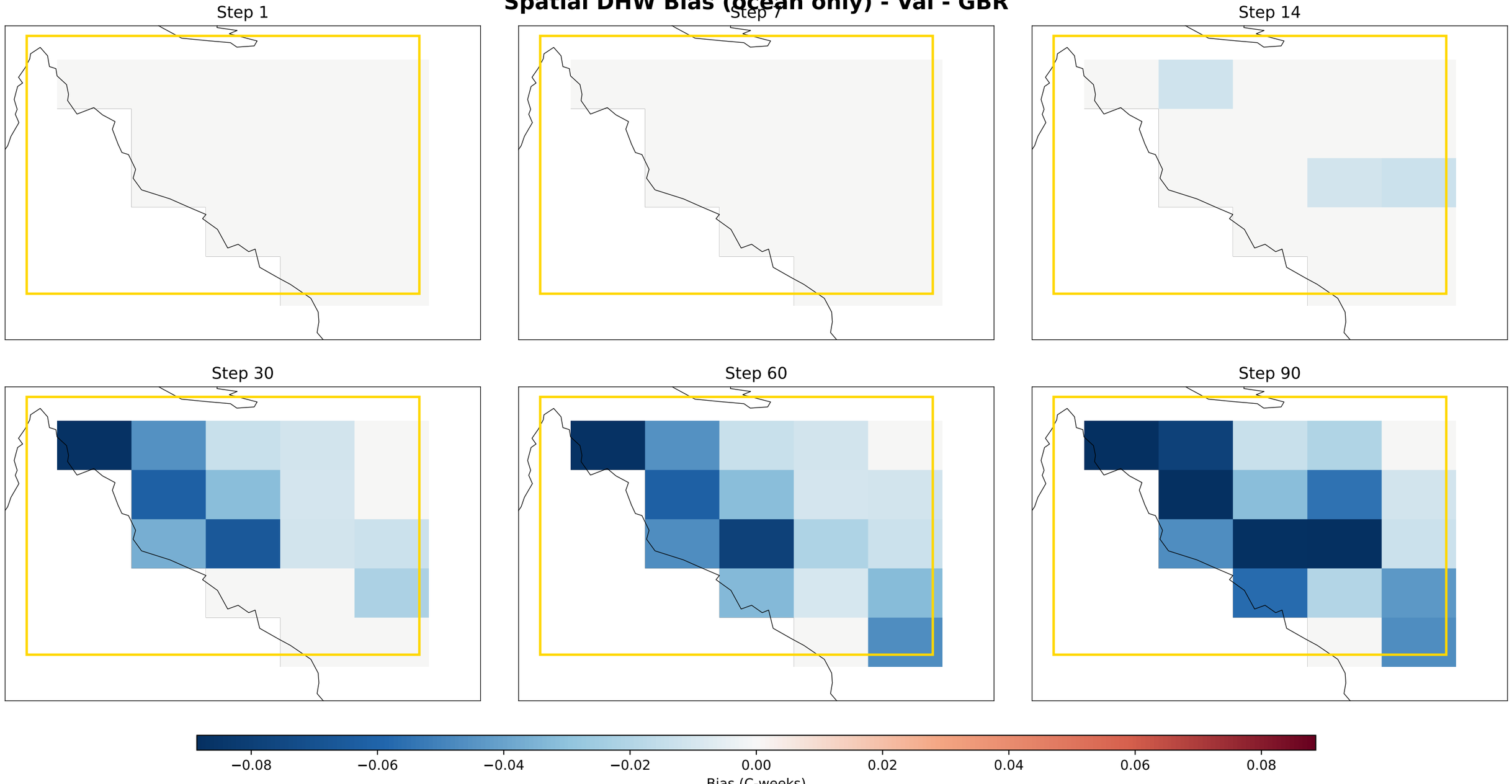





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

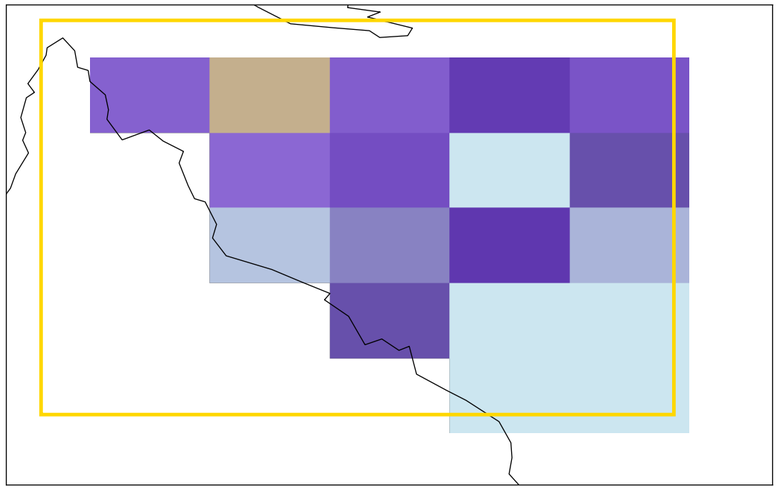

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

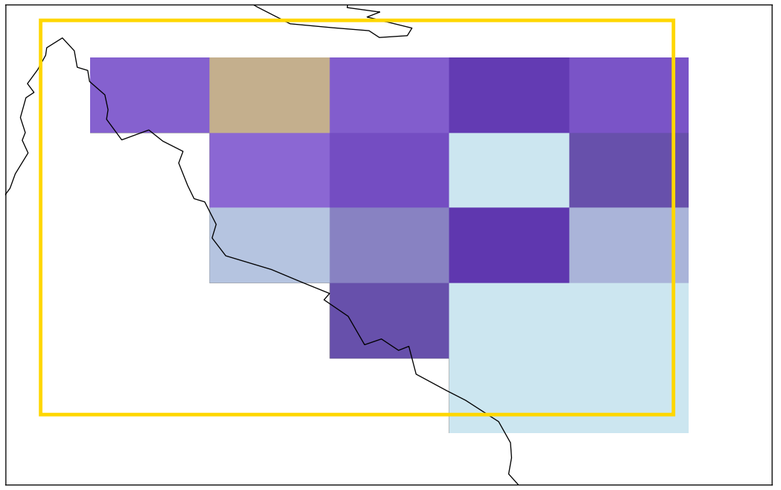

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

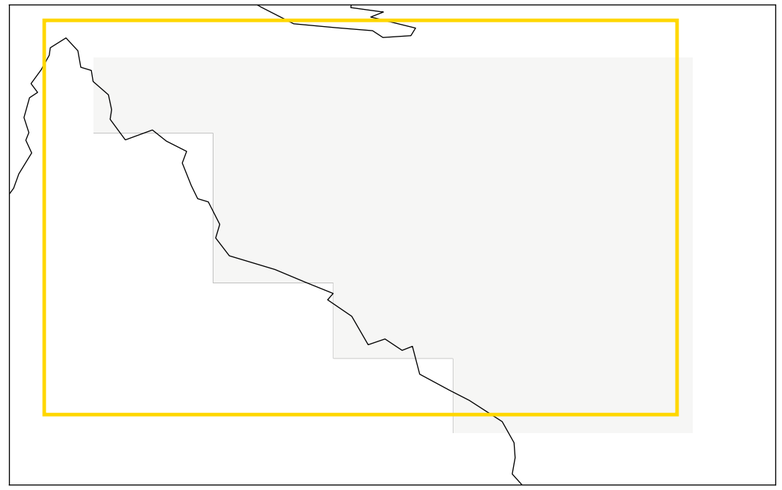
Actual DHW - Step 1



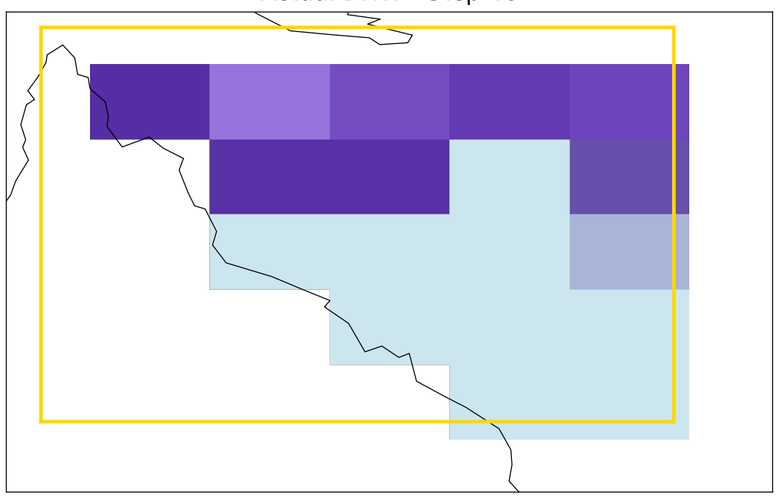
Predicted DHW - Step 1



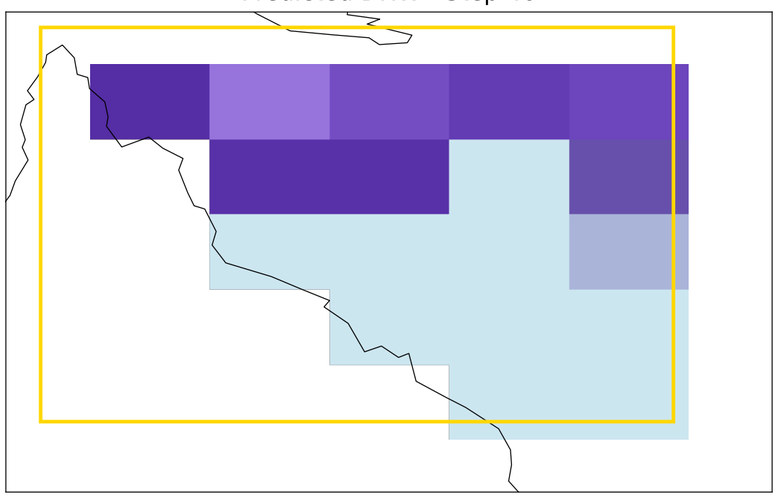
Error (Pred - Actual) - Step 1 [scale ±8]



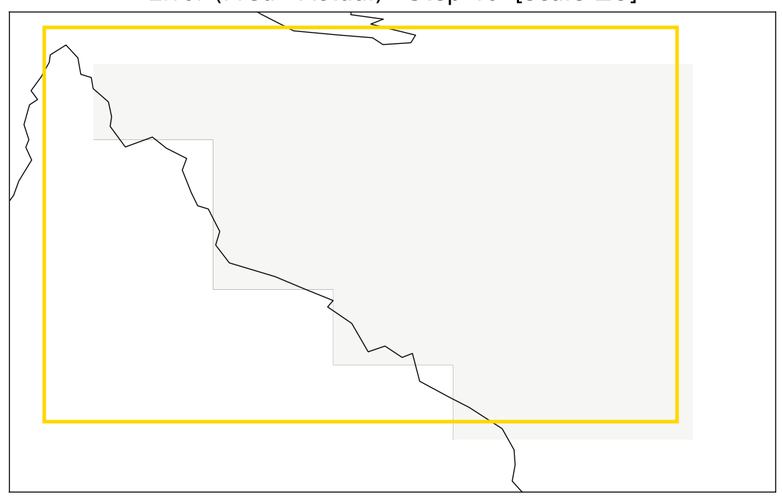
Actual DHW - Step 46



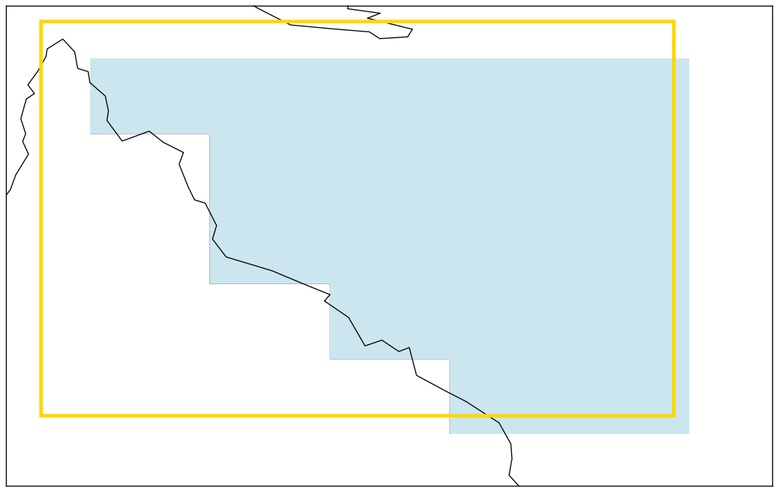
Predicted DHW - Step 46



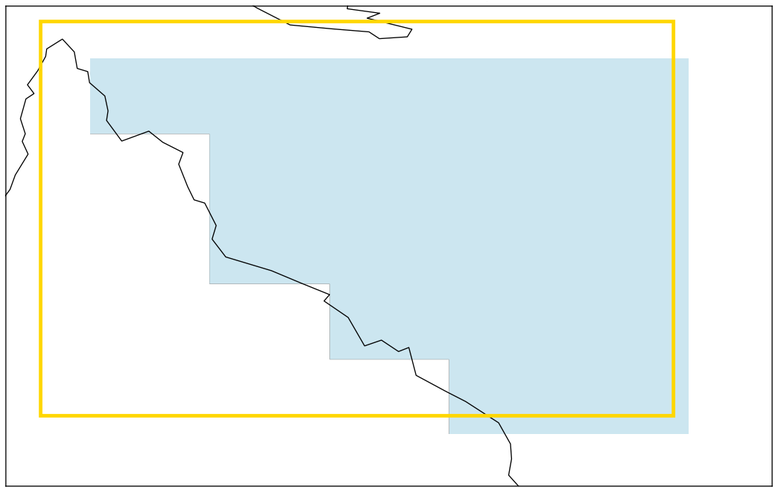
Error (Pred - Actual) - Step 46 [scale ±8]



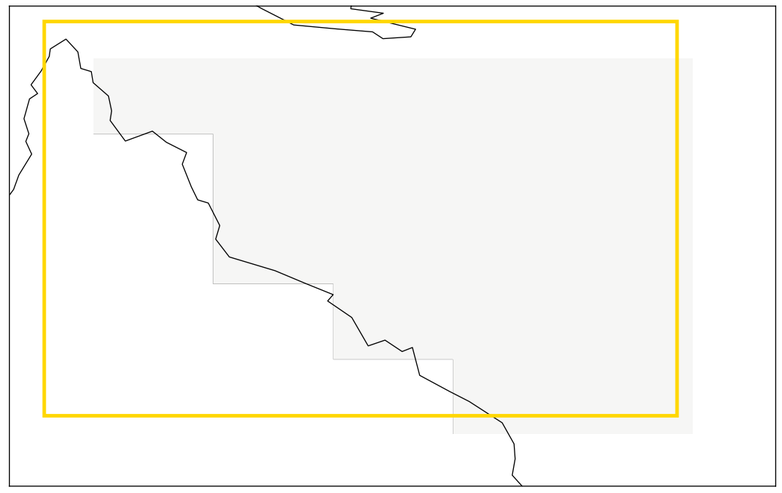
Actual DHW - Step 90

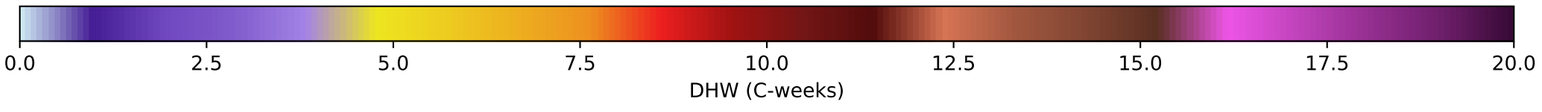


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

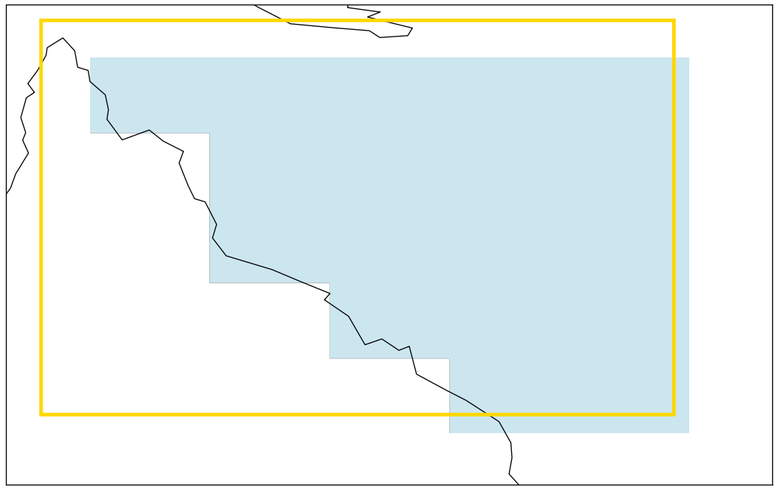




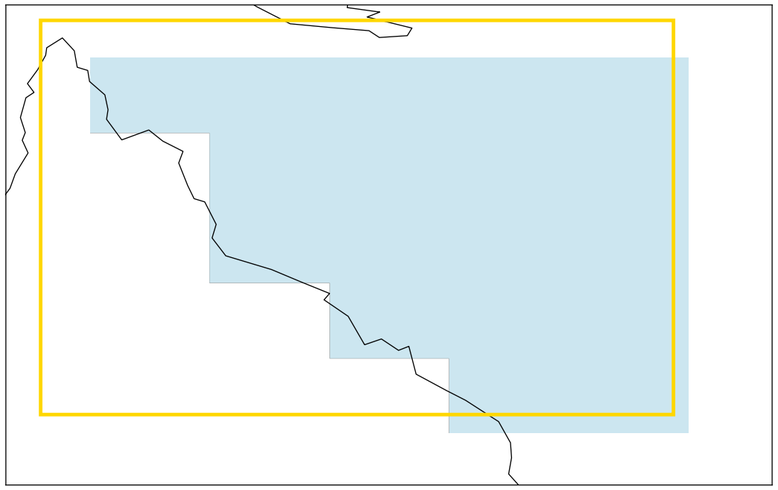


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

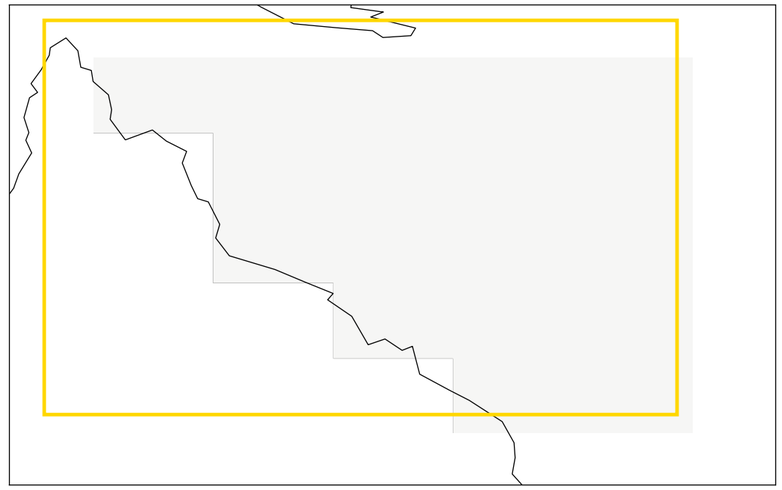
Actual DHW - Step 1



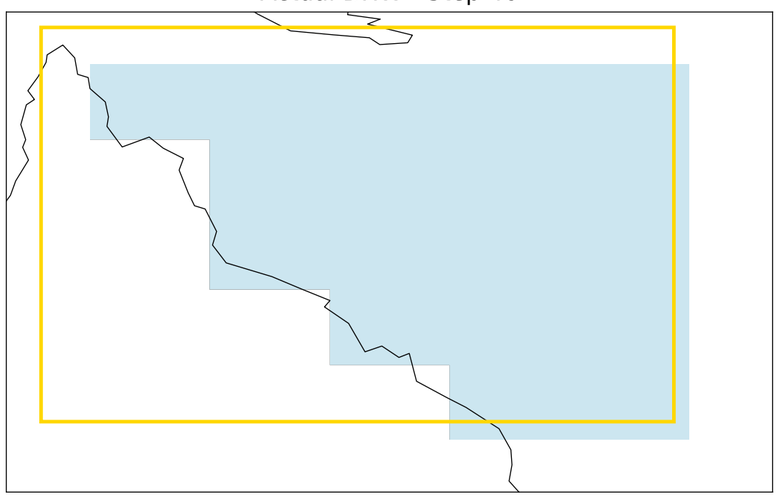
Predicted DHW - Step 1



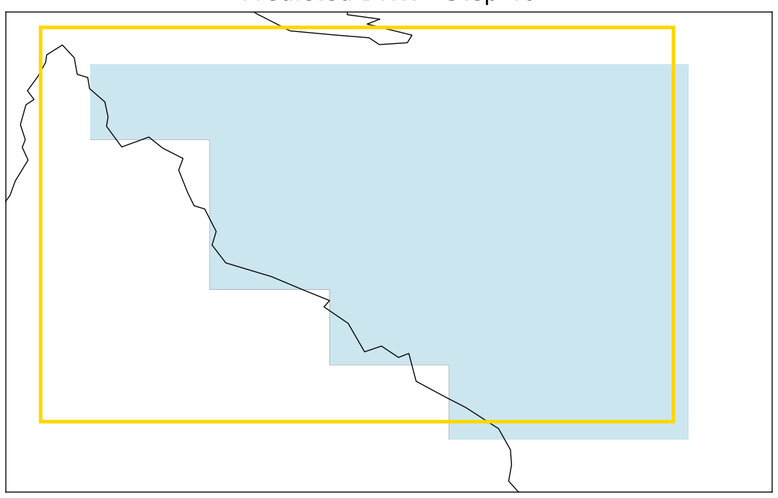
Error (Pred - Actual) - Step 1 [scale ±8]



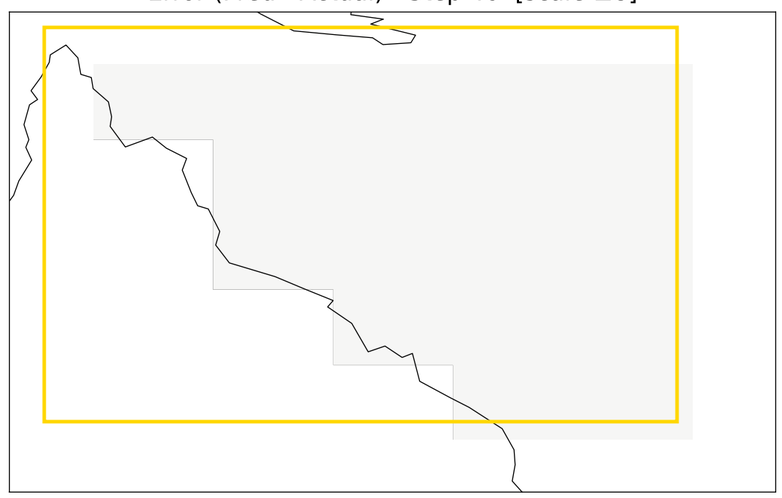
Actual DHW - Step 46



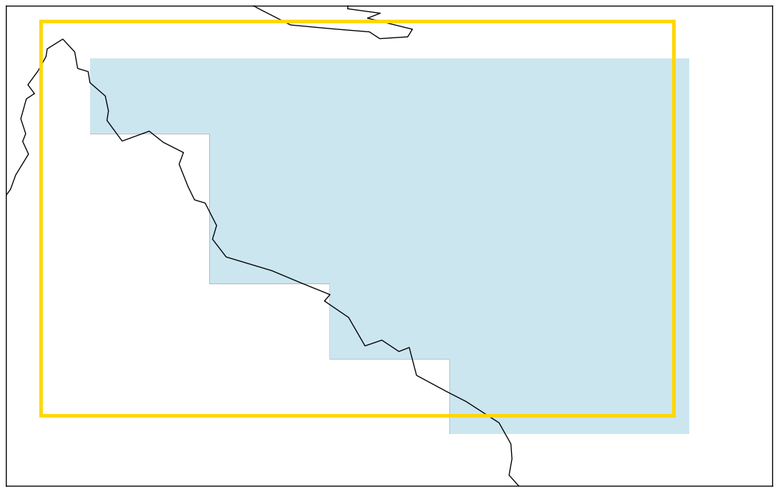
Predicted DHW - Step 46



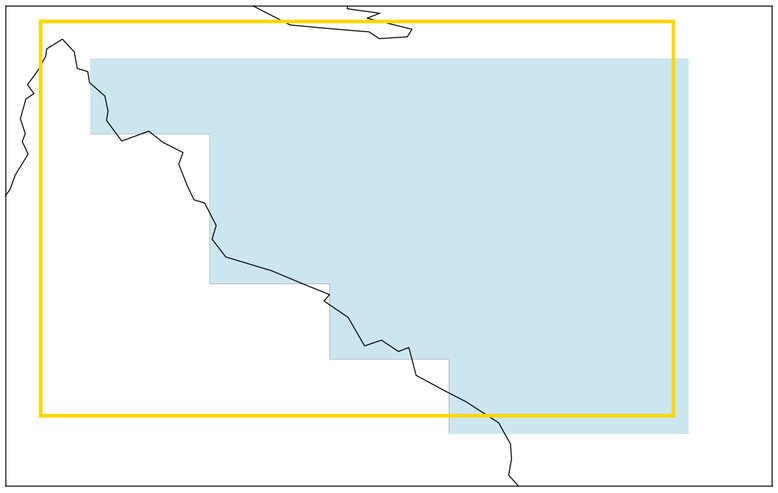
Error (Pred - Actual) - Step 46 [scale ±8]



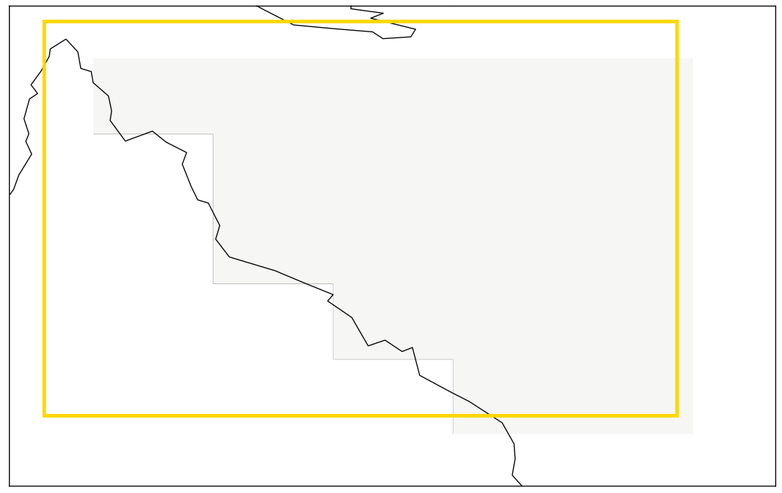
Actual DHW - Step 90

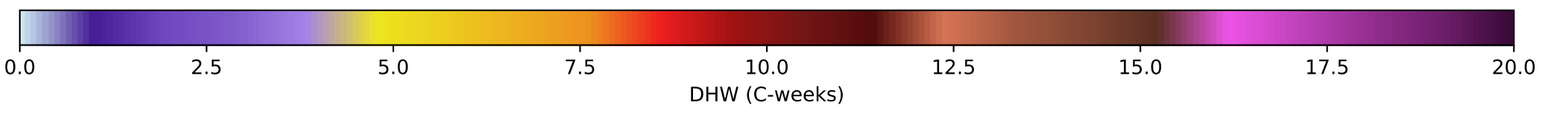


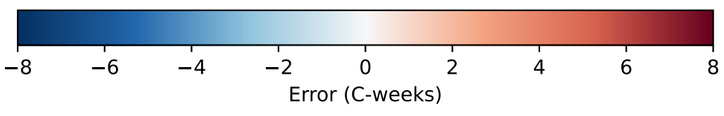
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

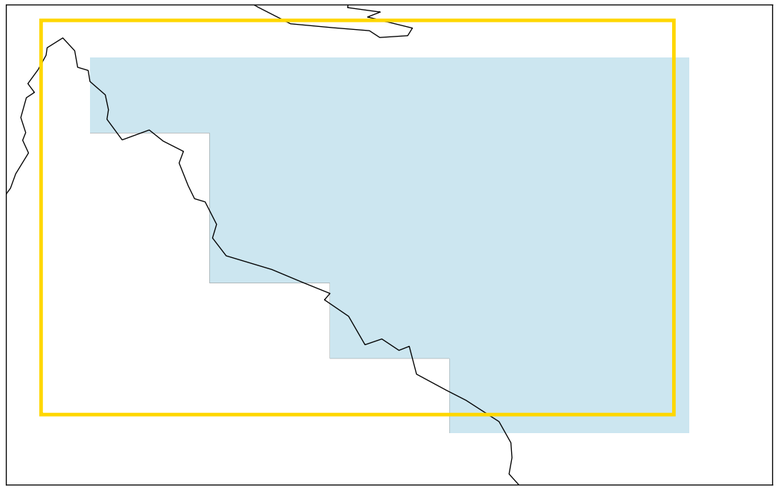




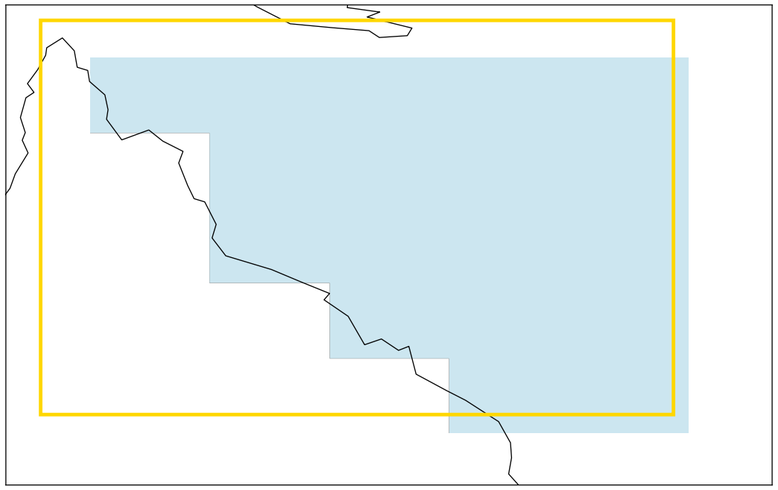


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

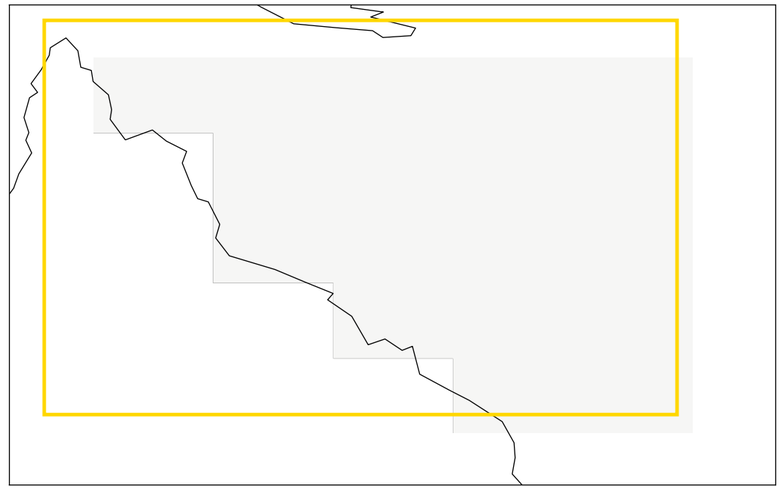
Actual DHW - Step 1



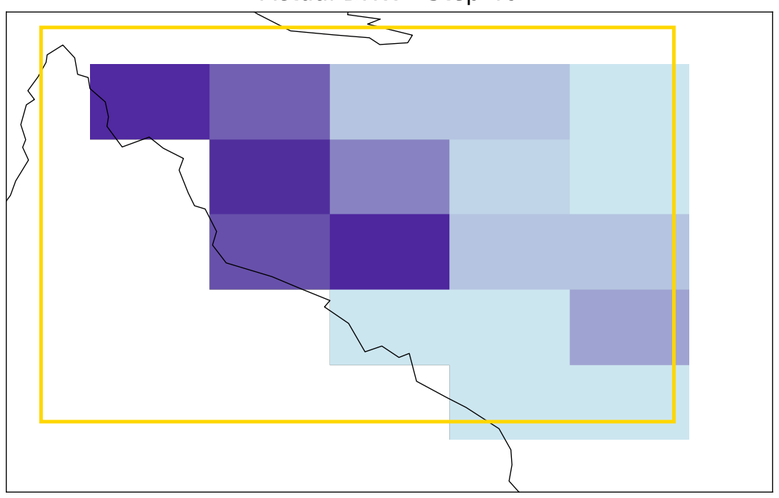
Predicted DHW - Step 1



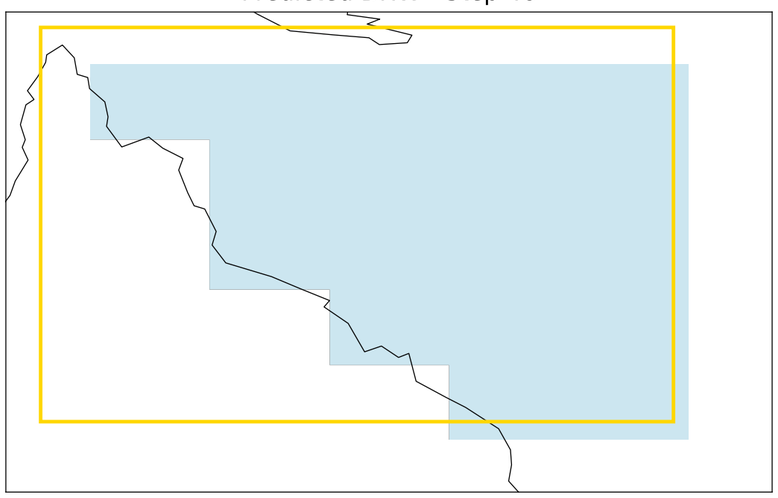
Error (Pred - Actual) - Step 1 [scale ±8]



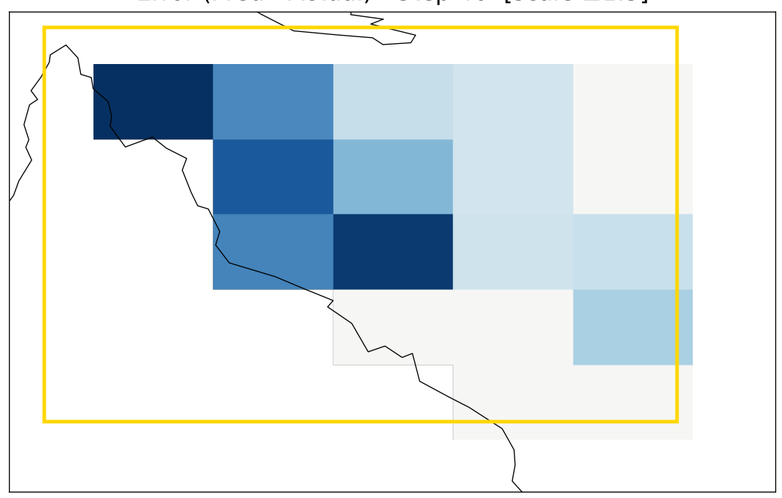
Actual DHW - Step 46



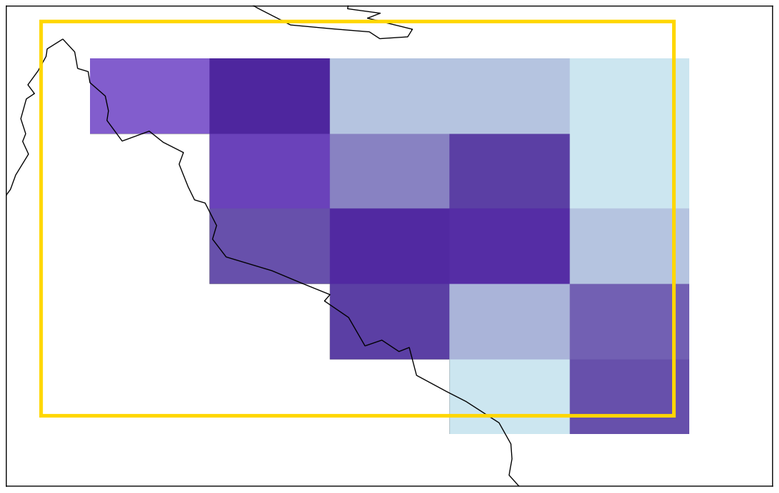
Predicted DHW - Step 46



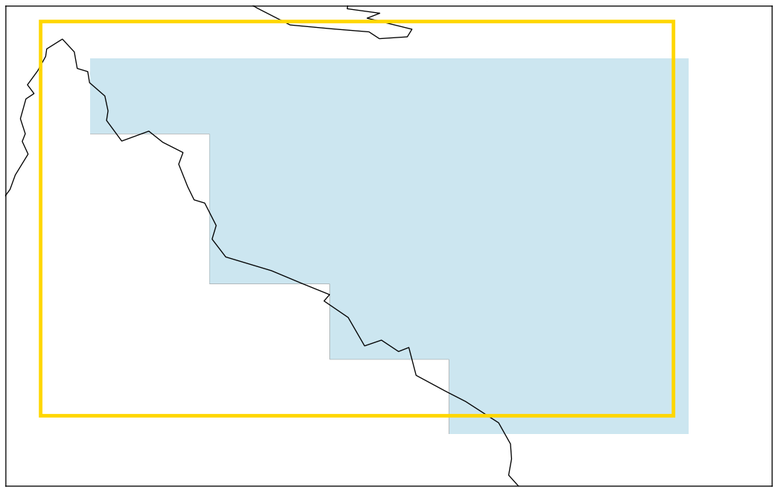
Error (Pred - Actual) - Step 46 [scale ±1.3]



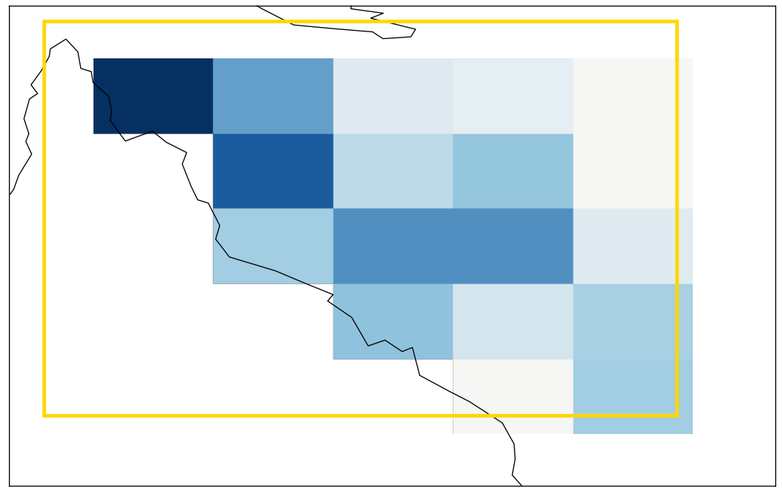
Actual DHW - Step 90

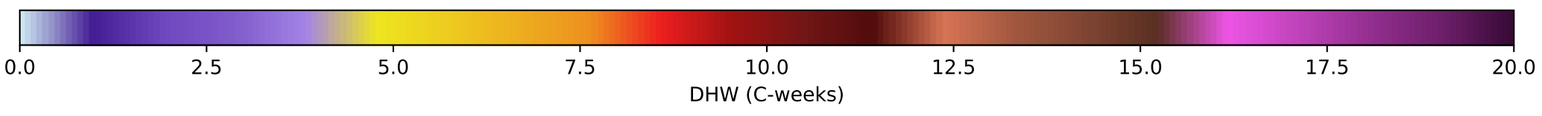


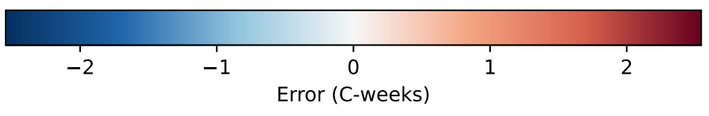
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.5]

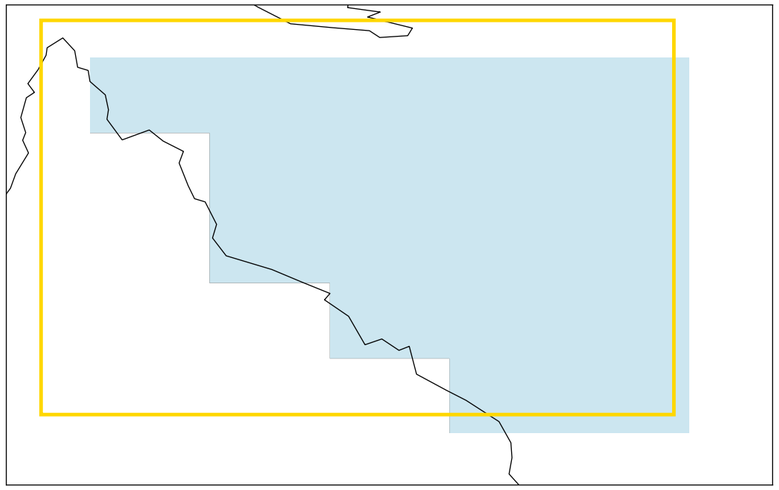




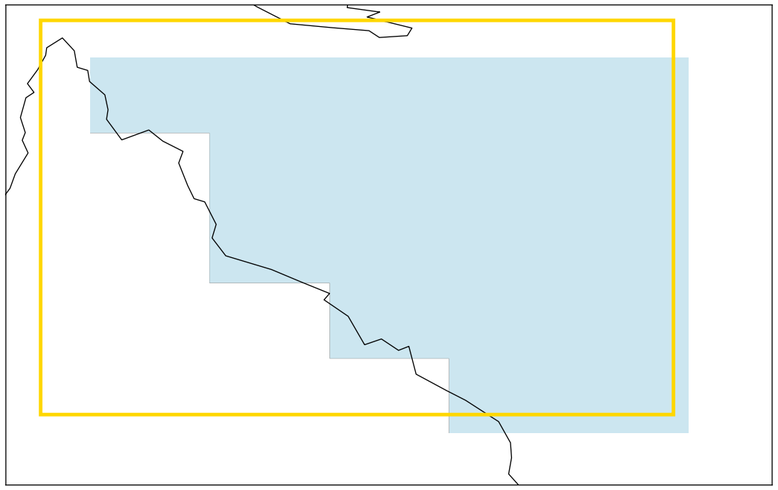


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

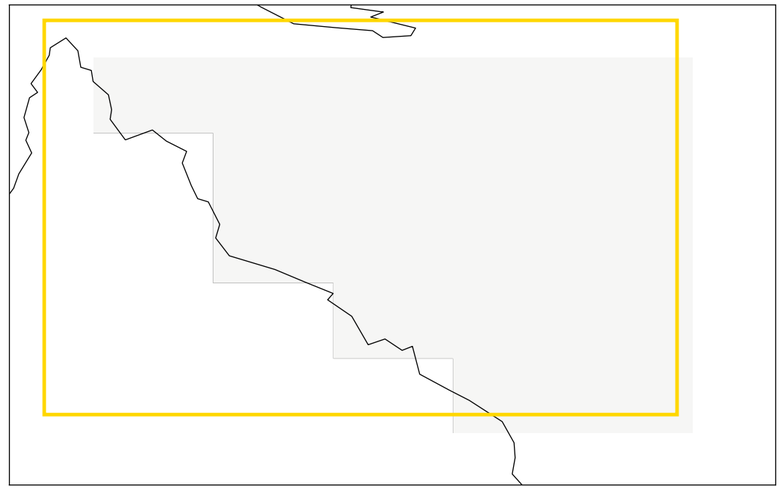
Actual DHW - Step 1



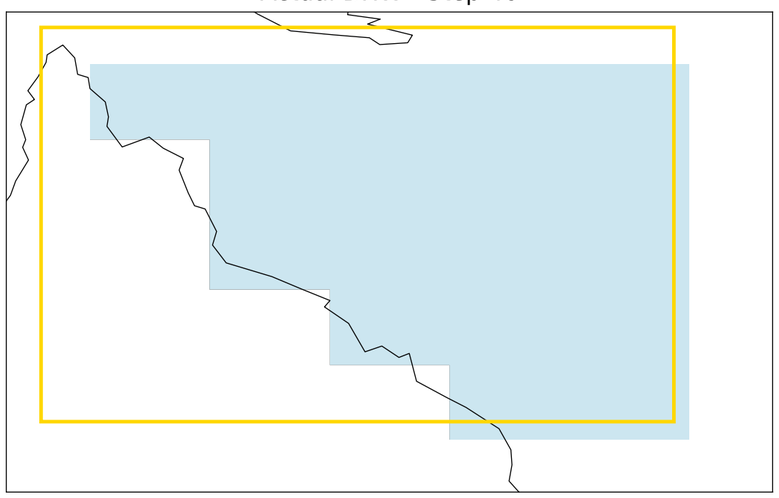
Predicted DHW - Step 1



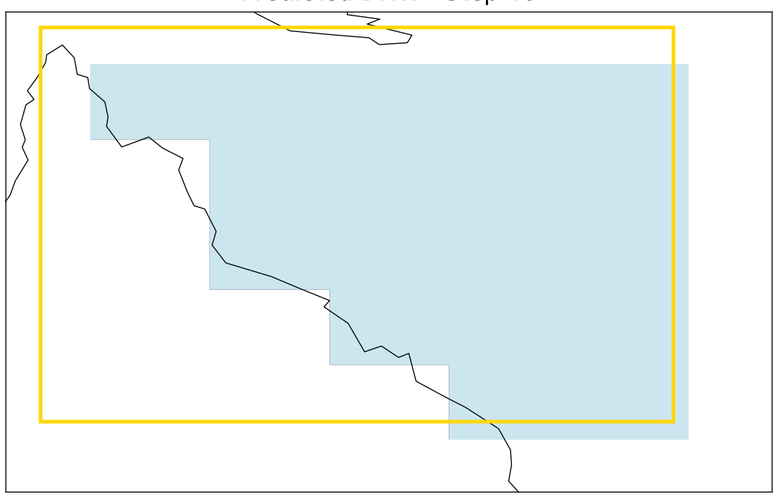
Error (Pred - Actual) - Step 1 [scale ±8]



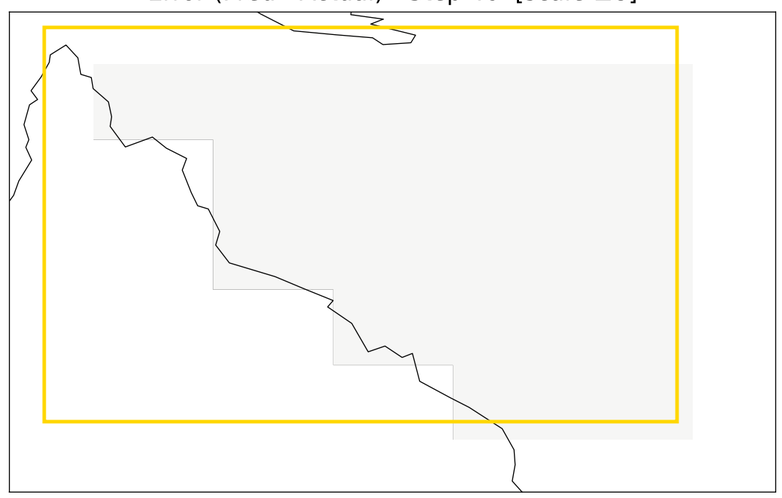
Actual DHW - Step 46



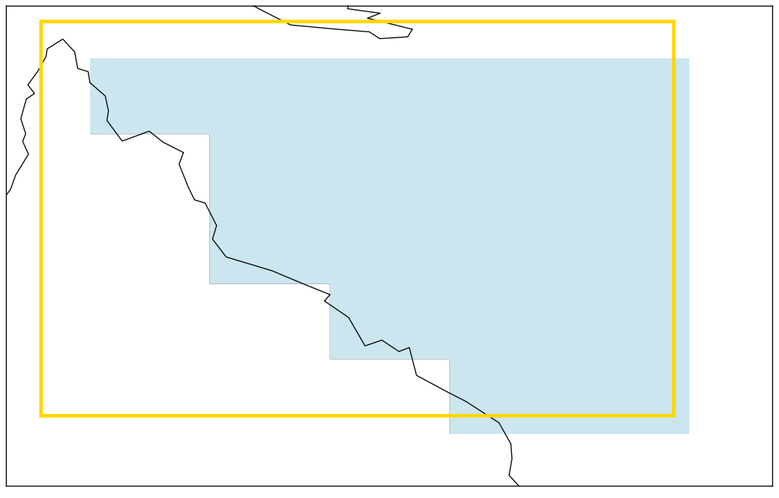
Predicted DHW - Step 46



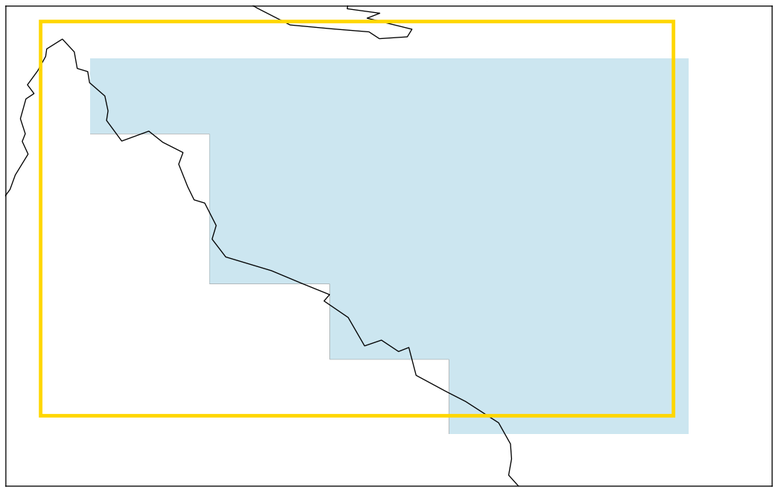
Error (Pred - Actual) - Step 46 [scale ±8]



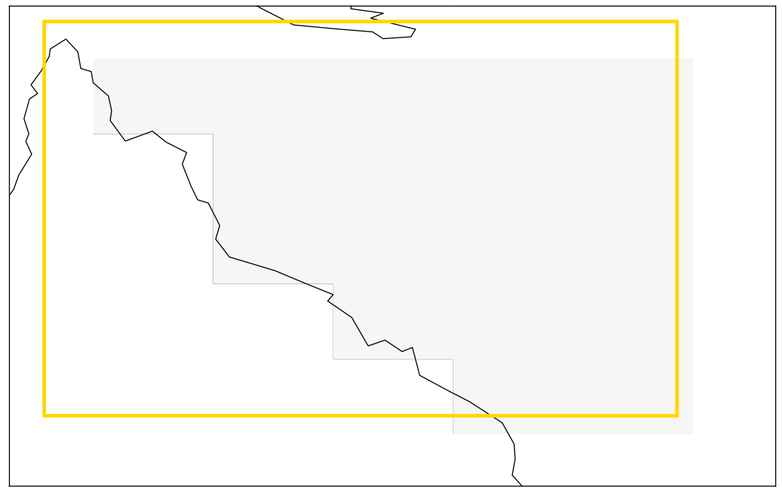
Actual DHW - Step 90

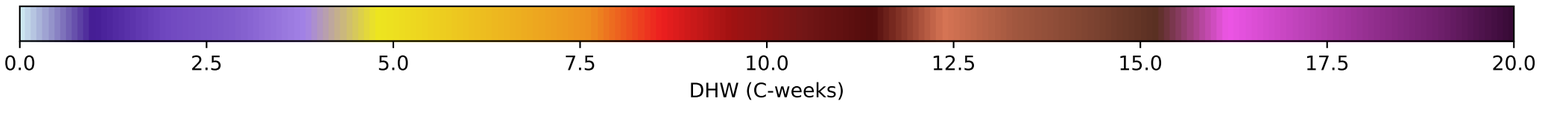


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

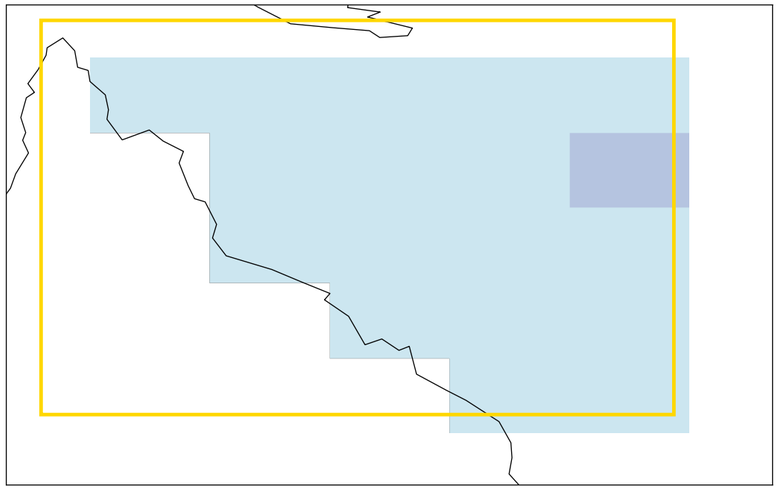




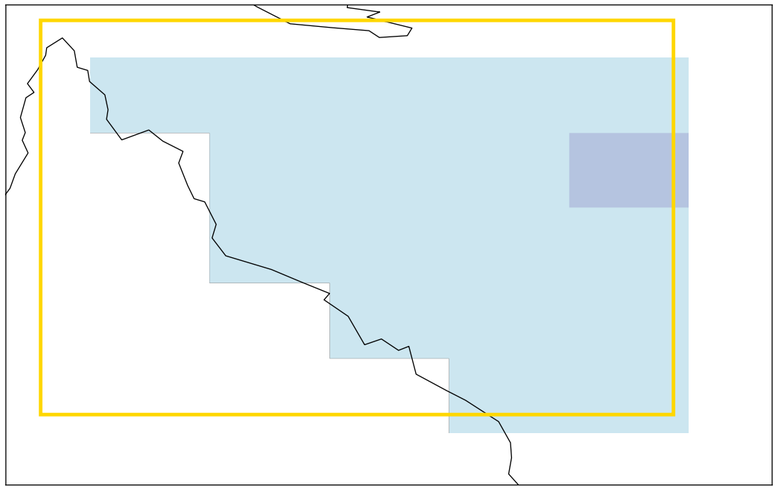


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

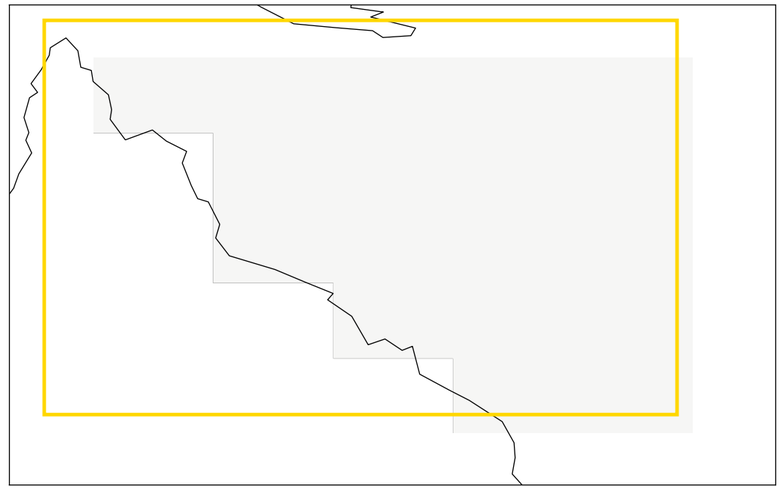
Actual DHW - Step 1



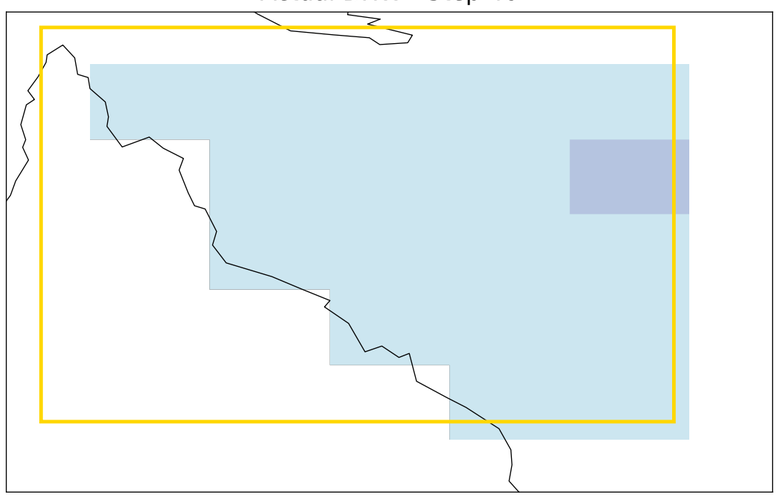
Predicted DHW - Step 1



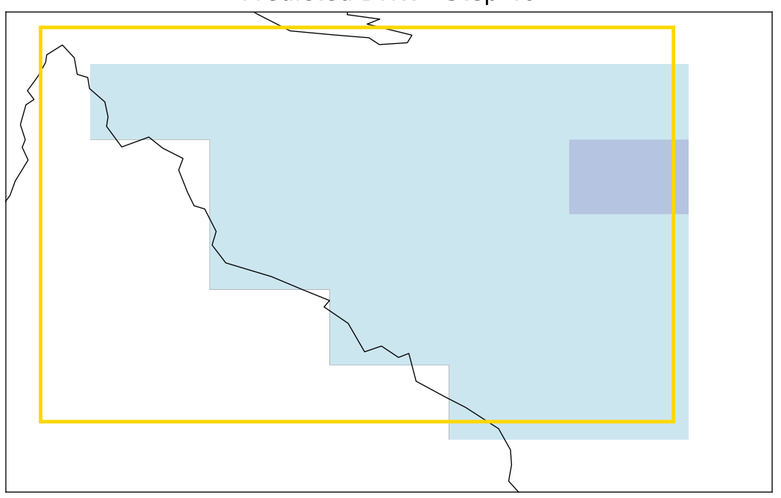
Error (Pred - Actual) - Step 1 [scale ±8]



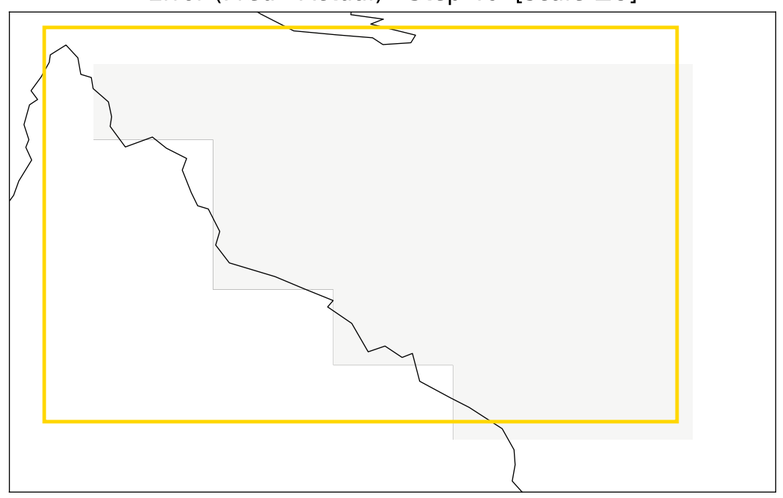
Actual DHW - Step 46



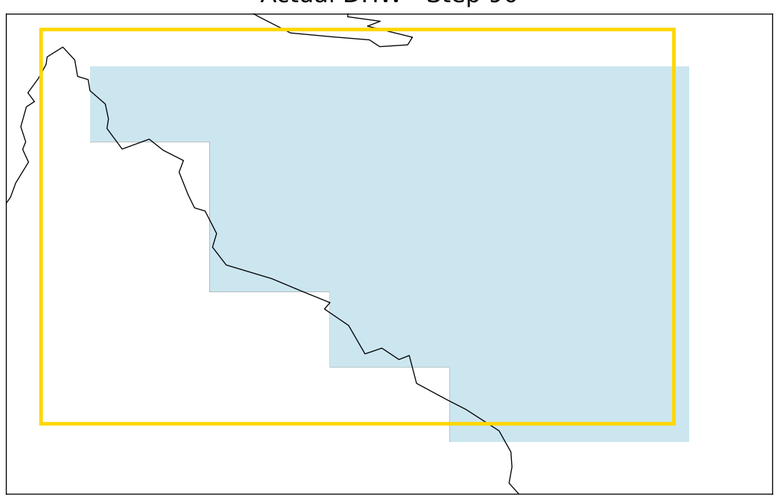
Predicted DHW - Step 46



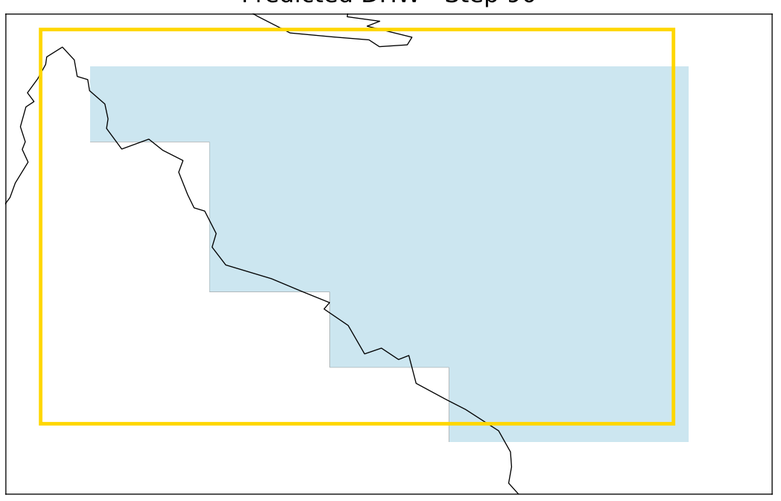
Error (Pred - Actual) - Step 46 [scale ±8]



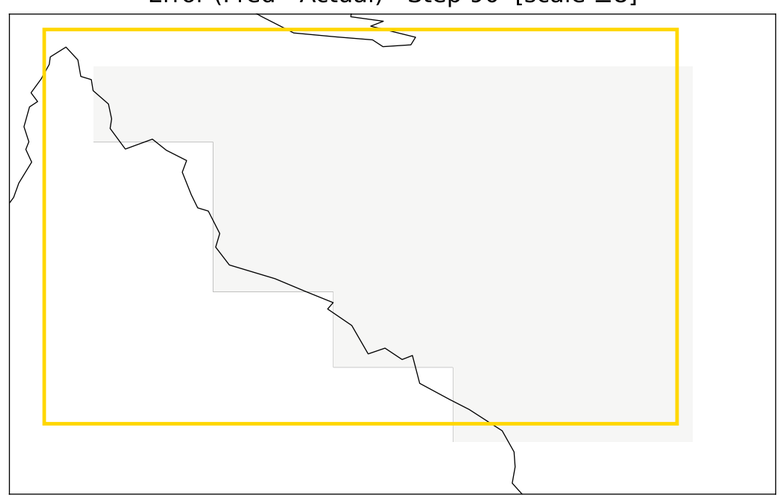
Actual DHW - Step 90

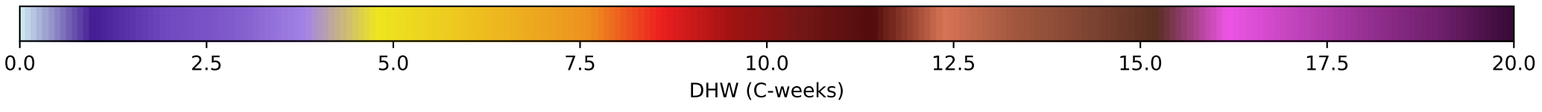


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

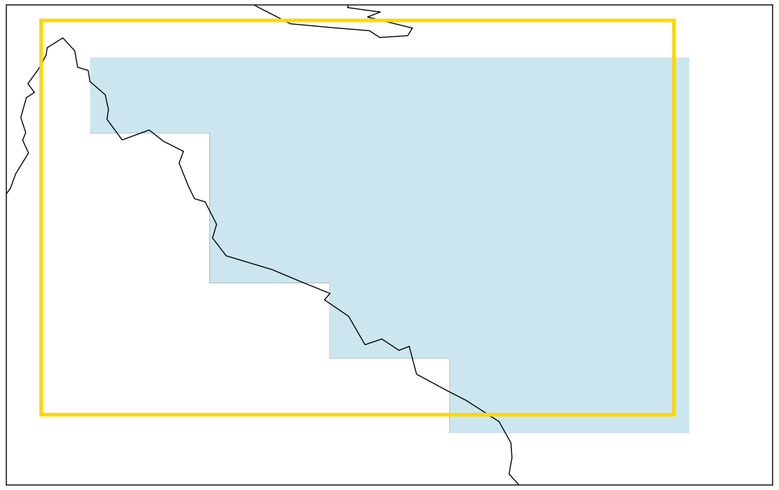




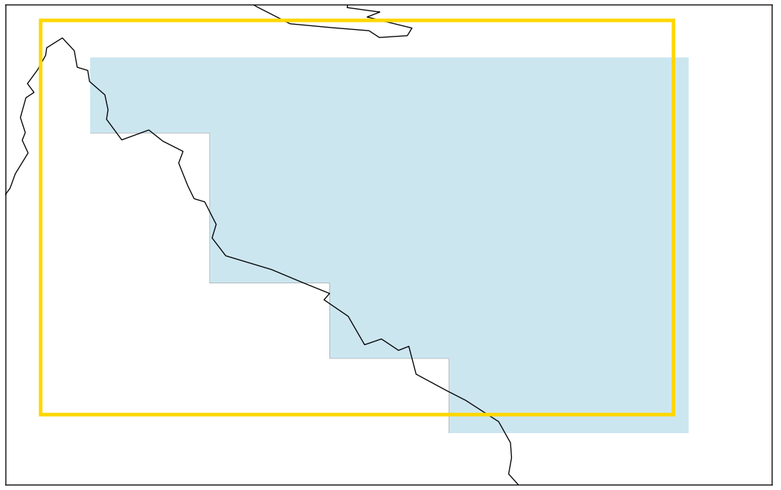


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

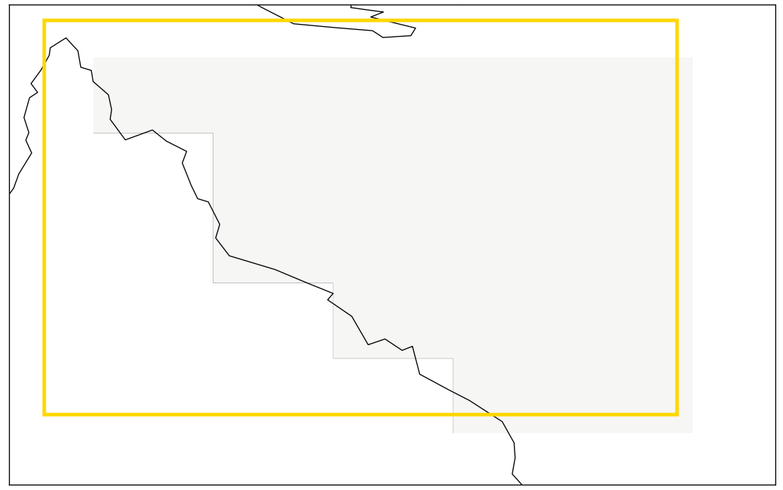
Actual DHW - Step 1



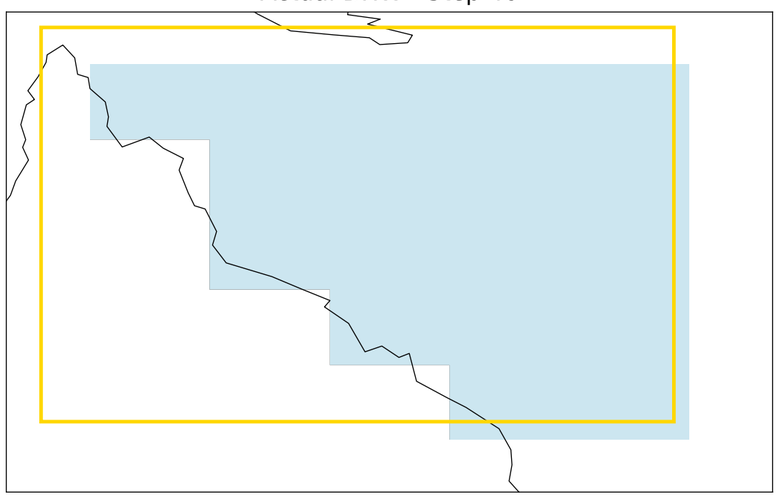
Predicted DHW - Step 1



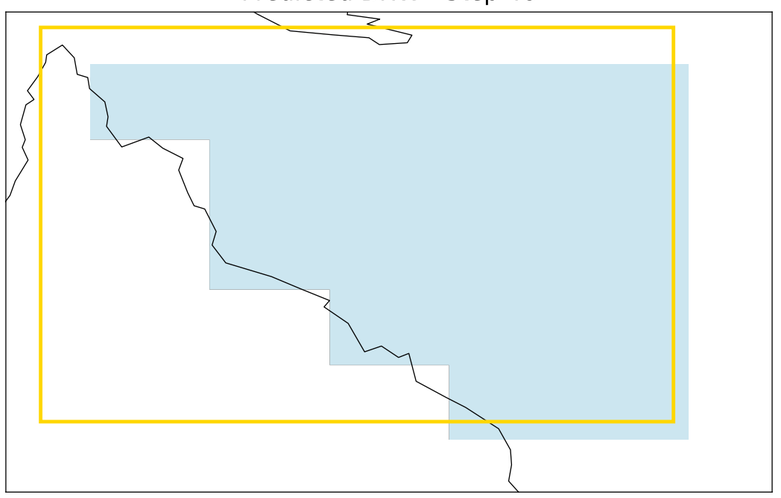
Error (Pred - Actual) - Step 1 [scale ±8]



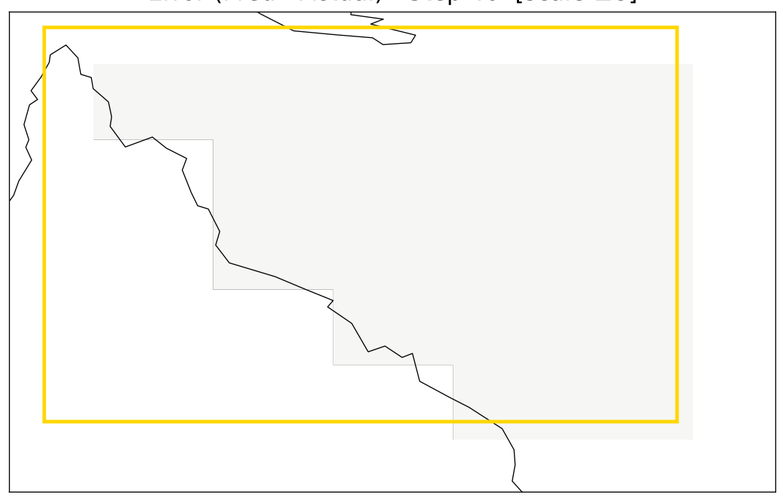
Actual DHW - Step 46



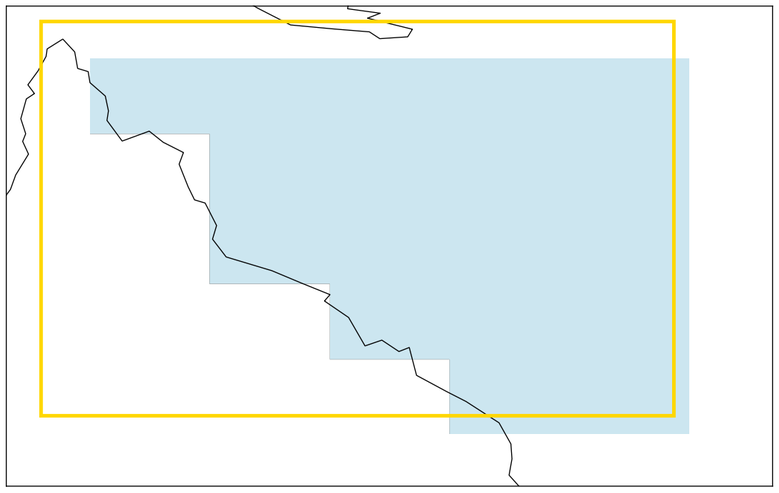
Predicted DHW - Step 46



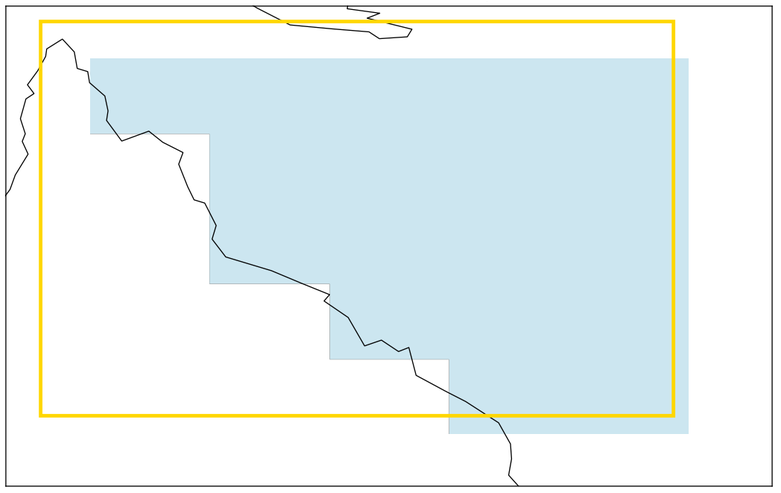
Error (Pred - Actual) - Step 46 [scale ±8]



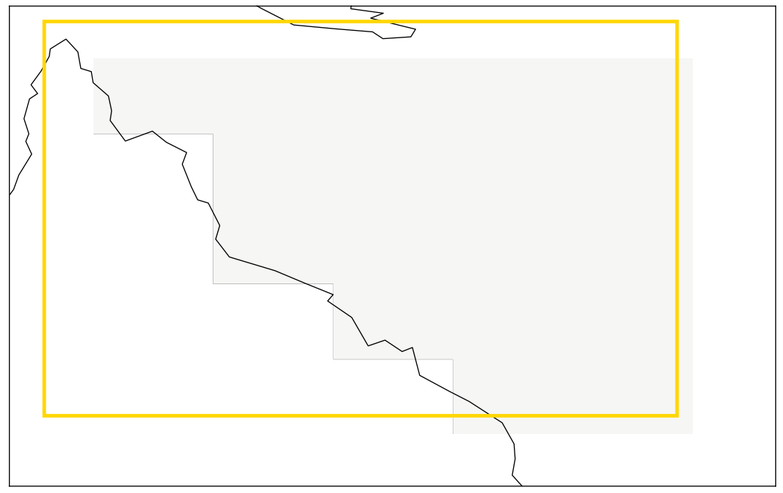
Actual DHW - Step 90

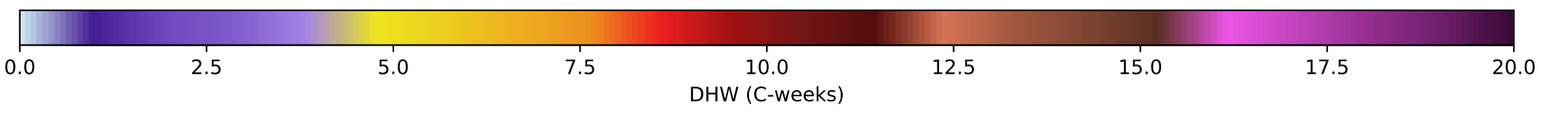


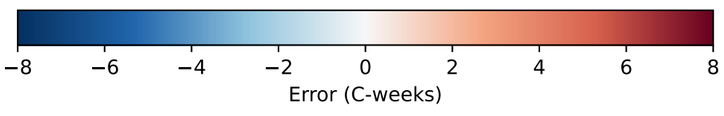
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

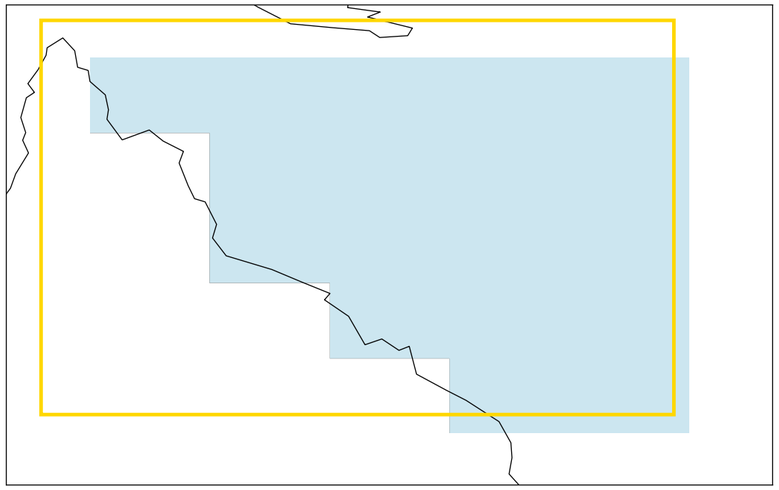




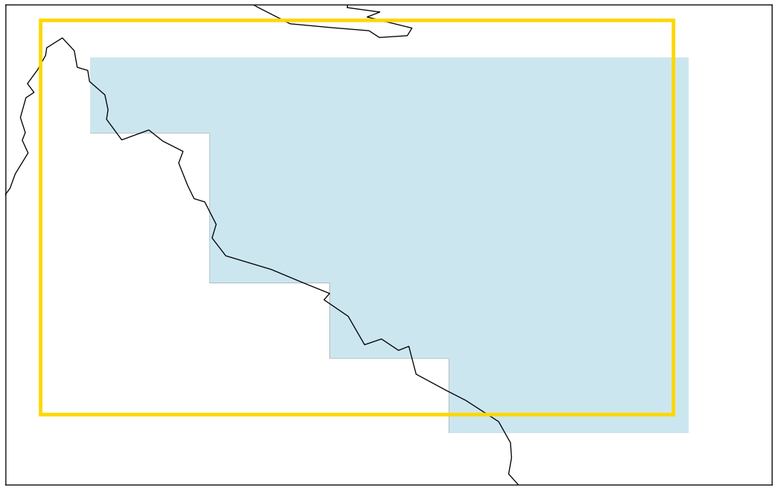


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

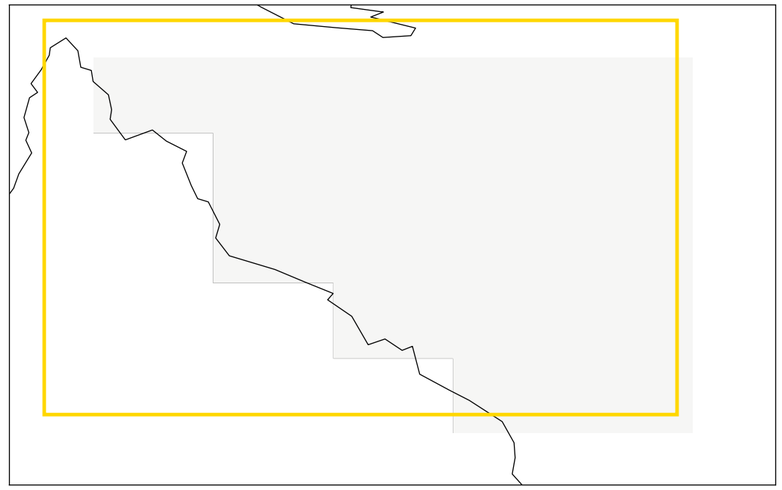
Actual DHW - Step 1



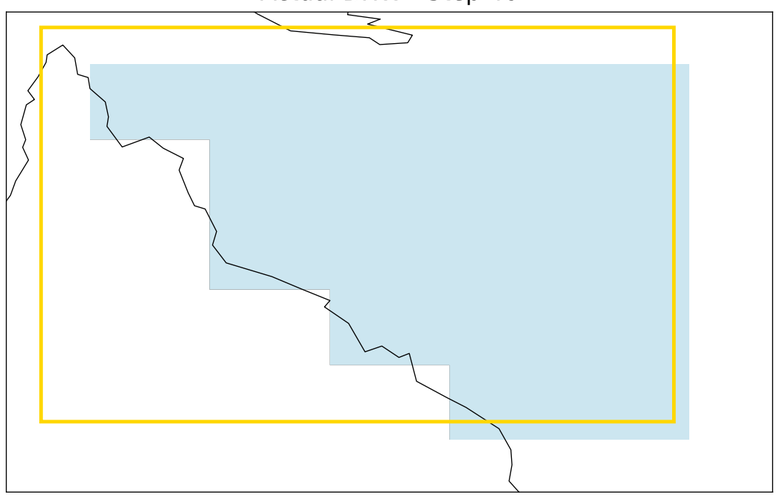
Predicted DHW - Step 1



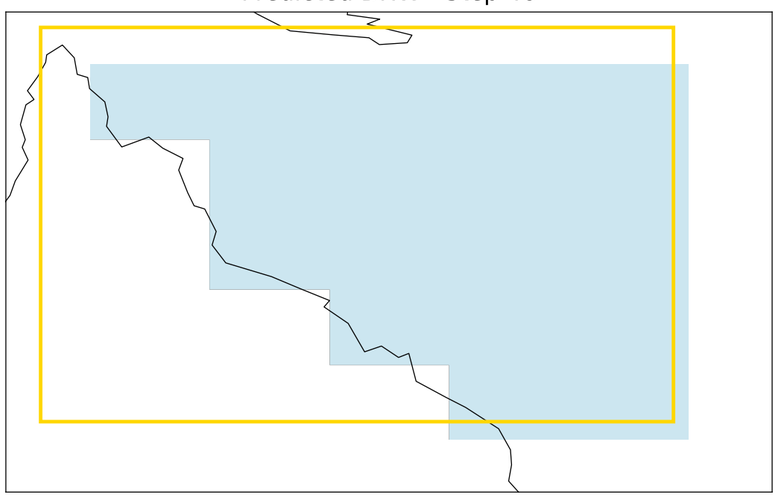
Error (Pred - Actual) - Step 1 [scale ±8]



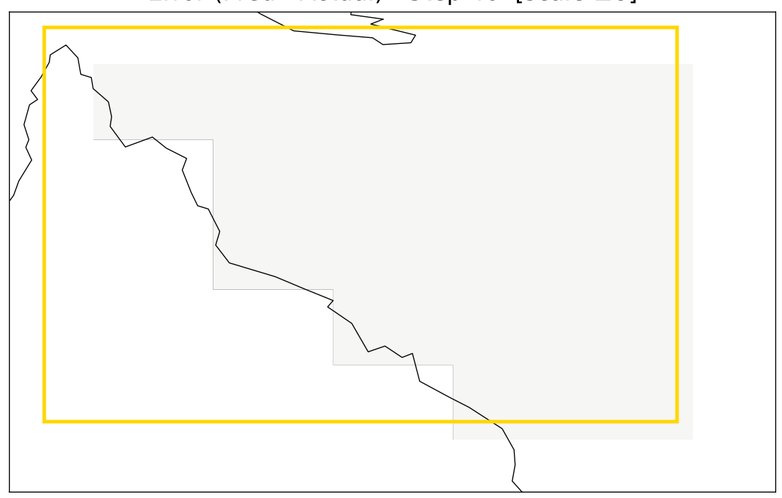
Actual DHW - Step 46



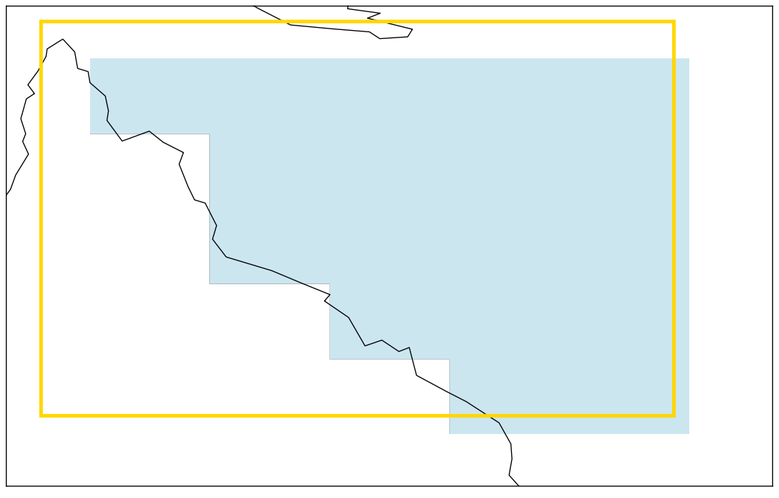
Predicted DHW - Step 46



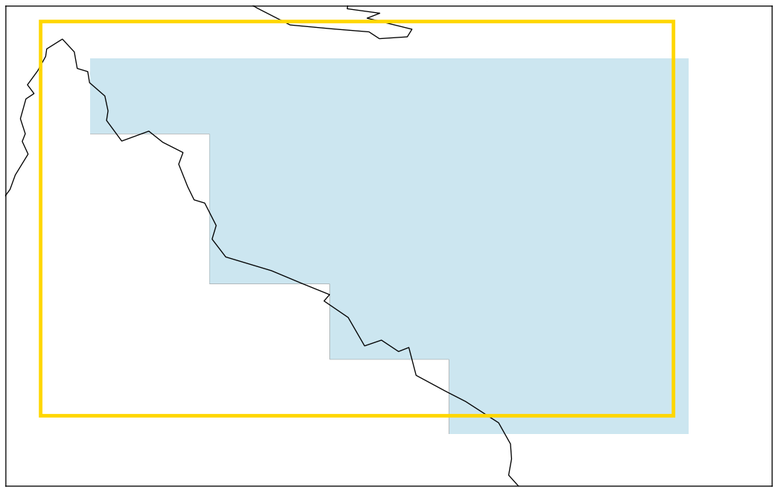
Error (Pred - Actual) - Step 46 [scale ±8]



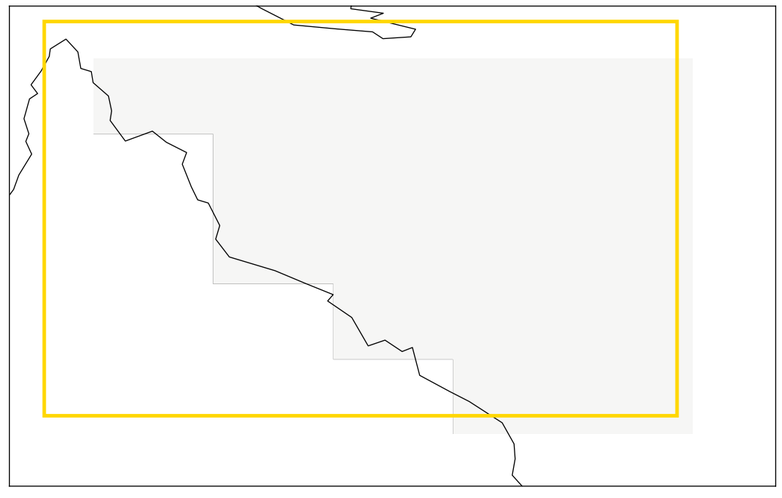
Actual DHW - Step 90

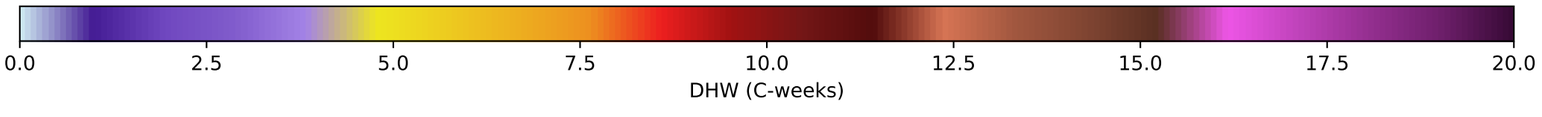


Predicted DHW - Step 90



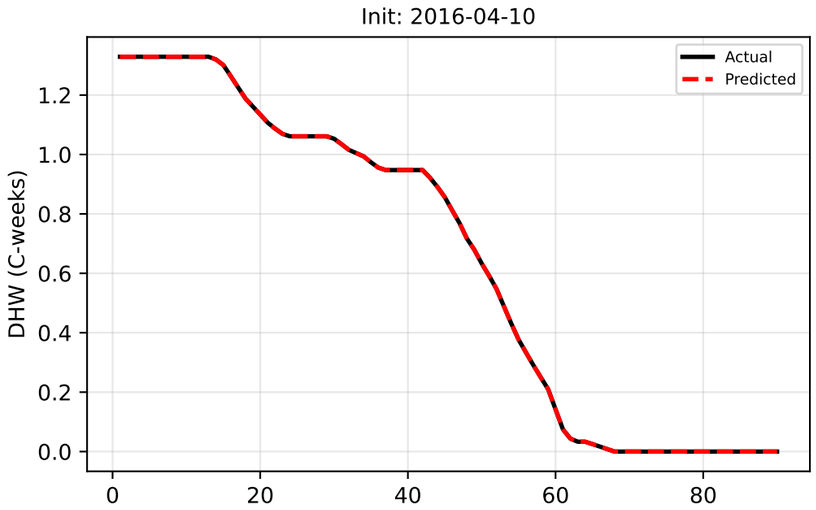
Error (Pred - Actual) - Step 90 [scale ±8]

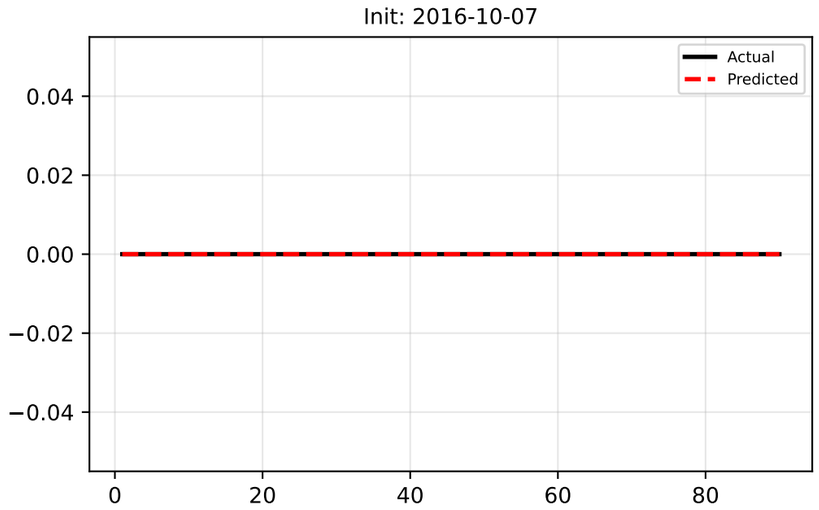


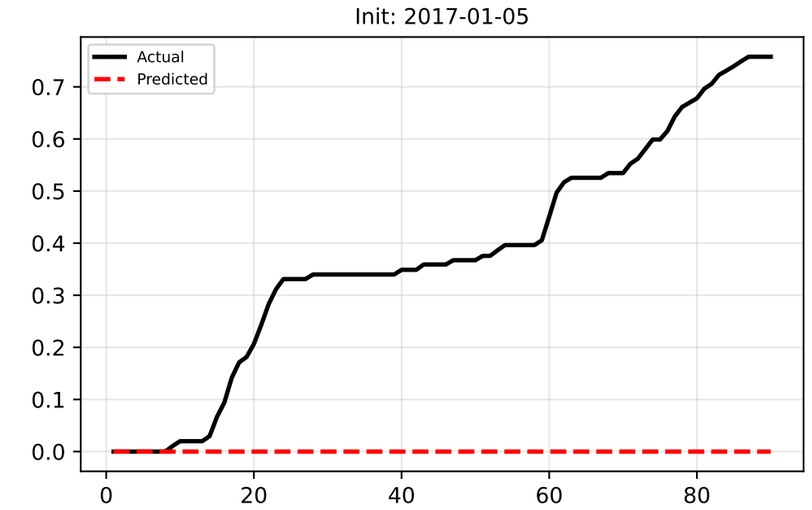


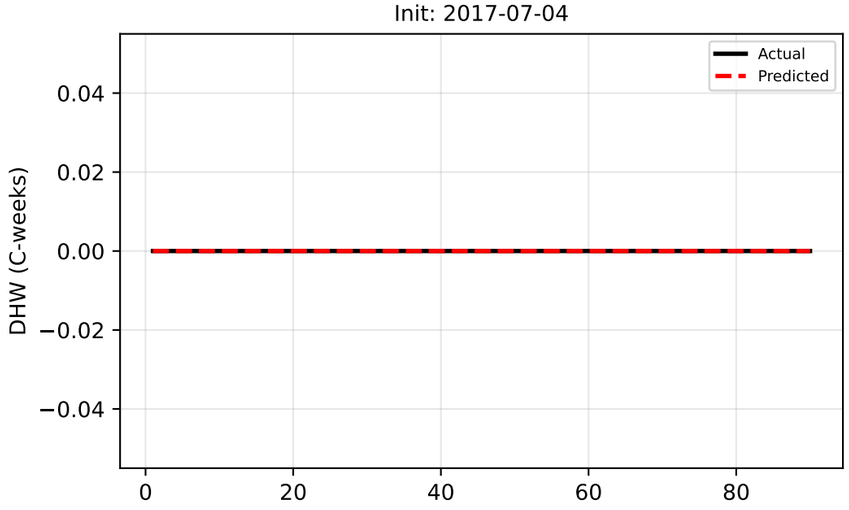


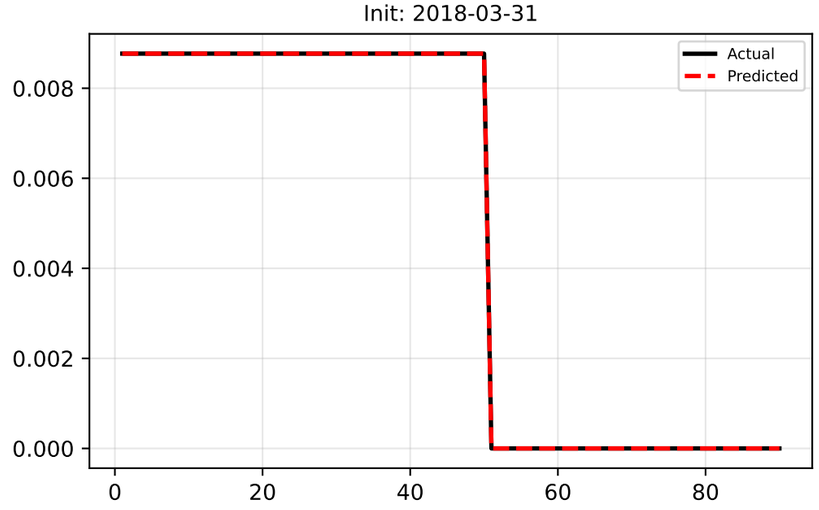
Global-Mean DHW - Val - GBR (C-weeks)

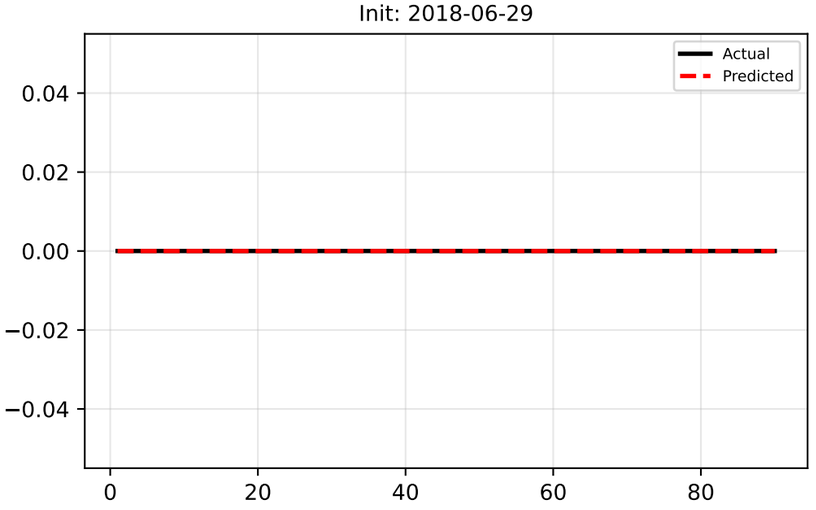


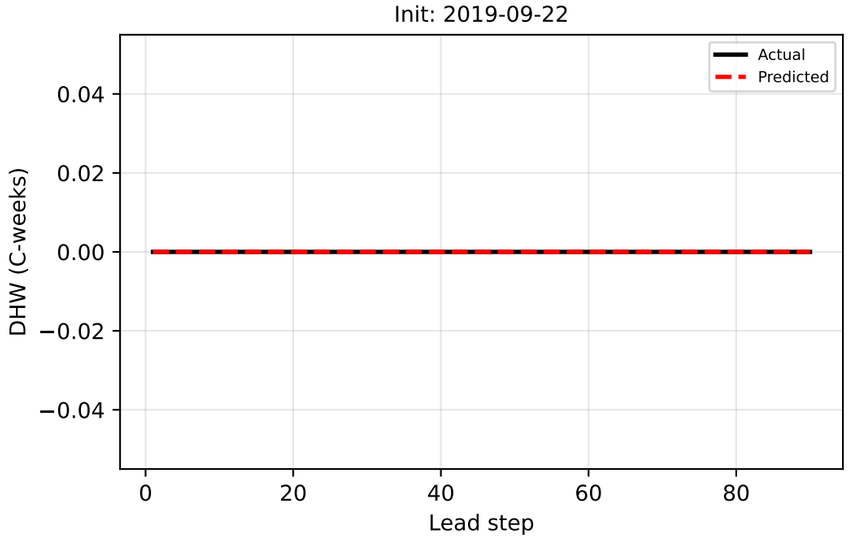


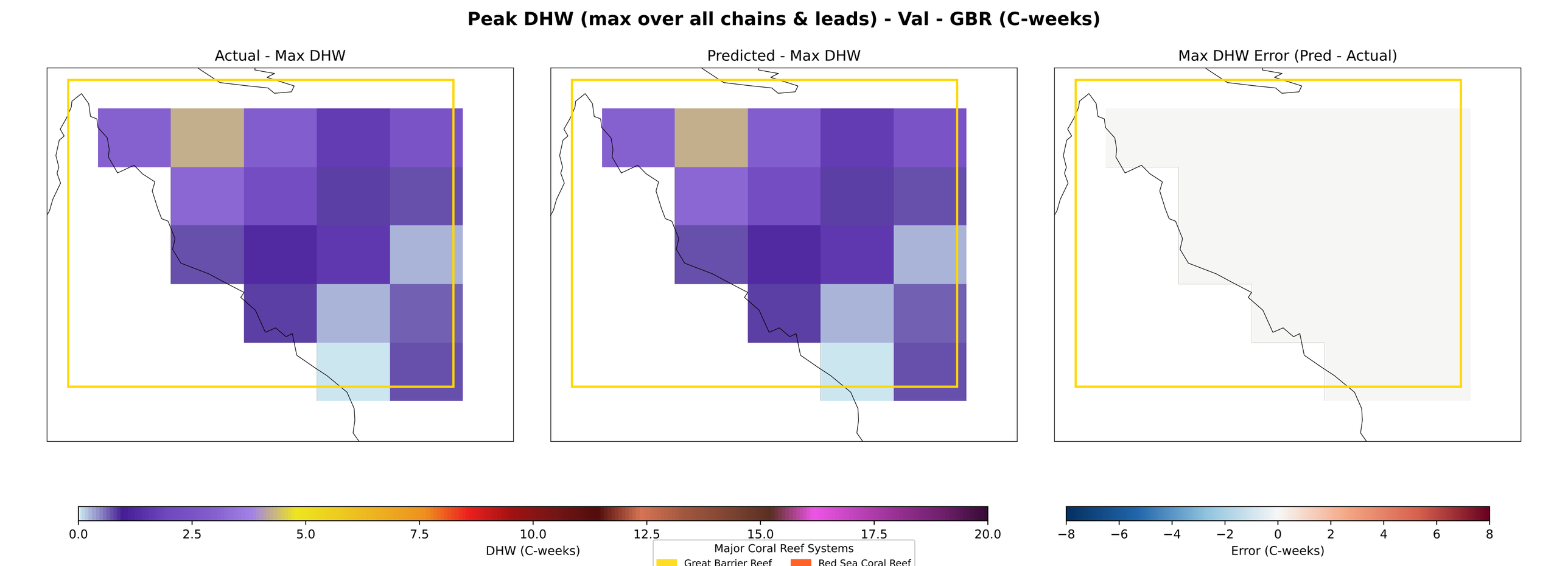


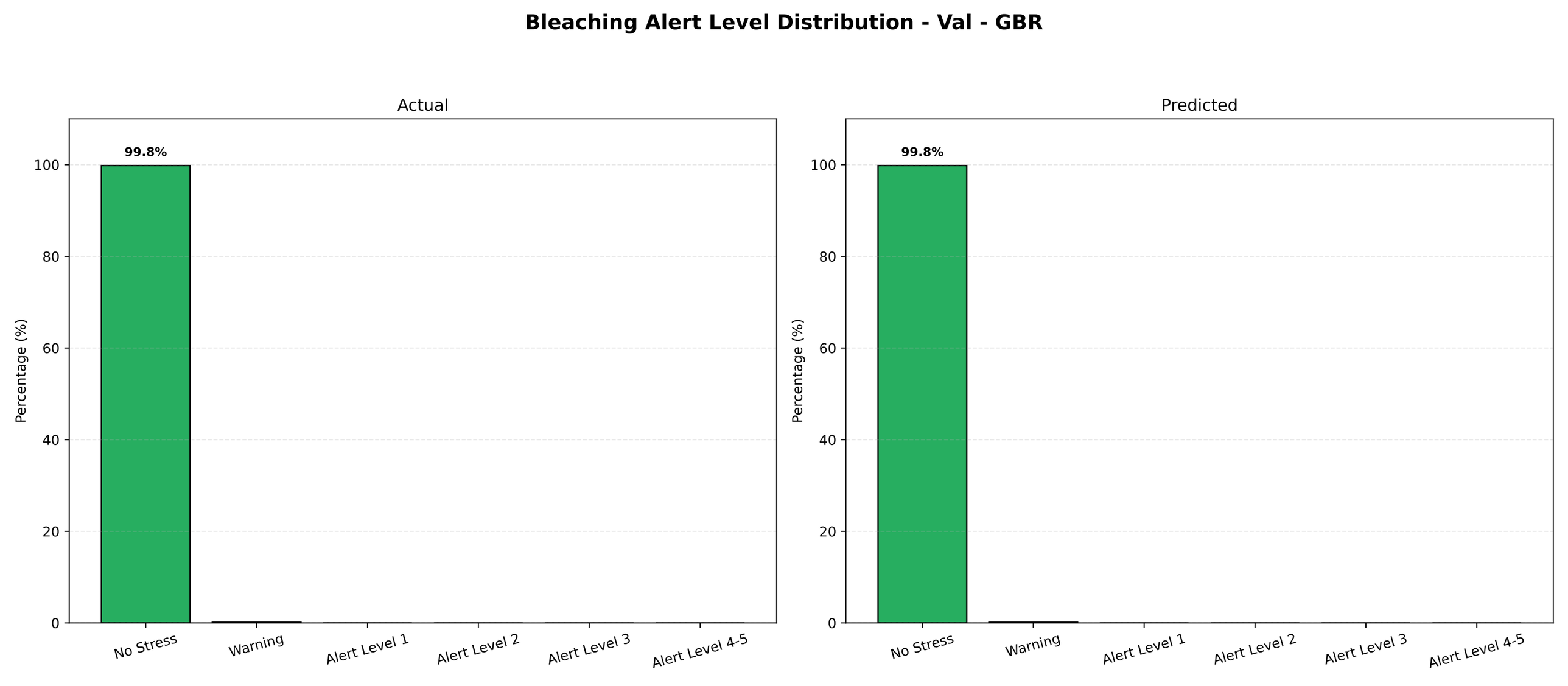






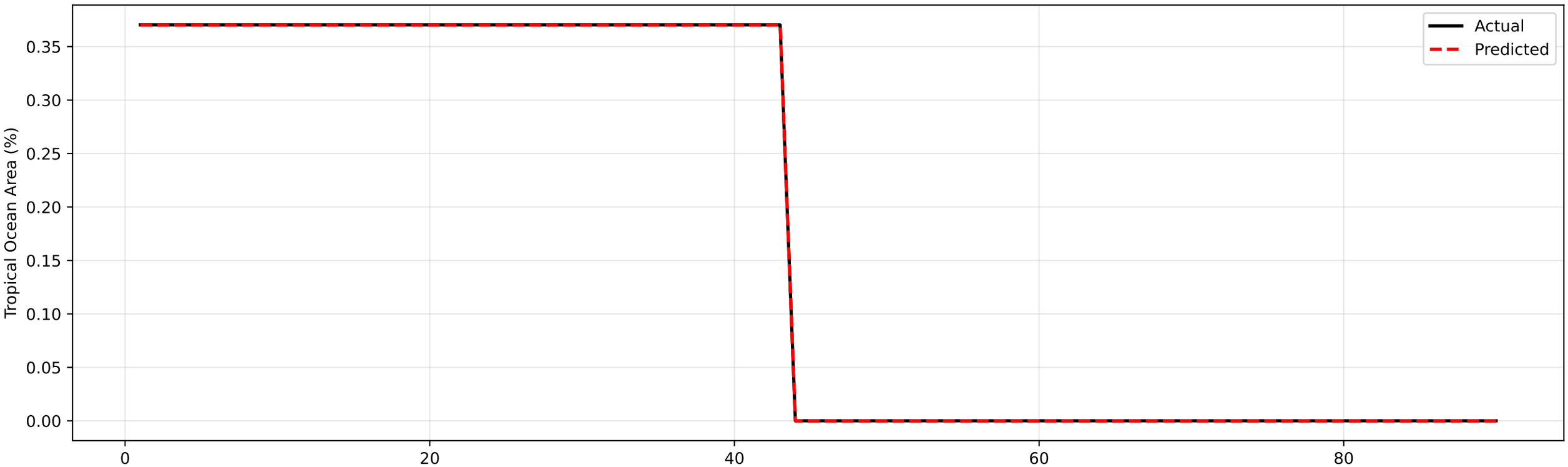



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

